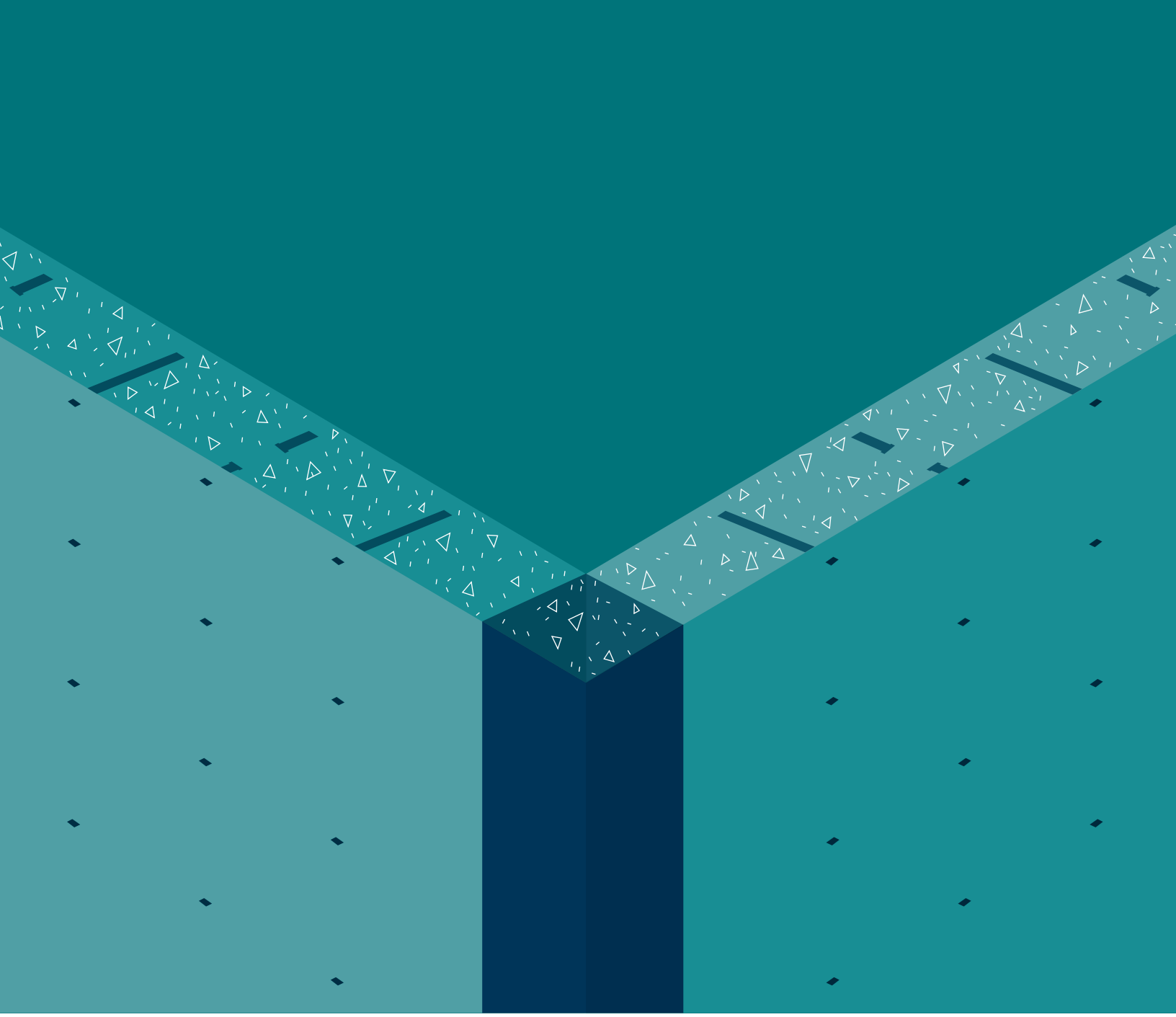


Design Guide 38

SpeedCore Systems for Steel Structures

Smarter. Stronger. Steel.



Design Guide 38

SpeedCore Systems for Steel Structures

Amit H. Varma, PhD
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Soheil Shafaei, PhD
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American Institute of Steel Construction

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by

American Institute of Steel Construction

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Preface

This Design Guide is intended to facilitate the design of composite plate shear walls—concrete filled (C-PSW/CF) and coupled composite plate shear walls—concrete filled (CC-PSW/CF), also referred to as uncoupled and coupled SpeedCore systems, for shear walls and elevator core walls in building structures. This Guide is to be used in conjunction with the 2022 AISC *Specification for Structural Steel Buildings* and the 2022 AISC *Seismic Provisions for Structural Steel Buildings*. The Design Guide discusses the behavior, design, and detailing of uncoupled and coupled SpeedCore systems for wind (Chapter 2), seismic (Chapter 3 for uncoupled and Chapter 4 for coupled), and fire (Chapter 6) loading combinations.

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Chapter 1
Composite Plate Shear Walls—Concrete Filled (SpeedCore Systems)

1.1 INTRODUCTION

Composite plate shear walls—concrete filled (C-PSW/CF), hereafter referred to as SpeedCore systems, are the subject matter of this Design Guide. SpeedCore systems are being considered and designed as an alternative to conventional reinforced concrete (RC) shear walls and core wall structures. Similar to RC walls, composite walls provide lateral load resistance to the building structure. In nonseismic design, composite wall systems provide the stiffness and strength needed to resist lateral forces resulting from wind loading. In seismic design, they provide the stiffness, strength, and deformation capacity needed to serve as the primary lateral force-resisting system.

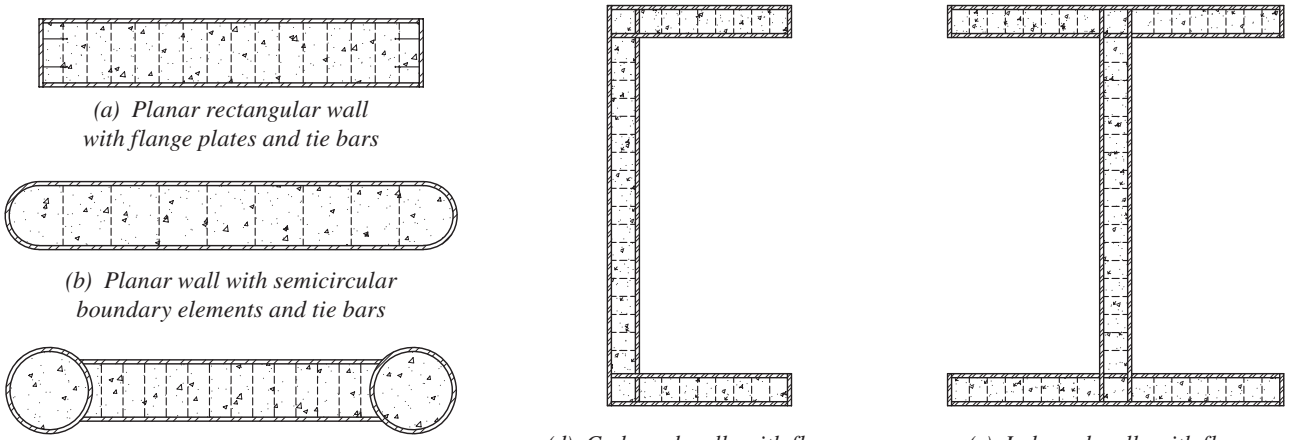
Similar to RC walls, SpeedCore systems may be used as the elevator core structure in high-rise buildings or as individual shear walls in low-to-midrise buildings. Two versions of SpeedCore systems are possible: uncoupled and coupled. For a given building structure, it is possible to have an uncoupled system in one direction and a coupled system in the orthogonal direction.

SpeedCore walls can be planar, C-shaped, or I-shaped, and may consist of linear wall segments connected together. Each linear wall segment consists of two steel web plates along the length of the segment and flange (or closure) plates, boundary elements, or other linear segments at the ends. For example, Figure 1-1 shows several potential SpeedCore walls. Boundary elements for planar walls can be semicircular or circular concrete-filled steel hollow structural sections (HSS). For C-shaped or I-shaped walls, the individual linear segments can be referred to as flange walls or web walls, depending on the direction of lateral loading.

Figure 1-2 shows the structural elements of a planar SpeedCore linear wall segment. As shown, the web plates define the length, and the concrete thickness plus the web plate thicknesses define the overall thickness of the wall. The web plates are connected to each other using regularly spaced tie bars in the horizontal and vertical directions. The steel web and flange plates can also have steel headed stud anchors (located on the internal surface) to anchor them to the concrete infill. The concrete infill can be normal weight concrete or self-consolidating concrete as needed.

Depending on the design, the flange plates can be a bit wider than the wall thickness (as shown in Figure 1-2) or fit in between the web plates with width equal to the concrete thickness. In nonseismic design, it is possible to replace the flange plates with I-shaped or C-shaped sections that fit within the web plates and serve as the boundary elements.

The coupling beams in the coupled version of SpeedCore can also be filled composite sections; for example, rectangular built-up box sections filled with concrete, as shown in Figure 1-3, may be used. Another possibility is to use steel coupling beams in nonseismic design.



(a) Planar rectangular wall with flange plates and tie bars

(b) Planar wall with semicircular boundary elements and tie bars

(c) Planar wall with circular boundary elements and tie bars

(d) C-shaped walls with flange (closure) plates and tie bars

(e) I-shaped walls with flange (closure) plates and tie bars

Fig. 1-1. SpeedCore system with boundary elements or flange (closure) plates.

It is important to note that there are no additional reinforcing bars needed in SpeedCore systems. The steel plates provide all the reinforcement needed to resist axial, flexural, and in-plane shear forces. The steel plates, tie bars, and steel headed stud anchors together form steel modules that can be prefabricated in the shop and shipped to the field for assembly and erection. The stiffness and stability of steel modules (before concrete filling) is governed by the diameter and spacing of the tie bars, and the slenderness of steel plates (after concrete filling) is governed by the maximum spacing of the tie bars and steel headed stud anchors (if used). After assembly and erection, the modular steel structure serves as falsework for construction activities and as formwork for placing the concrete infill. This enables the erection and construction activities for the gravity framing system to continue in parallel with the assembly and construction of the core and shear wall structure. The erection tolerances for the core and shear wall structure are comparable and compatible with those for the steel gravity framing system.

Commercial interest in the SpeedCore system stems from these potential advantages of modularity, reduction in construction schedule, and improvement of the overall project economy. The reduction in construction schedule results from (1) elimination of rebar cages and related activity, (2) elimination of falsework and formwork for the construction of the core walls, (3) simultaneous construction of the core walls and the gravity system, and (4) reduction in tolerance issues between the gravity system and core walls due to misaligned embedment plates. Reduction in construction schedule of about 30 to 40% can have a significant impact on the overall project economy.

For example, the Rainier Square project in downtown Seattle has used the SpeedCore system to achieve a 40% reduction in the construction schedule. The Rainier Square project is an extraordinary building and construction achievement that is discussed in more detail in the following section. Other projects, such as 200 Park Avenue in San Jose, California, are being constructed using the SpeedCore system to actualize construction schedule contraction and optimize overall project economy.

1.2 PRACTICAL APPLICATIONS

The Rainier Square tower is a 58-story office plus residential use building structure with seven levels of below grade parking. It totals 1.4 million ft^2, including 722,000 ft^2 of office space. A rendering of the elevation view of the project is shown in Figure 1-4. The structural floor plans for the lower, mid, and higher stories are shown in Figure 1-5(a), (b), and (c), respectively. In the structural floor plans, the core wall structures (shown in red shading) are SpeedCore walls. Figure 1-6(a) and (b) show photographs of the coupled core wall system during construction.

The composite wall thicknesses varied from 45 in. at the base to about 21 in. at the top. The tie bars were 1 in. in diameter and located at 12 in. spacing in both the horizontal and vertical directions throughout. No steel headed stud anchors were used in this design. The typical steel plate thickness was ½ in.

For this structure, the original RC construction schedule estimated topping out in 21 months after steel arrival on site. Switching to the SpeedCore system reduced the construction schedule to 12 months, that is, a 43% reduction in schedule for this first of its kind mega project. It is evident that the SpeedCore system can indeed deliver on its initial promise.

Figure 1-7 shows a 3D rendering and structural floor plan of one of the levels of the 200 Park Avenue project being constructed in San Jose, California. This is a 19-story structure located in a high seismic region. The lateral force-resisting

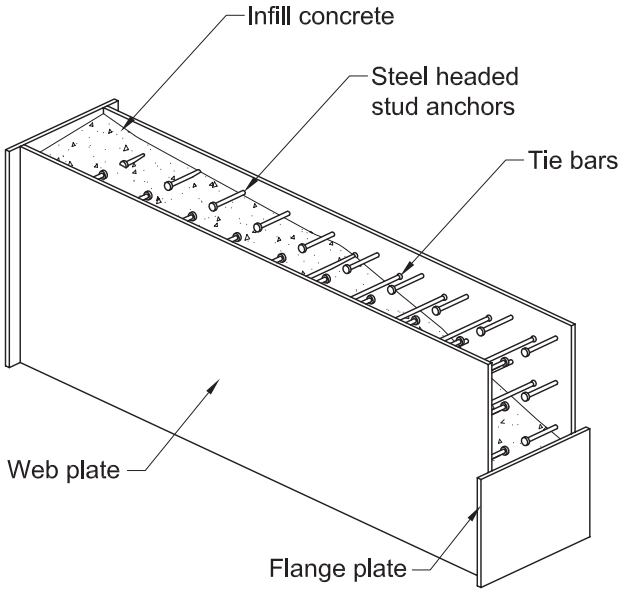


Fig. 1-2. A typical SpeedCore wall and its components (Shafaei et al., 2021b).

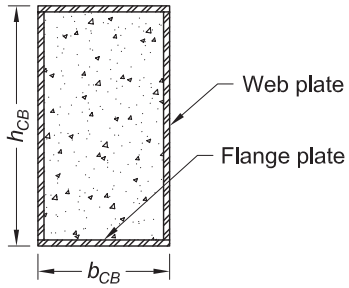
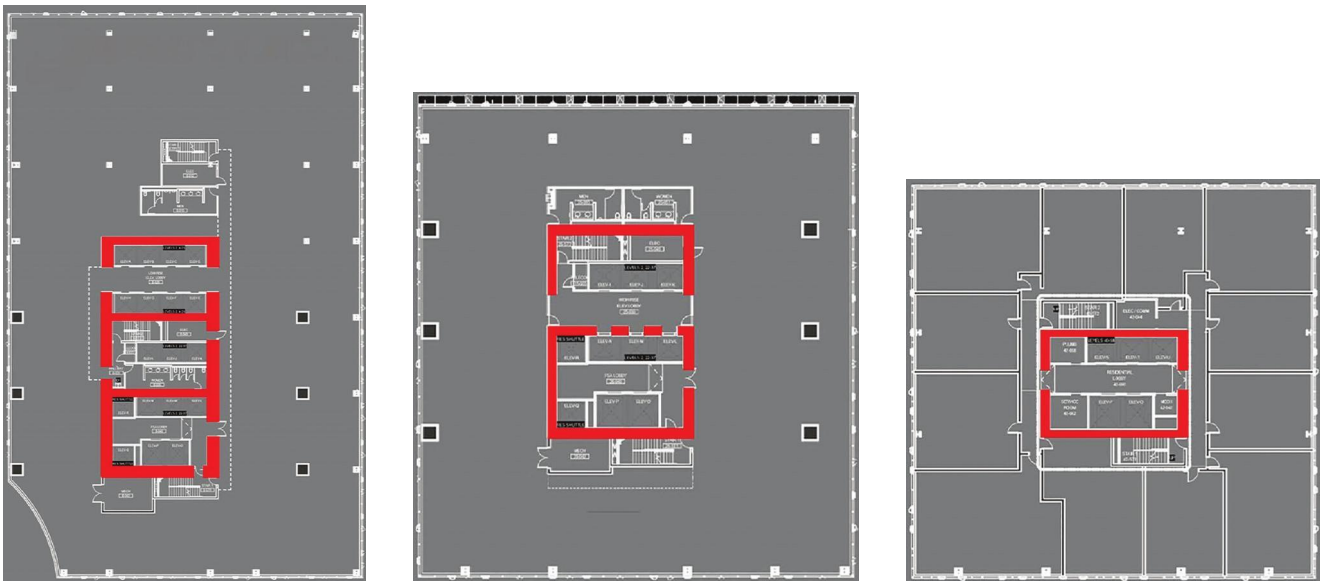


Fig. 1-3. Coupling beam cross section.

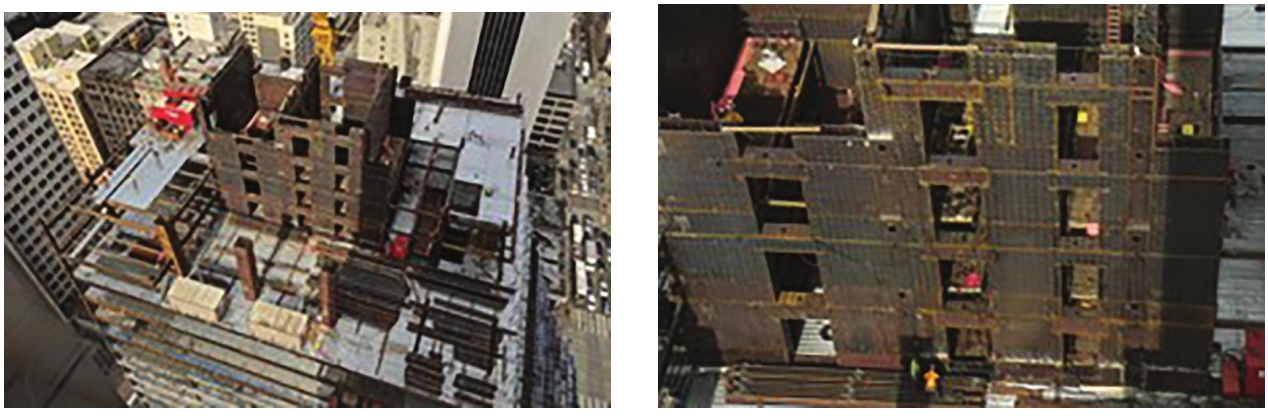


Fig. 1-4. 3D rendering view of Rainier Square (image courtesy of Wright Runstadt & Co.).



(a) Typical low-rise office *(b) Typical mid-rise office* *(c) Typical high-rise residential*

Fig. 1-5. Structural floor plans highlighting SpeedCore system in Rainier Square (image courtesy of Magnusson Klemencic Associates).



(a) SpeedCore wall core *(b) Close-up view of SpeedCore wall core*

Fig. 1-6. Photographs of coupled SpeedCore system under construction in Rainier Square (photos courtesy of Magnusson Klemencic Associates).

system is the coupled SpeedCore system shown in Figure 1-7(b). This building has 937,000 ft^2 of floor space at an average of 54,000 ft^2 per floor. The target completion date for this project is May 2023.

1.3 SAFETY-RELATED NUCLEAR STRUCTURES

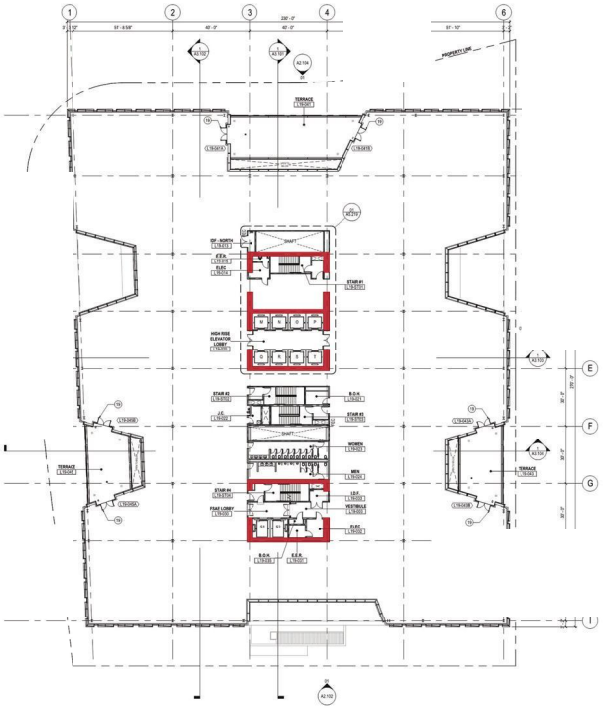
Concrete filled composite plate shear walls have also been used in the design of safety-related nuclear structures, where they are referred to as steel-plate composite or SC structures. However, there are significant differences between SC structures in nuclear design and SpeedCore in commercial building structures. The design of SC structures is governed by ANSI/AISC N690, *Specification for Safety-Related Steel Structures for Nuclear Facilities* (AISC, 2018), hereafter referred as the AISC *Nuclear Specification*. AISC Design Guide 32, *Design of Modular Steel Plate Composite (SC) Walls for Safety-Related Nuclear Facilities* (Bhardwaj and Varma, 2017), demonstrates the use of the AISC *Nuclear Specification* to design safety-related nuclear structures. Primary differences between safety-related nuclear facilities and commercial building structures are as follows:

1. Safety-related nuclear structures are labyrinthine in plan, and there are no clear gravity framing systems or lateral force-resisting systems. The entire structural plan is typically made of structural wall systems (no columns and no beams.)
2. The wall structures of safety-related nuclear structures are typically short, resulting in low height-to-length ratios (typically less than 2).
3. The large thickness of wall structures (typically greater than 36 in.) for safety-related nuclear applications may be governed by nonstructural considerations, such as radiation shielding, or by extreme events such as missile impact or pressure impulse loading.
4. The walls of safety-related nuclear structures are analyzed and designed for seismic plus accidental thermal or pressure loading combinations resulting in simultaneous application of three in-plane membrane forces, three membrane moments, and two out-of-plane shears.
5. Safety-related nuclear structures are designed to undergo the first onset of significant inelastic deformation at the safe shutdown earthquake level with no seismic response modification factors, R, or specific requirements for ductility.

Despite these differences, there are some elements of behavior and design that are applicable to SpeedCore walls



(a) 3D rendering



(b) Floor plan at one level

Fig. 1-7. 200 Park Avenue under construction (images courtesy of Gensler).

in commercial buildings. These include plate slenderness requirements for local buckling, axial compressive strength, and in-plane shear strength of composite walls. All these elements are included and discussed in this Design Guide as applicable. SC walls have been used extensively in the design and construction of AP1000® at Plant Vogtle in Georgia, United States. SC walls are being used for containment internal structures, such as the 1,100-ton CA20 module shown in Figure 1-8(a). This module is more than five stories tall and will house various plant components, including the used fuel storage area. SC wall modules were fabricated and assembled in a dedicated module assembly building, and Figure 1-8(b) shows a photograph of an SC submodule being assembled. SC walls were also used as the external shield building used to provide aircraft impact resistance (in addition to other functions) for AP1000. Figure 1-9 shows a photograph of the cylindrical shield building being assembled at Plant Vogtle.

1.4 BACKGROUND AND RESEARCH

Over the past few years, significant research has been conducted on the behavior, analysis, and design of composite plate shear walls. These include research projects focusing on:

1. Experimental and numerical evaluation of the cyclic in-plane shear behavior of composite walls (Varma et al., 2014; Seo et al., 2016; Ji et al., 2017; Booth et al., 2020).
2. Experimental and numerical evaluation of the local buckling behavior and axial compressive strength of composite walls (Zhang et al., 2014; Zhang et al., 2020).
3. Experimental and numerical evaluation of the stiffness and stability of empty steel modules during transportation, erection, and construction (Varma et al., 2019).
4. Experimental investigation of the cyclic behavior of planar composite walls with flange plates (Shafaei et al., 2021b).
5. Experimental investigation of the cyclic behavior of C-shaped composite walls (Kenarangi et al., 2021).
6. Development and benchmarking of numerical models for the cyclic behavior of composite walls (Alzeni and Bruneau, 2017; Shafaei et al., 2021a; Bruneau et al., 2019).
7. FEMA P-695, *Quantification of Building Seismic Performance* (FEMA, 2009), hereafter referred to as FEMA P-695, studies of the seismic behavior and design factors for coupled composite wall systems (Bruneau et al., 2019).



(a) Module erected at site *(b) Submodule wall in module assembly building*

Fig. 1-8. SC modules and submodules used for AP1000 construction at Plant Vogtle, Georgia (photos courtesy of Georgia Power).

8. FEMA P-695 studies of the seismic behavior and design factors for uncoupled composite wall systems (Agarwal et al., 2020; Kenarangi et al., 2021).
9. Structural fire engineering and fire-resistant design of composite walls (Anvari et al., 2020a, 2020b).

At the time of publication, research is still ongoing on the following topics:

10. Experimental evaluation of coupling beam-to-composite wall connections for seismic design.
11. Experimental evaluation of bolted splices and connections for composite walls for wind design.
12. Experimental evaluation of wall-to-foundation connections for seismic and wind design.
13. Experimental and numerical evaluation of composite floor-to-wall connections for fire design.
14. Experimental evaluation of T-shaped composite walls.

1.5 SPECIFICATIONS AND PROVISIONS

This Design Guide is based on (1) research results and findings available so far, (2) AISC design specifications and provisions, and (3) implementation of SpeedCore for building construction projects.

The design specification provisions relevant to this Design Guide are as follows:

1. Specifications for the design of uncoupled or coupled C-PSW/CF (SpeedCore) walls for nonseismic applications in the 2022 version of the AISC *Specification for Structural Steel Buildings* (AISC, 2022b), hereafter referred to as the AISC *Specification*. These provisions are in Chapter I of the 2022 AISC *Specification*.
2. Specifications for the design of C-PSW/CF (SpeedCore) walls for fire loading in the 2022 version of the AISC *Specification*. These provisions are in Appendix 4 of the 2022 AISC *Specification*.
3. Seismic design provisions for uncoupled C-PSW/CF (SpeedCore) wall systems in the 2022 version of the AISC *Seismic Provisions for Structural Steel Buildings* (AISC, 2022a), hereafter referred to as the AISC *Seismic Provisions*. These provisions are in Section H7 of the 2022 AISC *Seismic Provisions*.
4. Seismic design provisions for coupled C-PSW/CF (SpeedCore) wall systems in the 2022 version of the AISC *Seismic Provisions*. They are located in Section H8 of the 2022 AISC *Seismic Provisions*. These design provisions are referenced in the *NEHRP Recommended Seismic Provisions for New Buildings and*



Fig. 1-9. SC modules using in the AP1000 shield building at Plant Vogtle, Georgia (photo courtesy of Georgia Power).

Other Structures (FEMA, 2020). They are also referenced in the 2022 version of ASCE/SEI 7, *Minimum Design Loads and Associated Criteria for Buildings and Other Structures* (ASCE, 2022), hereafter referred to as ASCE/SEI 7.

1.6 DESIGN GUIDE OUTLINE

This Design Guide has been organized into five chapters:

- **Chapter 2** focuses on the nonseismic (wind) design of uncoupled and coupled SpeedCore wall systems.
- **Chapter 3** focuses on the seismic design of uncoupled SpeedCore wall systems.
- **Chapter 4** focuses on the seismic design of coupled SpeedCore wall systems.
- **Chapter 5** summarizes the results from the seismic performance evaluation of SpeedCore wall systems.
- **Chapter 6** focuses on the design of SpeedCore walls for fire resistance.

Each section follows the same general format:

1. Presentation of the design specifications and provisions for the system along with extended commentary, which includes discussions based on relevant research.
2. A detailed, step-by-step design procedure for the system along with flowcharts to assist with the design procedure.
3. Detailed design examples showcasing the use of the design specification and procedure. Each design example has been programmed in Mathcad (PTC, 2010, 2017) using the same variables for all the designs across the guide. These Mathcad files can be downloaded and used by engineers for future work at the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

Chapter 2
Wind Design of SpeedCore Coupled and Uncoupled Systems

2.1 DESIGN FOR NONSEISMIC LOADING COMBINATIONS

2.1.1 General

The SpeedCore system is used in building structures primarily to provide lateral force resistance. It may be used as the elevator core wall structure or as individual shear wall systems. It is important to note that the SpeedCore system may be uncoupled or coupled for lateral load resistance. In some cases, the SpeedCore system may be uncoupled in one direction and coupled in the orthogonal direction. The coupled SpeedCore system consists of composite (C-PSW/CF) walls connected by coupling beams, which may be composite concrete-filled box sections or steel beams, depending on the design.

As shown in Figure 2-1, SpeedCore walls may be planar, C-shaped, or I-shaped in cross section. In all of these cases, the wall cross section consists of one or more connected linear wall segments to make up the cross-sectional shape. Each linear wall segment consists of two steel web plates along its length and flange plates (or closure plates) at the ends. Planar walls can also be constructed using composite filled columns or steel I-shaped or C-shaped sections for boundary elements. For example, Alzeni and Bruneau (2017) have tested planar walls with semicircular or circular filled composite columns for boundary elements.

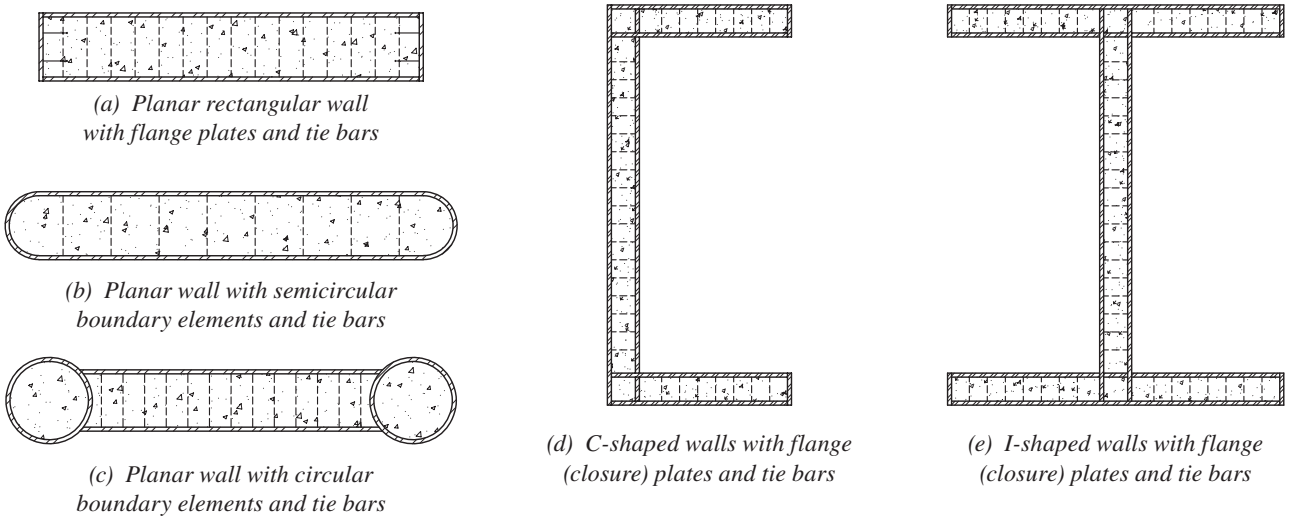
The behavior and design of SpeedCore systems for nonseismic, wind-governed loading combinations depend on a few key elements, such as section detailing, stiffness, strength, and connection design requirements.

2.1.2 Section Detailing and Geometric Considerations

SpeedCore walls do not utilize any additional steel reinforcing bars. The steel plates serve as the primary reinforcement for resisting in-plane shear and flexural demands. Consequently, section detailing is an important aspect of the design. Section detailing includes steel plate thickness and reinforcement ratio, tie bar spacing and size, and plate slenderness ratio.

The steel plate thicknesses used for composite walls are governed by practical fabrication, assembly, and erection considerations. Plate thickness less than ⅜ in. is impractical for most fabrication and assembly activities. In most design cases, particularly for nonseismic load combinations, plate thickness of ⅜ to ⅝ in. will be more than adequate to resist the calculated demands. This is particularly true because the geometric plan size and layout of the SpeedCore system is typically governed by architectural considerations, elevator core size requirements, and bay size limitations.

The reinforcement ratio of the composite wall cross section is defined as the area of the steel plates divided by the



(a) Planar rectangular wall with flange plates and tie bars

(b) Planar wall with semicircular boundary elements and tie bars

(c) Planar wall with circular boundary elements and tie bars

(d) C-shaped walls with flange (closure) plates and tie bars

(e) I-shaped walls with flange (closure) plates and tie bars

Fig. 2-1. SpeedCore systems with boundary elements or flange (closure) plates.

gross area of the composite wall, A_s/A_g. According to the AISC *Specification*, a minimum steel reinforcement ratio of 1% is required for the section to be classified as composite. An upper limit on the reinforcement ratio of 10% is specified based on the range of parameters included in the experimental investigations. Reinforcement ratios greater than 10% are rarely needed in design or construction and may indicate significant overdesign.

The tie bars connecting the steel web plates serve various important roles in the behavior and design of composite walls. In the pre-composite phase, before concrete casting, the tie bar size, d_{tie}, and spacing, S, govern the stiffness and stability of the empty modules during transportation, assembly, erection, and other construction-related activities. During concrete casting, the tie bar spacing governs the flexibility of the steel plates to resist local bending induced by the hydrostatic pressure during concrete casting. After the concrete sets—that is, it is in the composite phase—the maximum spacing of the tie bars, b (or tie bars and steel headed stud anchors, if they are included in the design), governs the plate slenderness ratio, b/t. Additionally, the effective diameter of the tie bar, d_{tie}, and largest clear spacing of the tie bar, S, govern the out-of-plane shear strength of the composite walls, which (if needed) can be calculated using AISC *Nuclear Specification* Section N9.3.5. This out-of-plane shear strength is rarely needed in building design.

The tie bar size and spacing requirements are based on research conducted by Varma et al. (2019) on the behavior, analysis, and stability of empty steel modules before and during concrete casting. The plate slenderness requirements, which lead to the maximum spacing of the tie bars (or tie bars and steel headed stud anchors), are based on research conducted by Zhang et al. (2014, 2020) on the local buckling behavior and axial compressive strength of composite walls. These are discussed in more detail in the following chapter along with the specific requirements.

In the coupled SpeedCore system, the coupling beams may be filled composite sections—that is, rectangular built-up box sections filled with plain concrete—or steel beams. Filled composite sections need to meet the requirements of AISC *Specification* Chapter I. They are classified as compact, noncompact, or slender for flexure in accordance with AISC *Specification* Section I1.4. The limitations for filled composite members are provided in AISC *Specification* Section I2.2a, which requires the area of steel in the cross section to be at least 1% of the total composite cross section.

2.1.3 Stiffness

The building structure is modeled and analyzed in accordance with the requirements of ASCE/SEI 7 and AISC *Specification* Chapter C to calculate the design demands for various load combinations, including lateral (wind) loading. The SpeedCore system provides lateral resistance to the overall building structure and is modeled appropriately when performing different analyses (for serviceability or strength load combinations).

The lateral stiffness of uncoupled SpeedCore walls is governed by the extent of concrete cracking and its influence on the effective flexural stiffness, EI_{eff}. The lateral stiffness of coupled walls is also governed by the extent of concrete cracking and its influence on both the effective flexural stiffness and effective axial stiffness, EA_{eff}. For coupled wall systems, the axial stiffness of the walls is just as relevant as the flexural stiffness due to the extensive contribution of the coupling action and moment to the overall lateral resistance.

The extent of concrete cracking can vary along the height of the structure. This is particularly true for high-rise buildings (greater than 15 stories in height). However, this may also be relevant for mid-rise buildings (less than or equal to 15 stories in height) depending on the extent of overdesign in the SpeedCore system due to the limitations and constraints imposed by architectural considerations, elevator core geometry, bay size, and minimum plate thickness requirements. In such cases, it may be important to estimate the cracking moment, M_{ck}, of the wall cross section and use it to define the appropriate portion of the wall height with uncracked (composite) stiffnesses.

For both coupled and uncoupled SpeedCore systems, the effective shear stiffness, $GA_{v.eff}$, of the composite walls is not a relevant parameter if the behavior is flexurally dominant (i.e., the overall wall height-to-length aspect ratio is greater than or equal to 3.0.) Consequently, the uncracked shear stiffness can be used to model flexurally dominant composite walls.

Walls with an aspect ratio less than 3.0 can be shear critical, but this depends on several parameters, including the relative ratio of the nominal flexural strength of the wall, M_n, to the nominal shear strength of the wall, V_n. Shear-critical walls are not recommended for design of building structures, but they are used prominently in the design of safety-related nuclear facilities (AISC, 2018), and the engineer can use other references, such as AISC Design Guide 32, to design them.

Additional details of the recommended lateral stiffness values are discussed in the following chapter along with the specific requirements.

The flexural stiffness of composite coupling beams can be estimated in accordance with AISC *Specification* Sections I1.5 and I2.2b. According to the User Note in Section I1.5, the flexural stiffness of filled composite members is estimated as $0.64EI_{eff}$ to account for the effects of concrete cracking due to flexure and steel partial yielding due to residual stresses. The flexural stiffness, EI_{eff}, can be calculated using Equation I2-12 in AISC *Specification* Section I2.2b, which accounts for the contributions of the steel section and

the cracked concrete section. Similarly, the flexural stiffness of steel beams can be determined in accordance with AISC *Specification* Section C2.3.

2.1.4 Strength

The design demands, in terms of the required flexural strength, M_r, required axial force, P_r, and required shear strength, V_r, are obtained from the analysis of the structure for the applicable load combinations. The flexural strength of the composite wall, M_n, can be calculated using the plastic stress distribution method, while accounting for the effects of axial force and using nominal steel and concrete material properties. This plastic stress distribution method is provided in AISC *Specification* Section I1.2a and has been verified by Shafaei et al. (2021b) and Kenarangi et al. (2021) for composite walls.

Other methods listed in the same section of the AISC *Specification* can also be used to estimate the flexural strength. For example, AISC *Specification* Section I1.2b describes the strain compatibility method similar to ACI 318, *Building Code Requirements for Structural Concrete* (ACI, 2019), and Section I1.2d describes the effective stress-strain method that can be used with section fiber analysis (Shafaei et al., 2021b).

The in-plane nominal shear strength of composite walls, V_n, can be calculated using equations developed and verified by Seo et al. (2016). These equations account for the orthotropic, cracked, composite behavior of the wall as shown in Varma et al. (2014). This in-plane nominal shear strength, V_n, includes the contribution of the steel web plates acting compositely with the cracked orthotropic concrete infill. It corresponds to the ductile limit state of von Mises yielding in the steel web plates. It is important to note that there is additional reserve capacity beyond the yielding of the steel plates. Booth et al. (2020) have developed and verified equations for calculating the ultimate shear strength of composite walls, which account for the failure of the concrete diagonal compression struts after von Mises yielding of the steel web plates. These equations provide ultimate strengths that are approximately 25 to 30% greater than the yield strength.

Additional details of the recommended flexural and strength equations are discussed in the following sections along with the specific requirements.

The flexural strength of filled composite coupling beams are calculated in accordance with AISC *Specification* Section I3.4. The shear strength of filled composite coupling beams can be calculated in accordance with the equations developed by Kenarangi et al. (2021) to account for the contributions of the steel web plates and concrete infill. These equations are included in the AISC *Specification* in Section I4.2. The steel webs of the filled composite section are assumed to develop their shear yield strength, and the concrete infill contribution is limited to the equivalent of $0.06A_c\sqrt{f_c'}$, where f_c' is the specified compressive strength of concrete in ksi, and A_c is the area of the concrete. The equations developed by Kenarangi et al. (2021) indicate that the concrete contribution can be much higher for rectangular filled composite members with shear span-to-depth ratios less than or equal to 0.75. However, this will rarely be the case for filled composite coupling beams.

2.1.5 Connections

The behavior and design of SpeedCore systems for nonseismic, wind-governed load combinations have no explicit ductility or performance requirements for connections beyond those of adequate strength for the design demands (required strengths) calculated from analyses conducted for the applicable factored load combinations. The different types of connections include:

- Connections at the section level; for example, tie bar to steel plate connections, flange-to-web plate connections, and wall segment connections to each other.
- Connections at the member level; for example, splices between wall modules and coupling beam-to-wall connections.
- Connections at the structure level; for example, floor system-to-wall connections and wall-to-foundation connections.

Welded and bolted connections are designed in accordance with the requirements of AISC *Specification* Chapter J. Connections at the section level may be designed to achieve tensile yielding of the gross section as the governing limit state and, thus, achieve local ductility. For example, tie bar-to-steel plate connections can be designed for the yield strength of the tie bar. Similarly, flange-to-web plate connections can be designed (if needed) to develop the yield strength of the weaker of the two connected plate elements. Connections at the member and structure levels can be designed in accordance with the calculated design demands (required strengths) at the level and location of the connections, and the applicable provisions of AISC *Specification* Chapter J.

2.2 NONSEISMIC DESIGN REQUIREMENTS FOR SPEEDCORE SYSTEMS

Nonseismic design of SpeedCore systems is performed according to the requirements of the AISC *Specification*.

2.2.1 Section Detailing

According to AISC *Specification* Section I1.6, the steel plates comprise at least 1% but no more than 10% of the total composite cross-sectional area. The opposing steel plates are connected to each other using tie bars consisting

of bars, structural shapes, or built-up members. The steel plates are also anchored to the concrete infill using tie bars or a combination of tie bars and steel anchors. Walls without flange (closure) plates or boundary elements are not included as part of this system and are not discussed further in this Design Guide.

The two steel plates in a SpeedCore wall must be connected using tie bars. These tie bars govern the structural behavior and stability of the empty steel modules before concrete placement. Additional steel headed stud anchors may be used along with tie bars to reduce the slenderness and improve the stability of steel plates after concrete placement.

2.2.1.1 Slenderness

According to AISC *Specification* Section I1.6a, the slenderness ratio of the steel plates, b/t_p, is limited as follows:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{2-1}$$

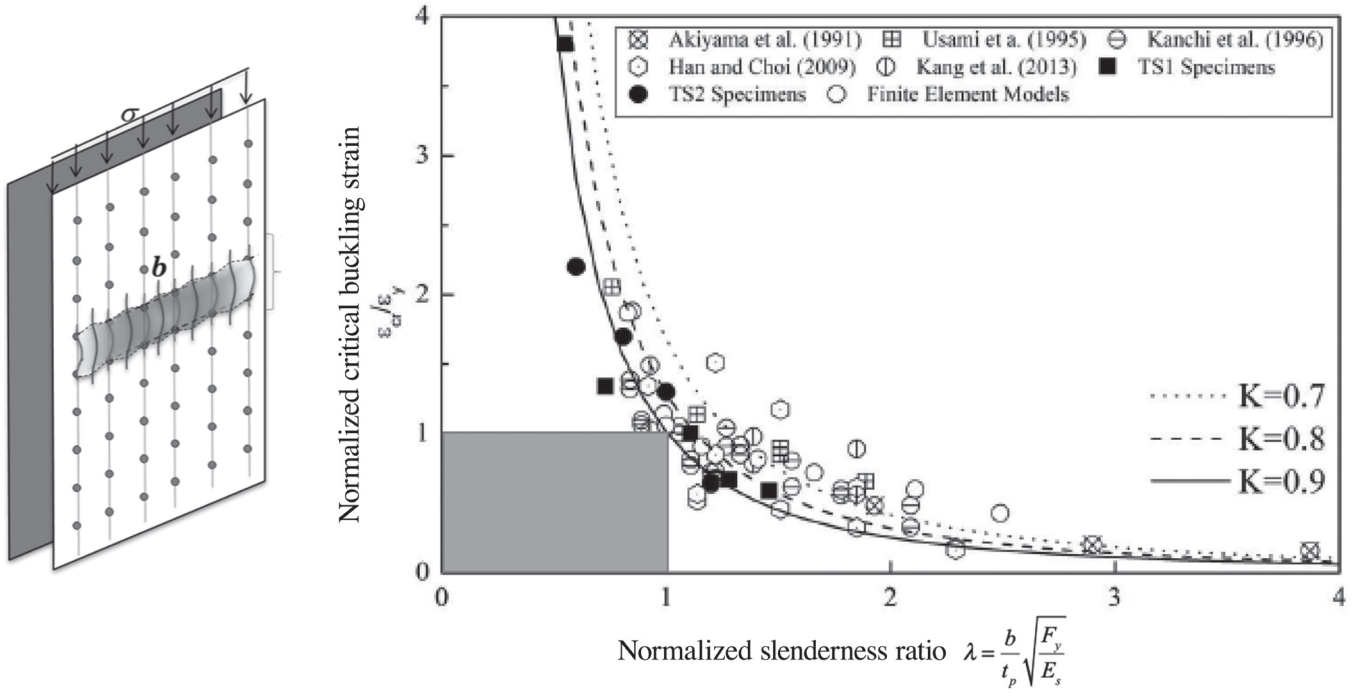
where

b = largest clear distance between rows of steel anchors or tie bars, in.

t_p = thickness of steel plate, in.

The steel plates in a SpeedCore wall are required to be nonslender (i.e., yielding in compression must occur before local buckling). When subjected to compressive stresses, the plate undergoes local buckling between the steel tie bars or anchors, as shown in Figure 2-2. The horizontal lines joining the steel anchors (or tie bars) act as fold lines, and local buckling occurs between them. The buckling mode indicates fixed ends along the vertical lines with steel anchors, and partial fixity along the vertical lines between steel anchors. Zhang et al. (2014, 2020) have summarized the results from experimental studies and conducted additional numerical analyses to confirm and expand the experimental database.

Figure 2-2 from Zhang et al. (2020) shows the relationship between the normalized critical buckling strain (buckling strain of steel/yield strain of steel, $\varepsilon_{cr}/\varepsilon_y$) and the normalized plate slenderness ratio $\left[(b/t_p)(F_y/E_s)\right]$. As shown, ε_{cr} is reasonably consistent with Euler's curve with a partially fixed ($K = 0.8$) end condition. This leads to Equation 2-1 for the



(a) Local buckling between rows of steel anchors or tie bars

(b) Normalized critical buckling strain vs. slenderness ratio

Fig. 2-2. Local buckling of steel plates and plot of normalized critical buckling strain vs. slenderness ratio (Zhang et al., 2014, 2020).

slenderness limit for nonseismic conditions. Because tie bars may also act as anchors, the equation considers the largest unsupported length between rows of steel anchors or tie bars, b. When slenderness exceeds the limit of Equation 2-1, local buckling will occur when the compressive stress reaches F_{cr}, which can be calculated using Equation 2-2, and the nominal compressive strength of the composite wall section can be estimated using Equation 2-3 (Zhang et al., 2020).

$$F_{cr} = \frac{\pi^2 E_s}{12(0.8)^2 (b/t_p)^2} \le F_y \tag{2-2}$$

$$P_{no} = A_s F_{cr} + 0.7 A_c f_c' \tag{2-3}$$

where

A_c = area of concrete, in.^{2}

A_s = area of steel plates, in.^{2}

f_c' = specified compressive strength of concrete, ksi

2.2.1.2 Tie Bars

According to AISC *Specification* Section I1.6b, tie bar spacing can be no greater than 1.0 times the wall thickness, t_{sc}. The tie bar spacing to plate thickness ratio, S/t_p, is also limited as follows:

$$\frac{S}{t_p} \le 1.0\sqrt{\frac{E_s}{2\alpha + 1}} \tag{2-4}$$

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$

where

S = largest clear spacing of the tie bars, in.

d_{tie} = effective diameter of the tie bar, in.

t_p = thickness of plate, in.

t_{sc} = thickness of SpeedCore wall, in.

The tie bar spacing requirement is based on the flexibility and shear buckling of empty steel modules before concrete placement, discussed in detail in Varma et al. (2019). The flexibility of the empty modules for transportation, shipping, and handling activities is dominated by their effective shear stiffness, $GA_{v.eff}$, which can be estimated accurately using numerical models as shown in Varma et al. (2019) or calculated conservatively for a unit cell of the module using Equation 2-6. In this equation, I_p and I_t represent the moments of inertia of the steel faceplates and steel tie bar. S and d_{tie} represent the tie bar spacing and diameter. Equation 2-7 defines α, which is the ratio of the flexural stiffness of the steel plate to the flexural stiffness of the tie bar and simplifies to the form of Equation 2-8.

$$GA_{v.eff} = 24\left(\frac{E_s I_p}{S^2}\right)\frac{1}{(2\alpha + 1)} \tag{2-6}$$

$$\alpha = 24\left(\frac{\frac{E_s I_p}{S}}{\frac{E_s I_t}{t_{sc} - 2t_p}}\right) = \frac{\frac{S t_p^3}{12S}}{\frac{\pi d_{tie}^4}{64(t_{sc} - 2t_p)}} \tag{2-7}$$

$$\sigma_{cr} = \left[\frac{E_s}{\left(\frac{S}{t_p}\right)^2}\right]\frac{1}{(2\alpha + 1)} \tag{2-8}$$

After assembly and before concrete casting, the empty modules provide structural support for construction activities, loads, and the steel framework connected to it. The buckling of the empty module subjected to compression loading is also governed by its effective shear stiffness, $GA_{v.eff}$, and can be estimated conservatively using Equation 2-6. If the value of α obtained from Equation 2-7 is equal to or greater than 25, it will result in a critical buckling stress equal to or greater than 1 ksi, which is equivalent to a distributed load of 12 kips per linear ft for walls with two ½-in.-thick steel plates. The stresses and deflections induced by concrete casting hydrostatic pressure can also be estimated as shown in Varma et al. (2019). Research by Bhardwaj et al. (2018) indicates that modules that meet the plate slenderness requirement of this section can be typically cast with concrete pour heights of up to 30 ft without significant influence of induced deflections and stresses on the compressive strength or buckling of the steel plates.

2.2.2 Stiffness

According to AISC *Specification* Section I1.5, the flexural, axial, and shear stiffnesses of SpeedCore walls account for the extent of concrete cracking corresponding to the required strength. The following equations estimate the stiffnesses:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \tag{2-9}$$

$$EA_{eff} = E_s A_s + 0.45 E_c A_c \tag{2-10}$$

$$GA_{v.eff} = G_s A_{sw} + G_c A_c \tag{2-11}$$

where

A_{sw} = area of steel plates in the direction of in-plane shear, in.^{2}

E_c = modulus of elasticity of concrete
$= w_c^{1.5}\sqrt{f_c'}$, ksi

E_s = modulus of elasticity of steel
= 29,000 ksi

G_c = shear modulus of elasticity of concrete
$= 0.4E_c$, ksi

G_s = shear modulus of elasticity of steel
= 11,200 ksi

I_c = moment of inertia of the concrete section about the elastic neutral axis of the composite section, in.^{4}

I_s = moment of inertia of steel shape about the elastic neutral axis of the composite section, in.^{4}

w_c = weight of concrete per unit volume, lb/ft^3 ($90 \le w_c \le 155$ lb/ft^3)

The lateral stiffness of uncoupled SpeedCore walls is governed by the extent of concrete cracking and its influence on the effective flexural stiffness, EI_{eff}. The lateral stiffness of coupled walls is also governed by the extent of concrete cracking and its influence on both the effective flexural stiffness and effective axial stiffness, EA_{eff}. For coupled wall systems, the axial stiffness is just as relevant as the flexural stiffness due to the extensive contribution of the coupling action and moment (force couple) to the overall lateral resistance.

The stiffness of composite plate shear walls must account for the extent of concrete cracking corresponding to the calculated required strengths (design demands). These stiffnesses can be estimated conservatively using equations developed in Agarwal et al. (2020) for uncoupled composite plate shear walls. The effective stiffness of composite plate shear walls can be approximated reasonably using Equation 2-9. This stiffness, EI_{eff}, is representative of the secant flexural stiffness corresponding to 60% of the nominal flexural strength, M_n, of walls subjected to axial compression of about 10% of the concrete capacity, $A_g f'_c$, where A_g is the gross area of the composite section. Equation 2-10 for axial stiffness was calibrated to match the concrete cracking associated with the flexural stiffness of Equation 2-11. Equation 2-11 corresponds to the use of uncracked shear stiffness of SpeedCore walls. SpeedCore walls with overall height-to-length ratio greater than or equal to 3.0 will be flexurally dominant and the shear stiffness can be approximated using uncracked values for ease.

When drift limits and stiffness requirements govern design, SpeedCore walls may be overdesigned with respect to the required strengths. Overdesigned walls may endure less concrete cracking and their stiffness may be underestimated using Equations 2-9 to 2-11. In such cases, it may be appropriate to use rational analysis methods to estimate the flexural stiffness of the walls, while accounting for the extent of concrete cracking corresponding to the calculated required strength (design demands). For example, the stiffness of SpeedCore walls can be estimated accurately using fiber-based analysis methods, as the secant flexural stiffness corresponding to 60% of the calculated plastic moment capacity of the wall cross section (Shafaei et al., 2021b). However, this procedure can be a bit cumbersome.

It is also important to note that the extent of concrete cracking can vary along the height of the structure. This is particularly true for high-rise buildings that are greater than 15 stories in height. However, this may also be relevant for mid-rise buildings, depending on the extent of overdesign in the SpeedCore system due to the limitations and constraints imposed by architectural considerations, elevator core geometry, bay size, and minimum plate thickness requirements. In such cases, it may be important to estimate the cracking moment, M_{ck}, of the wall cross section and use it to define the portion of the wall height with uncracked (composite) stiffnesses (Shafaei et al., 2022.)

2.2.3 Strength

The design compressive, tensile, flexural, combined axial force and flexural, and in-plane shear strength of SpeedCore systems are calculated according to the following subsections. These requirements and equations for calculating the axial tensile strength, compressive strength, flexural strength, in-plane shear strength, and combined axial force and flexural strength are based on the corresponding sections in AISC *Specification* Chapter I. The resistance factors for load and resistance factor design (LRFD) are equal to 0.90 for all these calculated strengths.

According to AISC *Specification* Section I2.3a, the nominal compressive strength of axially loaded composite plate shear walls is determined for the limit state of flexural buckling in accordance with AISC *Specification* Section I2.1b. The value of flexural stiffness, EI_{eff}, from Section 2.2.2 is used along with the section nominal compressive strength, P_{no}, calculated using Equation 2-12. The unsupported length for flexural buckling of composite walls is typically assumed to be equal to the story height.

$$P_{no} = F_y A_s + 0.85 f'_c A_c \tag{2-12}$$

The nominal tensile strength of axially loaded SpeedCore shear walls is determined for the limit state of yielding as follows:

$$P_n = F_y A_s \tag{2-13}$$

According to AISC *Specification* Section I3.5, the nominal flexural strength of the SpeedCore wall is determined in accordance with AISC *Specification* Section I1.2 using either the plastic stress distribution method or the effective stress-strain method. For the plastic stress distribution method, the nominal strength is computed assuming that steel components have reached a stress of F_y in either tension or compression, and concrete components in compression due to axial force and/or flexure have reached a stress of $0.85 f'_c$, where f'_c is the specified compressive strength of concrete in ksi. For the effective stress-strain method, the nominal strength is computed assuming strain compatibility and effective stress-strain relationships for steel and concrete components accounting for the effects of local buckling, yielding, interaction, and concrete confinement.

According to AISC *Specification* Section I4.4, the nominal in-plane shear strength, $V_{n.wall}$, of the SpeedCore wall is determined while accounting for the contributions of the steel section and concrete infill as follows:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \quad (2\text{-}14)$$

$$K_s = G_s A_{sw} \quad (2\text{-}15)$$

$$K_{sc} = \frac{0.7(E_c A_c)(E_s A_{sw})}{4E_s A_{sw} + E_c A_c} \quad (2\text{-}16)$$

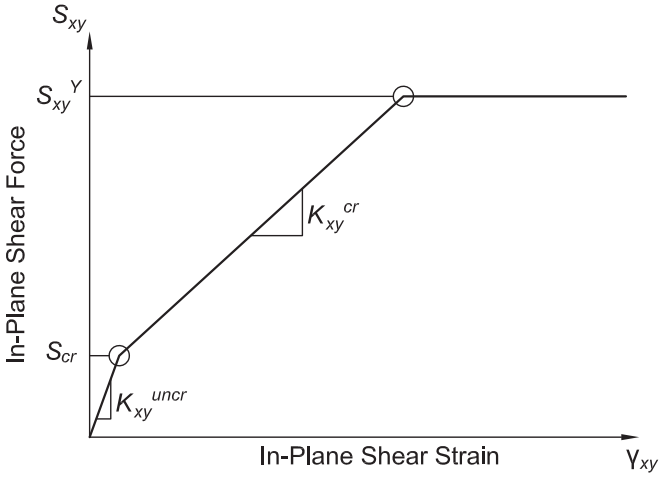
The in-plane shear behavior of the composite wall is governed by the plane stress behavior of the plates and the orthotropic elastic behavior of concrete cracked in principal tension. Varma et al. (2014) and Seo et al. (2016) discuss the fundamental mechanics-based model (MBM) for in-plane shear behavior of composite walls. The in-plane shear behavior can be estimated as the trilinear shear force-strain curve shown in Figure 2-3. The first part of the curve is before the concrete cracks. The second part is after the concrete cracking but before the plate yielding. The third part of the curve corresponds to the onset of plate von Mises yielding. The shear force corresponding to this onset of von Mises yielding is given by Equation 2-14. The corresponding principal compressive stress in the cracked (orthotropic) concrete is less than $0.7f_c'$ for typical composite walls with reinforcement ratios less than or equal to 10%. For walls with very high reinforcement ratios (i.e., walls with very thick steel plates compared to overall thickness), the concrete principal compressive stress can be the limiting failure criterion (Seo et al., 2016; Varma et al., 2014). However, recent research by Booth et al. (2020) quantified the post-yield ultimate shear strength of composite walls with boundary elements. This ultimate shear strength is governed by the failure of the concrete compression strut and is typically 25 to 30% greater than the shear yield strength calculated using Equation 2-14.

The out-of-plane shear strength of SpeedCore walls rarely governs the design. However, if perimeter SpeedCore walls are used on the exterior surfaces of the building, then out-of-plane shear strength may have to be checked for wind pressure. Equations for calculating out-of-plane strength are included in the AISC *Nuclear Specification* Section N9.3.5 (AISC, 2018). Their development is presented in Varma et al. (2014), and their use is demonstrated in AISC Design Guide 32 (Bhardwaj and Varma, 2017).

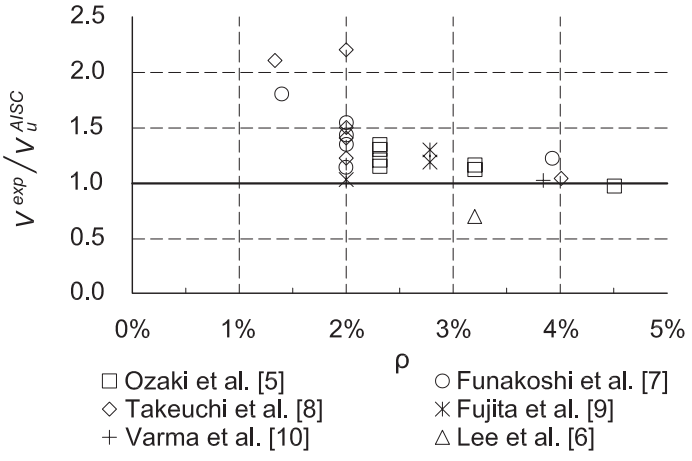
According to AISC *Specification* Section I5, for composite plate shear wall sections, the interaction between axial force and flexure is based on methods defined in Section I1.2. The flexural strength of composite plate shear walls subjected to axial compression can be estimated using the plastic stress distribution method of Section I1.2a or the effective stress-strain method of Section I1.2d (Bruneau et al., 2019; Agarwal et al., 2020). Bruneau et al. (2019) and Shafaei et al. (2021b) have reported the results of experimental investigations on planar walls subjected to different axial load and cyclic lateral loading up to failure.

2.3 COUPLING BEAM DESIGN REQUIREMENTS

For nonseismic applications, filled composite sections or steel beams can be used as the coupling beams for coupled SpeedCore systems. The composite coupling beams can be concrete-filled, built-up box sections, or rectangular hollow structural sections (HSS). The requirements for composite coupling beams are based on those in AISC *Specification* Chapter I as identified in the following subsections.



(a) In-plane shear force-strain response of composite walls



(b) Comparison of experimental results with shear strength calculated using Equation 2-14

Fig. 2-3. In-plane shear force-strain response of composite walls and comparison of experimental results with shear strength calculated using Equation 2-14 (Seo et al., 2016).

2.3.1 Section Detailing

AISC *Specification* Section I2.2a requires that the cross-sectional area of the steel section comprise at least 1% of the total composite cross section.

2.3.1.1 Slenderness

According to AISC *Specification* Table I1.1b, the composite coupling beams are classified as compact, noncompact, or slender for flexure based on the steel section width-to-thickness limits in AISC *Specification* Table I1.1b. There are no limitations on the use of noncompact or slender sections for nonseismic design.

2.3.2 Stiffness

As required by AISC *Specification* Section I2.2b, the effective flexural stiffness, EI_{eff}, of concrete-filled composite coupling beams is calculated using Equation 2-17. In this Equation, C_3 accounts for effects of cracking on the concrete rigidity contribution to EI_{eff} and is estimated using Equation 2-18 given in AISC *Specification* Section I2.2b. The stiffness reduction factors in AISC *Specification* Section I1.5 result in the use of $0.64EI_{eff}$ for the flexural stiffness to further account for the effects of concrete cracking at low axial load levels and steel inelasticity due to residual stresses. The stiffness reduction factors require the use of 0.8 times the nominal axial stiffness of filled composite members.

$$EI_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \tag{2-17}$$

$$C_3 = 0.45 + 3\left(\frac{A_s + A_{sr}}{A_g}\right) \le 0.9 \tag{2-18}$$

where

A_{sr} = area of continuous reinforcing bars, in.^{2}

C_3 = coefficient for calculation of effective rigidity of filled composite compression member

I_{sr} = moment of inertia of reinforcing bars about the elastic neutral axis of the composite section, in.^{4}

2.3.3 Strength

As required by AISC *Specification* Section I3.4b, the nominal flexural strength of a composite coupling beam is calculated using the plastic stress distribution method. For example, Figure 2-4 shows typical steel and concrete stress blocks for determining the nominal flexural strength of a compact filled rectangular box section. The nominal flexural strengths calculated using the plastic stress distribution method compare conservatively with experimental results (Lai et al., 2014; Lai and Varma, 2015). Longitudinal and transverse reinforcing bars are neither required nor needed for filled composite beams.

According to AISC *Specification* Section I4.2, the nominal shear strength of a filled composite coupling beam, $V_{n.CB}$, includes the contributions of the steel box section and concrete infill as shown in Equation 2-19:

$$V_{n.CB} = 0.6F_y A_v + 0.06K_c A_c \sqrt{f_c'} \tag{2-19}$$

where

A_v = Shear area of the steel portion of a composite member; the shear area for a circular section is equal to $2A_s/\pi$, and for a rectangular section is equal to the sum of the areas of webs in the direction of in-plane shear, in.^{2}

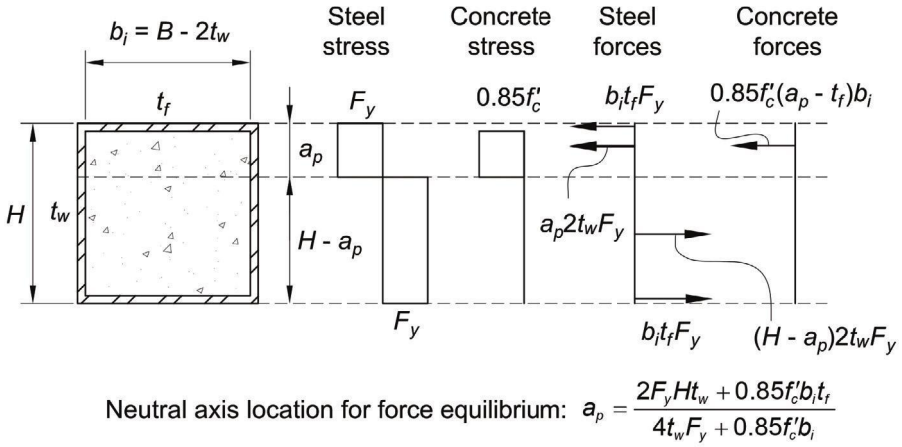


Fig. 2-4. Stress blocks for calculating nominal flexural strengths of compact filled rectangular box sections (Lai et al., 2014).

K_c = 1 for members with shear span-to-depth $(M_u/V_u)/d \geq 0.7$, where M_u and V_u are equal to the maximum moment and shear demands, respectively, along the member length, and d is equal to the member depth in the direction of bending

= 10 for members with rectangular compact cross sections and $(M_u/V_u)/d$ less than 0.5

= 1 for members having other than compact cross sections irrespective of $(M_u/V_u)/d$

Linear interpolation between the K_c values is used for members with compact cross sections and $(M_u/V_u)/d$ between 0.5 and 0.7. For most coupling beam design situations, K_c will be equal to 1.0. The nominal shear strength calculated using Equation 2-19 was compared with the shear test database from the existing literature and found to be safe; it accurately captures the contribution of the steel HSS or built-up box to the total shear strength and conservatively approximates that of the concrete (Kenarangi et al., 2021).

2.3.4 Steel Coupling Beams

Using steel coupling beams is an alternative option for coupled SpeedCore systems for nonseismic applications. The design of steel coupling beams for flexure and shear can be accomplished using AISC *Specification* Chapter F and Chapter G, respectively.

2.4 CONNECTION REQUIREMENTS

The behavior and design of SpeedCore systems for nonseismic or wind-governed load combinations have no explicit ductility or performance requirements for the connections beyond those of adequate strength. Welded and bolted connections are designed in accordance with the requirements of AISC *Specification* Chapter J.

Connections of the tie bars to the steel plate are designed to develop the yield strength of the tie bar in axial tension. This enables yielding of the tie bars before failure of the tie bar-to-plate connection. Samples of tie bar-to-plate connection details are shown in Figure 2-5 for round tie bars.

Connections at the member and structure levels are designed to withstand the required strengths at the corresponding levels and location. Some examples of the different types of connections in the systems include (1) coupling beam-to-composite wall connections, (2) composite wall-to-foundation or subgrade structure connections, and (3) splice connections in composite walls.

2.5 DIAPHRAGMS, COLLECTORS, AND CHORDS

SpeedCore wall systems act as the vertical elements of the overall lateral force-resisting system. The composite floor systems serve as the horizontal elements or diaphragms that collect and transfer the lateral forces at each story level to the SpeedCore wall systems. The framing elements (girders or beams) that are directly connected to the SpeedCore wall system eventually transfer these lateral forces to the SpeedCore wall system as shown in Figures 2-6 and 2-7. The composite floor systems typically consist of trapezoidal metal decks, steel beams, and girders, all composite with the concrete floor slab. The gravity loads and lateral forces resisted by the diaphragms can be calculated using applicable ASCE/SEI 7 provisions. The diaphragms are designed for the calculated in-plane shear and bending stresses. The chords are designed for the induced tension and compression forces. The collectors (girders) are designed to collect the inertial forces from the floor beams and transfer them to the SpeedCore system. The connections between the collector (girders) and the SpeedCore system are designed to transfer the lateral forces and the gravity loads as applicable.

There are limited studies and guidelines on the seismic design of diaphragm components. Sabelli et al. (2011) discussed the design of composite steel deck diaphragms for seismic loads. Rational analysis methods can be used to distribute lateral forces and calculate their effects (stresses, forces, etc.) on various elements of the diaphragms. Collectors and chords can be conservatively designed as noncomposite members to resist the axial forces induced by the diaphragm forces (AISC, 2018). The connections between collectors (girders) and SpeedCore are designed to transfer gravity loads and diaphragm lateral loads to SpeedCore walls. Currently, there are no research studies on the specific design of SpeedCore walls to floor connections. However, single-plate shear connections can be designed and used to transfer forces from the collectors to the SpeedCore walls. Single-plate shear connections, also referred to as shear tabs, should be designed for the generated shear forces (gravity loads), axial loads (diaphragm forces), imposed bending moments (eccentricity), and the interaction of the loads.

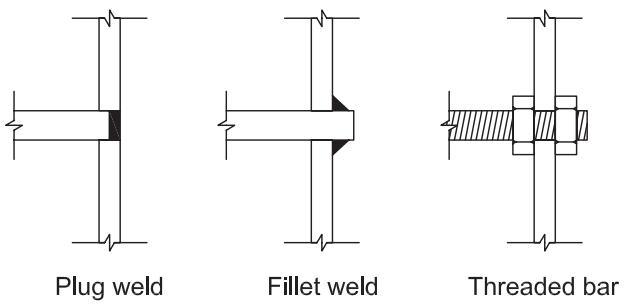


Fig. 2-5. Round tie bar-to-plate connection detail samples.

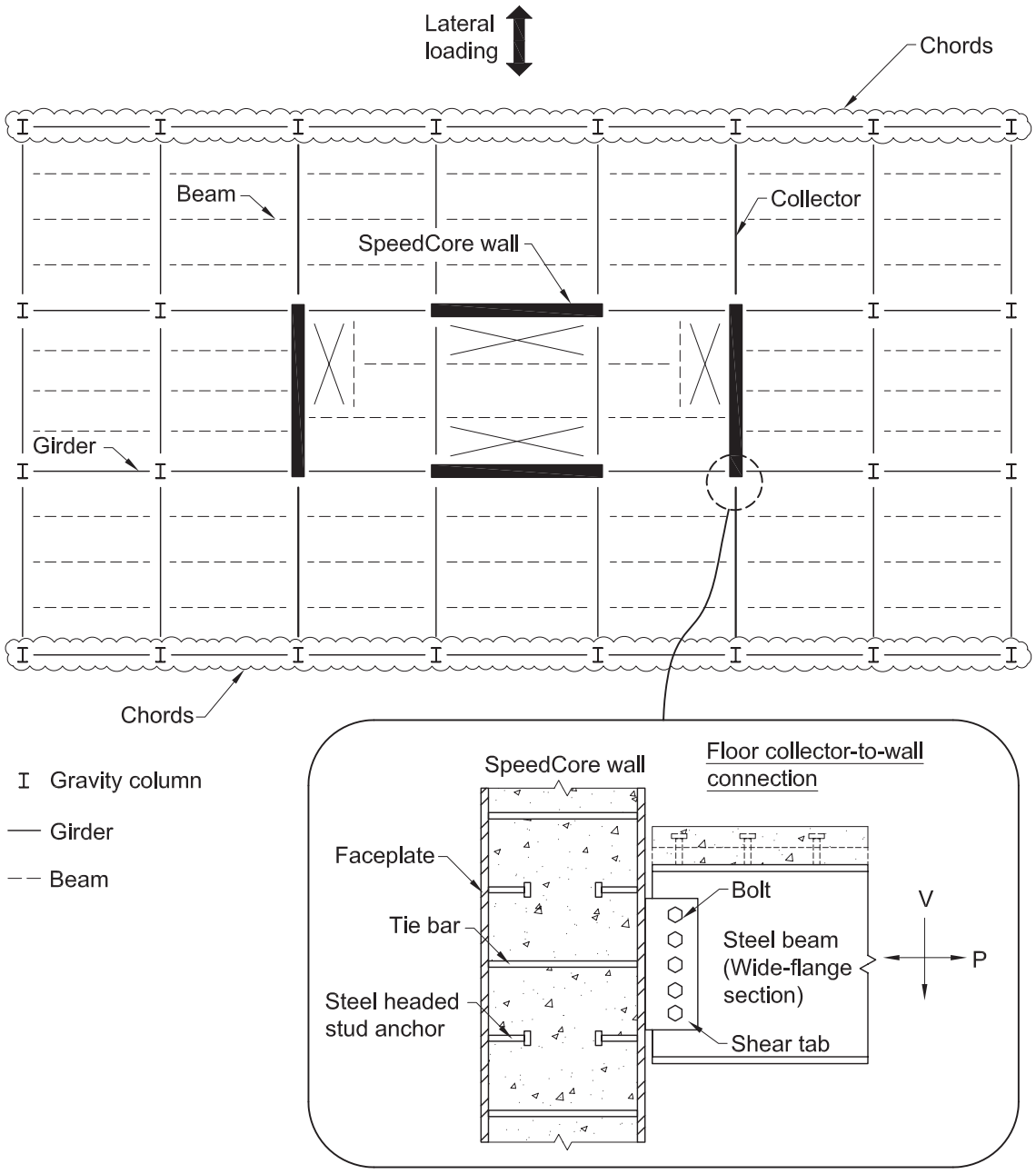


Fig. 2-6. Components of a diaphragm for a building with SpeedCore systems and a floor collector-to-wall connection.

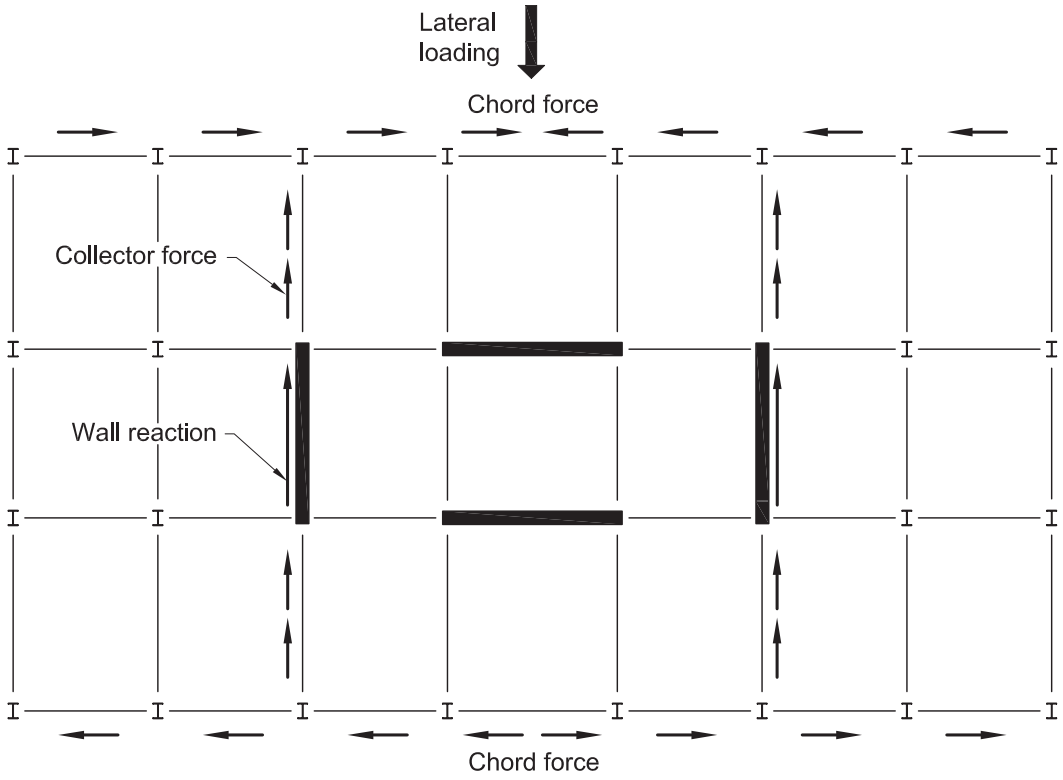


Fig. 2-7. Direction of the generated force in collectors and chords due to diaphragm loads.

2.6 WIND DESIGN PROCEDURE FOR SPEEDCORE SYSTEMS

This section describes the wind design procedure for uncoupled and coupled SpeedCore walls. This design procedure considers the requirements discussed in the previous sections and applicable building codes.

2.6.1 Wind Design Procedure for Uncoupled SpeedCore Walls

Figure 2-8 summarizes the general wind design procedure for planar uncoupled SpeedCore walls.

Step 1. General Information of the Considered Building

In this step, the initial design information such as building location, geometry and form of the building, number of stories, story height, floor dimension, building importance, and material properties are collected.

Step 2. Calculation of the Wind Loads

SpeedCore systems are designed to resist the code-specified wind loads. Design wind loads are determined based on the location and geometry of the buildings according to ASCE/SEI 7 in the absence of any wind tunnel testing or other special requirements for the given structure. Wind loads at mean

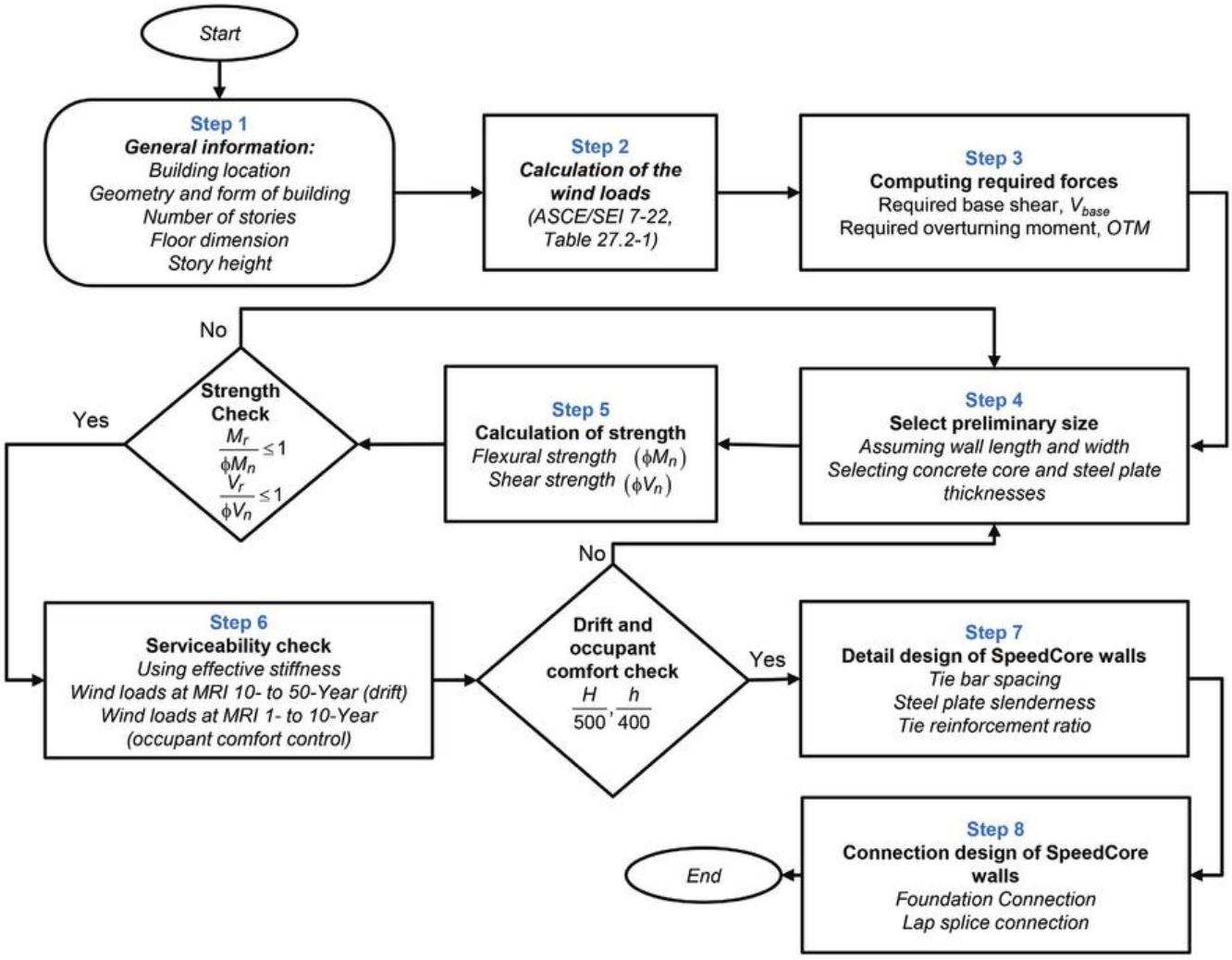


Fig. 2-8. Flowchart showing general wind design procedure for uncoupled SpeedCore walls.

recurrence interval (MRI) of 10-year or 25-year wind speed are used to check the drift limit state. Generally, an MRI of 10-year wind speed is appropriate to check the lateral drift of SpeedCore systems.

Step 3. Calculation of Base Shear and Overturning Moment

The design wind loads determined in Step 2 for the specific risk category are used to calculate the demand base shear and overturning moment (OTM). For uncoupled SpeedCore systems, the required base shear and OTM can be simply calculated by hand (if needed) without detailed computational analysis.

Step 4. Select Preliminary Size for the Uncoupled SpeedCore Walls

The size of uncoupled SpeedCore walls is selected in this step considering the floor layout, architectural design, and required base shear and moment. It includes selecting the total wall length, L_{wall}, thickness, t_{sc}, and steel plate thickness, t_p. Engineering experience and judgment are needed for selecting initial sizes.

Step 5. Strength Check of Uncoupled SpeedCore Walls

The flexural strength of the selected planar SpeedCore wall is calculated according to Section 2.2.3 and compared with the required OTM. The shear strength is calculated using Equation 2-14 and compared with the required base shear. If the strength check is not satisfied, the size of the uncoupled SpeedCore wall is revised, and the procedure restarts from Step 4. In the design of mid- or high-rise buildings, the strength of planar SpeedCore walls can be considerably higher than the demand forces because drift limits (wall stiffness considerations) govern the design.

Step 6. Serviceability Check of Uncoupled SpeedCore Walls

The lateral deflection is checked to evaluate the serviceability of the uncoupled SpeedCore system. The deflection limit for wind loads in common usage for building design is on the order of 1/600 to 1/400 of the building or story height (Griffis, 1993). In AISC Design Guide 3, *Serviceability Design Considerations for Steel Buildings* (West et al., 2003), it is recommended that the lateral deflection and story drift due to 10-year mean recurrence interval (MRI) wind loading be limited to $H/500$ and $h/400$, where H is the total height of the building and h is the story height.

A commercial structural analysis software program can be used to calculate the roof displacement and interstory drift (ID) of uncoupled SpeedCore walls. The effective stiffnesses of planar walls, as discussed in Section 2.2.2, are used in the computer models to generate conservative but reasonable estimates. Additionally, the effect of rotational stiffness of wall-to-foundation connections may need to be considered in the calculation of lateral displacements. If the drift check is not satisfied, the size of uncoupled SpeedCore walls is revised and the procedure restarts from Step 4.

Step 7. Detail Design of Uncoupled SpeedCore Systems

Tie bar size (diameter) and spacing of planar SpeedCore walls are designed to satisfy the slenderness limits and tie bar reinforcement ratio requirements. Slenderness limits and tie bar requirements are presented in Section 2.2.1.1 and 2.2.1.2.

Step 8. Connection Design of Uncoupled SpeedCore Systems

The composite wall-to-foundation (or subgrade structure) connections are designed based on the calculated demands (required strengths). The composite wall splices are also designed based on the calculated demands at the corresponding locations.

2.6.2 Wind Design Procedure for Coupled SpeedCore Systems

Figure 2-9 summarizes the general wind design procedure for coupled SpeedCore systems.

Step 1. General Information of the Considered Building

This step is similar to Step 1 of uncoupled SpeedCore systems. The building information is collected and compiled, which includes the building location, floor dimension, number of stories, and story height.

Step 2. Calculation of the Wind Loads

Step 2 for coupled SpeedCore systems is similar to Step 2 for uncoupled SpeedCore systems.

Step 3. Select Preliminary Sizes for SpeedCore Walls and Coupling Beams

The SpeedCore walls and coupling beams are sized in this step considering floor layout and architectural design. This includes selecting the wall length, L_{wall}, wall thickness, t_{sc}, steel plate thicknesses, t_p, and infill concrete core thickness. In addition, the coupling beam depth, h_{CB}, width, b_{CB}, web plate thickness, $t_{pw.CB}$, flange plate thickness, $t_{pf.CB}$, and length, L, are also selected in this step. Engineering experience and judgment are needed for selecting initial sizes.

Step 4. Calculation of Base Shear and Overturning Moment

The distribution of forces along the height of a coupled SpeedCore wall is directly influenced by the stiffnesses of the SpeedCore wall and coupling beams. A finite element model of the coupled SpeedCore system is developed using

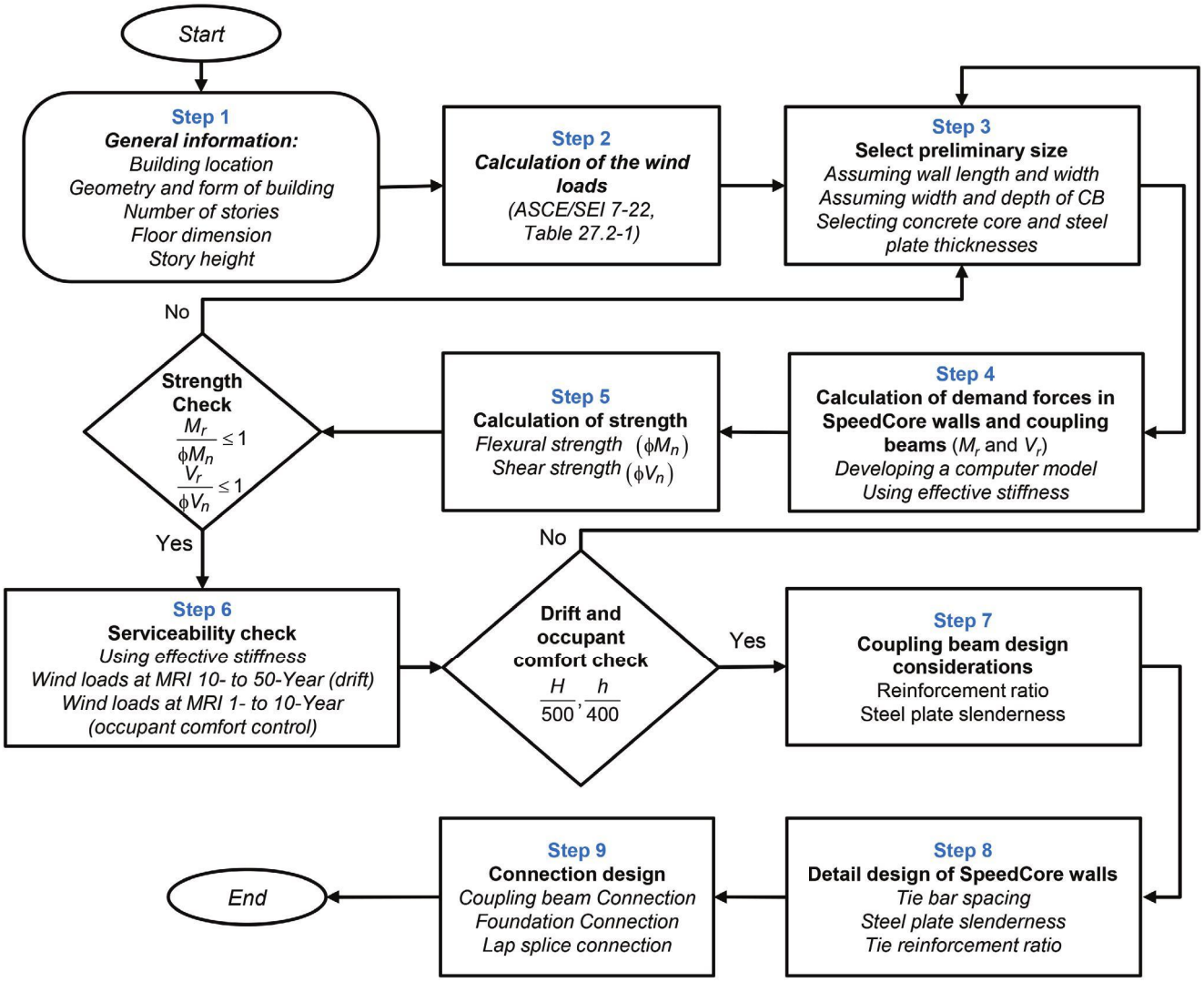


Fig. 2-9. Flowchart showing general wind design procedure of coupled SpeedCore systems.

effective stiffnesses for the composite walls and coupling beams in accordance with Sections 2.2.2 and 2.3.2, respectively. A commercial structural analysis program can be used to conduct the analysis and calculate the required strengths.

Step 5. Strength Check of SpeedCore Walls and Coupling Beams

The flexural strength of the selected SpeedCore wall is calculated according to Section 2.2.3 and compared with the required OTM. The shear strength is calculated using Equation 2-14 and compared with the required base shear. Additionally, the design flexural and shear strengths of coupling beams are calculated according to Section 2.3.3. If the strength check is not satisfied, the size of coupling beams or SpeedCore walls are revised, and the procedure restarts from Step 4. In the design of mid- to high-rise buildings, the coupling beam strength typically governs the design, and the strengths of SpeedCore walls are considerably higher than the demand forces. In some cases, the lateral drift limits for wind loading govern the design of coupled SpeedCore walls.

Step 6. Serviceability Check of Coupled SpeedCore Walls

A structural analysis is conducted in this step using an MRI of 10-year wind loads to check the drift limits. Appropriate effective stiffnesses of the composite walls and coupling beams are used in the structural analysis models of the coupled SpeedCore walls. The lateral deflection due to wind loading is limited to $H/500$, where H is the total height of the building, and the story drift is limited to $h/400$, where h is the story height (West at al., 2003). Although the rotational stiffness of the wall-to-foundation connection has a small effect on coupled systems due to the benefits of coupling action, the connection stiffness can be included to improve the estimation of lateral displacements. If the roof displacement or ID requirements are not met, the sizes of members (SpeedCore walls or coupling beams) are revised and the procedure restarts from Step 3.

Step 7. Detail Design of Coupling Beams

The section slenderness requirements and limits for the coupling beams are checked. These slenderness limits were identified in Section 2.3.1. The geometric size and details of the coupling beams are used to assess coupling beam-to-wall connection possibilities.

Step 8. Detail Design of SpeedCore Walls

Tie bar size (diameter) and spacing of planar SpeedCore walls are designed to satisfy the slenderness limits and tie bar reinforcement ratio requirements. Slenderness limits and tie bar requirements are presented in Section 2.2.1.1 and 2.2.1.2.

Step 9. Connection Design

The composite wall-to-foundation (or subgrade structure) connections are designed based on the calculated demands (required strengths). The coupling beam-to-composite wall connections are designed based on the calculated demands at the corresponding levels. The composite wall splices are also designed based on the calculated demands at the corresponding locations.

2.7 DESIGN EXAMPLES

Example 2.1—Wind Design of 15-Story Structure Using Uncoupled SpeedCore Systems

This example presents the wind design of a 15-story office building located in Chicago with typical design loads, floor geometry, and wind loads (ASCE/SEI 7, Category B). Figure 2-10 shows the floor plan of the building with 200 ft length and 120 ft width (a total of 24,000 ft^2 of area). Two uncoupled SpeedCore walls in the east-west direction and two coupled SpeedCore walls in the north-south direction are used to resist the wind loads. Example 2.1 and Example 2.2 present the wind design of uncoupled and coupled walls of this building, respectively.

Design the uncoupled, planar SpeedCore walls shown in Figure 2-10 using the given geometry, material properties, and loads.

Given:

Material properties and calculated wind loads for this example are given in Steps 1 and 2, respectively. The self-weight of the walls (axial compression force) is not considered in this example.

ASTM A572/A572M Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi

Step 1. General Information of the Considered Building

Building geometry:

H_{wall} = wall height
= 213 ft
L_f = building length
= 200 ft
W_f = building width
= 120 ft
h_{typ} = typical story height
= 14 ft
h_1 = first-story height
= 17 ft
n = number of stories
= 15

Floor load:

DL = floor dead load
= 0.12 ksf

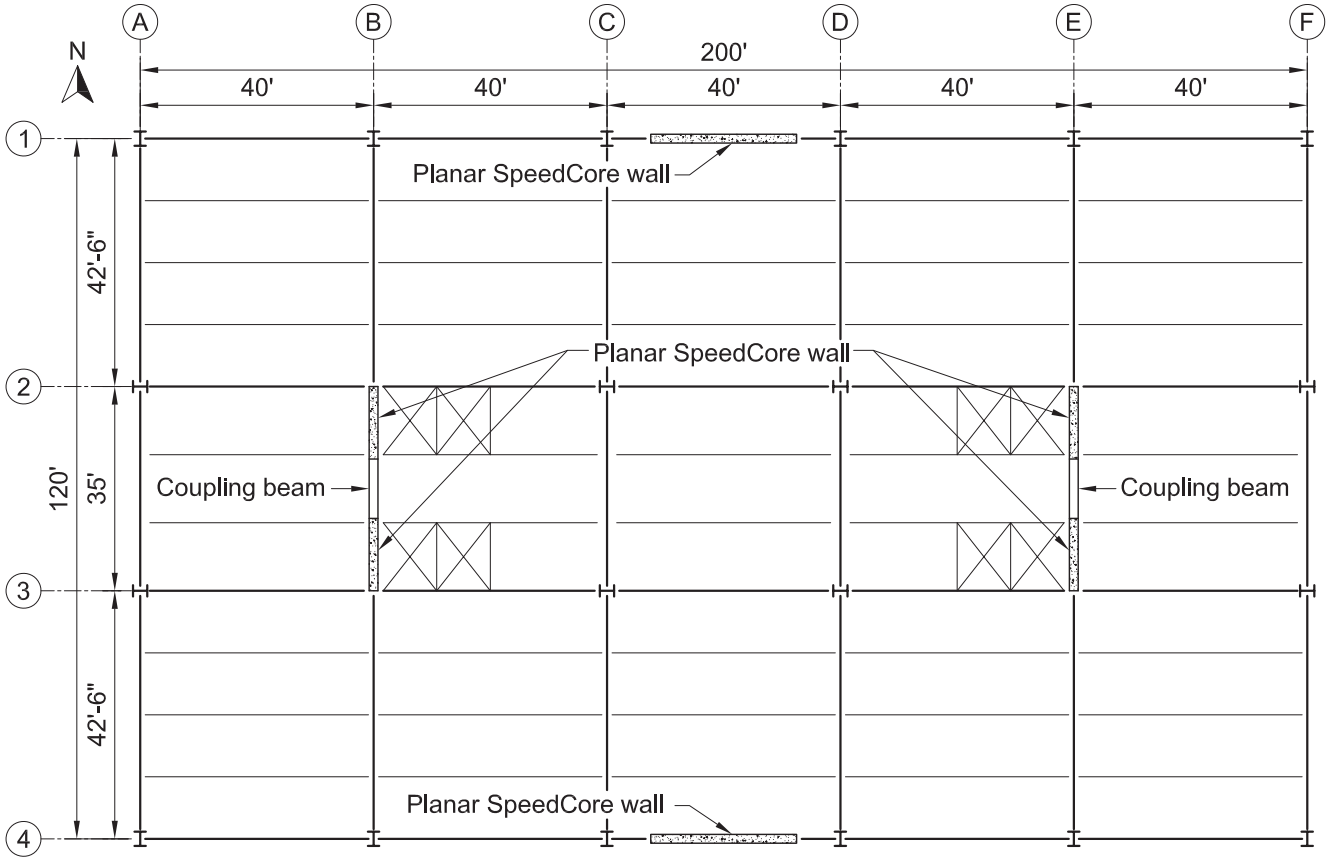


Fig. 2-10. Building floor plan for Examples 2.1 and 2.2.

Table 2-1. Wind Speeds for Chicago (ASCE/SEI 7, Chapter 26, Appendix CC)	
Wind Hazard Basic	**Wind Speed (mph)**
MRI 10-Year	74
MRI 25-Year	80
MRI 50-Year	85
MRI 100-Year	92
Risk Category I	100
Risk Category II	107
Risk Category III	114
Risk Category IV	119

For ASTM A572/A572M Grade 50 steel, from the AISC *Steel Construction Manual* (AISC, 2017), hereafter referred to as the AISC *Manual*, Table 2-4, the material properties are as follows:

$F_y = 50$ ksi
$F_u = 65$ ksi

Step 2. Calculation of the Wind Loads

The building is located in Chicago and wind speeds for this city are shown in Table 2-1. Wind speeds at ASCE/SEI 7, Risk Category II, and a 10-year mean recurrence interval (MRI), are used to calculate wind loads for the design and serviceability checks, respectively. The calculated wind loads for the design of each uncoupled wall in the east-west direction are shown in Table 2-2.

Solution:

Step 3. Calculation of Base Shear and Overturning Moment

The wind loads given in Step 2 are used to calculate the required base shear and overturning moment (OTM), as shown in Table 2-2. The Risk Category II values are larger and will be used as the required values for the remainder of this example.

Required base shear for each planar SpeedCore wall (Risk Category II):

$V_{r.wall} = 490$ kips

Required moment for each planar SpeedCore wall (Risk Category II):

$M_{r.wall} = (57{,}900 \text{ kip-ft})(12 \text{ in./ft})$
$= 695{,}000$ kip-in.

Step 4. Select Preliminary Size for the Uncoupled SpeedCore Walls

The size of the uncoupled SpeedCore walls including wall length, L_{wall}, wall thickness, t_{sc}, and steel plate thickness, t_p, as shown in Figure 2-11, are selected to resist the required base shear and OTM. The wall and plate dimensions selected are as follows:

L_{wall} = wall length
= 300 in.
t_p = wall plate thickness
= ½ in.
t_{sc} = wall thickness
= 18 in.

Table 2-2. Wind Loads in East-West Direction							
		Wind Loads East-West Direction (Risk Category II)			Wind Loads East-West Direction (MRI 10-Year)		
Story No.	Story Elevation (ft)	Windward Wall (kips)	Leeward Wall (kips)	Moment (kip-ft)	Windward Wall (kips)	Leeward Wall (kips)	Moment (kip-ft)
1	17	19.0	–13.5	553	8.90	–6.30	258
2	31	17.3	–11.1	880	8.20	–5.20	415
3	45	18.5	–11.1	1,330	8.70	–5.20	626
4	59	19.4	–11.1	1,800	9.10	–5.20	844
5	73	20.1	–11.1	2,280	9.50	-5.20	1,070
6	87	20.7	–11.1	2,770	9.80	–5.20	1,310
7	101	21.3	–11.1	3,270	10.0	–5.20	1,540
8	115	21.8	–11.1	3,780	10.2	–5.20	1,770
9	129	22.2	–11.1	4,300	10.5	–5.20	2,030
10	143	22.6	–11.1	4,820	10.6	–5.20	2,260
11	157	23.0	–11.1	5,350	10.8	–5.20	2,510
12	171	23.3	–11.1	5,880	11.0	–5.20	2,770
13	185	23.6	–11.1	6,420	11.1	–5.20	3,020
14	199	23.9	–11.1	6,970	11.3	–5.20	3,280
15	213	24.2	–11.1	7,520	11.4	–5.20	3,540
Sum		321	–169	57,900	150	–79.1	27,200
		Base Shear	490		Base Shear	229	

Step 5. Strength Check of Uncoupled SpeedCore Walls

In this step, design flexural and shear strengths of the selected uncoupled SpeedCore walls are calculated and compared with the required base shear and OTM. Figure 2-12 illustrates the plastic neutral axis location and the compression and tension force regions in the wall.

Design Flexural Strength of the Uncoupled Walls

Plastic neutral axis location:

$$C = \frac{2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$

$$= \frac{2(½\text{ in.})(300\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[18\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})}{4(½\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[18\text{ in.} - 2(½\text{ in.})\right]}$$

$$= 80.6\text{ in.}$$

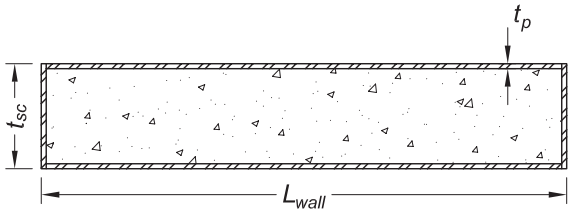


Fig. 2-11. Wall cross-section dimensions.

The compression force in the flange plate is:

$$C_1 = (t_{sc} - 2t_p)t_p F_y$$
$$= [18\text{ in.} - 2(½\text{ in.})](½\text{ in.})(50\text{ ksi})$$
$$= 425\text{ kips}$$

The compression force in the web plate is:

$$C_2 = 2t_p C F_y$$
$$= 2(½\text{ in.})(80.6\text{ in.})(50\text{ ksi})$$
$$= 4{,}030\text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85f_c'(t_{sc} - 2t_p)(C - t_p)$$
$$= 0.85(6\text{ ksi})[18.0\text{ in.} - 2(½\text{ in.})](80.6\text{ in.} - ½\text{ in.})$$
$$= 6{,}940\text{ kips}$$

The tension force in the web plate is:

$$T_1 = (t_{sc} - 2t_p)t_p F_y$$
$$= [18\text{ in.} - 2(½\text{ in.})](½\text{ in.})(50\text{ ksi})$$
$$= 425\text{ kips}$$

The tension force in the flange plate is:

$$T_2 = 2t_p(L_{wall} - C)F_y$$
$$= 2(½\text{ in.})(300\text{ in.} - 80.6\text{ in.})(50\text{ ksi})$$
$$= 11.000\text{ kips}$$

The plastic flexural strength of the planar SpeedCore wall is:

$$M_{P.wall} = C_1\left(C - \frac{t_p}{2}\right) + C_2\left(\frac{C}{2}\right) + C_3\left(\frac{C - t_p}{2}\right) + T_1\left(L_{wall} - C - \frac{t_p}{2}\right) + T_2\left(\frac{L_{wall} - C}{2}\right)$$
$$= (425\text{ kips})\left(80.6\text{ in.} - \frac{½\text{ in.}}{2}\right) + (4{,}030\text{ kips})\left(\frac{80.6\text{ in.}}{2}\right) + (6{,}940\text{ kips})\left(\frac{80.6\text{ in.} - ½\text{ in.}}{2}\right)$$
$$+ (425\text{ kips})\left(300\text{ in.} - 80.6\text{ in.} - \frac{½\text{ in.}}{2}\right) + (11{,}000\text{ kips})\left(\frac{300\text{ in.} - 80.6\text{ in.}}{2}\right)$$
$$= 1{,}770{,}000\text{ kip-in.}$$

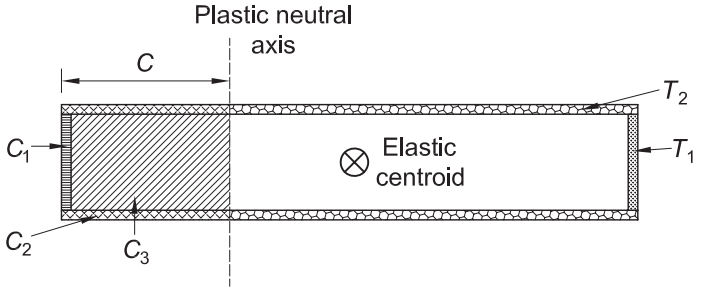


Fig. 2-12. Cross section with labeled regions for plastic moment calculation of tension wall.

The nominal flexural strength of the planar SpeedCore wall is:

$$M_{n.wall} = M_{P.wall} = 1{,}770{,}000 \text{ kip-in.}$$

Using $\phi_b = 0.90$, the design flexural strength of the uncoupled SpeedCore wall is:

$$\phi_b M_{n.wall} = 0.90(1{,}770{,}000 \text{ kip-in.})$$
$$= 1{,}590{,}000 \text{ kip-in.} > M_{r.wall} = 695{,}000 \text{ kip-in.}$$

Design Shear Strength of the Uncoupled Wall

The area of steel in the direction of shear is:

$$A_{sw} = 2L_{wall}\, t_p$$
$$= 2(300 \text{ in.})(½ \text{ in.})$$
$$= 300 \text{ in.}^2$$

The area of steel in the wall is:

$$A_s = t_p\left[2(L_{wall} - 2t_p) + 2t_{sc}\right]$$
$$= (½ \text{ in.})\left\{2\left[300 \text{ in.} - 2(½ \text{ in.})\right] + 2(18 \text{ in.})\right\}$$
$$= 317 \text{ in.}^2$$

The area of concrete in the wall is:

$$A_c = L_{wall} t_{sc} - A_s$$
$$= (300 \text{ in.})(18 \text{ in.}) - 317 \text{ in.}^2$$
$$= 5{,}080 \text{ in.}^2$$

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200 \text{ ksi})(300 \text{ in.}^2)$$
$$= 3{,}360{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_c)(E_s A_{sw})}{4E_s A_{sw} + E_c A_c} \tag{2-16}$$
$$= \frac{0.7(4{,}500 \text{ ksi})(5{,}080 \text{ in.}^2)(29{,}000 \text{ ksi})(300 \text{ in.}^2)}{4(29{,}000 \text{ ksi})(300 \text{ in.}^2) + (4{,}500 \text{ ksi})(5{,}080 \text{ in.}^2)}$$
$$= 2{,}410{,}000 \text{ kips}$$

The nominal shear strength of the uncoupled SpeedCore wall is calculated using Equation 2-14 of this Design Guide:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} A_{sw} F_y \tag{2-14}$$
$$= \frac{(3{,}360{,}000 \text{ kips}) + (2{,}410{,}000 \text{ kips})}{\sqrt{3(3{,}360{,}000 \text{ kips})^2 + (2{,}410{,}000 \text{ kips})^2}} (300 \text{ in.}^2)(50 \text{ ksi})$$
$$= 13{,}700 \text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the uncoupled SpeedCore wall is:

$$\phi_v V_{n.wall} = 0.90(13{,}700 \text{ kips})$$
$$= 12{,}300 \text{ kips} > V_{r.wall} = 490 \text{ kips}$$

Step 6. Serviceability Check of Uncoupled SpeedCore Walls

In this step, the lateral deflection of the uncoupled wall is checked to evaluate serviceability. A 10-year MRI wind load and the effective stiffness of the uncoupled wall are used in a finite element model to check the drift limits.

Effective Flexural and Shear Stiffnesses of the Uncoupled Wall

The area of steel in the wall was calculated in Step 5 as $A_s = 317 \text{ in.}^2$

The area of concrete in the wall was calculated in Step 5 as $A_c = 5{,}080 \text{ in.}^2$

The effective axial stiffness of the wall is:

$$EA_{eff} = E_s A_s + 0.45 E_c A_c \tag{2-10}$$
$$= (29{,}000 \text{ ksi})(317 \text{ in.}^2) + \left[0.45(4{,}500 \text{ ksi})(5{,}080 \text{ in.}^2)\right]$$
$$= 1.95 \times 10^7 \text{ ksi-in.}^2$$

The steel area in the direction of shear was calculated in Step 5 as $A_{sw} = 300 \text{ in.}^2$

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_s A_{sw} + G_c A_c \tag{2-11}$$
$$= (11{,}200 \text{ ksi})(300 \text{ in.}^2) + (1{,}800 \text{ ksi})(5{,}080 \text{ in.}^2)$$
$$= 1.25 \times 10^7 \text{ ksi-in.}^2$$

The moment of inertia of steel in the wall is:

$$I_s = 2\left[\frac{t_p (L_{wall} - 2t_p)^3}{12}\right] + 2\left[\left(\frac{t_{sc} t_p^3}{12}\right) + t_{sc} t_p \left(\frac{L_{wall}}{2} - \frac{t_p}{2}\right)^2\right]$$
$$= 2\left\{\frac{(½ \text{ in.})\left[300 \text{ in.} - 2(½ \text{ in.})\right]^3}{12}\right\} + 2\left[\frac{(18 \text{ in.})(½ \text{ in.})^3}{12} + (18 \text{ in.})(½ \text{ in.})\left(\frac{300 \text{ in.}}{2} - \frac{½ \text{ in.}}{2}\right)^2\right]$$
$$= 2.63 \times 10^6 \text{ in.}^4$$

The moment of inertia of concrete in the wall is:

$$I_c = \frac{(t_{sc} - 2t_p)(L_{wall} - 2t_p)^3}{12}$$
$$= \frac{\left[18 \text{in.} - 2(½ \text{ in.})\right]\left[300 \text{ in.} - 2(½ \text{ in.})\right]^3}{12}$$
$$= 3.79 \times 10^7 \text{ in.}^4$$

The effective flexural stiffness of the wall is:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \tag{2-9}$$
$$= (29{,}000 \text{ ksi})(2.63 \times 10^6 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(3.79 \times 10^7 \text{ in.}^4)$$
$$= 1.36 \times 10^{11} \text{ kip-in.}^2$$

Because the rotational stiffness of the wall-to-foundation connection has a considerable effect on the lateral deflection of the uncoupled wall, it is included in the finite element model to check the serviceability. It is assumed the wall-to-foundation connection provides a fully restrained connection. The rotational stiffness of a fully restrained wall-to-foundation connection, $K_{s.con}$, is calculated according to AISC *Specification* Commentary Figure C-B3.3:

$$K_{s.con} = \frac{20 EI_{eff}}{H_{wall}}$$
$$= \frac{20(1.36 \times 10^{11} \text{ kip-in.}^2)}{(213 \text{ ft})(12 \text{ in./ft})}$$
$$= 1.06 \times 10^9 \text{ kip-in./rad}$$

The lateral deflection due to a 10-year MRI wind load is limited to a roof displacement of $H_{wall}/500$, where H_{wall} is the total height of the building, or an ID of $h/400$, where h is the story height.

Allowable roof displacement:

$$\frac{H_{wall}}{500} = \frac{(213 \text{ ft})(12 \text{ in./ft})}{500}$$
$$= 5.11 \text{ in.}$$

Allowable ID for a typical floor:

$$\frac{h}{400} = \frac{(14 \text{ ft})(12 \text{ in./ft})}{400}$$
$$= 0.420 \text{ in.}$$
$$= 0.25\%$$

The ID ratio normalizes the ID by the height of the story. This gives a consistent value to check against for different story heights. The ID ratio of each story can be calculated as the change in lateral deflection along the story divided by the height of the story.

A commercial structural analysis software program is used to check roof displacement and ID. Figure 2-13 shows the lateral deformation shape, lateral displacement, and ID of the uncoupled SpeedCore wall. Additionally, Table 2-3 summarizes lateral displacement and ID.

Step 7. Detail Design of Uncoupled SpeedCore Walls

The required steel reinforcement ratio of the uncoupled wall is checked first. From Section 2.2.1, the steel plates must comprise at least 1% but no more than 10% of the total composite cross-sectional area.

The minimum steel required is:

$$A_{s.min} = 0.01 L_{wall}\, t_{sc}$$
$$= 0.01(300 \text{ in.})(18.0 \text{ in.})$$
$$= 54.0 \text{ in.}^2$$

$$A_s = 317 \text{ in.}^2 > A_{s.min} = 54.0 \text{ in.}^2$$

The maximum steel required is:

$$A_{s.max} = 0.10 L_{wall}\, t_{sc}$$
$$= 0.10(300 \text{ in.})(18.0 \text{ in.})$$
$$= 540 \text{ in.}^2$$

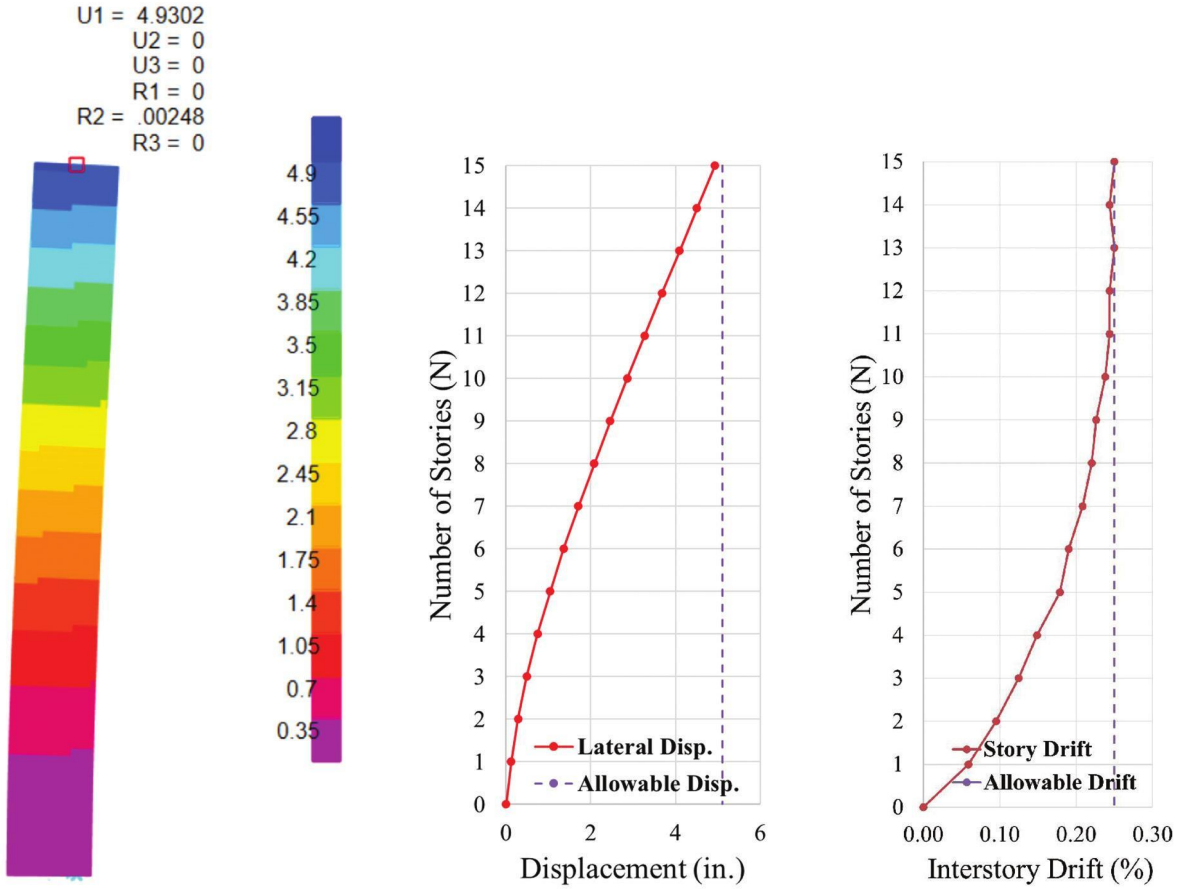
$$A_s = 317 \text{ in.}^2 < A_{s.max} = 540 \text{ in.}^2$$

The tie bar spacing, S, is selected to be 12 in. for the uncoupled wall, and the slenderness requirement is checked as follows:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{2-1}$$

where b is the maximum tie bar spacing = S = 12 in.

$$\frac{b}{t_p} = \frac{12 \text{in.}}{½ \text{ in.}}$$
$$= 24.0$$



(a) Deformed shape *(b) Lateral displacement* *(c) ID*

Fig. 2-13. Uncoupled SpeedCore wall deformed shape, lateral displacement, and ID.

Table 2-3. Lateral Displacement and ID Summary			
Story No.	**Story Height (ft)**	**Displacement (in.)**	**ID (%)**
15	14	4.93	0.25
14	14	4.51	0.24
13	14	4.10	0.25
12	14	3.68	0.24
11	14	3.27	0.24
10	14	2.86	0.24
9	14	2.46	0.23
8	14	2.08	0.22
7	14	1.71	0.21
6	14	1.36	0.19
5	14	1.04	0.18
4	14	0.74	0.15
3	14	0.49	0.13
2	14	0.28	0.10
1	14	0.12	0.06
Base	17	0.00	0.00

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000\,\text{ksi}}{50\,\text{ksi}}}$$
$$= 28.9$$

And therefore, the slenderness requirement is satisfied.

A tie bar diameter of ⅝ in. is selected for the uncoupled wall, and the bar requirement considering the stability of the empty steel module during construction is checked as follows:

$$\frac{S}{t_p} \le 1.0\sqrt{\frac{E_s}{2\alpha+1}} \tag{2-4}$$

$$\frac{S}{t_p} = \frac{12\text{ in.}}{½\text{ in.}}$$
$$= 24.0$$

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$
$$= 1.7\left(\frac{18.0\text{ in.}}{½\text{ in.}} - 2\right)\left(\frac{½\text{ in.}}{⅝\text{ in.}}\right)^4$$
$$= 23.7$$

$$1.0\sqrt{\frac{E_s}{2\alpha+1}} = 1.0\sqrt{\frac{29{,}000\text{ ksi}}{2(23.7)+1}}$$
$$= 24.5$$

Therefore, the bar requirement is satisfied.

Step 8. Connection Design of Uncoupled SpeedCore Walls

Connections are designed in accordance with the requirements of AISC *Specification* Chapter J. Either welded or bolted connections are permitted. The composite wall-to-foundation (or subgrade structure) connections are designed based on the calculated demands (required strengths). The composite wall splices are also designed based on the calculated demands at the corresponding locations.

Example 2.2—Wind Design of 15-Story Structure Using Coupled SpeedCore Walls

This example presents the design of coupled SpeedCore walls in the north-south direction of the same building presented in Example 2.1. The coupled north-south walls are shown in Figure 2-10.

Design the coupled, planar SpeedCore walls shown in Figure 2-10 using the given geometry, material properties, and loads.

Given:

Step 1. General Information of the Considered Building

Material properties, building geometry, and floor loads are shown in Step 1 of Example 2.1.

Step 2. Calculation of the Wind Loads

Calculated wind loads for this example are given in Table 2-4.

Solution:

Step 3. Select Preliminary Sizes for SpeedCore Walls and Coupling Beams

The size of the coupled SpeedCore walls and coupling beams are selected in this section as shown in Figures 2-14 and 2-15.

Wall dimensions:

L_{cb} = coupling beam length
= (10 ft)(12 in./ft)
= 120 in.
L_{wall} = wall length
= (12.5 ft)(12 in./ft)
= 150 in.
t_p = wall plate thickness
= ½ in.
t_{sc} = wall thickness
= 18 in.

Coupling beam dimensions:

b_{CB} = coupling beam width
= 18 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf.CB}$ = coupling beam flange plate thickness
= ¾ in.
$t_{pw.CB}$ = coupling beam web plate thickness
= ½ in.

Step 4. Calculation of Base Shear and Overturning Moment

The effective stiffnesses for composite coupling beams and planar walls are considered in a software analysis program to calculate the moment and force distribution in the members.

Table 2-4. Wind Loads in North-South Direction for Each Coupled Wall					
Story No.	Story Elevation (ft)	Wind Loads North-South Direction (Risk Category II)		Wind Loads North-South Direction (MRI 10-Year)	
		Windward Wall (kips)	Leeward Wall (kips)	Windward Wall (kips)	Leeward Wall (kips)
1	17	31.0	–30.1	14.7	–14.3
2	31	28.4	–24.8	13.4	–11.7
3	45	30.3	–24.8	14.3	–11.7
4	59	31.7	–24.8	15.0	–11.7
5	73	32.9	–24.8	15.6	–11.7
6	87	34.0	–24.8	16.1	–11.7
7	101	34.8	–24.8	16.5	–11.7
8	115	35.6	–24.8	16.9	–11.7
9	129	36.4	–24.8	17.2	–11.7
10	143	37.0	–24.8	17.5	–11.7
11	157	37.6	–24.8	17.8	–11.7
12	171	38.2	–24.8	18.1	–11.7
13	185	38.7	–24.8	18.3	–11.7
14	199	39.2	–24.8	18.6	–11.7
15	213	39.7	–24.8	18.8	–11.7
Sum		526	–377	249	–178
		Base Shear	903	Base Shear	427

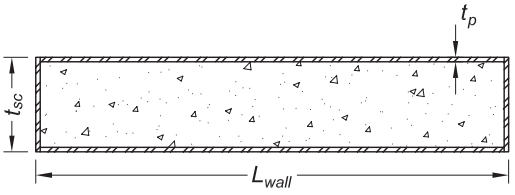


Fig. 2-14. Wall cross-section dimensions.

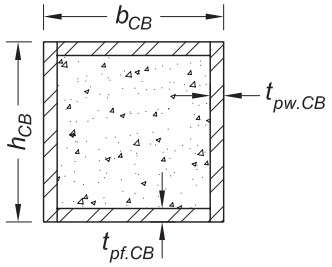


Fig. 2-15. Coupling beam cross-section dimensions.

Effective Flexural and Shear Stiffnesses of Composite Coupling Beams and Planar SpeedCore Walls

The area of steel in each wall is:

$$A_s = t_p\left[2(L_{wall} - 2t_p) + 2t_{sc}\right]$$
$$= (½\text{ in.})\left\{2\left[150\text{ in.} - 2(½\text{ in.})\right] + 2(18\text{ in.})\right\}$$
$$= 167\text{ in.}^2$$

The area of concrete in each wall:

$$A_c = L_{wall}t_{sc} - A_s$$
$$= (150\text{ in.})(18\text{ in.}) - 167\text{ in.}^2$$
$$= 2{,}530\text{ in.}^2$$

The effective axial stiffness of each wall is:

$$EA_{eff} = E_sA_s + 0.45E_cA_c \tag{2-10}$$
$$= (29{,}000\text{ ksi})(167\text{ in.}^2) + 0.45(4{,}500\text{ ksi})(2{,}530\text{ in.}^2)$$
$$= 9{,}970{,}000\text{ kips}$$

The area of steel in the direction of shear is:

$$A_{sw} = 2L_{wall}t_p$$
$$= 2(150\text{ in.})(½\text{ in.})$$
$$= 150\text{ in.}^2$$

The effective shear stiffness of each wall is:

$$GA_{v.eff} = G_sA_{sw} + G_cA_c \tag{2-11}$$
$$= (11{,}200\text{ ksi})(150\text{ in.}^2) + (1{,}800\text{ ksi})(2{,}530\text{ in.}^2)$$
$$= 6{,}230{,}000\text{ kips}$$

The moment of inertia of the steel in each wall is:

$$I_s = 2\left[\frac{t_p(L_{wall} - 2t_p)^3}{12}\right] + 2\left[\left(\frac{t_{sc}t_p^3}{12}\right) + t_{sc}t_p\left(\frac{L_{wall}}{2} - \frac{t_p}{2}\right)^2\right]$$
$$= 2\left\{\frac{(½\text{ in.})\left[150\text{ in.} - 2(½\text{ in.})\right]^3}{12}\right\} + 2\left[\frac{(18\text{ in.})(½\text{ in.})^3}{12} + (18\text{ in.})(½\text{ in.})\left(\frac{150\text{ in.}}{2} - \frac{½\text{ in.}}{2}\right)^2\right]$$
$$= 376{,}000\text{ in.}^4$$

The moment of inertia of concrete in each wall is:

$$I_c = \frac{(t_{sc} - 2t_p)(L_{wall} - 2t_p)^3}{12}$$
$$= \frac{\left[18\text{ in.} - 2(½\text{ in.})\right]\left[150\text{ in.} - 2(½\text{ in.})\right]^3}{12}$$
$$= 4{,}690{,}000\text{ in.}^4$$

The effective flexural stiffness of each wall is:

$$EI_{eff} = E_s I_s + 0.35E_c I_c \tag{2-9}$$
$$= (29{,}000 \text{ ksi})(376{,}000 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(4{,}690{,}000 \text{ in.}^4)$$
$$= 1.83 \times 10^{10} \text{ kip-in.}^2$$

Coupling Beam Properties

The area of steel in the coupling beam is:

$$A_{s.CB} = 2t_{pw.CB}h_{CB} + 2(b_{CB} - 2t_{pw.CB})t_{pf.CB}$$
$$= \left[2(½ \text{ in.})(24 \text{ in.})\right] + \left\{2\left[18 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})\right\}$$
$$= 49.5 \text{ in.}^2$$

The area of concrete in the coupling beam is:

$$A_{c.CB} = (b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB})$$
$$= \left[18 \text{ in.} - 2(½ \text{ in.})\right]\left[24 \text{ in.} - 2(¾ \text{ in.})\right]$$
$$= 383 \text{ in.}^2$$

The uncracked axial stiffness of concrete in the coupling beam is:

$$EA_{uncr.CB} = E_s A_{s.CB} + E_c A_{c.CB}$$
$$= (29{,}000 \text{ ksi})(49.5 \text{ in.}^2) + (4{,}500 \text{ ksi})(383 \text{ in.}^2)$$
$$= 3{,}160{,}000 \text{ kips}$$

The moment of inertia of the steel section about the elastic neutral axis in the coupling beam is:

$$I_{s.CB} = 2\left[\frac{t_{pw.CB}h_{CB}^3}{12} + \frac{(b_{CB} - 2t_{pw.CB})t_{pf.CB}^3}{12} + (b_{CB} - 2t_{pw.CB})t_{pf.CB}\left(\frac{h_{CB}}{2} - \frac{t_{pf.CB}}{2}\right)^2\right]$$
$$= 2\left\{\frac{(½ \text{ in.})(24 \text{ in.})^3}{12} + \frac{\left[18 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})^3}{12} + \left[18 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})\left(\frac{24 \text{ in.}}{2} - \frac{¾ \text{ in.}}{2}\right)^2\right\}$$
$$= 4{,}600 \text{ in.}^4$$

The moment of inertia of the concrete section about the elastic neutral axis in the coupling beam is:

$$I_{c.CB} = \frac{(b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB})^3}{12}$$
$$= \frac{\left[18 \text{ in.} - 2(½ \text{ in.})\right]\left[24 \text{ in.} - 2(¾ \text{ in.})\right]^3}{12}$$
$$= 16{,}100 \text{ in.}^4$$

From AISC *Specification* Section I2.2b, Equation I2-13, the effective rigidity coefficient is:

$$C_{3.CB} = 0.45 + 3\left(\frac{A_{s.CB}}{b_{CB}h_{CB}}\right) \le 0.9 \quad \text{(from Eq. 2-18)}$$

$$= 0.45 + 3\left[\frac{49.5 \text{ in.}^2}{(18 \text{ in.})(24 \text{ in.})}\right]$$

$$= 0.794$$

The effective flexural stiffness of the coupling beam is:

$$EI_{eff.CB} = E_s I_{s.CB} + C_{3.CB} E_c I_{c.CB} \quad \text{(from Eq. 2-17)}$$

$$= (29{,}000 \text{ ksi})(4{,}600 \text{ in.}^4) + 0.794(4{,}500 \text{ ksi})(16{,}100 \text{ in.}^4)$$

$$= 1.91 \times 10^8 \text{ kip-in.}^2$$

The area of steel in the direction of shear is:

$$A_{sw.CB} = 2h_{CB}t_{pw.CB}$$

$$= 2(24 \text{ in.})(½ \text{ in.})$$

$$= 24.0 \text{ in.}^2$$

$$GA_{v.CB} = G_s A_{sw.CB} + G_c A_{c.CB} \quad \text{(from Eq. 2-11)}$$

$$= (11{,}200 \text{ ksi})(24.0 \text{ in.}^2) + (1{,}800 \text{ ksi})(383 \text{ in.}^2)$$

$$= 958{,}000 \text{ kips}$$

The effective flexural stiffness of the coupling beam, from Section 2.3.2, is:

$$0.64EI_{eff.CB} = 0.64(1.91 \times 10^8 \text{ kip-in.}^2)$$

$$= 1.22 \times 10^8 \text{ kip-in.}^2$$

The reduced axial stiffness of the coupling beam, from Section 2.3.2, is:

$$0.8EA_{uncr.CB} = 0.8(3{,}160{,}000 \text{ kips})$$

$$= 2{,}530{,}000 \text{ kips}$$

Because the rotational stiffness of the wall-to-foundation connection has effects on the moment and force distribution and lateral deflection of the coupled SpeedCore wall, it is included in the finite element model.

Rotational Stiffness of the Wall-to-Foundation Connection

It is assumed that the wall-to-foundation connection provides a fully restrained connection. The rotational stiffness of a fully restrained wall-to-foundation connection, $K_{s.con}$, is calculated according to AISC *Specification* Figure C-B3.3:

$$K_{s.con} = \frac{20EI_{eff}}{H_{wall}}$$

$$= \frac{20(1.83 \times 10^{10} \text{ kip-in.}^2)}{(213 \text{ ft})(12 \text{ in./ft})}$$

$$= 1.43 \times 10^8 \text{ kip-in./rad}$$

From a structural analysis program, the required forces and moments in each of the two SpeedCore walls and coupling beam for Risk Category II are as follows:

$M_{r.CB}$ = required flexural strength for coupling beams
= 22,800 kip-in.
$M_{r.wall}$ = required flexural strength for each planar SpeedCore wall
= 147,000 kip-in.
$P_{r.wall}$ = required compression force in the compression SpeedCore wall
= 6,110 kips
$T_{r.wall}$ = required tensile force in the tension SpeedCore wall
= 1,310 kips
$V_{r.CB}$ = required shear force for coupling beams
= 381 kips
$V_{r.wall}$ = required shear strength for each planar SpeedCore wall
= 458 kips

Step 5. Strength Check of SpeedCore Walls and Coupling Beams

In this step, the design flexural and shear strengths of coupling beams and uncoupled SpeedCore walls are calculated and compared with the required shear forces and moments.

Flexural Strength of the Coupling Beam

The width of concrete in the coupling beam is:

$$t_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 18 \text{ in.} - 2(½ \text{ in.})$$
$$= 17.0 \text{ in.}$$

The plastic neutral axis location is:

$$C_{CB} = \frac{2t_{pw.CB}h_{CB}F_y + 0.85f'_c t_{c.CB} t_{pf.CB}}{4t_{pw.CB}F_y + 0.85f'_c t_{c.CB}}$$
$$= \frac{2(½ \text{ in.})(24 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(17.0 \text{ in.})(¾ \text{ in.})}{4(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(17.0 \text{ in.})}$$
$$= 6.78 \text{ in.}$$

(Note: The plastic neutral axis location, C_{CB}; compression forces, C_1, C_2, C_3; and tension forces, T_1, T_2, for a rectangular coupling beam are depicted in Appendix A, Figure A-13.)

The compression force in the flange is:

$$C_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$
$$= \left[18 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})(50 \text{ ksi})$$
$$= 638 \text{ kips}$$

The compression force in the web is:

$$C_2 = 2t_{pw.CB}C_{CB}F_y$$
$$= 2(½ \text{ in.})(6.78 \text{ in.})(50 \text{ ksi})$$
$$= 339 \text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85 f'_c t_{c.CB} (C_{CB} - t_{pf.CB})$$
$$= 0.85(6 \text{ ksi})(17.0 \text{ in.})(6.78 \text{ in.} - ¾ \text{ in.})$$
$$= 523 \text{ kips}$$

The tension force in the flange is:

$$T_1 = t_{pf.CB} (b_{CB} - 2t_{pw.CB}) F_y$$
$$= (¾ \text{ in.})\left[18 \text{ in.} - 2(½ \text{ in.})\right](50 \text{ ksi})$$
$$= 638 \text{ kips}$$

The tension force in the web is:

$$T_2 = 2t_{pw.CB} (h_{CB} - C_{CB}) F_y$$
$$= 2(½ \text{ in.})(24 \text{ in.} - 6.78 \text{ in.})(50 \text{ ksi})$$
$$= 861 \text{ kips}$$

The plastic flexural strength of the coupling beam is:

$$M_{Pn.CB} = C_1\left(C_{CB} - \frac{t_{pf.CB}}{2}\right) + C_2\left(\frac{C_{CB}}{2}\right) + C_3\left(\frac{C_{CB} - t_{pf.CB}}{2}\right) + T_1\left(h_{CB} - C_{CB} - \frac{t_{pf.CB}}{2}\right) + T_2\left(\frac{h_{CB} - C_{CB}}{2}\right)$$
$$= (638 \text{ kips})\left(6.78 \text{ in.} - \frac{¾ \text{ in.}}{2}\right) + (339 \text{ kips})\left(\frac{6.78 \text{ in.}}{2}\right) + (523 \text{ kips})\left(\frac{6.78 \text{ in.} - ¾ \text{ in.}}{2}\right)$$
$$+ (638 \text{ kips})\left(24 \text{ in.} - 6.78 \text{ in.} - \frac{¾ \text{ in.}}{2}\right) + (861 \text{ kips})\left(\frac{24 \text{ in.} - 6.78 \text{ in.}}{2}\right)$$
$$= 25{,}000 \text{ kip-in.}$$

The nominal flexural strength of the coupling beam is:

$$M_{n.CB} = M_{Pn.CB} = 25{,}000 \text{ kip-in.}$$

Using $\phi_b = 0.90$, the design flexural strength of the coupling beam is:

$$\phi_b M_{n.CB} = 0.90(25{,}000 \text{ kip-in.})$$
$$= 22{,}500 \text{ kip-in.} < M_{r.CB} = 22{,}800 \text{ kip-in.}$$

The ratio of required flexural strength to design flexural strength exceeds 1.0. The coupling beam design would be adjusted so that the beam's design strength is greater than the required strength. Increasing the coupling beam flange plate thickness, $t_{pf.CB}$, to ⅞ in. would satisfy the required flexural strength. The iterative design process is not fully shown in this example.

Shear Strength of the Coupling Beam

The area of the steel web in shear in the coupling beam is:

$$A_{sw.CB} = 2h_{CB} t_{pw.CB}$$
$$= 2(24 \text{ in.})(½ \text{ in.})$$
$$= 24.0 \text{ in.}^2$$

The nominal shear strength of the coupling beam is:

$$V_{n.CB} = 0.6F_y A_{sw.CB} + 0.06A_{c.CB}\sqrt{f'_c} \quad \text{(from Eq. 2-19)}$$
$$= 0.6(50\text{ ksi})(24.0\text{ in.}) + 0.06(383\text{ in.}^2)\sqrt{6\text{ ksi}}$$
$$= 776\text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the coupling beam is:

$$\phi_v V_{n.CB} = 0.90(776\text{ kips})$$
$$= 698\text{ kips} > V_{r.CB} = 381\text{ kips}$$

Flexural Strength of the SpeedCore Wall in Tension

It is assumed that the tensile axial force acts at the elastic centroid of the walls. Figure 2-16 illustrates the plastic neutral axis location and the compression and tension force regions in the wall.

As stated in the Given, the tensile force in the wall is:

$$T_{r.wall} = 1{,}310\text{ kips}$$

Plastic neutral axis location:

$$C_T = \frac{-T_{r.wall} + 2t_p L_{wall} F_y + 0.85f'_c(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85f'_c(t_{sc} - 2t_p)}$$
$$= \frac{-1{,}310\text{ kips} + 2(½\text{ in.})(150\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[18\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})}{4(½\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[18\text{ in.} - 2(½\text{ in.})\right]}$$
$$= 33.4\text{ in.}$$

The compression force in the flange plate is:

$$C_{1.T} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[18\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})(50\text{ ksi})$$
$$= 425\text{ kips}$$

The compression force in the web plate is:

$$C_{2.T} = 2t_p C_T F_y$$
$$= 2(½\text{ in.})(33.4\text{ in.})(50\text{ ksi})$$
$$= 1{,}670\text{ kips}$$

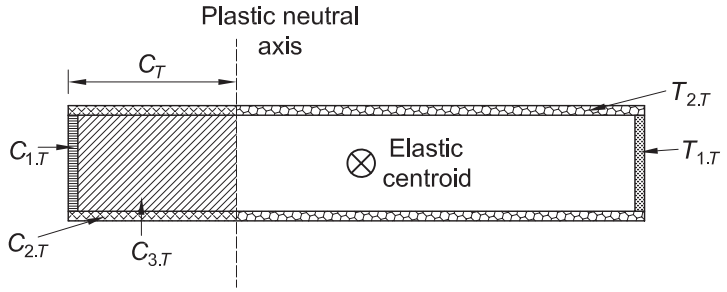


Fig. 2-16. Cross section with labeled regions for plastic moment calculation of wall in tension.

The concrete compression force is:

$$C_{3.T} = 0.85 f_c'(t_{sc} - 2t_p)(C_T - t_p)$$
$$= 0.85(6 \text{ ksi})\left[18 \text{ in.} - 2(½ \text{ in.})\right](33.4 \text{ in.} - ½ \text{ in.})$$
$$= 2{,}850 \text{ kips}$$

The tension force in the flange plate is:

$$T_{1.T} = t_p(t_{sc} - 2t_p)\, F_y$$
$$= (½ \text{ in.})\left[18 \text{ in.} - 2(½ \text{ in.})\right](50 \text{ ksi})$$
$$= 425 \text{ kips}$$

The tension force in the web plate is:

$$T_{2.T} = 2t_p(L_{wall} - C_T)F_y$$
$$= 2(½ \text{ in.})(150 \text{ in.} - 33.4 \text{ in.})(50 \text{ ksi})$$
$$= 5{,}830 \text{ kips}$$

The plastic flexural strength of the planar SpeedCore wall in tension is:

$$M_{PT.wall} = C_{1.T}\left(C_T - \frac{t_p}{2}\right) + C_{2.T}\left(\frac{C_T}{2}\right) + C_{3.T}\left(\frac{C_T - t_p}{2}\right) + T_{1.T}\left(L_{wall} - C_T - \frac{t_p}{2}\right) + T_{2.T}\left(\frac{L_{wall} - C_T}{2}\right)$$
$$+ T_{r.wall}\left(C_T - \frac{L_{wall}}{2}\right)$$
$$= (425 \text{ kips})\left(33.4 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (1{,}670 \text{ kips})\left(\frac{33.4 \text{ in.}}{2}\right) + (2{,}850 \text{ kips})\left(\frac{33.4 \text{ in.} - ½ \text{ in.}}{2}\right)$$
$$+ (425 \text{ kips})\left(150 \text{ in.} - 33.4 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (5{,}830 \text{ kips})\left(\frac{150 \text{ in.} - 33.4 \text{ in.}}{2}\right)$$
$$+ (1{,}310 \text{ kips})\left(33.4 \text{ in.} - \frac{150 \text{ in.}}{2}\right)$$
$$= 424{,}000 \text{ kip-in.}$$

The nominal flexural strength of the planar SpeedCore wall in tension is:

$$M_{nT.wall} = M_{PT.wall} = 424{,}000 \text{ kip-in.}$$

Flexural Strength of the SpeedCore Wall in Compression

Figure 2-17 illustrates the plastic neutral axis location and the compression and tension force regions in the wall.

As stated in the Given, the compression force in the compression wall is:

$$P_{r.wall} = 6{,}110 \text{ kips}$$

The location of the plastic neutral axis is:

$$C_C = \frac{P_{r.wall} + 2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$
$$= \frac{6{,}110 \text{ kips} + 2(½ \text{ in.})(150 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})\left[18 \text{ in.} - 2(½ \text{ in.})\right](½ \text{ in.})}{4(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})\left[18 \text{ in.} - 2(½ \text{ in.})\right]}$$
$$= 73.1 \text{ in.}$$

The compression force in the flange plate is:

$$C_{1.C} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[18 \text{ in.} - 2(½ \text{ in.})\right](½ \text{ in.})(50 \text{ ksi})$$
$$= 425 \text{ kips}$$

The compression force in the web plate is:

$$C_{2.C} = 2t_p C_C F_y$$
$$= 2(½ \text{ in.})(73.1 \text{ in.})(50 \text{ ksi})$$
$$= 3{,}660 \text{ kips}$$

The compression force in the concrete is:

$$C_{3.C} = 0.85 f_c'(t_{sc} - 2t_p)(C_C - t_p)$$
$$= 0.85(6 \text{ ksi})\left[18 \text{ in.} - 2(½ \text{ in.})\right](73.1 \text{ in.} - ½ \text{ in.})$$
$$= 6{,}290 \text{ kips}$$

The tensile force in the flange plate is:

$$T_{1.C} = (t_{sc} - 2t_p)\, t_p F_y$$
$$= \left[18 \text{ in.} - 2(½ \text{ in.})\right](½ \text{ in.})(50 \text{ ksi})$$
$$= 425 \text{ kips}$$

The tensile force in the web plates is:

$$T_{2.C} = 2t_p(L_{wall} - C_C)F_y$$
$$= 2(½ \text{ in.})(150 \text{ in.} - 73.1 \text{ in.})(50 \text{ ksi})$$
$$= 3{,}850 \text{ kips}$$

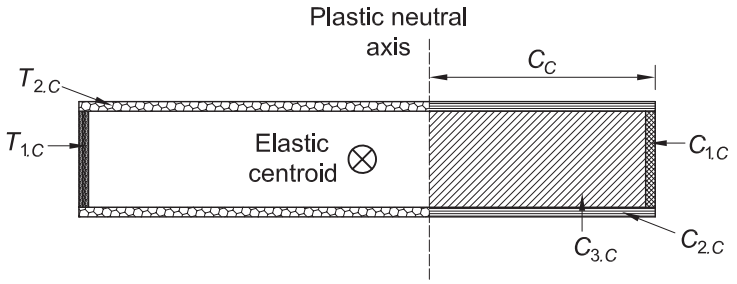


Fig. 2-17. Cross section with labeled regions for plastic moment calculation of wall in compression.

The plastic flexural strength of the planar SpeedCore wall in compression is:

$$M_{PC.wall} = C_{1.C}\left(C_C - \frac{t_p}{2}\right) + C_{2.C}\left(\frac{C_C}{2}\right) + C_{3.C}\left(\frac{C_C - t_p}{2}\right) + T_{1.C}\left(L_{wall} - C_C - \frac{t_p}{2}\right) + T_{2.C}\left(\frac{L_{wall} - C_C}{2}\right) + P_{r.wall}\left(\frac{L_{wall}}{2} - C_C\right)$$

$$= (425 \text{ kips})\left(73.1 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (3{,}660 \text{ kips})\left(\frac{73.1 \text{ in.}}{2}\right) + (6{,}290 \text{ kips})\left(\frac{73.1 \text{ in.} - ½ \text{ in.}}{2}\right) + (425 \text{ kips})\left(150 \text{ in.} - 73.1 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (3{,}850 \text{ kips})\left(\frac{150 \text{ in.} - 73.1 \text{ in.}}{2}\right) + (6{,}110 \text{ kips})\left(\frac{150 \text{ in.}}{2} - 73.1 \text{ in.}\right)$$

$$= 585{,}000 \text{ kip-in.}$$

The nominal flexural strength of the planar SpeedCore wall in compression is:

$$M_{nC.wall} = M_{PC.wall} = 585{,}000 \text{ kip-in.}$$

The design flexural strength of the tension wall is:

$$\phi_b M_{nT.wall} = 0.90(424{,}000 \text{ kip-in.})$$
$$= 382{,}000 \text{ kip-in.} > M_{r.wall} = 147{,}000 \text{ kip-in.}$$

The design flexural strength of the compression wall is:

$$\phi_b M_{nC.wall} = 0.90(585{,}000 \text{ kip-in.})$$
$$= 527{,}000 \text{ kip-in.} > M_{r.wall} = 147{,}000 \text{ kip-in.}$$

The ratio of demand to capacity is:

$$\frac{M_{r.wall}}{\phi_b M_{nT.wall}} = \frac{147{,}000 \text{ kip-in.}}{382{,}000 \text{ kip-in.}}$$
$$= 0.385$$

$$\frac{M_{r.wall}}{\phi_b M_{nC.wall}} = \frac{147{,}000 \text{ kip-in.}}{527{,}000 \text{ kip-in.}}$$
$$= 0.279$$

Shear Strength of the SpeedCore Wall

The steel area in the direction of shear is:

$$A_{sw} = 2L_{wall}t_p$$
$$= 2(150 \text{ in.})(½ \text{ in.})$$
$$= 150 \text{ in.}^2$$

$$A_c = (L_{wall} - 2t_p)(t_{sc} - 2t_p)$$
$$= \left[150 \text{ in.} - 2(½ \text{ in.})\right]\left[18 \text{ in.} - 2(½ \text{ in.})\right]$$
$$= 2{,}530 \text{ in.}^2$$

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200 \text{ ksi})(150 \text{ in.}^2)$$
$$= 1{,}680{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_c)(E_s A_{sw})}{4E_s A_{sw} + E_c A_c} \tag{2-16}$$
$$= \frac{0.7(4{,}500 \text{ ksi})(2{,}530 \text{ in.}^2)(29{,}000 \text{ ksi})(150 \text{ in.}^2)}{4(29{,}000 \text{ ksi})(150 \text{ in.}^2) + (4{,}500 \text{ ksi})(2{,}530 \text{ in.}^2)}$$
$$= 1{,}200{,}000 \text{ kips}$$

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \tag{2-14}$$
$$= \frac{1{,}680{,}000 \text{ kips} + 1{,}200{,}000 \text{ kips}}{\sqrt{3(1{,}680{,}000 \text{ kips})^2 + (1{,}200{,}000 \text{ kips})^2}} (50 \text{ ksi})(150 \text{ in.}^2)$$
$$= 6{,}860 \text{ kips}$$

The design shear strength of the SpeedCore wall is:

$$\phi_v V_{n.wall} = 0.90(6{,}860 \text{ kips})$$
$$= 6{,}170 \text{ kips} > V_{r.wall} = 458 \text{ kips}$$

Ratio of demand-to-capacity is:

$$\frac{V_{r.wall}}{\phi_v V_{n.wall}} = \frac{458 \text{ kips}}{6{,}170 \text{ kips}}$$
$$= 0.0742$$

The tensile strength of the SpeedCore wall is:

$$P_{nT} = F_y A_s \qquad \text{(from Eq. 2-13)}$$
$$= (50 \text{ ksi})(167 \text{ in.}^2)$$
$$= 8{,}350 \text{ kips}$$

Using $\phi_T = 0.90$, the design tensile strength of the SpeedCore wall is:

$$\phi_T P_{nT} = 0.90(8{,}350 \text{ kips})$$
$$= 7{,}520 \text{ kips} > T_{r.wall} = 1{,}310 \text{ kips}$$

The compression strength of the SpeedCore wall is:

$$P_{no} = F_y A_s + 0.85 f'_c A_c \tag{2-12}$$
$$= (50 \text{ ksi})(167 \text{ in.}^2) + 0.85(6 \text{ ksi})(2{,}530 \text{ in.}^2)$$
$$= 21{,}300 \text{ kips}$$

The moment of inertia of the steel about the minor axis is:

$$I_{s.min} = 2\left[\frac{(L_{wall} - 2t_p)t_p^3}{12} + (L_{wall} - 2t_p)\,t_p\left(\frac{t_{sc} - t_p}{2}\right)^2\right] + 2\left(\frac{t_p t_{sc}^3}{12}\right)$$

$$= 2\left\{\frac{[150\text{ in.} - 2(½\text{ in.})](½\text{ in.})^3}{12} + [150\text{ in.} - 2(½\text{ in.})](½\text{ in.})\left(\frac{18\text{ in.} - ½\text{ in.}}{2}\right)^2\right\} + 2\left[\frac{(½\text{ in.})(18\text{ in.})^3}{12}\right]$$

$$= 11{,}900\text{ in.}^4$$

The moment of inertia of concrete about the minor axis is:

$$I_{c.min} = \frac{(L_{wall} - 2t_p)(t_{sc} - 2t_p)^3}{12}$$

$$= \frac{[150\text{ in.} - 2(½\text{ in.})][18\text{ in.} - 2(½\text{ in.})]^3}{12}$$

$$= 61{,}000\text{ in.}^4$$

The effective flexural stiffness of the wall about the minor axis is:

$$EI_{eff.min} = E_s I_{s.min} + 0.35E_c I_{c.min} \qquad \text{(from Eq. 2-9)}$$

$$= (29{,}000\text{ ksi})(11{,}900\text{ in.}^4) + 0.35(4{,}500\text{ ksi})(61{,}000\text{ in.}^4)$$

$$= 4.41\times10^8\text{ kip-in.}^2$$

The critical unsupported length for buckling of the wall is:

$$L_c = \max(h_1, h_{typ})$$

$$= \max(17\text{ ft}, 14\text{ ft})$$

$$= 17\text{ ft}$$

where h_1 is the first story height and h_{typ} is the typical story height.

The Euler buckling load from AISC *Specification* Equation I2-4 is:

$$P_e = \frac{\pi^2 EI_{eff.min}}{L_c^2}$$

$$= \frac{\pi^2(4.41\times10^8\text{ kip-in.}^2)}{\left[(17\text{ ft})(12\text{ in./ft})\right]^2}$$

$$= 105{,}000\text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{21{,}300\text{ kips}}{105{,}000\text{ kips}}$$

$$= 0.203 < 2.25$$

The nominal compressive strength of the wall from AISC *Specification* Equation I2-2 is:

$$P_{nC} = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right)$$

$$= (21{,}300\text{ kips})\left(0.658^{\frac{21{,}300\text{ kips}}{105{,}000\text{ kips}}}\right)$$

$$= 19{,}600\text{ kips}$$

The design compressive strength of the SpeedCore wall is:

$$\phi_c P_{nC} = 0.90(19{,}600 \text{ kips})$$
$$= 17{,}600 \text{ kips} > P_{r.wall} = 6{,}110 \text{ kips}$$

Step 6. Serviceability Check of Coupled SpeedCore Walls

In this step, the lateral deflection of the coupled SpeedCore wall is checked to evaluate the serviceability. Wind loads at a 10-year MRI are used in a finite element model to check the drift limits. Effective flexural and shear stiffnesses of the coupling beams and SpeedCore walls are used in the model.

The lateral deflection due to wind loads at a 10-year MRI is limited to a roof displacement of $H_{wall}/500$, where H_{wall} is the total height of the wall, or an ID of $h/400$, where h is the story height. In this design example, allowable roof displacement and ID are 5.11 in. and 0.25%, respectively. The interstory ratio is determined as lateral displacement between two consecutive floors divided by story height.

Allowable roof displacement:

$$\frac{H_{wall}}{500} = \frac{(213 \text{ ft})(12 \text{ in./ft})}{500}$$
$$= 5.11 \text{ in.}$$

Allowable ID ratio for $h/400$:

$$\text{ID} = 0.25\%$$

A commercial structural analysis software program is used to check roof displacement and ID. Figure 2-18 shows the deformed shape, lateral displacement, and ID of the coupled SpeedCore wall. Table 2-5 summarizes lateral displacement and ID.

Step 7. Detail Design of Coupling Beams

The clear width of the coupling beam flange plate is:

$$b_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 18 \text{ in.} - 2(\tfrac{1}{2} \text{ in.})$$
$$= 17.0 \text{ in.}$$

The clear width of the coupling beam web plate is:

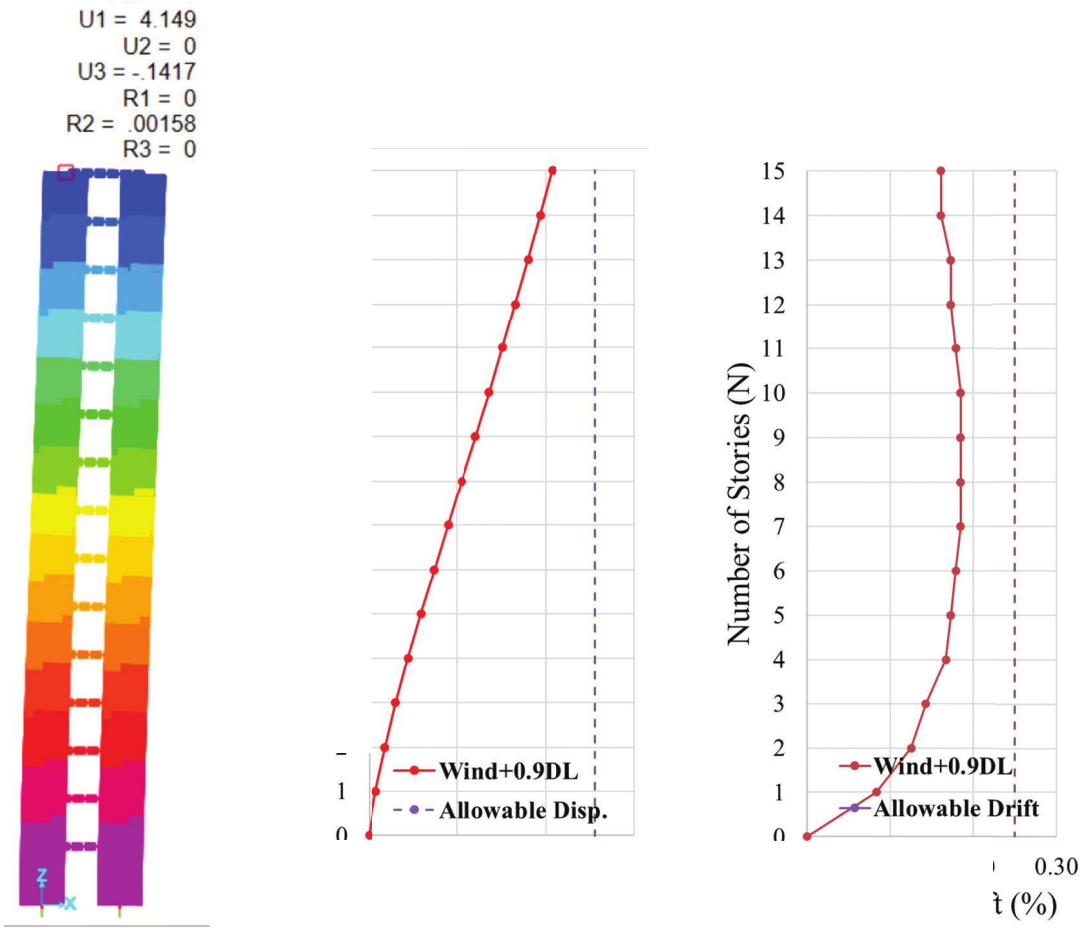
$$h_{c.CB} = h_{CB} - 2t_{pf.CB}$$
$$= 24 \text{ in.} - 2(\tfrac{3}{4} \text{ in.})$$
$$= 22.5 \text{ in.}$$

As stated in Section 2.3.1.1 of this Design Guide, the coupling beam is classified as compact, noncompact, or slender based on the limits given in AISC *Specification* Table I1.1b.

$$\frac{b_{c.CB}}{t_{pf.CB}} = \frac{17.0 \text{ in.}}{\tfrac{3}{4} \text{ in.}}$$
$$= 22.7$$

$$2.26\sqrt{\frac{E_s}{F_y}} = 2.26\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 54.4$$

$$\frac{b_{c.CB}}{t_{pf.CB}} < 2.26\sqrt{\frac{E_s}{F_y}}$$



(a) Deformed shape *(b) Lateral displacement* *(c) ID*

Fig. 2-18. Coupled SpeedCore walls deformed shape, lateral displacement, and ID.

Table 2-5. Lateral Displacement and ID Summary			
Story No.	Story Height (ft)	Displacement (in.)	ID (%)
15	14	4.15	0.16
14	14	3.88	0.16
13	14	3.61	0.17
12	14	3.32	0.17
11	14	3.03	0.18
10	14	2.73	0.18
9	14	2.42	0.18
8	14	2.11	0.18
7	14	1.80	0.18
6	14	1.49	0.18
5	14	1.19	0.17
4	14	0.90	0.17
3	14	0.62	0.14
2	14	0.38	0.13
1	14	0.17	0.08
Base	17	0.00	0

$$\frac{h_{c.CB}}{t_{pw.CB}} = \frac{22.5 \text{ in.}}{\frac{1}{2} \text{ in.}}$$
$$= 45.0$$
$$3.00\sqrt{\frac{E_s}{F_y}} = 3.00\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 72.2$$
$$\frac{h_{c.CB}}{t_{pw.CB}} < 3.00\sqrt{\frac{E_s}{F_y}}$$

The coupling beam section is compact. Note that compact sections are not required for wind design.

Step 8. Detail Design of SpeedCore Walls

The required steel reinforcement ratio of the uncoupled SpeedCore wall is checked as shown in the following.

The minimum steel area required according to AISC *Specification* Section I1.6 is:

$$A_{s.min} = 0.01 L_{wall} t_{sc}$$
$$= 0.01(150 \text{ in.})(18.0 \text{ in.})$$
$$= 27.0 \text{ in.}^2$$

$A_s = 167 \text{ in.}^2 > A_{s.\min} = 27.0 \text{ in.}^2$ **o.k.**

The maximum steel area required according to AISC *Specification* Section I1.6 is:

$$A_{s.max} = 0.1 L_{wall}\, t_{sc}$$
$$= 0.1(150 \text{ in.})(18.0 \text{ in.})$$
$$= 270 \text{ in.}^2$$

$$A_s = 167 \text{ in.}^2 < A_{s.\max} = 270 \text{ in.}^2 \quad \textbf{o.k.}$$

A tie bar spacing, S_{tie}, of 12 in. is selected for the SpeedCore wall, and the slenderness requirement is checked as follows:

$$\frac{S_{tie}}{t_p} = \frac{12 \text{ in.}}{½ \text{ in.}}$$
$$= 24.0$$

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 28.9$$

$$\frac{S_{tie}}{t_p} < 1.2\sqrt{\frac{E_s}{F_y}} \quad \textbf{o.k.} \qquad \text{(from Eq. 2-1)}$$

The tie bar diameter, d_{tie}, is selected as ⅝ in. for the SpeedCore wall, and the bar requirement considering the stability of the empty steel module is checked as follows:

Tie bar requirement for empty steel module stability:

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \qquad (2\text{-}5)$$
$$= 1.7\left(\frac{18 \text{ in.}}{½ \text{ in.}} - 2\right)\left(\frac{½ \text{ in.}}{⅝ \text{ in.}}\right)^4$$
$$= 23.7$$

$$\frac{S_{tie}}{t_p} = 24.0$$

$$1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 1.0\sqrt{\frac{29{,}000 \text{ ksi}}{2(23.7) + 1}}$$
$$= 24.5$$

$$\frac{S_{tie}}{t_p} < 1.0\sqrt{\frac{E_s}{2\alpha + 1}} \quad \textbf{o.k.} \qquad \text{(from Eq. 2-4)}$$

Step 9. Connection Design

Connections are designed in accordance with the requirements of the AISC *Specification* Chapter J. Either welded or bolted connections are permitted. The composite wall-to-foundation (or subgrade structure) connections are designed based on the calculated demands (required strengths). The coupling beam-to-composite wall connections are designed based on the calculated demands at the corresponding levels. The composite wall splices are also designed based on the calculated demands at the corresponding locations.

EXAMPLE 2.3—Wind Design of 22-Story Structure Using Coupled C-Shaped SpeedCore Walls

Example 2.3 presents the wind design of a 22-story office building located in Chicago. This example presents the wind design of the coupled C-shaped SpeedCore walls of the building in the north-south direction, as shown in Figure 2-19.

Design the coupled, C-shaped SpeedCore walls shown in Figure 2-19 using the given geometry, material properties, and loads.

Given:

Material properties and building information are given in Step 1.

Calculated wind loads for this example are given in Step 2. The gravity load (axial compression force) of the C-shaped SpeedCore walls is not considered in this example.

Step 1. General Information of the Considered Building

Building geometry:

H_{wall} = total wall height
= 311 ft
L_f = building length
= 200 ft
W_f = building width
= 120 ft
h_{typ} = typical story height
= 14 ft
h_1 = first-story height
= 17 ft
n = number of stories
= 22

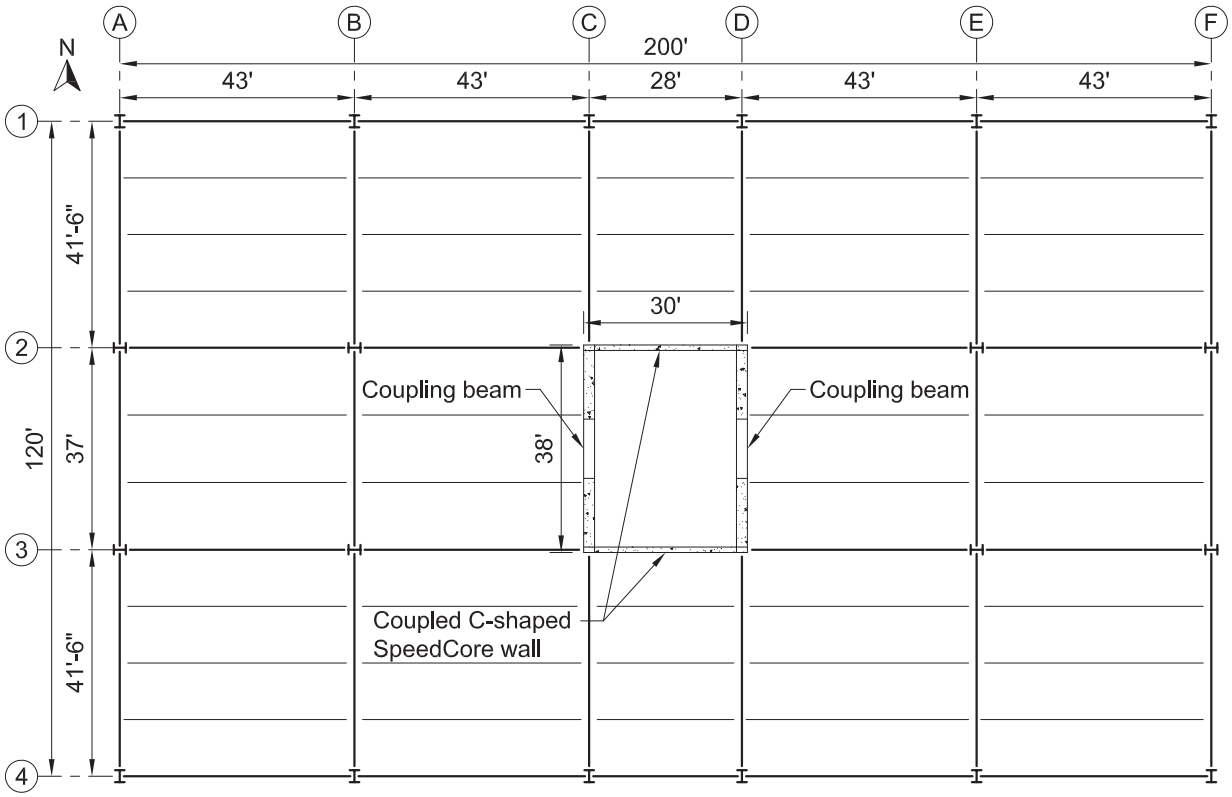


Fig. 2-19. Building floor plan for Example 2.3.

Floor load:

DL = floor dead load
= 0.12 ksf

Material properties are as follows:

ASTM A572/A572M Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi

For ASTM A572/A572M Grade 50 steel, from AISC *Manual* Table 2-4, the steel material properties are as follows:

F_y = 50 ksi
F_u = 65 ksi

Step 2. Calculation of the Wind Loads

The building is located in Chicago and wind speeds for this city (ASCE/SEI 7, Chapter 26) are shown in Table 2-1. Wind speeds at Risk Category II and a 10-year MRI are used to calculate wind loads for the design and serviceability checks, respectively. The calculated wind loads for the design of coupled C-shaped SpeedCore walls in the north-south direction are given in Table 2-6.

Solution:

Step 3. Select Preliminary Sizes for SpeedCore Walls and Coupling Beams

The size of the C-shaped SpeedCore wall and coupling beam are selected in this section. C-shaped SpeedCore walls and coupling beam dimensions are as follows (see Figures 2-20 and 2-21):

L_{cb} = clear span coupling beam length
= 10 ft (12 in./ft)
= 120 in.
L_{wall} = web wall length
= 30 ft (12 in./ft)
= 360 in.
W_{wall} = flange wall length
= 14 ft (12 in./ft)
= 168 in.
t_p = wall plate thickness
= ½ in.
$t_{sc.w}$ = web wall thickness
= 14 in.
$t_{sc.f}$ = flange wall thickness
= 24 in.

Coupling beam dimensions:

b_{CB} = coupling beam width
= 24 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf.CB}$ = coupling beam flange plate thickness
= ¾ in.
$t_{pw.CB}$ = coupling beam web plate thickness
= ½ in.

Table 2-6. Wind Loads in North-South Direction					
Story No.	Story Elevation (ft)	Wind Loads North-South Direction (Risk Category II)		Wind Loads North-South Direction (MRI 10-Year)	
		Windward Wall (kips)	Leeward Wall (kips)	Windward Wall (kips)	Leeward Wall (kips)
1	17	62.4	–64.6	29.4	–30.5
2	31	57.0	–53.2	26.9	–25.1
3	45	60.8	–53.2	28.7	–25.1
4	59	63.8	–53.2	30.1	–25.1
5	73	66.2	–53.2	31.2	–25.1
6	87	68.2	–53.2	32.2	–25.1
7	101	70.0	–53.2	33.0	–25.1
8	115	71.6	–53.2	33.8	–25.1
9	129	73.1	–53.2	34.4	–25.1
10	143	74.4	–53.2	35.1	–25.1
11	157	75.6	–53.2	35.6	–25.1
12	171	76.7	–53.2	36.2	–25.1
13	185	77.8	–53.2	36.7	–25.1
14	199	78.8	–53.2	37.1	–25.1
15	213	79.7	–53.2	37.6	–25.1
16	227	80.6	–53.2	38.0	–25.1
17	241	81.4	–53.2	38.4	–25.1
18	255	82.2	–53.2	38.8	–25.1
19	269	83.0	–53.2	39.1	–25.1
20	283	83.7	–53.2	39.5	–25.1
21	297	84.5	–53.2	39.8	–25.1
22	311	85.1	–53.2	40.1	–25.1

Step 4. Calculation of Base Shear and Overturning Moment

The effective stiffnesses for composite coupling beams and C-shaped SpeedCore walls are considered in a structural analysis software program to calculate the moment and force distribution in the members.

Effective Flexural and Shear Stiffnesses of Composite Coupling Beams and C-Shaped Walls

Coupling Beam Properties

The area of steel in the coupling beam is:

$$A_{s.CB} = 2t_{pw.CB}h_{CB} + 2(b_{CB} - 2t_{pw.CB})t_{pf.CB}$$
$$= 2(\text{½ in.})(24\text{ in.}) + 2\left[24\text{ in.} - 2(\text{½ in.})\right](\text{¾ in.})$$
$$= 58.5\text{ in.}^2$$

The area of concrete in the coupling beam is:

$$A_{c.CB} = (b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB})$$
$$= \left[24 \text{ in.} - 2(½ \text{ in.})\right]\left[24 \text{ in.} - 2(¾ \text{ in.})\right]$$
$$= 518 \text{ in.}^2$$

The uncracked axial stiffness of the concrete in the coupling beam is:

$$EA_{uncr.CB} = E_s A_{s.CB} + E_c A_{c.CB}$$
$$= (29{,}000 \text{ ksi})(58.5 \text{ in.}^2) + (4{,}500 \text{ ksi})(518 \text{ in.}^2)$$
$$= 4{,}030{,}000 \text{ kips}$$

The moment of inertia of the steel section about the elastic neutral axis in the coupling beam is:

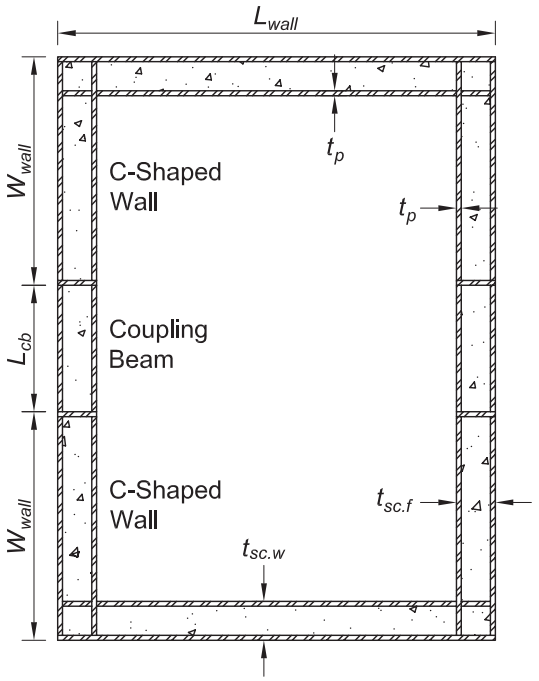


Fig. 2-20. Wall cross-sectional dimensions.

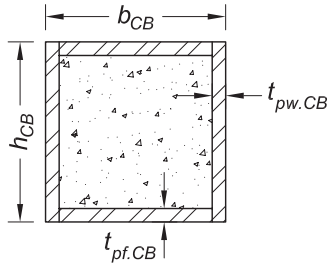


Fig. 2-21. Coupling beam cross-sectional dimensions.

$$I_{s.CB} = \left(\frac{2t_{pw.CB}h_{CB}^3}{12}\right) + \left[\frac{2(b_{CB} - 2t_{pw.CB})t_{pf.CB}^3}{12}\right] + 2\left[(b_{CB} - 2t_{pw.CB})t_{pf.CB}\right]\left(\frac{h_{CB}}{2} - \frac{t_{pf.CB}}{2}\right)^2$$

$$= \left[\frac{2(½\text{ in.})(24\text{ in.})^3}{12}\right] + \left\{\frac{2\left[24\text{ in.} - 2(½\text{ in.})\right](¾\text{ in.})^3}{12}\right\} + 2\left[24\text{ in.} - 2(½\text{ in.})\right](¾\text{ in.})\left(\frac{24\text{ in.}}{2} - \frac{¾\text{ in.}}{2}\right)^2$$

$$= 5{,}820\text{ in.}^4$$

The moment of inertia of the concrete section about the elastic neutral axis in the coupling beam is:

$$I_{c.CB} = \frac{(b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB})^3}{12}$$

$$= \frac{\left[24\text{ in.} - 2(½\text{ in.})\right]\left[24\text{ in.} - 2(¾\text{ in.})\right]^3}{12}$$

$$= 21{,}800\text{ in.}^4$$

From AISC *Specification* Section I2.2b, Equation I2-13, the effective rigidity coefficient is:

$$C_{3.CB} = 0.45 + 3\left(\frac{A_{s.CB}}{b_{CB}h_{CB}}\right) \le 0.9$$

$$= 0.45 + 3\left[\frac{58.5\text{ in.}^2}{(24\text{ in.})(24\text{ in.})}\right]$$

$$= 0.755$$

The effective stiffness of the coupling beam from AISC *Specification* Equation I2-12 is:

$$EI_{eff.CB} = E_sI_{s.CB} + C_3E_cI_{c.CB}$$

$$= (29{,}000\text{ ksi})(5{,}820\text{ in.}^4) + 0.755(4{,}500\text{ ksi})(21{,}800\text{ in.}^4)$$

$$= 2.43 \times 10^8\text{ kip-in.}^2$$

The area of the steel web of the coupling beam in the direction of shear is:

$$A_{sw.CB} = 2h_{CB}t_{pw.CB}$$

$$= 2(24\text{ in.})(½\text{ in.})$$

$$= 24\text{ in.}^2$$

$$GA_{v.CB} = G_sA_{sw.CB} + G_cA_{c.CB} \qquad \text{(from Eq. 2-11)}$$

$$= (11{,}200\text{ ksi})(24.0\text{ in.}^2) + (1{,}800\text{ ksi})(518\text{ in.}^2)$$

$$= 1{,}200{,}000\text{ kips}$$

From Section 2.3.2, the effective flexural stiffnesses of the coupling beam is:

$$0.64EI_{eff.CB} = 0.64(2.43 \times 10^8\text{ kip-in.}^2)$$

$$= 1.56 \times 10^8\text{ kip-in.}^2$$

The reduced axial stiffness of the coupling beam, from Section 2.3.2, is:

$$0.8EA_{uncr.CB} = 0.8(4.03 \times 10^6\text{ kips})$$

$$= 3.22 \times 10^6\text{ kips}$$

C-Shaped SpeedCore Wall Properties

The area of steel in the wall is:

$$A_s = t_p(4W_{wall} + 2L_{wall} + 2t_{sc.f} - 12t_p)$$
$$= (½\text{ in.})\left[4(168\text{ in.}) + 2(360\text{ in.}) + 2(24\text{ in.}) - 12(½\text{ in.})\right]$$
$$= 717\text{ in.}^2$$

The area of concrete in the wall is:

$$A_c = 2W_{wall}t_{sc.f} + (L_{wall} - 2t_{sc.f})t_{sc.w} - A_s$$
$$= 2(168\text{ in.})(24\text{ in.}) + \left[(360\text{ in.}) - 2(24\text{ in.})\right](14\text{ in.}) - 717\text{ in.}^2$$
$$= 11{,}700\text{ in.}^2$$

The effective axial stiffness of the wall is:

$$EA_{eff} = E_sA_s + 0.45E_cA_c \qquad (2\text{-}10)$$
$$= (29{,}000\text{ ksi})(717\text{ in.}^2) + 0.45(4{,}500\text{ ksi})(11{,}700\text{ in.}^2)$$
$$= 4.45\times10^7\text{ kips}$$

The area of steel in the direction of shear is:

$$A_{sw} = 4W_{wall}t_p$$
$$= 4(168\text{ in.})(½\text{ in.})$$
$$= 336\text{ in.}^2$$

The area of concrete in the direction of shear is:

$$A_{cw} = 2(W_{wall} - 3t_p)(t_{sc.f} - 2t_p)$$
$$= 2\left[168\text{ in.} - 3(½\text{ in.})\right]\left[24\text{ in.} - 2(½\text{ in.})\right]$$
$$= 7{,}660\text{ in.}^2$$

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_sA_{sw} + G_cA_{cw} \qquad \text{(from Eq. 2-11)}$$
$$= (11{,}200\text{ ksi})(336\text{ in.}^2) + (1{,}800\text{ ksi})(7{,}660\text{ in.}^2)$$
$$= 1.76\times10^7\text{ kips}$$

The center of area in the Y-direction of the C-shaped wall is:

$$\overline{y} = \frac{\sum Ay}{A_s + \frac{E_c}{E_s}A_c}$$

$$\sum Ay = 4W_{wall}t_p\frac{W_{wall}}{2} + (L_{wall}-4t_p)t_p\left(t_{sc.w}-\frac{t_p}{2}\right) + (L_{wall}-4t_p)t_p\left(\frac{t_p}{2}\right) + 2(t_{sc.f}-2t_p)t_p\left(W_{wall}-\frac{t_p}{2}\right)$$
$$+\frac{E_c}{E_s}\left[2(W_{wall}-t_{sc.w}-t_p)(t_{sc.f}-2t_p)\left(\frac{W_{wall}-t_{sc.w}-t_p}{2}+t_{sc.w}\right)+(L_{wall}-4t_p)(t_{sc.w}-2t_p)\left(\frac{t_{sc.w}}{2}\right)\right]$$
$$=4(168\text{ in.})(½\text{ in.})\left(\frac{168\text{ in.}}{2}\right)+[360\text{ in.}-4(½\text{ in.})](½\text{ in.})\left(14\text{ in.}-\frac{½\text{ in.}}{2}\right)+[360\text{ in.}-4(½\text{ in.})](½\text{ in.})\left(\frac{½\text{ in.}}{2}\right)$$
$$+2[24\text{ in.}-2(½\text{ in.})](½\text{ in.})\left(168\text{ in.}-\frac{½\text{ in.}}{2}\right)$$
$$+\frac{4{,}500\text{ ksi}}{29{,}000\text{ ksi}}\left\{\begin{array}{l}2(168\text{ in.}-14\text{ in.}-½\text{ in.})[24\text{ in.}-2(½\text{ in.})]\left(\frac{168\text{ in.}-14\text{ in.}-½\text{ in.}}{2}+14\text{ in.}\right)\\+[360\text{ in.}-4(½\text{ in.})][14\text{ in.}-2(½\text{ in.})]\left(\frac{14\text{ in.}}{2}\right)\end{array}\right\}$$
$$=139{,}000\text{ in.}^3$$

$$\bar{y}=\frac{\sum Ay}{A_s+\frac{E_c}{E_s}A_c}$$
$$=\frac{139{,}000\text{ in.}^3}{717\text{ in.}^2+\left(\frac{4{,}500\text{ ksi}}{29{,}000\text{ ksi}}\right)(11{,}700\text{ in.}^2)}$$
$$=54.9\text{ in.}$$

The moment of inertia of the steel in the wall is:

$$I_s=4\left[\frac{t_pW_{wall}^3}{12}+(W_{wall}t_p)\left(\frac{W_{wall}}{2}-\bar{y}\right)^2\right]+2\left[\frac{(t_{sc.f}-2t_p)t_p^3}{12}+(t_{sc.f}-2t_p)t_p\left(W_{wall}-\frac{t_p}{2}-\bar{y}\right)^2\right]$$
$$+\left[\frac{(L_{wall}-4t_p)t_p^3}{12}\right]+(L_{wall}-4t_p)t_p\left(\bar{y}-\frac{t_p}{2}\right)^2+\left[\frac{(L_{wall}-4t_p)t_p^3}{12}\right]+(L_{wall}-4t_p)t_p\left[\bar{y}-\left(t_{sc.w}-\frac{t_p}{2}\right)\right]^2$$
$$=4\left[\frac{(½\text{ in.})(168\text{ in.})^3}{12}+(168\text{ in.})(½\text{ in.})\left(\frac{168\text{ in.}}{2}-54.9\text{ in.}\right)^2\right]$$
$$+2\left\{\frac{[24\text{ in.}-2(½\text{ in.})](½\text{ in.})^3}{12}+[24\text{ in.}-2(½\text{ in.})](½\text{ in.})\left(168\text{ in.}-\frac{½\text{ in.}}{2}-54.9\text{ in.}\right)^2\right\}$$
$$+\frac{[360\text{ in.}-4(½\text{ in.})](½\text{ in.})^3}{12}+[360\text{ in.}-4(½\text{ in.})](½\text{ in.})\left(54.9\text{ in.}-\frac{½\text{ in.}}{2}\right)^2$$
$$+\frac{[360\text{ in.}-4(½\text{ in.})](½\text{ in.})^3}{12}+[360\text{ in.}-4(½\text{ in.})](½\text{ in.})\left[54.9\text{ in.}-\left(14\text{ in.}-\frac{½\text{ in.}}{2}\right)\right]^2$$
$$=2{,}210{,}000\text{ in.}^4$$

The moment of inertia of concrete in the wall is:

$$I_c = 2\left[\frac{(t_{sc.f}-2t_p)(W_{wall}-t_{sc.w}-t_p)^3}{12}\right]+2(t_{sc.f}-2t_p)(W_{wall}-t_{sc.w}-t_p)\left(\frac{W_{wall}-t_{sc.w}-t_p}{2}+t_{sc.w}-\bar{y}\right)^2$$
$$+\frac{(L_{wall}-4t_p)(t_{sc.w}-2t_p)^3}{12}+(t_{sc.w}-2t_p)(L_{wall}-4t_p)\left(\bar{y}-\frac{t_{sc.w}}{2}\right)^2$$
$$=2\left\{\frac{\left[24\text{ in.}-2(\text{½ in.})\right](168\text{ in.}-14\text{ in.}-\text{½ in.})^3}{12}\right\}$$
$$+2\left[24\text{ in.}-2(\text{½ in.})\right](168\text{ in.}-14\text{ in.}-\text{½ in.})\left(\frac{168\text{ in.}-14\text{ in.}-\text{½ in.}}{2}+14\text{ in.}-54.9\text{ in.}\right)^2$$
$$+\frac{\left[480\text{ in.}-4(\text{½ in.})\right]\left[14\text{ in.}-2(\text{½ in.})\right]^3}{12}+\left[14\text{ in.}-2(\text{½ in.})\right]\left[360\text{ in.}-4(\text{½ in.})\right]\left(54.9\text{ in.}-\frac{14\text{ in.}}{2}\right)^2$$
$$=3.37\times10^7\text{ in.}^4$$

The effective flexural stiffness of the wall is:

$$EI_{eff} = E_sI_s + 0.35E_cI_c \tag{2-9}$$
$$=(29{,}000\text{ ksi})(2{,}210{,}000\text{ in.}^4)+0.35(4{,}500\text{ ksi})(3.37\times10^7\text{ in.}^4)$$
$$=1.17\times10^{11}\text{ kip-in.}^2$$

Because the rotational stiffness of the wall-to-foundation connection has an effect on the moment and force distribution and the lateral deflection of the coupled C-shaped wall, it is included in the finite element model. It is assumed that the wall-to-foundation connection provides a fully restrained connection. The rotational stiffness of a fully restrained wall-to-foundation connection, $K_{s.con}$, is calculated according to AISC *Specification* Figure C-B3.3:

$$K_{s.con}=\frac{20EI_{eff}}{H_{wall}}$$
$$=\frac{20(1.17\times10^{11}\text{ kip-in.}^2)}{(311\text{ ft})(12\text{ in./ft})}$$
$$=6.27\times10^8\text{ kip-in./rad}$$

From a structural analysis program, the required flexural and shear strengths of the SpeedCore wall and coupling beams are:

$V_{r.wall}$ = required shear strength of each planar wall (Risk Category II)
= 1,410 kips
$M_{r.wall}$ = required moment for each planar wall (Risk Category II)
= 722,000 kip-in.
$P_{r.wall}$ = compression force in the compression wall (Risk Category II)
= 12,300 kips
$T_{r.wall}$ = tensile force in the tension wall (Risk Category II)
= 12,300 kips
$V_{r.CB}$ = required shear force for coupling beams (Risk Category II)
= 438 kips
$M_{r.CB}$ = required moment for coupling beams (Risk Category II)
= 26,300 kip-in.

Step 5. Strength Check of SpeedCore Walls and Coupling Beams

In this step, the design flexural and shear strengths of the coupling beams and C-shaped SpeedCore walls are calculated and compared with the required strengths.

Design Flexural Strength of the Coupling Beam

The width of concrete in the coupling beam is:

$$t_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 24 \text{ in.} - 2(½ \text{ in.})$$
$$= 23.0 \text{ in.}$$

The plastic neutral axis location is:

$$C_{CB} = \frac{2t_{pw.CB}h_{CB}F_y + 0.85f_c't_{c.CB}t_{pf.CB}}{4t_{pw.CB}F_y + 0.85f_c't_{c.CB}}$$
$$= \frac{2(½ \text{ in.})(24 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(23.0 \text{ in.})(¾ \text{ in.})}{4(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(23.0 \text{ in.})}$$
$$= 5.93 \text{ in.}$$

(Note: The plastic neutral axis location, C_{CB}; compression forces, C_1, C_2, C_3; and tension forces, T_1, T_2, for a rectangular coupling beam are depicted in Appendix A, Figure A-13.)

The compression force in the flange is:

$$C_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$
$$= \left[24 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})(50 \text{ ksi})$$
$$= 863 \text{ kips}$$

The compression force in the web is:

$$C_2 = 2t_{pw.CB}C_{CB}F_y$$
$$= 2(½ \text{ in.})(5.93 \text{ in.})(50 \text{ ksi})$$
$$= 297 \text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85f_c't_{c.CB}(C_{CB} - t_{pf.CB})$$
$$= 0.85(6 \text{ ksi})(23.0 \text{ in.})(5.93 \text{ in.} - ¾ \text{ in.})$$
$$= 608 \text{ kips}$$

The tension force in the flange is:

$$T_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$
$$= \left[24 \text{ in.} - 2(½ \text{ in.})\right](¾ \text{ in.})(50 \text{ ksi})$$
$$= 863 \text{ kips}$$

The tensile force in the web is:

$$T_2 = 2t_{pw.CB}(h_{CB} - C_{CB})F_y$$
$$= 2(½ \text{ in.})(24 \text{ in.} - 5.93 \text{ in.})(50 \text{ ksi})$$
$$= 904 \text{ kips}$$

The plastic moment of the coupling beam is:

$$M_{Pn.CB} = C_1\left(C_{CB} - \frac{t_{pf.CB}}{2}\right) + C_2\left(\frac{C_{CB}}{2}\right) + C_3\left(\frac{C_{CB} - t_{pf.CB}}{2}\right) + T_1\left(h_{CB} - C_{CB} - \frac{t_{pf.CB}}{2}\right) + T_2\left(\frac{h_{CB} - C_{CB}}{2}\right)$$

$$= 863 \text{ kips}\left(5.93 \text{ in.} - \frac{¾ \text{ in.}}{2}\right) + 297 \text{ kips}\left(\frac{5.93 \text{ in.}}{2}\right) + 608 \text{ kips}\left(\frac{5.93 \text{ in.} - ¾ \text{ in.}}{2}\right)$$

$$+ 863 \text{ kips}\left(24 \text{ in.} - 5.93 \text{ in.} - \frac{¾ \text{ in.}}{2}\right) + 904 \text{ kips}\left(\frac{24 \text{ in.} - 5.93 \text{ in.}}{2}\right)$$

$$= 30{,}700 \text{ kip-in.}$$

The nominal flexural strength of the coupling beam is:

$$M_{n.CB} = M_{Pn.CB}$$

Using $\phi_b = 0.90$, the design flexural strength of the coupling beam is:

$$\phi_b M_{n.CB} = 0.90(30{,}700 \text{ kip-in.})$$

$$= 27{,}600 \text{ kip-in.} > M_{r.CB} = 26{,}300 \text{ kip-in.}$$

Shear Strength of the Coupling Beam

The area of the steel web in shear in the coupling beam is:

$$A_{sw.CB} = 2h_{CB}t_{pw.CB}$$

$$= 2(24 \text{ in.})(½ \text{ in.})$$

$$= 24.0 \text{ in.}^2$$

The nominal shear strength of the coupling beam is:

$$V_{n.CB} = 0.6F_yA_{sw.CB} + 0.06A_{c.CB}\sqrt{f'_c} \quad \text{(from Eq. 2-19)}$$

$$= 0.6(50 \text{ ksi})(24.0 \text{ in.}^2) + 0.06(518 \text{ in.}^2)\sqrt{6 \text{ ksi}}$$

$$= 796 \text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the coupling beam is:

$$\phi_v V_{n.CB} = 0.90(796 \text{ kips})$$

$$= 716 \text{ kips} > V_{r.CB} = 438 \text{ kips}$$

Flexural Strength of the C-Shaped SpeedCore Wall in Tension

Figure 2-22 illustrates the plastic neutral axis location and the compression and tension force regions in the tension wall.

It is assumed that the axial force acts at the elastic centroid of the walls.

As stated in the Given, the tensile force in the tension wall is:

$$T_{r.wall} = 12{,}300 \text{ kips}$$

$$t_{c.f} = t_{sc.f} - 2t_p$$

$$= 24 \text{ in.} - 2(½ \text{ in.})$$

$$= 23.0 \text{ in.}$$

$$t_{c.w} = t_{sc.w} - 2t_p$$
$$= 14 \text{ in.} - 2(½ \text{ in.})$$
$$= 13.0 \text{ in.}$$

The plastic neutral axis location is:

$$C_T = \frac{T_{r.wall} + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y + 0.85 f_c' 2t_{c.f} (W_{wall} - t_p) - 2t_p (L_{wall} - 4t_p) F_y}{8t_p F_y + 0.85 f_c' 2t_{c.f}}$$
$$= \frac{\begin{array}{l}12{,}300 \text{ kips} + 4(½ \text{ in.})(168 \text{ in.})(50 \text{ ksi}) + 2(½ \text{ in.})(23.0 \text{ in.})(50 \text{ ksi}) \\ + 0.85(6 \text{ ksi})[2(23.0 \text{ in.})](168 \text{ in.} - ½ \text{ in.}) - 2(½ \text{ in.})[360 \text{ in.} - 4(½ \text{ in.})](50 \text{ ksi})\end{array}}{8(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(2)(23 \text{ in.})}$$
$$= 119 \text{ in.}$$

The compression force in the flange wall edge plate is:

$$C_{1.T} = 4(W_{wall} - C_T) t_p F_y$$
$$= 4(168 \text{ in.} - 119 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 4{,}900 \text{ kips}$$

The compression force in the flange wall end plate is:

$$C_{2.T} = 2t_{c.f} t_p F_y$$
$$= 2(23.0 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 1{,}150 \text{ kips}$$

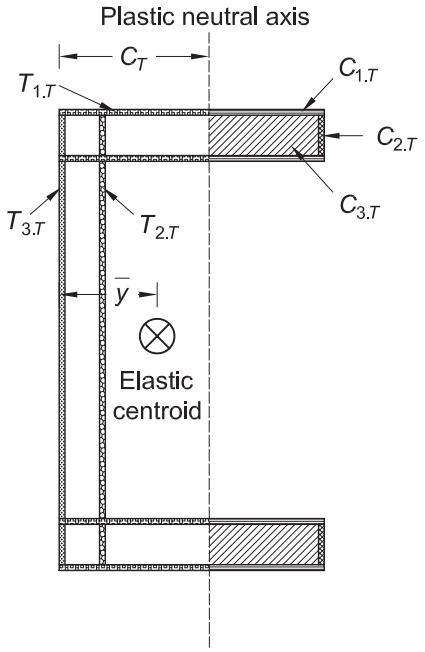


Fig. 2-22. Cross section with labeled regions for plastic moment calculation of tension wall.

The compression force in the concrete is:

$$C_{3.T} = 0.85 f_c' 2 t_{c.f} (W_{wall} - C_T - t_p)$$
$$= 0.85(6 \text{ ksi})[2(23.0 \text{ in.})](168 \text{ in.} - 119 \text{ in.} - ½ \text{ in.})$$
$$= 11{,}400 \text{ kips}$$

The tensile force in the flange wall edge plate is:

$$T_{1.T} = 4 C_T t_p F_y$$
$$= 4(119 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 11{,}900 \text{ kips}$$

The tensile force in the web wall inside steel plate is:

$$T_{2.T} = (L_{wall} - 4 t_p) t_p F_y$$
$$= [360 \text{ in.} - 4(½ \text{ in.})](½ \text{ in.})(50 \text{ ksi})$$
$$= 8{,}950 \text{ kips}$$

The tensile force in the web wall outside steel plate is:

$$T_{3.T} = (L_{wall} - 4 t_p) t_p F_y$$
$$= [360 \text{ in.} - 4(½ \text{ in.})](½ \text{ in.})(50 \text{ ksi})$$
$$= 8{,}950 \text{ kips}$$

The plastic flexural strength of the C-shaped SpeedCore wall in tension is:

$$M_{PT.wall} = C_{1.T}\left(\frac{W_{wall} - C_T}{2}\right) + C_{2.T}\left(W_{wall} - C_T - \frac{t_p}{2}\right) + C_{3.T}\left(\frac{W_{wall} - C_T - t_p}{2}\right) + T_{1.T}\left(\frac{C_T}{2}\right)$$
$$+ T_{2.T}\left(C_T - t_{sc.w} + \frac{t_p}{2}\right) + T_{3.T}\left(C_T - \frac{t_p}{2}\right) + T_{r.wall}(\bar{y} - C_T)$$
$$= (4{,}900 \text{ kips})\left(\frac{168 \text{ in.} - 119 \text{ in.}}{2}\right) + (1{,}150 \text{ kips})\left(168 \text{ in.} - 119 \text{ in.} - \frac{½ \text{ in.}}{2}\right)$$
$$+ (11{,}400 \text{ kips})\left(\frac{168 \text{ in.} - 119 \text{ in.} - ½ \text{ in.}}{2}\right) + (11{,}900 \text{ kips})\left(\frac{119 \text{ in.}}{2}\right)$$
$$+ (8{,}950 \text{ kips})\left(119 \text{ in.} - 14 \text{ in.} + \frac{½ \text{ in.}}{2}\right) + (8{,}950 \text{ kips})\left(119 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (12{,}300 \text{ kips})(54.9 \text{ in.} - 119 \text{ in.})$$
$$= 2{,}380{,}000 \text{ kip-in.}$$

The nominal flexural strength of the C-shaped SpeedCore wall in tension is:

$$M_{nT.wall} = M_{PT.wall} = 2{,}380{,}000 \text{ kip-in.}$$

Flexural Strength of the C-Shaped SpeedCore Wall in Compression

Figure 2-23 illustrates the plastic neutral axis location and the compression and tension force regions in the compression wall.

The compression force in the compression wall is:

$$P_{r.wall} = 12{,}300 \text{ kips}$$

The plastic neutral axis location is:

$$C_C = \frac{\begin{array}{l} P_{r.wall} - t_p(L_{wall} - 4t_p)F_y + (t_{c.w} + t_p)(L_{wall} - 4t_p)F_y - t_{c.w}(L_{wall} - 4t_p)0.85f'_c + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y \\ + t_{sc.w}(L_{wall} - 4t_p)F_y \end{array}}{8t_p F_y + 2(L_{wall} - 4t_p)F_y}$$

$$= \frac{\begin{array}{l} 12{,}300 \text{ kips} - (\text{½ in.})[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi}) + (13.0 \text{ in.} + \text{½ in.})[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi}) \\ - (13.0 \text{ in.})[360 \text{ in.} - 4(\text{½ in.})]0.85(6 \text{ ksi}) + 4(\text{½ in.})(168 \text{ in.})(50 \text{ ksi}) + 2(\text{½ in.})(23.0 \text{ in.})(50 \text{ ksi}) \\ + (14 \text{ in.})[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi}) \end{array}}{8(\text{½ in.})(50 \text{ ksi}) + 2[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi})}$$

$$= 13.6 \text{ in.}$$

The compression force in the flange wall edge plate is:

$$C_{1.C} = 4C_C t_p F_y$$
$$= 4(13.6 \text{ in.})(\text{½ in.})(50 \text{ ksi})$$
$$= 1{,}360 \text{ kips}$$

The compression force in the web wall exterior plate is:

$$C_{2.C} = t_p(L_{wall} - 4t_p)F_y$$
$$= (\text{½ in.})[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi})$$
$$= 8{,}950 \text{ kips}$$

The compression force in the web wall interior plate is:

$$C_{3.C} = (C_C - t_{c.w} - t_p)(L_{wall} - 4t_p)F_y$$
$$= (13.6 \text{ in.} - 13.0 \text{ in.} - \text{½ in.})[360 \text{ in.} - 4(\text{½ in.})](50 \text{ ksi})$$
$$= 1{,}790 \text{ kips}$$

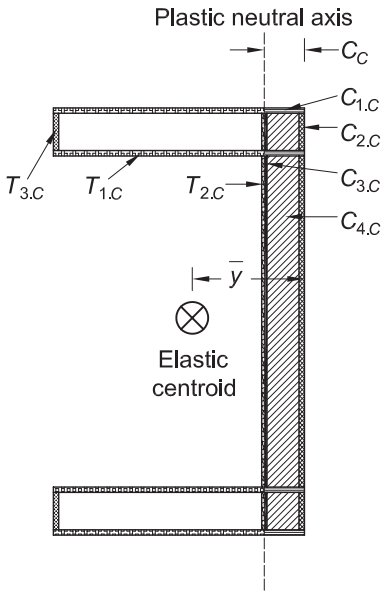


Fig. 2-23. Cross section with labeled regions for plastic moment calculation of compression wall.

The compression force in the concrete is:

$$C_{4.C} = 0.85 f_c' t_{c.w} (L_{wall} - 4t_p)$$
$$= 0.85(6 \text{ ksi})(13.0 \text{ in.})[360 \text{ in.} - 4(½ \text{ in.})]$$
$$= 23{,}700 \text{ kips}$$

The tensile force in the flange wall edge plate is:

$$T_{1.C} = 4(W_{wall} - C_C) t_p F_y$$
$$= 4(168 \text{ in.} - 13.6 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 15{,}400 \text{ kips}$$

The tensile force in the web wall interior plate is:

$$T_{2.C} = (t_{sc.w} - C_C)(L_{wall} - 4t_p) F_y$$
$$= (14 \text{ in.} - 13.6 \text{ in.})[360 \text{ in.} - 4(½ \text{ in.})](50 \text{ ksi})$$
$$= 7{,}160 \text{ kips}$$

The tensile force in the flange wall end plate is:

$$T_{3.C} = 2 t_p t_{c.f} F_y$$
$$= 2(½ \text{ in.})(23.0 \text{ in.})(50 \text{ ksi})$$
$$= 1{,}150 \text{ kips}$$

The plastic flexural strength of the C-shaped SpeedCore wall in compression is:

$$M_{PC.wall} = C_{1.C}\left(\frac{C_C}{2}\right) + C_{2.C}\left(C_C - \frac{t_p}{2}\right) + C_{3.C}\left(\frac{C_C - t_{c.w} - t_p}{2}\right) + C_{4.C}\left(C_C - t_p - \frac{t_{c.w}}{2}\right) + T_{1.C}\left(\frac{W_{wall} - C_C}{2}\right)$$
$$+ T_{2.C}\left(\frac{t_{sc.w} - C_C}{2}\right) + T_{3.C}\left(W_{wall} - C_C - \frac{t_p}{2}\right) + P_{r.wall}\left(\bar{y} - C_C\right)$$
$$= (1{,}360 \text{ kips})\left(\frac{13.6 \text{ in.}}{2}\right) + (8{,}950 \text{ kips})\left(13.6 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (1{,}790 \text{ kips})\left(\frac{13.6 \text{ in.} - 13.0 \text{ in.} - ½ \text{ in.}}{2}\right)$$
$$+ (23{,}700 \text{ kips})\left(13.6 \text{ in.} - ½ \text{ in.} - \frac{13.0 \text{ in.}}{2}\right) + (15{,}400 \text{ kips})\left(\frac{168 \text{ in.} - 13.6 \text{ in.}}{2}\right)$$
$$+ (7{,}160 \text{ kips})\left(\frac{14 \text{ in.} - 13.6 \text{ in.}}{2}\right) + (1{,}150 \text{ kips})\left(168 \text{ in.} - 13.6 \text{ in.} - \frac{½ \text{ in.}}{2}\right)$$
$$+ (12{,}300 \text{ kips})(54.9 \text{ in.} - 13.6 \text{ in.})$$
$$= 2{,}160{,}000 \text{ kip-in.}$$

The nominal flexural strength of the C-shaped SpeedCore wall in compression is:

$$M_{nC.wall} = M_{PC.wall} = 2{,}160{,}000 \text{ kip-in.}$$

The design flexural strength in the tension wall:

$$\phi_b M_{nT.wall} = 0.90(2{,}380{,}000 \text{ kip-in.})$$
$$= 2{,}140{,}000 \text{ kip-in.} > M_{r.wall} = 722{,}000 \text{ kip-in.}$$

The design flexural strength in the compression wall is:

$$\phi_b M_{nC.wall} = 0.90(2{,}160{,}000 \text{ kip-in.})$$
$$= 1{,}940{,}000 \text{ kip-in.} > M_{r.wall} = 722{,}000 \text{ kip-in.}$$

Shear Strength of the SpeedCore Wall

The area of steel in the direction of shear is:

$$A_{sw} = 4W_{wall}t_p$$
$$= 4(168 \text{ in.})(½ \text{ in.})$$
$$= 336 \text{ in.}^2$$

The area of concrete in the direction of shear is:

$$A_{cw} = 2(W_{wall} - 3t_p)(t_{sc.f} - 2t_p)$$
$$= 2\left[168 \text{ in.} - 3(½ \text{ in.})\right]\left[24 \text{ in.} - 2(½ \text{ in.})\right]$$
$$= 7{,}660 \text{ in.}^2$$

$$K_s = G_s A_{sw} \quad (2\text{-}15)$$
$$= (11{,}200 \text{ ksi})(336 \text{ in.}^2)$$
$$= 3{,}760{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_{cw})(E_s A_{sw})}{(4E_s A_{sw}) + (E_c A_{cw})} \quad \text{(from Eq. 2-16)}$$
$$= \frac{0.7\left[(4{,}500 \text{ ksi})(7{,}660 \text{ in.}^2)\right]\left[(29{,}000 \text{ ksi})(336 \text{ in.}^2)\right]}{4(29{,}000 \text{ ksi})(336 \text{ in.}^2) + (4{,}500 \text{ ksi})(7{,}660 \text{ in.}^2)}$$
$$= 3{,}200{,}000 \text{ kips}$$

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} A_{sw} F_y \quad (2\text{-}14)$$
$$= \frac{3{,}760{,}000 \text{ kips} + 3{,}200{,}000 \text{ kips}}{\sqrt{3(3{,}760{,}000 \text{ kips})^2 + (3{,}200{,}000 \text{ kips})^2}}(336 \text{ in.}^2)(50 \text{ ksi})$$
$$= 16{,}100 \text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the coupling beam is:

$$\phi_v V_{n.wall} = 0.90(16{,}100 \text{ kips})$$
$$= 14{,}500 \text{ kips} > V_{r.wall} = 1{,}410 \text{ kips}$$

The nominal tensile strength of the SpeedCore wall is:

$$P_{n.T} = A_s F_y \quad \text{(from Eq. 2-13)}$$
$$= (717 \text{ in.}^2)(50 \text{ ksi})$$
$$= 35{,}900 \text{ kips}$$

Using $\phi_t = 0.90$, the design tensile strength of the SpeedCore wall is:

$$\phi_t P_{n.T} = 0.90(35{,}900 \text{ kips})$$
$$= 32{,}300 \text{ kips} > T_{r.wall} = 12{,}300 \text{ kips}$$

The compression strength of the SpeedCore wall is:

$$P_{no} = A_s F_y + 0.85 f_c' A_c \quad (2\text{-}12)$$
$$= (717 \text{ in.}^2)(50 \text{ ksi}) + \left[0.85(6 \text{ ksi})(11{,}700 \text{ in.}^2)\right]$$
$$= 95{,}500 \text{ kips}$$

The moment of inertia of steel about the minor axis is:

$$I_{s.min} = 2\left[\frac{(W_{wall}-3t_p)t_p^3}{12}+(W_{wall}-3t_p)t_p\left(\frac{L_{wall}}{2}-t_{sc.f}+\frac{t_p}{2}\right)^2\right]+2\left[\frac{t_p t_{sc.f}^3}{12}+t_p t_{sc.f}\left(\frac{L_{wall}}{2}-\frac{t_{sc.f}}{2}\right)^2\right]$$
$$+2\left[\frac{(W_{wall}-3t_p)t_p^3}{12}+(W_{wall}-3t_p)t_p\left(\frac{L_{wall}}{2}-\frac{t_p}{2}\right)^2\right]+2\left(\frac{t_p L_{wall}^3}{12}\right)$$
$$=2\left\{\frac{\left[168\text{ in.}-3(\text{½ in.})\right](\text{½ in.})^3}{12}+\left[168\text{ in.}-3(\text{½ in.})\right](\text{½ in.})\left(\frac{360\text{ in.}}{2}-24\text{ in.}+\frac{\text{½ in.}}{2}\right)^2\right\}$$
$$+2\left[\frac{(\text{½ in.})(24\text{ in.})^3}{12}+(\text{½ in.})(24\text{ in.})\left(\frac{360\text{ in.}}{2}-\frac{24\text{ in.}}{2}\right)^2\right]$$
$$+2\left\{\frac{\left[168\text{ in.}-3(\text{½ in.})\right](\text{½ in.})^3}{12}+\left[168\text{ in.}-3(\text{½ in.})\right](\text{½ in.})\left(\frac{360\text{ in.}}{2}-\frac{\text{½ in.}}{2}\right)^2\right\}+2\left[\frac{(\text{½ in.})(360\text{ in.})^3}{12}\right]$$
$$=1.40\times10^7\text{ in.}^4$$

The moment of inertia of concrete about the minor axis is:

$$I_{c.min}=\left[\frac{(t_{sc.w}-2t_p)(L_{wall}-2t_{sc.f})^3}{12}\right]+2\left[\frac{(W_{wall}-3t_p)(t_{sc.f}-2t_p)^3}{12}+(W_{wall}-3t_p)(t_{sc.f}-2t_p)\left(\frac{L_{wall}}{2}-\frac{t_{sc.f}}{2}\right)^2\right]$$
$$=\left\{\frac{\left[14\text{ in.}-2(\text{½ in.})\right]\left[360\text{ in.}-2(24\text{ in.})\right]^3}{12}\right\}$$
$$+2\left\{\begin{array}{l}\dfrac{\left[168\text{ in.}-3(\text{½ in.})\right]\left[24\text{ in.}-2(\text{½ in.})\right]^3}{12}\\ +\left[168\text{ in.}-3(\text{½ in.})\right]\left[24\text{ in.}-2(\text{½ in.})\right]\left(\dfrac{360\text{ in.}}{2}-\dfrac{24\text{ in.}}{2}\right)^2\end{array}\right\}$$
$$=2.49\times10^8\text{ in.}^4$$

The effective stiffness of the wall about the minor axis is:

$$EI_{eff.min}=(E_s I_{s.min})+(0.35E_c I_{c.min}) \qquad \text{(from Eq. 2-9)}$$
$$=(29{,}000\text{ ksi})(1.40\times10^7\text{ in.}^4)+0.35(4{,}500\text{ ksi})(2.49\times10^8\text{ in.}^4)$$
$$=7.98\times10^{11}\text{ kip-in.}^2$$

The critical unsupported length for buckling of the wall is:

$$L_c=\max(h_1,h_{typ})$$
$$=\max(17\text{ ft},14\text{ ft})$$
$$=17\text{ ft}$$

The Euler buckling load, from AISC *Specification* Equation I2-5 is:

$$P_e = \frac{\pi^2 EI_{eff.min}}{L_c^2}$$
$$= \frac{\pi^2 (7.98\times10^{11}\ \text{kip-in.}^2)}{\left[(17\,\text{ft})(12\ \text{in./ft})\right]^2}$$
$$= 1.89\times10^{8}\ \text{kips}$$

$$\frac{P_{no}}{P_e} = \frac{95{,}500\ \text{kips}}{1.89\times10^{8}\ \text{kips}}$$
$$= 0.001 < 2.25$$

The nominal compressive strength from AISC *Specification* Equation I2-2:

$$P_{n.C} = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right)$$
$$= (95{,}500\ \text{kips})\left(0.658^{\frac{95{,}500\ \text{kips}}{1.89\times10^{8}\ \text{kips}}}\right)$$
$$= 95{,}500\ \text{kips}$$

Using $\phi_c = 0.90$, the design compression strength of the SpeedCore wall is:

$$\phi_c P_{n.C} = 0.90(95{,}500\ \text{kips})$$
$$= 86{,}000\ \text{kips} > P_{r.wall} = 12{,}300\ \text{kips}$$

Step 6. Serviceability Check of Coupled SpeedCore Walls

In this step, the lateral deflection of coupled C-shaped SpeedCore walls is checked to evaluate serviceability. Wind loads at a 10-year MRI and the effective stiffness of the uncoupled walls are used in a finite element model to check the drift limits. Effective flexural and shear stiffnesses of the coupling beams and C-shaped SpeedCore walls are used in the model.

The lateral deflection due to wind loads at a 10-year MRI is limited to a roof displacement of $H_{wall}/500$, where H_{wall} is the total height of the building, or an ID of $h/400$, where h is the story height. In this design example, the allowable roof displacement and ID are 7.46 in. and 0.25%, respectively.

Allowable roof displacement:

$$\frac{H_{wall}}{500} = \frac{(311\ \text{ft})(12\ \text{in./ft})}{500}$$
$$= 7.46\ \text{in.}$$

Allowable ID ratio:

$$\text{ID} = 25\%$$

A commercial structural analysis software program is used to check roof displacement and ID. Figure 2-24 shows the lateral deformation shape, lateral displacement, and ID of the coupled walls. Table 2-7 summarizes lateral displacement and ID.

Step 7. Detail Design of Coupling Beams

The clear width of the coupling beam flange plate:

$$b_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 24\ \text{in.} - 2(½\ \text{in.})$$
$$= 23.0\ \text{in.}$$

Table 2-7. Lateral Displacement and ID Summary			
Story No.	Story Height (ft)	Displacement (in.)	ID (%)
22	14	4.85	0.13
21	14	4.63	0.13
20	14	4.41	0.14
19	14	4.18	0.14
18	14	3.94	0.14
17	14	3.70	0.14
16	14	3.46	0.15
15	14	3.21	0.15
14	14	2.96	0.15
13	14	2.70	0.15
12	14	2.45	0.15
11	14	2.19	0.15
10	14	1.93	0.15
9	14	1.68	0.15
8	14	1.43	0.14
7	14	1.19	0.14
6	14	0.95	0.13
5	14	0.73	0.13
4	14	0.52	0.11
3	14	0.34	0.10
2	14	0.18	0.07
1	14	0.07	0.03
Base	17	0.00	0.00

The clear width of the coupling beam web plate:

$$h_{c.CB} = h_{CB} - 2t_{pf.CB}$$
$$= 24 \text{ in.} - 2(¾ \text{ in.})$$
$$= 22.5 \text{ in.}$$

The limiting width-to-thickness ratios from AISC *Specification* Table I1.1b are:

$$\frac{b_{c.CB}}{t_{pf.CB}} = \frac{23.0 \text{ in.}}{¾ \text{ in.}}$$
$$= 30.7$$

$$2.26\sqrt{\frac{E_s}{F_y}} = 2.26\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 54.4$$

$$\frac{b_{c.CB}}{t_{pf.CB}} < 2.26\sqrt{\frac{E_s}{F_y}}$$

$$\frac{h_{c.CB}}{t_{pw.CB}} = \frac{22.5 \text{ in.}}{\frac{1}{2} \text{ in.}}$$
$$= 45$$
$$3.00\sqrt{\frac{E_s}{F_y}} = 3.00\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 72.2$$
$$\frac{h_{c.CB}}{t_{pw.CB}} < 3.00\sqrt{\frac{E_s}{F_y}}$$

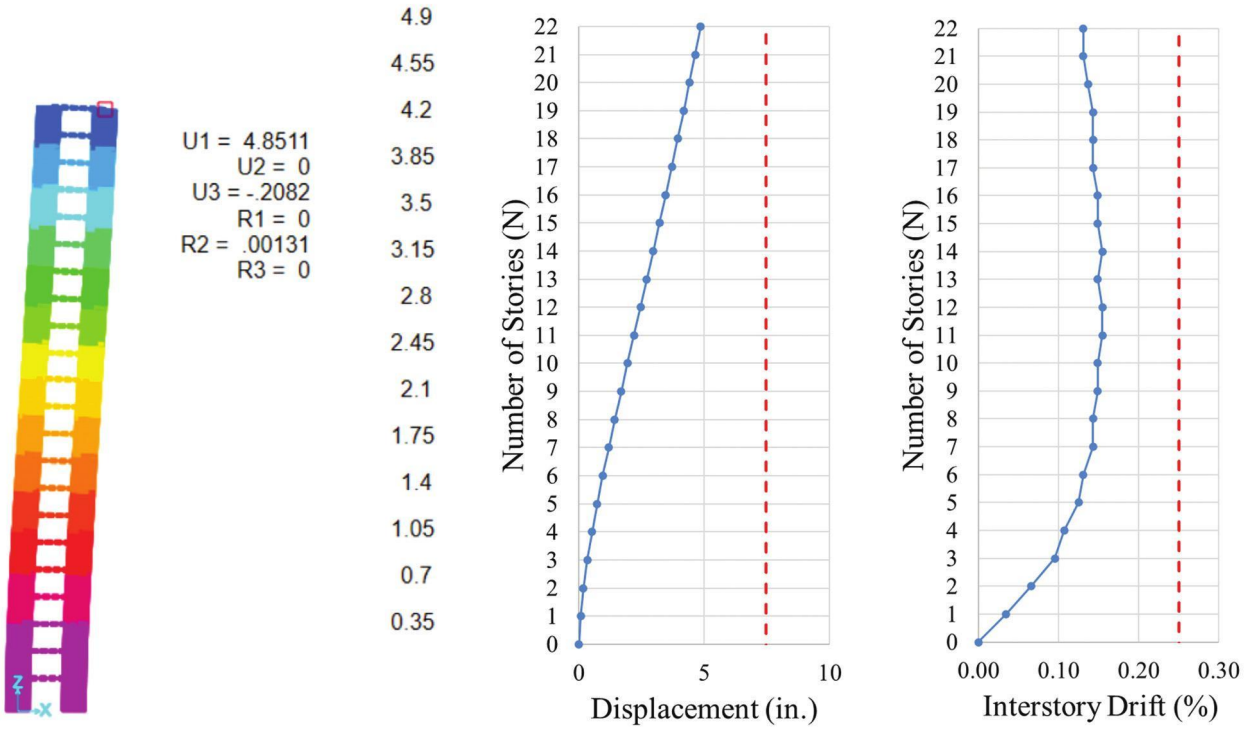
The coupling beam section is compact; however, noncompact/slender sections can be selected for the wind design.

Step 8. Detail Design of SpeedCore Walls

Required steel reinforcement ratio of the C-shaped walls is checked and shown in the following:

The gross area of the C-shaped wall is:

$$A_g = (L_{wall} t_{sc.w}) + 2(W_{wall} - t_{sc.w}) t_{sc.f}$$
$$= (360 \text{ in.})(14 \text{ in.}) + 2(168 \text{ in.} - 14 \text{ in.})(24 \text{ in.})$$
$$= 12{,}400 \text{ in.}^2$$



(a) Deformed shape *(b) Lateral displacement* *(c) ID*

Fig. 2-24. Coupled SpeedCore walls deformed shape, lateral displacement, and ID.

The minimum steel area required according to AISC *Specification* Section I1.6 is:

$$A_{s.min} = 0.01A_g$$
$$= 0.01(12{,}400 \text{ in.}^2)$$
$$= 124 \text{ in.}^2$$
$$A_s = 717 \text{ in.}^2 > A_{s.min} = 124 \text{ in.}^2$$

The maximum steel area required according to AISC *Specification* Section I1.6 is:

$$A_{s.max} = 0.1A_g$$
$$= 0.1(12{,}400 \text{ in.}^2)$$
$$= 1{,}240 \text{ in.}^2$$
$$A_s = 717 \text{ in.}^2 < A_{s.max} = 1{,}240 \text{ in.}^2$$

A tie bar spacing, S_{tie}, of 12 in. is selected for the C-shaped SpeedCore wall, and the slenderness requirement is checked as follows:

$$\frac{S_{tie}}{t_p} = \frac{12 \text{ in.}}{½ \text{ in.}}$$
$$= 24.0$$
$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 28.9$$
$$\frac{S_{tie}}{t_p} < 1.2\sqrt{\frac{E_s}{F_y}} \quad \textbf{o.k.} \qquad \text{(from Eq. 2-1)}$$

The tie bar diameter, d_{tie}, of ¾ in. is selected for the C-shaped wall, and the bar requirement considering the stability of the empty steel module is checked as follows:

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \qquad (2\text{-}5)$$
$$= 1.7\left(\frac{24 \text{ in.}}{½ \text{ in.}} - 2\right)\left(\frac{½ \text{ in.}}{¾ \text{ in.}}\right)^4$$
$$= 15.5$$
$$\frac{S_{tie}}{t_p} = \frac{12 \text{ in.}}{½ \text{ in.}}$$
$$= 24.0$$
$$1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 1.0\sqrt{\frac{29{,}000 \text{ ksi}}{2(15.4) + 1}}$$
$$= 30.2$$

$$\frac{S_{tie}}{t_p} < 1.0\sqrt{\frac{E_s}{2\alpha+1}} \quad \textbf{o.k.} \qquad \text{(from Eq. 2-4)}$$

Step 9. Connection Design

Connections are designed in accordance with the requirements of the AISC *Specification* Chapter J. Either welded or bolted connections are permitted. The composite wall-to-foundation (or subgrade structure) connections are designed based on the calculated demands (required strengths). The coupling beam-to-composite wall connections are designed based on the calculated demands at the corresponding levels. The composite wall splices are also designed based on the calculated demands at the corresponding locations.

Chapter 3
Seismic Design of Uncoupled SpeedCore Walls

3.1 OVERVIEW

This chapter covers the design of composite plate shear walls—concrete filled (C-PSW/CF), hereafter referred to as SpeedCore walls, with rectangular flange plates for seismic regions. Seismic response modification factors, R, Ω_0, and C_d for C-PSW have been included in ASCE/SEI 7, Table 12.2-1, for several years (ASCE, 2022). Due to fundamental differences in behavior, the AISC *Seismic Provisions* (AISC, 2016), divided the system into two separate systems: (1) composite plate shear walls—concrete encased and (2) composite plate shear walls—concrete filled. The 2016 AISC *Seismic Provisions* Section H7 included seismic design provisions for uncoupled C-PSW/CF. However, these provisions were limited to planar walls with (1) semicircular or circular concrete-filled steel tube boundary elements or (2) walls without any boundary elements. These provisions were based on research studies conducted by Alzeni and Bruneau (2017), Kurt et al. (2016), and Seo et al. (2016), and they included provisions for section detailing, including plate slenderness and tie bar size and spacing, member stiffness, calculating strength, and requirements for wall-to-foundation connections.

Over the past few years, significant research has been conducted on the seismic behavior, analysis, and design of uncoupled SpeedCore walls. This work included experimental investigations of the cyclic lateral loading behavior of planar SpeedCore walls with flange plates acting as boundary elements (Shafaei et al., 2021b) and the cyclic lateral loading behavior of C-shaped SpeedCore walls (Kenarangi et al., 2021). Analytical investigations were also conducted to develop and calibrate numerical models for estimating the cyclic lateral loading behavior of SpeedCore walls (Shafaei et al., 2021a; Bruneau et al., 2019). Additionally, detailed studies following FEMA P-695, *Quantification of Building Seismic Performance Factors: Component Equivalency Methodology* (FEMA, 2009), were conducted by Agarwal et al. (2020) to verify the seismic response modification factors, R, Ω_0, and C_d, for uncoupled walls.

Agarwal et al. (2020) used the seismic design criteria presented in detail (along with extended commentary) in the following chapter to design archetype structures. Four archetype structures with planar walls with flange plates (3-, 6-, 9-, and 12-story), and three archetype structures with C-shaped SpeedCore walls (15-, 18-, and 22-story) were designed using the detailed design procedure presented in Section 3.3. The nonlinear inelastic numerical models were developed and analyzed using OpenSEES and the modeling recommendations of Shafaei et al. (2021a) and Bruneau et al. (2019). The results from the numerical analyses confirmed that the seismic response was governed by the lateral stiffness and strength of uncoupled SpeedCore walls and the eventual formation of flexural plastic hinges at the base of the walls. Inelastic behavior occurs primarily at the base of the walls because they are mostly responding as vertical cantilever structural systems. All the SpeedCore walls in the archetype structure were flexure dominant with overall structure height-to-length ratios greater than or equal to 3. The in-plane shear strengths of the walls were much greater than the calculated demands. The analytical results of the 6- and 18-story uncoupled SpeedCore structures from this FEMA study are included in Chapter 5.

The behavior and design of uncoupled SpeedCore wall systems for seismic loading combinations depends on a few key elements, including (1) system limitations and section detailing, (2) stiffness, (3) strength, and (4) connection design requirements. These topics are presented in detail in the remainder of this chapter, along with an extended commentary. This is followed by the detailed design procedure in steps and flowcharts. Two detailed design examples are presented, including a 6-story structure with uncoupled planar SpeedCore walls and an 18-story structure with uncoupled C-shaped SpeedCore walls.

3.2 DESIGN REQUIREMENTS BASED ON 2022 AISC *SEISMIC PROVISIONS*

This section details the seismic design provisions in the 2022 version of the AISC *Seismic Provisions* Section H7 for the design of uncoupled C-PSW/CF. However, the design provisions and examples included in this Design Guide are limited to C-PSW/CF with rectangular flange plates.

3.2.1 Scope

As specified in AISC *Seismic Provisions* Section H7.1, composite plate shear walls—concrete filled (C-PSW/CF), which are referred to in this Design Guide as SpeedCore walls, consist of planar, C-shaped, or I-shaped walls, where each wall element consists of two planar steel plates with concrete infill between them. Composite action between the plates and concrete infill is achieved using either tie bars or a combination of tie bars and steel headed stud anchors.

SpeedCore walls are an alternative to reinforced concrete shear walls or special plate shear walls (SPSW), especially when relatively large seismic demand on the walls leads to dense reinforcement and large thicknesses in conventional

concrete shear walls, or to relatively large wall thicknesses of the web infill and boundary elements in SPSW. SpeedCore walls can also be provided with concrete-filled HSS boundary elements in lieu of flange (closure) plates to address high seismic demands.

In each wall element, the two steel plates are of equal nominal thickness and connected using tie bars. The steel plates comprise at least 1% but no more than 10% of the wall element cross section.

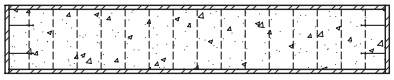
Flange or closure plates or boundary elements are required at the open ends of the wall elements. The boundary elements can be (1) half-circular sections of diameter equal to the distance between the two web plates, or (2) circular filled composite members.

Examples of the types of wall cross sections addressed by AISC *Seismic Provisions* Section H7 are shown in Figure 3-1. These include planar walls with flange or closure plates, planar walls with circular or half-circular boundary elements, and C- or I-shaped walls with flange and web walls with closure plates. Previous editions of the AISC *Seismic Provisions* permitted walls without any flange or closure plates or boundary elements. However, the 2022 AISC *Seismic Provisions* do not permit such designs, as recent studies (Agrawal et al., 2020) have shown that flange or closure plates enhance the ductility of these systems.

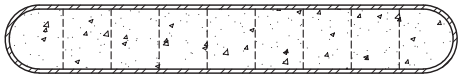
The wall height-to-length ratio, H_{wall}/L_{wall}, of the composite wall are required to be greater than or equal to 3. This height-to-length ratio limit is based on the range of available test data for flexure-critical composite walls, and the range of H_{wall}/L_{wall} values considered in the archetype structures (Agrawal et al., 2020). The limits in the scope are generally associated with the range of parameters considered in the investigations conducted on uncoupled walls (Alzeni and Bruneau, 2017; Agarwal et al., 2020).

3.2.2 Basis of Design

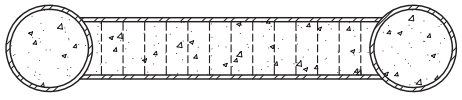
As required in AISC *Seismic Provisions* Section H7.2, SpeedCore walls are expected to provide significant inelastic deformation capacity through developing the plastic moment strength of the composite wall cross section, by yielding of the steel plate and the concrete attaining its compressive strength. The cross section is detailed such that the section is able to attain its plastic moment strength. Shear yielding of the steel web plates is not the mechanism governing system behavior. This behavior was seen in the experimental investigations of Shafaei et al. (2021b) detailed herein. Shafaei et al. (2021b) conducted experimental investigations on planar C-PSW subjected to different axial load and cyclic lateral loading up to failure. The parameters included in the experimental investigations were the axial load level, tie bar spacing, and steel plate slenderness. All the wall specimens developed and exceeded the plastic flexural strength corresponding to the axial load level. Shafaei et al. (2021a) have also developed and recommended effective stress-strain curves that can be used to model the cyclic behavior and calculate the flexural strength of SpeedCore walls. The typical cyclic lateral load-displacement of a planar wall is shown in Figure 3-2. The peak moment was reached due to yielding followed by local buckling of the flanges and webs of the walls. Strength degradation occurred due to fracture initiation in the flange plates of the wall. After fracture initiation, cyclic loading caused stiffness and strength degradation along with propagation of fracture through the flanges and webs of the wall cross section. The specimen exceeded a drift ratio of 3% before cyclic loading reduced the flexural strength of the wall to $0.8M_p$. This occurred during the first cycle with a drift ratio of five times the yield drift ratio corresponding to a plastic rotation of 0.02 rad in the plastic hinge at the base of the wall.



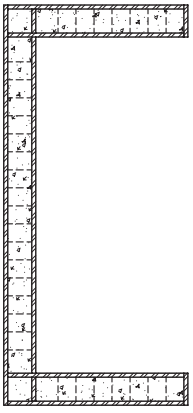
(a) Planar rectangular wall with flange plates and tie bars



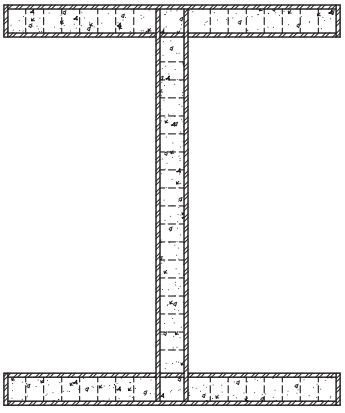
(b) Planar wall with semicircular boundary elements and tie bars



(c) Planar wall with circular boundary elements and tie bars



(d) C-shaped walls with flange (closure) plates and tie bars



(e) I-shaped walls with flange (closure) plates and tie bars

Fig. 3-1. SpeedCore walls with boundary elements or flange (closure) plates.

3.2.3 Analysis

According to AISC *Seismic Provisions* Section H7.3, the effective stiffness of composite walls is calculated in accordance with AISC *Specification* Section I1.5. This section of the AISC *Specification* is discussed in Section 2.2.2 of this Design Guide.

3.2.4 System Requirements

SpeedCore walls with flange plates are designed to meet the criteria outlined in Section 2.2.1 of this Design Guide. Additional provisions specific to seismic design are also required as follows.

3.2.4.1 Slenderness Requirement

According to AISC *Seismic Provisions* Section H7.5a, in regions of flexural yielding (at the base), the steel plate slenderness ratio, b/t_p, is limited as shown in Equation 3-1.

$$\frac{b}{t_p} \le 1.05\sqrt{\frac{E_s}{R_y F_y}} \tag{3-1}$$

where

E_s = modulus of elasticity of steel
= 29,000 ksi

F_y = specified minimum yield stress of steel, ksi

R_y = ratio of the expected yield stress to the specified minimum yield stress, F_y

b = largest unsupported length of plate between rows of steel anchors or tie bars, in.

t_p = thickness of plate, in.

In regions that are not subjected to flexural yielding, the steel plate slenderness ratio, b/t_p, is limited per AISC *Specification* Section I1.6a as follows:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{3-2}$$

For seismic applications, the plates dissipate energy through yielding and consequently need to meet the more stringent requirement of Equation 3-1 (in regions where flexural yielding is expected). Outside zones of flexural yielding, the slenderness limit for nonseismic walls as presented in Section 2.2.1.1 and replicated in Equation 3-2 is sufficient. Because tie bars may also act as anchors, the equation considers the largest unsupported length between rows of steel anchors or tie bars, b.

When the plate slenderness exceeds the limit of Equation 3-1, local buckling will occur when the compressive stress in the steel plate reaches F_{cr}, which can be calculated using Equation 3-3, and the compressive strength of the composite wall section can be estimated using Equation 3-4 (Zhang et al., 2014). This is identical to the discussion presented in Section 2.2.1.1 of this Design Guide.

$$F_{cr} = \frac{\pi^2 E_s}{12(0.8)^2 (b/t_p)^2} \le F_y \tag{3-3}$$

$$P_{no} = A_s F_{cr} + 0.7 A_c f_c' \tag{3-4}$$

where

A_c = area of concrete in SpeedCore wall, in.^{2}

A_s = area of steel section, in.^{2}

F_{cr} = critical buckling stress for steel elements of filled composite members, ksi

f_c' = specified compressive strength of concrete, ksi

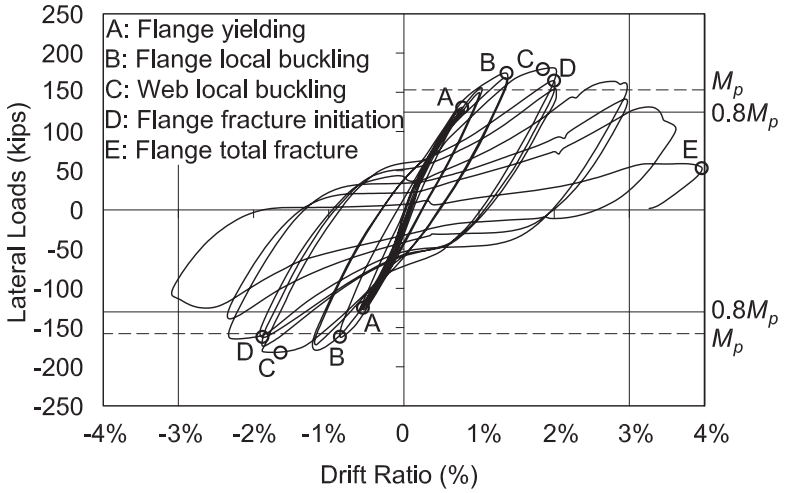


Fig. 3-2. Typical cyclic lateral load-displacement response of SpeedCore walls (Shafaei et al., 2021b).

3.2.4.2 Tie Bar Requirement

Tie bar spacing is selected to meet the requirements of Section 2.2.1.2.

3.2.5 Members

3.2.5.1 Flexural Strength

According to AISC *Seismic Provisions* Section H7.6a, the available plastic moment strength of the SpeedCore wall is determined in accordance with AISC *Specification* Section I1.2a.

3.2.5.2 Shear Strength

According to AISC *Seismic Provisions* Section H7.6b, the available shear strength of the wall is determined in accordance with AISC *Specification* Section I4.4.

3.2.5.3 Required Flexural Strength

The required flexural strength for composite walls is calculated in accordance with applicable seismic design requirements and analyses.

3.2.5.4 Required Shear Strength

The required shear strength for composite walls is calculated as the shear force obtained from the seismic analysis amplified by a factor of 4.0.

A base shear amplification factor of 4.0 is used to account for amplification due to higher mode effects in tall structures. This amplification factor is used only to calculate shear demand and is not used for overturning moment calculation. For reinforced concrete walls, ACI 318, Section 18.10.3.1 (ACI, 2019), prescribes a base shear amplification factor based on the number of stories and the ratio of the probable flexural strength of the member, M_{pr}, to the required flexural strength, M_u. In ACI 318, this amplification is limited to 3.0. However, the 4.0 factor used in this design is to conservatively account for similar phenomenon without developing and validating an equation.

3.2.6 Connection Requirements

3.2.6.1 Connection between Tie Bars and Steel Plates

According to AISC *Seismic Provisions* Section H7.7a, connections of tie bars to the steel plate are designed to develop the full tensile strength of the tie bar.

3.2.6.2 Connection between SpeedCore Wall Steel Components

According to AISC *Seismic Provisions* Section H7.7b, welds between the steel web plates and flange or closure plates or boundary elements are complete-joint-penetration (CJP) groove welds.

3.2.6.3 Composite Wall-to-Foundation Connections

According to AISC *Seismic Provisions* Section H7.7c, where the composite walls are connected directly to the foundation at the point of maximum moment in the walls, the composite wall-to-foundation connections are detailed such that the connection is able to transfer the base shear force and the axial force acting together with the overturning moment, corresponding to 1.1 times the plastic composite flexural strength of the wall. The plastic flexural composite strength of the wall is obtained by the plastic stress distribution method described in AISC *Specification* Section I1.2a. Applicable R_y and R_c factors are used for different elements of the cross section while establishing section force equilibrium and calculating the flexural strength.

When connected directly to the foundation at the point of maximum moment, the wall-to-foundation connections are designed for the expected flexural strength of the composite wall accounting for axial effects and the amplified shear force demand (amplification of 4). Some preliminary examples of wall-to-foundation connections are shown in Figure 3-3. Details of wall-to-foundation connections vary from project to project, and projects may incorporate walls that are not connected to the foundation at the point of maximum moment.

3.2.7 Protected Zones

According to AISC *Seismic Provisions* Section H7.8, regions subjected to inelastic straining at the base of the composite walls are designated as protected zones. The requirements for protected zones are in accordance with AISC *Seismic Provisions* Section D1.3 and Section I2.1.

For the SpeedCore system, the protected zones are designated as the regions of the composite walls undergoing significant inelastic straining and plastic hinging. The extent of the plastic hinge region undergoing significant inelastic strains (and the protected zone) can depend on wall cross-section geometry, web plate and flange (closure) plate thickness and lengths, and the height-to-length ratios of the walls. The extent of the protected zone can be determined from analysis.

3.2.7.1 Splices

According to AISC *Seismic Provisions* Section H7.8a, steel plate and boundary element splices located in the designated protected zones are designed to develop the full strength of the weaker of the two connected elements.

3.2.8 Demand Critical Welds in Connections

According to AISC *Seismic Provisions* Section H7.9, where located within protected zones, the following welds are demand critical and are designed to satisfy the applicable requirements:

(1) Welds connecting the composite wall closure plates to the faceplates

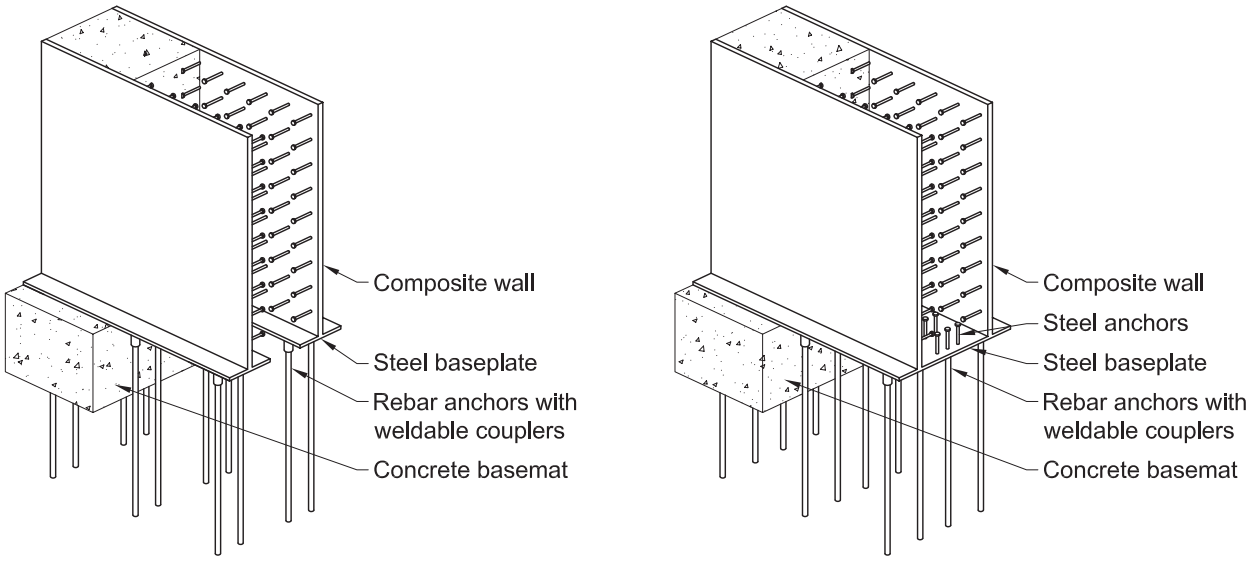
(2) Welds in the composite wall steel plate splices

(3) Welds at composite wall steel plate-to-base plate connections

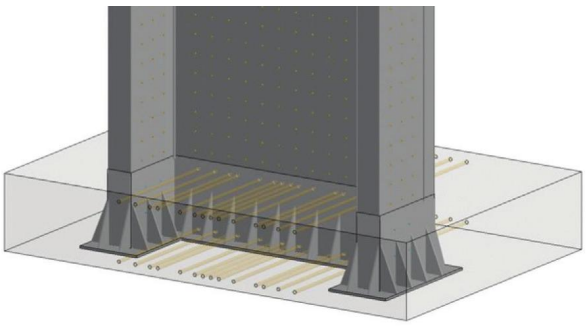
Demand critical welds are defined in the AISC *Seismic Provisions*, and the requirements are specified in Sections A3.4b and I2.3. These include requirements for the filler metals in terms of minimum levels of Charpy V-notch (CVN) toughness using two different test temperatures and specified test protocols, unless exempted from testing. Demand critical welds are generally CJP groove welds because they are subjected to yield level or higher stress demand and located in joints where failure can result in significant degradation in strength or stiffness. Welds that meet these criteria and are within the protected zone are designated as demand critical and, therefore, required to meet the corresponding requirements. As applicable, these may include CJP groove welds connecting the composite wall flange (closure) plates to the web plates, CJP groove welds used in composite wall steel plate splices, and CJP groove welds used in composite wall steel plate-to-base plate connections.

3.3 GENERAL DESIGN PROCEDURE FOR UNCOUPLED WALLS

This chapter describes the design approach for uncoupled SpeedCore walls. This design approach considers the guidance discussed in Section 3.2 and applicable building codes.



(a) Wall with welded base plate and rebar couplers (Bhardwaj and Varma, 2017)



(b) Wall embedded into the concrete foundation (Bruneau et al., 2019)

Fig. 3-3. Composite wall-to-foundation connections.

This design procedure has four main steps: (1) collecting design inputs, (2) analysis for design, (3) design of composite walls, and (4) design of connections. Design charts are presented in Figures 3-4 to 3-7. This design procedure is implemented for a 6- and 18-story uncoupled SpeedCore system in Examples 3.1 and 3.2, respectively.

3.3.1 Step 1. Design Inputs

To begin the design procedure, design input information is collected. These design inputs include the governing building code, floor loading, location considerations (i.e., site class, soil conditions, and seismic hazard level), building importance factor, and steel and concrete material properties. Additional information from the project architect is also collected at this stage such as the floor layout, length of walls, and story heights.

3.3.2 Step 2. Analysis for Design

The seismic forces acting on the composite walls are calculated following ASCE/SEI 7 guidelines. ASCE/SEI 7 offers multiple options for determining the seismic force demand including equivalent lateral force (ELF) and response spectra analysis. From this analysis, the overturning moment, base shear, and story forces values are calculated. Using these values, the amplified base shear is calculated, preliminary sizes are chosen for wall elements, and additional analysis is performed to check that the ID meets prescribed limits.

3.3.2.1 Base Shear and Amplified Base Shear

For the determination of shear demand, the base shear, V_{base}, is amplified to 4.0 times the base shear calculated by analysis. This base shear amplification factor is used to account for amplification due to higher mode effects in tall structures. This amplification factor is used only to calculate shear demand and is not used for the overturning moment calculation:

$$V_{amp} = 4.0V_{base} \tag{3-5}$$

3.3.2.2 Preliminary Section Sizes and Wall Stiffness

Preliminary sizes are selected for wall length, total wall thickness, and steel plate thickness. Selection of these elements is often iterative. These sections will need to meet drift requirements (Section 3.3.2.3) and wall section checks (Section 3.3.3). These dimensions are used to calculate the axial, flexural, and shear stiffnesses.

3.3.2.3 Interstory Drift Check

Story drift values from the elastic analysis are amplified by C_d, which is given in ASCE/SEI 7. The ID ratio (IDR) for each level is calculated as the ratio of the difference in the story displacement of two adjacent levels to the story height at that level. The maximum IDR is to be checked against the limits provided in ASCE/SEI 7, Table 12.12-1.

3.3.3 Step 3. Design of Composite Walls

The composite wall design check includes steel plate slenderness requirements, tie bar reinforcement requirements, empty module requirements, shear and flexural strength checks, and an ID check. The required design checks are summarized in this section.

3.3.3.1 Minimum and Maximum Area of Steel

The minimum and maximum reinforcement ratios are checked per Section 3.2.1.

3.3.3.2 Wall Steel Plate Slenderness Requirement

The composite wall steel plate slenderness requirement must satisfy tie bar spacing, S, requirement before and after placement of concrete using the guidelines provided in Section 3.2.4.1 and 3.2.4.2.

3.3.3.3 Wall Shear Strength

The in-plane shear strength required for the composite walls is calculated according to Section 3.2.5.4. The amplified base shear is then distributed to all walls to determine the required shear strength. The design shear strength is calculated according to Section 3.2.5.2.

3.3.3.4 Wall Flexural Strength

The required flexural strength of the composite wall is calculated as the overturning moment (OTM) distributed to the walls according to their relative stiffness. The design flexural strength of composite walls is calculated according to Section 3.2.5.1.

3.3.4 Step 4. Design of Connections

Three connections should be considered: (1) tie bars to steel plates, (2) steel plate to steel plate, and (3) wall-to-foundation connections. The requirements for these connections are established in Section 3.2.6.

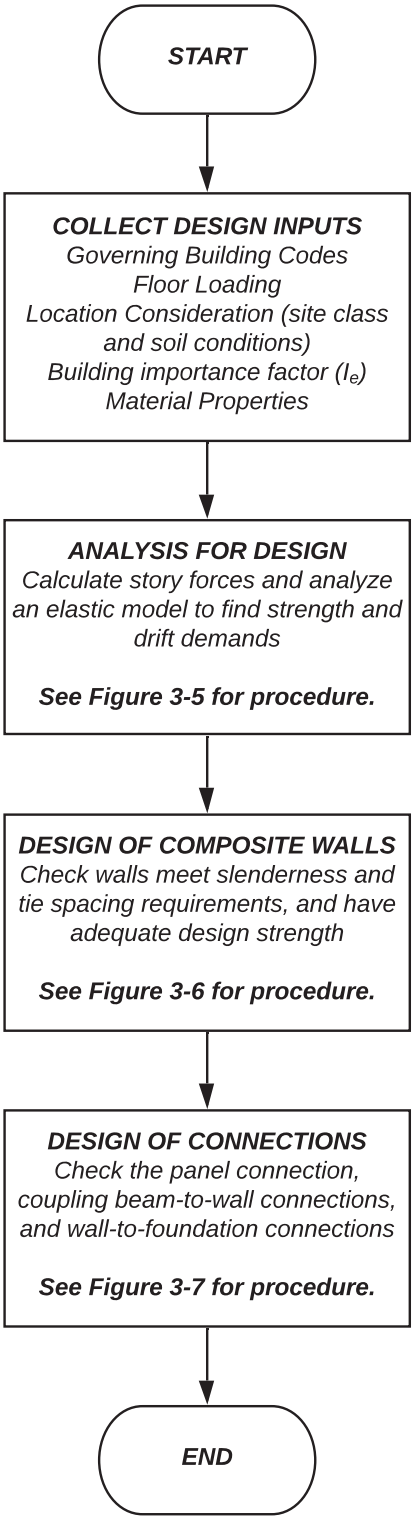


Fig. 3-4. Overview of uncoupled SpeedCore wall design steps.

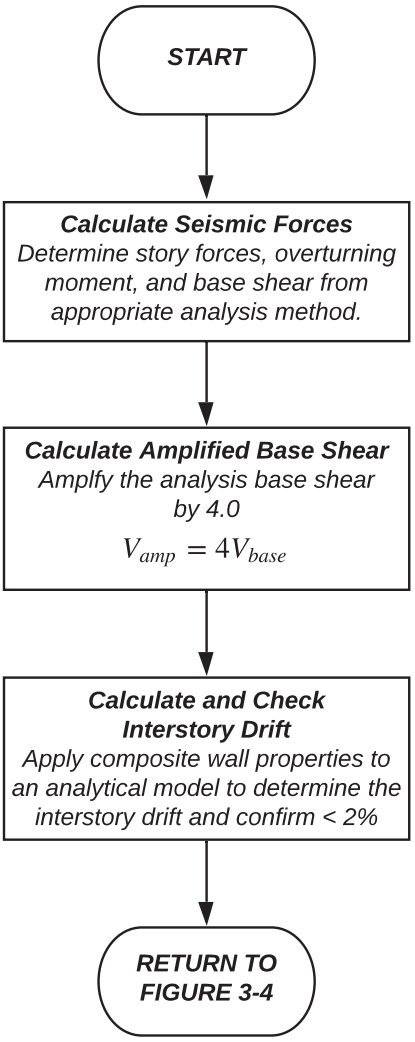


Fig. 3-5. Detailed chart for Analysis for Design step.

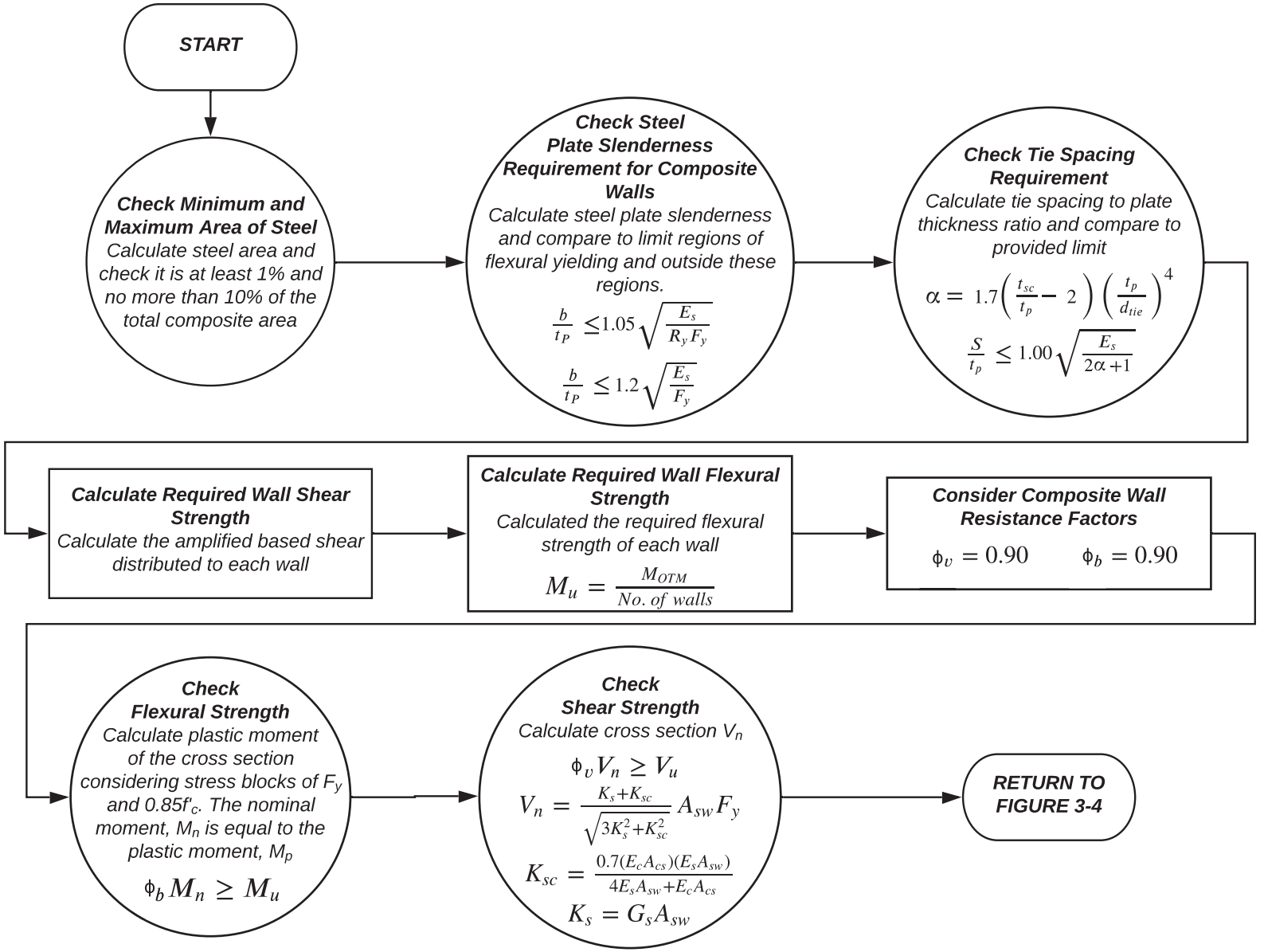


Fig. 3-6. Detailed flowchart for Design of Composite Walls step.

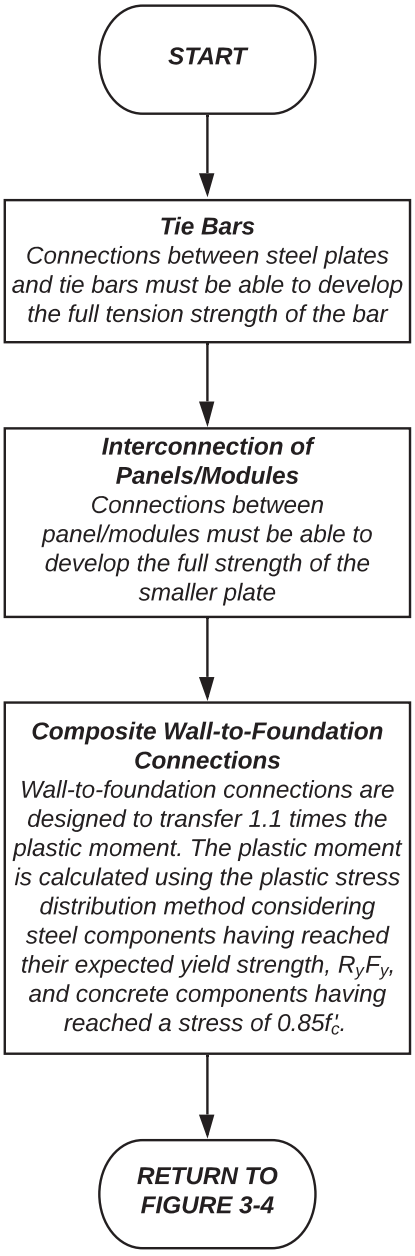


Fig. 3-7. Detailed chart for Design of Connections step.

3.4 DESIGN EXAMPLES

EXAMPLE 3.1—Seismic Design of 6-Story Structure Using Uncoupled SpeedCore Walls

This example presents the seismic design of a 6-story building using uncoupled SpeedCore walls with typical design loads, floor geometry, and high seismic design loads. The steps followed in this design mirror the design procedure laid out in Chapter 3. For simplicity, this design example does not consider accidental eccentricity and assumes a seismic redundancy factor of 1.0.

Design the uncoupled, planar SpeedCore walls shown in Figure 3-8 using the given geometry, material properties, and loads. In this example, the wall configuration is equivalent in both perpendicular directions; therefore, the design for both sets of walls will be the same.

The Mathcad file for this design example can be downloaded from the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

Given:

Building geometry:

L_f = building length
= 205 ft
W_f = building width
= 105 ft
h_{typ} = typical story height
= 14 ft
h_1 = first story height
= 17 ft
n = number of stories
= 6

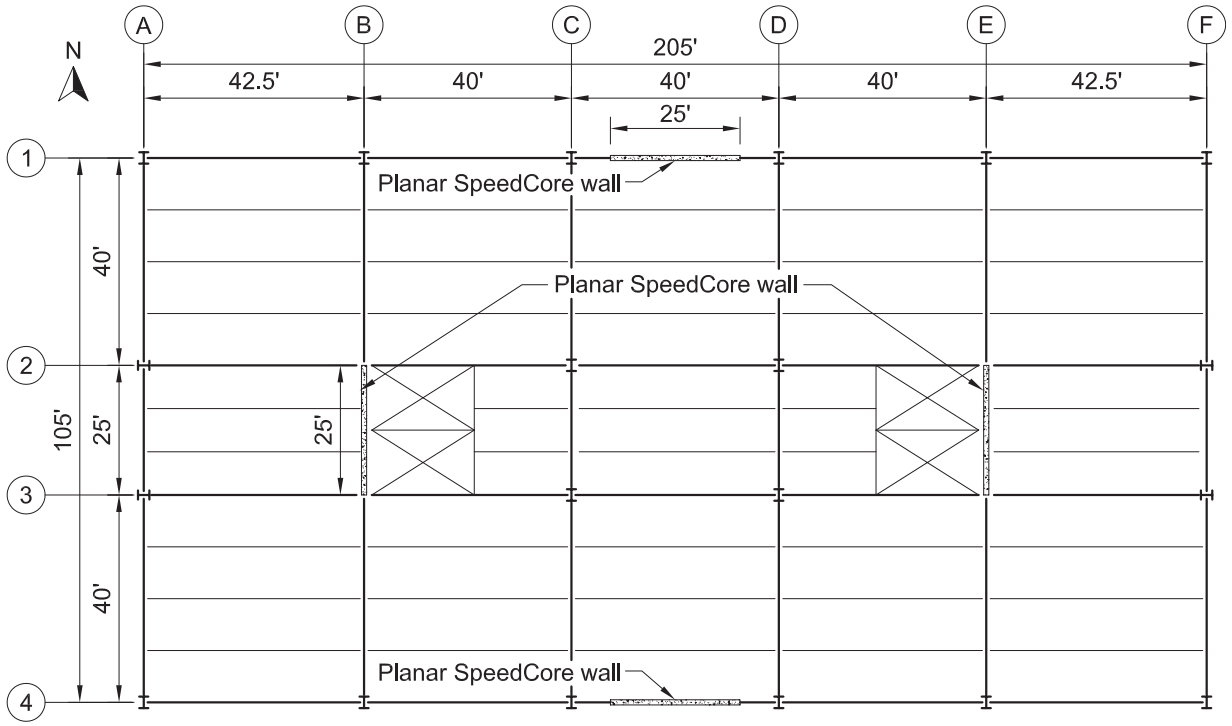


Fig. 3-8. Building floor plan for Example 3.1.

Table 3-1. Seismic Design Loads Calculated from ASCE/SEI 7			
Story No.	**Story Height (ft)**	**Story Force (kips)**	**Story Shear (kips)**
6	14	530	530
5	14	433	963
4	14	339	1,300
3	14	248	1,550
2	14	161	1,710
1	17	81.2	1,790
Total base shear, V_{base}			1,790 kips
Overturning moment, OTM			1,380,000 kip-in.

Floor load:

DL = floor dead load
= 0.12 ksf

Seismic design loads are given in Table 3-1.

Step 1. General Information of the Considered Building

The steel and concrete material properties are as follows:

ASTM A572/A572M Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi
R_c = 1.3

Seismic design coefficients:

C_d = deflection amplification factor (ASCE/SEI 7, Table 12.2-1)
= 5.5
I_e = importance factor (ASCE/SEI 7, Section 11.5.1, for an office building)
= 1
R = seismic response modification coefficient (from ASCE/SEI 7, Table 12.2-1)
= 6.5

Risk Category = II (ASCE/SEI 7, Table 1.5-1)

Ω_0 = overstrength factor (from ASCE/SEI 7, Table 12.2-1)
= 2.5
ρ = seismic redundancy factor (ASCE/SEI 7, Section 12.3.4)
= 1

Solution:

For ASTM A572/A572M Grade 50 steel, from AISC *Manual* Table 2-4 and AISC *Seismic Provisions* Table A3.2, the material properties are as follows:

F_y = 50 ksi
F_u = 65 ksi
R_y = 1.1

SpeedCore members are sized based on initial estimates of loads and then refined through iteration. Acceptable dimensions are presented here so that this example can focus on the appropriate limit states to check.

Wall dimensions as shown in Figure 3-9:

L_{wall} = wall length
= 300 in.
t_p = wall plate thickness
= 5/16 in.
t_{sc} = wall thickness
= 16 in.

Step 2. Analysis for Design

In this step an elastic computer model is built and analyzed to determine the force on system members and overall system deflection.

Planar Wall Geometric Properties

The effective axial, shear, and flexural stiffnesses are calculated for SpeedCore elements for elastic analysis.

The area of steel in the wall is:

$$A_s = t_p\left[2(L_{wall} - 2t_p) + 2t_{sc}\right]$$
$$= (5/16 \text{ in.})\left\{2\left[300 \text{ in.} - 2(5/16 \text{ in.})\right] + 2(16 \text{ in.})\right\}$$
$$= 197 \text{ in.}^2$$

The area of concrete in the wall is:

$$A_c = L_{wall}t_{sc} - A_s$$
$$= (300 \text{ in.})(16 \text{ in.}) - 197 \text{ in.}^2$$
$$= 4{,}600 \text{ in.}^2$$

The effective axial stiffness of the wall is:

$$EA_{eff} = E_sA_s + 0.45E_cA_c \quad (2\text{-}10)$$
$$= (29{,}000 \text{ ksi})(197 \text{ in.}^2) + 0.45(4{,}500 \text{ ksi})(4{,}600 \text{ in.}^2)$$
$$= 1.50 \times 10^7 \text{ kips}$$

The steel area in the direction of shear is:

$$A_{sw} = 2L_{wall}t_p$$
$$= 2(300 \text{ in.})(5/16 \text{ in.})$$
$$= 188 \text{ in.}^2$$

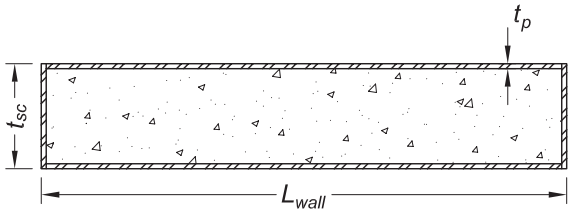


Fig. 3-9. Wall cross-section dimensions.

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_s A_{sw} + G_c A_c \tag{2-11}$$
$$= (11{,}200 \text{ ksi})(188 \text{ in.}^2) + (1{,}800 \text{ ksi})(4{,}600 \text{ in.}^2)$$
$$= 1.04 \times 10^7 \text{ kips}$$

The moment of inertia of the steel in the wall is:

$$I_s = 2\left[\frac{t_p (L_{wall} - 2t_p)^3}{12}\right] + 2\left[\left(\frac{t_{sc} t_p^3}{12}\right) + t_{sc} t_p \left(\frac{L_{wall}}{2} - \frac{t_p}{2}\right)^2\right]$$
$$= 2\left\{\frac{(5/16 \text{ in.})\left[300 \text{ in.} - 2(5/16 \text{ in.})\right]^3}{12}\right\} + 2\left[\frac{(16 \text{ in.})(5/16 \text{ in.})^3}{12} + (16 \text{ in.})(5/16 \text{ in.})\left(\frac{300 \text{ in.}}{2} - \frac{5/16 \text{ in.}}{2}\right)^2\right]$$
$$= 1{,}620{,}000 \text{ in.}^4$$

The moment of inertia of the concrete in the wall is:

$$I_c = \frac{(t_{sc} - 2t_p)(L_{wall} - 2t_p)^3}{12}$$
$$= \frac{\left[16 \text{ in.} - 2(5/16 \text{ in.})\right]\left[300 \text{ in.} - 2(5/16 \text{ in.})\right]^3}{12}$$
$$= 3.44 \times 10^7 \text{ in.}^4$$

The effective moment of inertia of the wall is:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \tag{2-9}$$
$$= (29{,}000 \text{ ksi})(1{,}620{,}000 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(3.44 \times 10^7 \text{ in.}^4)$$
$$= 1.01 \times 10^{11} \text{ kip-in.}^2$$

Numerical Model

To determine the ID ratio, an analysis model was built using commercial software. First-order, linear elastic analysis was performed on a model consisting of frame elements for the SpeedCore wall. This model was subjected to the earthquake loading previously defined, and all mass was applied at the story level. A representation of the model is shown in Figure 3-10.

Lateral displacement values are amplified by the C_d factor to obtain the amplified displacement. These amplified values are then used to calculate the ID ratio. The ID is the difference in displacement between two floors normalized by the story height. The maximum design ID is limited according to ASCE/SEI 7, Section 12.12.1. In this case, the maximum design ID is 2%. The story displacement, amplified displacement, and ID are presented in Table 3-2.

From Table 3-2, the maximum ID ratio is 1.64%, which is less than 2%.

Table 3-2. Lateral Displacement and ID Summary				
Story No.	Story Height (ft)	Displacement (in.)	Amplified Displacement (in.)	ID (%)
6	14	2.26	12.4	1.64
5	14	1.76	9.68	1.57
4	14	1.28	7.04	1.51
3	14	0.82	4.51	1.24
2	14	0.44	2.42	0.95
1	14	0.15	0.83	0.40
Base	17	0.00	0.00	0.00

Required Shear and Flexural Strength

Required shear strength of each planar wall:

The base shear, shown in Table 3-1, is amplified by a shear amplification factor of 4.0, following the recommendations of Section 3.3.2.1:

$$V_{amp} = 4.0V_{base} \tag{3-5}$$
$$= 4.0(1{,}790 \text{ kips})$$
$$= 7{,}160 \text{ kips}$$

$$V_{r.wall} = \frac{V_{amp}}{2}$$
$$= \frac{7{,}160 \text{ kips}}{2}$$
$$= 3{,}580 \text{ kips}$$

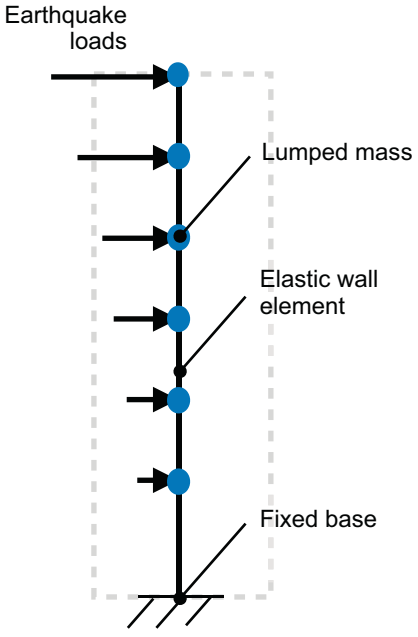


Fig. 3-10. Analysis model.

The required flexural strength of each planar wall using the overturning moment, OTM, given in Table 3-1 is:

$$M_{r.wall} = \frac{\text{OTM}}{2}$$
$$= \frac{1{,}380{,}000 \text{ kip-in.}}{2}$$
$$= 690{,}000 \text{ kip-in.}$$

Step 3. Design of Composite Walls

Step 3-1. Minimum and maximum area of steel

The minimum steel area required by AISC *Specification* Section I1.6 is:

$$A_{s.min} = 0.01 L_{wall} t_{sc}$$
$$= 0.01(300 \text{ in.})(16 \text{ in.})$$
$$= 48.0 \text{ in.}^2$$

$$A_s = 197 \text{ in.}^2 > A_{s.min} = 48.0 \text{ in.}^2 \quad \textbf{o.k.}$$

The maximum steel area required by AISC *Specification* Section I1.6 is:

$$A_{s.max} = 0.1 L_{wall} t_{sc}$$
$$= 0.1(300 \text{ in.})(16 \text{ in.})$$
$$= 480 \text{ in.}^2$$

$$A_s = 197 \text{ in.}^2 < A_{s.max} = 480 \text{ in.}^2 \quad \textbf{o.k.}$$

Step 3-2. Steel plate slenderness requirements for composite walls

The largest unsupported length between rows of steel anchors or tie bars is selected assuming that all support is through the tie bars and not the steel anchors, and therefore, plate slenderness and tie bar spacing requirements must be met. As discussed in Section 3.2.4.1 of the Design Guide, tie bars and steel anchors can be spaced farther apart outside of flexural yielding zones. The extent of the flexural yielding zones is addressed in depth in the following.

A tie bar spacing of 6 in. is selected for the first story of the uncoupled SpeedCore wall, and a tie bar spacing of 9 in. is selected for the other stories:

$$S_{tie} = 6 \text{ in.}$$

$$S_{tie.top} = 9 \text{ in.}$$

From Section 3.2.4.1, the steel plate slenderness ratio, b/t_p, is limited to:

$$\frac{b}{t_p} \le 1.05\sqrt{\frac{E_s}{R_y F_y}} \tag{3-1}$$

where b is the largest unsupported length of the plate between rows of steel anchors or tie bars, which in this case is S_{tie}.

$$\frac{b}{t_p} = \frac{S_{tie}}{t_p}$$
$$= \frac{6 \text{ in.}}{5/16 \text{ in.}}$$
$$= 19.2$$

$$1.05\sqrt{\frac{E_s}{R_y F_y}} = 1.05\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}}$$
$$= 24.1$$

$$\frac{b}{t_p} = 19.2 < 1.05\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}} = 24.1 \quad \textbf{o.k.}$$

From Section 3.2.4.1, the steel plate slenderness ratio, b/t_p, in the remainder of the stories is limited to:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{3-2}$$

where b is the largest unsupported length of the plate between rows of steel anchors or tie bars, which in this case is $S_{tie.top}$.

$$\frac{b}{t_p} = \frac{S_{tie.top}}{t_p}$$
$$= \frac{9 \text{ in.}}{5/16 \text{ in.}}$$
$$= 28.8$$

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 28.9$$

$$\frac{b}{t_p} = 28.8 < 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}} = 28.9 \quad \textbf{o.k.}$$

The slenderness ratio requirements for the steel plates are met.

Flexural Yielding Zone

Tie bar or stud spacing is intended to meet a more stringent requirement in zones of the wall where compression buckling is likely to occur or in the plastic hinge region.

Compression buckling of the faceplate is more likely to occur when the stress in the faceplate exceeds F_y. This load level will be called the yield moment in compression, M_{yc}. In addition to this limit, the plastic hinge region also requires more stringent tie bar or stud spacing. This region is assumed to have a moment greater than $0.8M_p$. The lesser of these two limits (M_{yc} or $0.8M_p$) is used to determine where the tighter tie bar or stud spacing can be relaxed in the SpeedCore walls.

M_p, M_{yc}, and the yield moment in tension, M_{yt}, are calculated and compared to determine the height to which the more stringent tie bar spacing requirement should be applied. To calculate this height, the moment distribution is assumed to vary linearly with a magnitude of M_p at the base of the wall and 0 at the top of the structure.

Figure 3-11 illustrates the plastic neutral axis location and the compression and tension regions in the wall.

The location of the plastic neutral axis of the uncoupled SpeedCore wall is:

$$C = \frac{2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$

$$= \frac{2(5/16 \text{ in.})(300 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})}{4(5/16 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right]}$$

$$= 66.7 \text{ in.}$$

The compression force in the flange is:

$$C_1 = (t_{sc} - 2t_p)t_p F_y$$

$$= \left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})(50 \text{ ksi})$$

$$= 240 \text{ kips}$$

The compression force in the web is:

$$C_2 = 2t_p C F_y$$

$$= 2(5/16 \text{ in.})(66.7 \text{ in.})(50 \text{ ksi})$$

$$= 2{,}080 \text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85 f_c'(t_{sc} - 2t_p)(C - t_p)$$

$$= 0.85(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right](66.7 \text{ in.} - 5/16 \text{ in.})$$

$$= 5{,}210 \text{ kips}$$

The tension force in the flange is:

$$T_1 = (t_{sc} - 2t_p)t_p F_y$$

$$= \left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})(50 \text{ ksi})$$

$$= 240 \text{ kips}$$

The tension force in the web is:

$$T_2 = 2t_p(L_{wall} - C)F_y$$

$$= 2(5/16 \text{ in.})(300 \text{ in.} - 66.7 \text{ in.})(50 \text{ ksi})$$

$$= 7{,}290 \text{ kips}$$

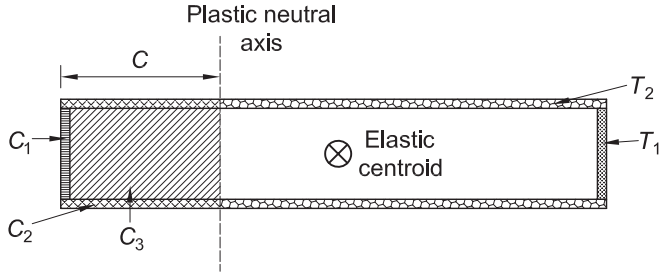


Fig. 3-11. Cross section with labeled regions for plastic moment calculation.

The plastic moment of the uncoupled SpeedCore wall is:

$$M_{P.wall} = C_1\left(C - \frac{t_p}{2}\right) + C_2\left(\frac{C}{2}\right) + C_3\left(\frac{C - t_p}{2}\right) + T_1\left(L_{wall} - C - \frac{t_p}{2}\right) + T_2\left(\frac{L_{wall} - C}{2}\right)$$

$$= (240 \text{ kips})\left(66.7 \text{ in.} - \frac{5/16 \text{ in.}}{2}\right) + (2{,}080 \text{ kips})\left(\frac{66.7 \text{ in.}}{2}\right) + (5{,}210 \text{ kips})\left(\frac{66.7 \text{ in.} - 5/16 \text{ in.}}{2}\right)$$

$$+ (240 \text{ kips})\left(300 \text{ in.} - 66.7 \text{ in.} - \frac{5/16 \text{ in.}}{2}\right) + (7{,}290 \text{ kips})\left(\frac{300 \text{ in.} - 66.7 \text{ in.}}{2}\right)$$

$$= 1{,}160{,}000 \text{ kip-in.}$$

$$0.8M_{P.wall} = 0.8(1{,}160{,}000 \text{ kip-in.})$$

$$= 928{,}000 \text{ kip-in.}$$

Yield Moments M_{yt} *and* M_{yc}

The yield moment is calculated from section moment curvature analysis considering nominal properties, see Figure 3-12 for the moment curvature plot. This case did not consider axial load.

From analysis:

$M_{yt} = 665{,}000$ kip-in.

$M_{yc} = 1{,}010{,}000$ kip-in.

The ratio of the tensile yield moment to the plastic moment of the uncoupled wall is:

$$\frac{M_{yt}}{M_{P.wall}} = \frac{665{,}000 \text{ kip-in.}}{1{,}160{,}000 \text{ kip-in}}$$

$$= 0.573$$

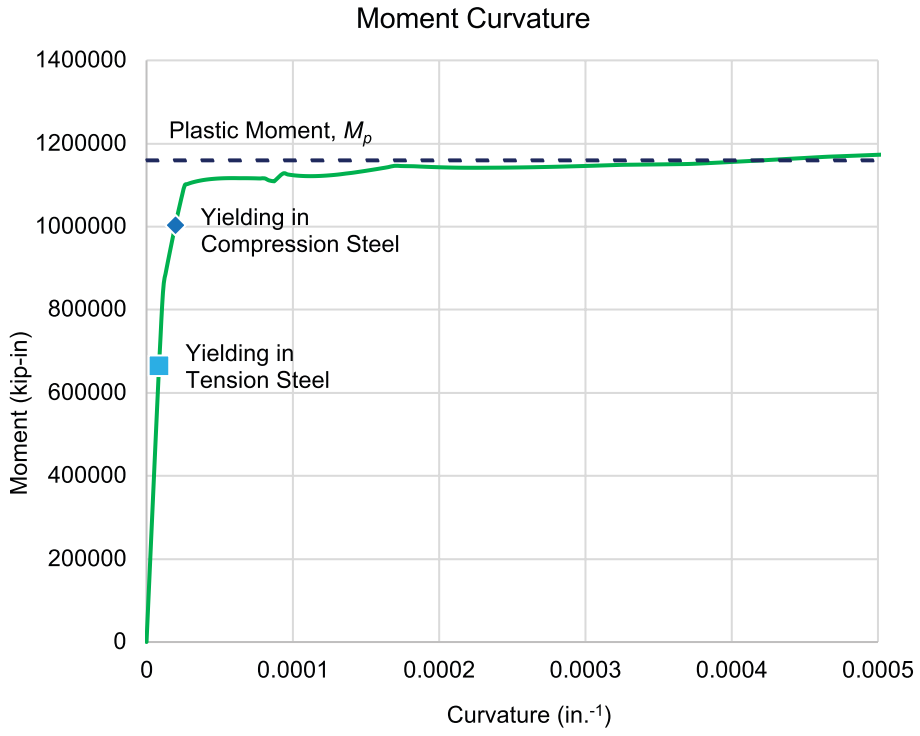


Fig. 3-12. Moment curvature plot.

The ratio of the compressive yield moment to the plastic moment of the uncoupled wall is:

$$\frac{M_{yc}}{M_{P.wall}} = \frac{1{,}010{,}000 \text{ kip-in.}}{1{,}160{,}000 \text{ kip-in.}}$$
$$= 0.871$$

The ratio of the limiting moment to $M_{p.wall}$ is:

$$r_1 = \frac{\min(M_{yc}, 0.8M_{P.wall})}{M_{P.wall}}$$
$$= \frac{\min(1{,}010{,}000 \text{ kip-in.}, 932{,}000 \text{ kip-in.})}{1{,}160{,}000 \text{ kip-in.}}$$
$$= 0.803$$

The portion of the wall with demand higher than the moment limit is:

$$(1-r_1)\left[h_1 + (n-1)h_{typ}\right] = (1-0.803)\left[17 \text{ ft} + (6-1)14 \text{ ft}\right]$$
$$= 17.1 \text{ ft}$$

Based on these calculations, the steel anchor and tie bar spacing would be 6 in. o.c. to just above the first story and then could transition to 9 in. o.c. for the remainder of the height.

Step 3-3. Tie bar spacing requirements for composite walls

A tie bar diameter of ½ in. is selected for the uncoupled SpeedCore wall. The bar requirement considering the stability of the empty steel module is checked as follows:

d_{tie} = ½ in.

S_{tie} = 6 in.

$S_{tie.top}$ = 9 in.

From Section 2.2.1.2, the tie bar spacing to plate thickness ratio, S/t_p, is limited as stated in Equation 2-4:

$$\frac{S}{t_p} \le 1.0\sqrt{\frac{E_s}{2\alpha + 1}} \tag{2-4}$$

The value of α is calculated using Equation 2-5:

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$
$$= 1.7\left(\frac{16 \text{ in.}}{5/16 \text{ in.}} - 2\right)\left(\frac{5/16 \text{ in.}}{1/2 \text{ in.}}\right)^4$$
$$= 12.8$$

$$1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 1.0\sqrt{\frac{29{,}000 \text{ ksi}}{2(12.8) + 1}}$$
$$= 33.0$$

For the first-floor tie bars:

$$\frac{S_{tie}}{t_p} = \frac{6 \text{ in.}}{5/16 \text{ in.}}$$
$$= 19.2$$

$$\frac{S_{tie}}{t_p} = 19.2 < 1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 33.0 \quad \textbf{o.k.}$$

For the remainder of the floor tie bars:

$$\frac{S_{tie.top}}{t_p} = \frac{9 \text{ in.}}{5/16 \text{ in.}}$$
$$= 28.8$$

$$\frac{S_{tie.top}}{t_p} = 28.8 < 1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 33.0 \quad \textbf{o.k.}$$

The tie bar spacing requirements are met.

Step 3-4. Required wall shear strength

As previously determined, the required shear strength of each planar wall is:

$V_{r.wall} = 3{,}580$ kips

Step 3-5. Required wall flexural strength

As previously determined, the required flexural strength of each planar wall is:

$M_{r.wall} = 690{,}000$ kip-in.

Step 3-6. Composite wall resistance factor

From AISC *Specification* Section I4.4, the resistance factor for a composite wall in shear is:

$\phi_v = 0.90$

From AISC *Specification* Section I3.5, the resistance factor for a composite wall in flexure is:

$\phi_b = 0.90$

Step 3-7. Composite wall flexural strength check

The plastic flexural strength of the wall was calculated in Step 3-2:

$M_{n.wall} = M_{P.wall} = 1{,}160{,}000$ kip-in.

The design flexural strength of the wall is:

$$\phi_b M_{n.wall} = 0.90(1{,}160{,}000 \text{ kip-in.})$$
$$= 1{,}040{,}000 \text{ kip-in.}$$
$$> M_{r.wall} = 690{,}000 \text{ kip-in.} \quad \textbf{o.k.}$$

Step 3-8. Composite wall shear strength check

The design shear strength of the uncoupled SpeedCore wall is calculated as follows.

The area of steel in the direction of in-plane shear is:

$$A_{sw} = 2L_{wall}t_p$$
$$= 2(300\text{ in.})(5/16\text{ in.})$$
$$= 188\text{ in.}^2$$

The area of concrete in the wall is:

$$A_c = L_{wall}t_{sc} - A_s$$
$$= (300\text{ in.})(16\text{ in.}) - 197\text{ in.}^2$$
$$= 4{,}600\text{ in.}^2$$

The K factors for shear calculation are as follows:

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200\text{ ksi})(188\text{ in.}^2)$$
$$= 2{,}110{,}000\text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_c)(E_s A_{sw})}{4E_s A_{sw} + E_c A_c} \tag{2-16}$$
$$= \frac{0.7(4{,}500\text{ ksi})(4{,}600\text{ in.}^2)(29{,}000\text{ ksi})(188\text{ in.}^2)}{4(29{,}000\text{ ksi})(188\text{ in.}^2) + (4{,}500\text{ ksi})(4{,}600\text{ in.}^2)}$$
$$= 1{,}860{,}000\text{ kips}$$

The nominal shear strength of the wall is:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \tag{2-14}$$
$$= \frac{2{,}110{,}000\text{ kips} + 1{,}860{,}000\text{ kips}}{\sqrt{3(2{,}110{,}000\text{ kips})^2 + (1{,}860{,}000\text{ kips})^2}}(50\text{ ksi})(188\text{ in.}^2)$$
$$= 9{,}100\text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the uncoupled SpeedCore wall is:

$$\phi_v V_{n.wall} = 0.90(9{,}100\text{ kips})$$
$$= 8{,}190\text{ kips}$$
$$> V_{r.wall} = 3{,}580\text{ kips}$$

The ratio of demand-to-capacity is:

$$\frac{V_{r.wall}}{\phi_v V_{n.wall}} = \frac{3{,}580\text{ kips}}{8{,}190\text{ kips}}$$
$$= 0.437$$

Step 4. Design of Connections

No connection design details are included in this design example. This section determines the length of the protected zone and the demands for the wall-to-foundation connection.

Protected zones extend from the base of the structure until no yielding in the cross section is expected; in other words, from the base to where the moment demand is the yield moment in tension, M_{yt}. The moment at the base is considered to be the expected moment, $M_{p.exp}$, because this is the moment associated with the structure's plastic mechanism. This moment is calculated and compared to the yield moment (from Step 3-2) to determine the height of the protected zone.

Figure 3-13 illustrates the plastic neutral axis location and the compression and tension force regions in the wall.

The location of the expected plastic neutral axis of the uncoupled SpeedCore wall:

$$C_{exp} = \frac{2t_p L_{wall} R_y F_y + R_c 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p R_y F_y + R_c 0.85 f_c'(t_{sc} - 2t_p)}$$

$$= \frac{2(5/16 \text{ in.})(300 \text{ in.})(1.1)(50 \text{ ksi}) + 1.3(0.85)(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})}{4(5/16 \text{ in.})(1.1)(50 \text{ ksi}) + 1.3(0.85)(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right]}$$

$$= 60.6 \text{ in.}$$

The compression force in the flange is:

$$C_{1.exp} = (t_{sc} - 2t_p)t_p R_y F_y$$

$$= \left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})(1.1)(50 \text{ ksi})$$

$$= 264 \text{ kips}$$

The compression force in the web is:

$$C_{2.exp} = 2t_p C_{exp} R_y F_y$$

$$= 2(5/16 \text{ in.})(60.6 \text{ in.})(1.1)(50 \text{ ksi})$$

$$= 2{,}080 \text{ kips}$$

The compression force in the concrete is:

$$C_{3.exp} = R_c 0.85 f_c'(t_{sc} - 2t_p)(C_{exp} - t_p)$$

$$= (1.3)(0.85)(6 \text{ ksi})\left[16 \text{ in.} - 2(5/16 \text{ in.})\right](60.6 \text{ in.} - 5/16 \text{ in.})$$

$$= 6{,}150 \text{ kips}$$

The tension force in the flange is:

$$T_{1.exp} = (t_{sc} - 2t_p)t_p R_y F_y$$

$$= \left[16 \text{ in.} - 2(5/16 \text{ in.})\right](5/16 \text{ in.})(1.1)(50 \text{ ksi})$$

$$= 264 \text{ kips}$$

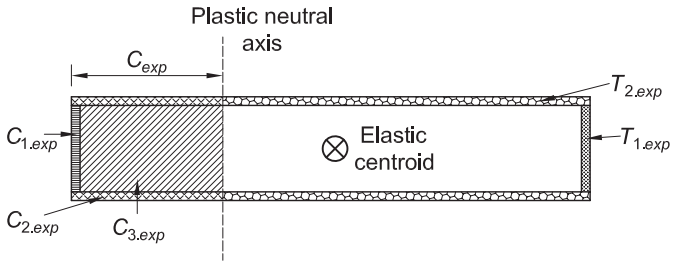


Fig. 3-13. Cross section with labeled regions for plastic moment calculation of tension wall.

The tension force in the web is:

$$T_{2.exp} = 2t_p(L_{wall} - C_{exp})R_y F_y$$
$$= 2(5/16 \text{ in.})(300 \text{ in.} - 60.6 \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 8{,}230 \text{ kips}$$

The plastic flexural strength of the uncoupled SpeedCore wall is:

$$M_{P.exp.wall} = C_{1.exp}\left(C_{exp} - \frac{t_p}{2}\right) + C_{2.exp}\left(\frac{C_{exp}}{2}\right) + C_{3.exp}\left(\frac{C_{exp} - t_p}{2}\right) + T_{1.exp}\left(L_{wall} - C_{exp} - \frac{t_p}{2}\right) + T_{2.exp}\left(\frac{L_{wall} - C_{exp}}{2}\right)$$
$$= (264 \text{ kips})\left(60.6 \text{ in.} - \frac{5/16 \text{ in.}}{2}\right) + (2{,}080 \text{ kips})\left(\frac{60.6 \text{ in.}}{2}\right) + (6{,}150 \text{ kips})\left(\frac{60.6 \text{ in.} - 5/16 \text{ in.}}{2}\right)$$
$$+ (264 \text{ kips})\left(300 \text{ in.} - 60.6 \text{ in.} - \frac{5/16 \text{ in.}}{2}\right) + (8{,}230 \text{ kips})\left(\frac{300 \text{ in.} - 60.6 \text{ in.}}{2}\right)$$
$$= 1{,}310{,}000 \text{ kip-in.}$$

The ratio of the limiting moment to $M_{p.exp.wall}$ is:

$$r_2 = \frac{M_{yt}}{M_{p.exp.wall}}$$
$$= \frac{665{,}000 \text{ kip-in.}}{1{,}310{,}000 \text{ kip-in.}}$$
$$= 0.508$$

The portion of the wall with demand higher than the moment limit is:

$$(1 - r_2)\left[h_1 + (n-1)h_{typ}\right] = (1 - 0.508)\left[17 \text{ ft} + (6-1)14 \text{ ft}\right]$$
$$= 42.8 \text{ ft}$$

Based on these calculations, the protected zone extends to the 3rd floor (45 ft).

Wall-to-Foundation Connection Demands

The flexural demand for a connection at the base of the wall is based on the system's failure mechanism (i.e., hinging at the base of the walls):

$$1.1M_{p.exp.wall} = 1.1(1{,}310{,}000 \text{ kip-in.})$$
$$= 1{,}440{,}000 \text{ kip-in.}$$

The shear demand on a connection at the base of the wall is the amplified equivalent lateral force (ELF) shear:

$$V_{r.wall} = 3{,}580 \text{ kips}$$

EXAMPLE 3.2—Seismic Design of 18-Story Structure Using Uncoupled C-Shaped SpeedCore Walls

This example presents the seismic design of an 18-story building using uncoupled SpeedCore walls with typical design loads, floor geometry, and high seismic design loads. The steps followed in this design follow the design procedure presented in Chapter 3. For simplicity, this design example does not consider accidental eccentricity and assumes a seismic redundancy factor of 1.0.

Design the uncoupled, C-shaped SpeedCore walls shown in Figure 3-14 using the given geometry, material properties, and loads.

Given:

Step 1. General Information of the Considered Building

The steel and concrete material properties are as follows:

ASTM A572/A572M Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi
R_c = 1.3

Building geometry:

L_f = building length
= 120 ft
W_f = building width
= 210 ft
h_{typ} = typical story height
= 14 ft
h_1 = first-story height
= 17 ft
n = number of stories
= 18

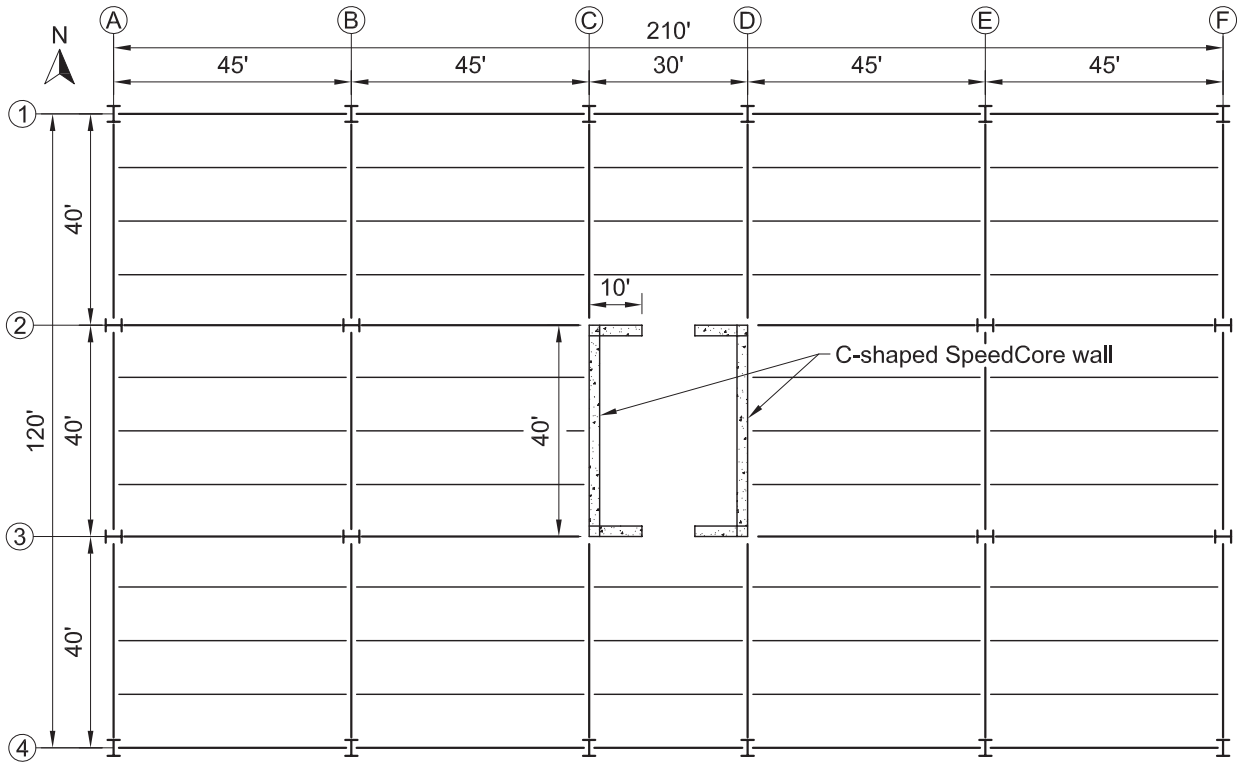


Fig. 3-14. Building floor plan for Example 3.2.

Table 3-3. Seismic Design Loads from ASCE/SEI 7			
Story No.	**Story Elevation (ft)**	**Story Force (kips)**	**Story Shear (kips)**
18	255	398	398
17	241	363	761
16	227	329	1,090
15	213	296	1,390
14	199	265	1,650
13	185	235	1,890
12	171	206	2,090
11	157	180	2,270
10	143	154	2,430
9	129	130	2,560
8	115	107	2,660
7	101	86.9	2,750
6	87	68.0	2,820
5	73	51.0	2,870
4	59	35.9	2,900
3	45	23.0	2,930
2	31	12.5	2,940
1	17	4.6	2,940
Total Base Shear, V_{base}			2,940 kips
Overturning Moment, OTM			6,720,000 kip-in.

Floor loads:

DL = floor dead load
= 0.12 ksf

Seismic design loads along the major axis of the C-shaped SpeedCore wall are listed in Table 3-3.

Seismic design coefficients:

C_d = deflection amplification factor (ASCE/SEI 7, Table 12.2-1)
= 5.5
I_e = importance factor (ASCE/SEI 7, Section 11.5.1, for an office building)
= 1
R = seismic response modification coefficient (from ASCE/SEI 7, Table 12.2-1)
= 6.5

Risk Category = II (ASCE/SEI 7, Table 1.5-1)

Ω_0 = overstrength factor from ASCE/SEI 7, Table 12.2-1
= 2.5
ϕ = seismic redundancy factor (ASCE/SEI 7, Section 12.3.4)
= 1

For ASTM A572/A572M Grade 50 steel, from AISC *Manual* Table 2-4 and AISC *Seismic Provisions* Table A3.2, the material properties are as follows:

F_y = 50 ksi
F_u = 65 ksi
R_y = 1.1

Solution:

SpeedCore members are sized based on initial estimates of loads and are then refined through iteration. Acceptable dimensions are presented here so that this example can focus on the appropriate limit state checks.

The wall and plate dimensions, shown in Figure 3-15, are selected as follows:

L_{cb} = clear span coupling beam length
= 120 in.
L_{wall} = web wall length
= 480 in.
W_{wall} = flange wall length
= 120 in.
t_p = wall plate thickness
= ½ in.
$t_{sc.f}$ = flange wall thickness
= 18 in.
$t_{sc.w}$ = web wall thickness
= 18 in.

Step 2. Analysis for Design

In this step, an elastic computer model is built and analyzed to determine the forces on system members and overall system deflection.

C-Shaped SpeedCore Wall Geometric Properties

The effective axial, shear, and flexural stiffnesses are calculated for SpeedCore elements for elastic analysis.

The area of steel in the wall is:

$$A_s = t_p(4W_{wall} + 2L_{wall} + 2t_{sc.f} - 12t_p)$$
$$= (½\text{ in.})\left[4(120\text{ in.}) + 2(480\text{ in.}) + 2(18\text{ in.}) - 12(½\text{ in.})\right]$$
$$= 735\text{ in.}^2$$

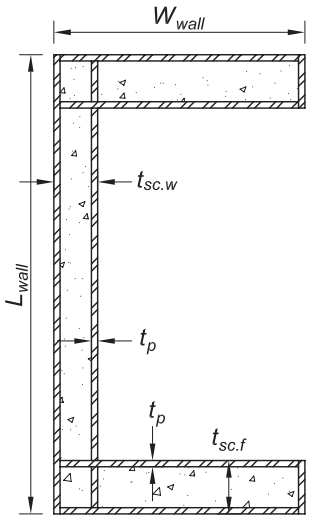


Fig. 3-15. Wall cross-section dimensions.

The area of concrete in the wall is:

$$A_c = 2W_{wall}t_{sc.f} + (L_{wall} - 2t_{sc.f})t_{sc.w} - A_s$$
$$= 2(120\text{ in.})(18\text{ in.}) + \left[480\text{ in.} - 2(18\text{ in.})\right](18\text{ in.}) - 735\text{ in.}^2$$
$$= 11{,}600\text{ in.}^2$$

The effective axial stiffness of the wall is:

$$EA_{eff} = E_sA_s + 0.45E_cA_c \quad (2\text{-}10)$$
$$= (29{,}000\text{ ksi})(735\text{ in.}^2) + 0.45(4{,}500\text{ ksi})(11{,}600\text{ in.}^2)$$
$$= 4.48\times10^7\text{ kips}$$

The area of steel in the direction of shear is:

$$A_{sw} = 2L_{wall}t_p$$
$$= 2(480\text{ in.})(½\text{ in.})$$
$$= 480\text{ in.}^2$$

The area of concrete in the direction of shear is:

$$A_{cw} = (L_{wall} - 4t_p)(t_{sc.w} - 2t_p)$$
$$= \left[480\text{ in.} - 4(½\text{ in.})\right]\left[18\text{ in.} - 2(½\text{ in.})\right]$$
$$= 8{,}130\text{ in.}^2$$

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_sA_{sw} + G_cA_{cw} \quad \text{(from Eq. 2-11)}$$
$$= (11{,}200\text{ ksi})(480\text{ in.}^2) + (1{,}800\text{ ksi})(8{,}130\text{ in.}^2)$$
$$= 2.00\times10^7\text{ kips}$$

The moment of inertia of steel in the wall is:

$$I_s = 2\left[\frac{t_p(L_{wall} - 2t_p)^3}{12} + \frac{W_{wall}t_p^3}{12} + (W_{wall}t_p)\left(\frac{L_{wall}}{2} - \frac{t_p}{2}\right)^2 + \frac{t_p(t_{sc.f} - 2t_p)^3}{12} + (t_{sc.f} - 2t_p)t_p\left(\frac{L_{wall}}{2} - \frac{t_{sc.f}}{2}\right)^2\right]$$
$$+ 2\left[\frac{(W_{wall} - 2t_p)t_p^3}{12} + (W_{wall} - 2t_p)t_p\left(\frac{L_{wall}}{2} - t_{sc.f} - \frac{t_p}{2}\right)^2\right]$$
$$= 2\left\{\frac{(½\text{ in.})\left[480\text{ in.} - 2(½\text{ in.})\right]^3}{12} + \frac{(120\text{ in.})(½\text{ in.})^3}{12} + (120\text{ in.})(½\text{ in.})\left(\frac{480\text{ in.}}{2} - \frac{½\text{ in.}}{2}\right)^2\right\}$$
$$+ 2\left\{\frac{(½\text{ in.})\left[18\text{ in.} - 2(½\text{ in.})\right]^3}{12} + \left[18\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})\left(\frac{480\text{ in.}}{2} - \frac{18\text{ in.}}{2}\right)^2\right\}$$
$$+ 2\left\{\frac{\left[120\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})^3}{12} + \left[120\text{ in.} - 2(½\text{ in.})\right](½\text{ in.})\left(\frac{480\text{ in.}}{2} - 18\text{ in.} - \frac{½\text{ in.}}{2}\right)^2\right\}$$
$$= 2.28\times10^7\text{ in.}^4$$

The moment of inertia of concrete in the wall is:

$$I_c = \frac{(t_{sc.w} - 2t_p)(L_{wall} - 2t_{sc.f})^3}{12} + 2\left[\frac{(W_{wall} - 3t_p)(t_{sc.f} - 2t_p)^3}{12} + (W_{wall} - 3t_p)(t_{sc.f} - 2t_p)\left(\frac{L_{wall}}{2} - \frac{t_{sc.f}}{2}\right)^2\right]$$

$$= \frac{[18\text{ in.} - 2(½\text{ in.})][480\text{ in.} - 2(18\text{ in.})]^3}{12}$$

$$+ 2\left\{\frac{[120\text{ in.} - 3(½\text{ in.})][18\text{ in.} - 2(½\text{ in.})]^3}{12} + [120\text{ in.} - 3(½\text{ in.})][18\text{ in.} - 2(½\text{ in.})]\left(\frac{480\text{ in.}}{2} - \frac{18\text{ in.}}{2}\right)^2\right\}$$

$$= 3.39\times10^8\text{ in.}^4$$

Therefore, the effective moment of inertia of the wall is:

$$EI_{eff} = E_s I_s + 0.35E_c I_c \quad (2\text{-}9)$$

$$= (29{,}000\text{ ksi})(2.28\times10^7\text{ in.}^4) + 0.35(4{,}500\text{ ksi})(3.39\times10^8\text{ in.}^4)$$

$$= 1.20\times10^{12}\text{ kip-in.}^2$$

Numerical Model

To determine the ID ratio and shear demand on the wall, an analysis model was built using commercial software. First-order, linear elastic analysis was performed on a model consisting of frame elements for the SpeedCore wall. This model was subjected to the earthquake loading previously defined and all mass was applied at the story level. The model is shown in Figure 3-16.

Wall elements were assigned section properties that were previously calculated:

$EA_{eff} = 4.48\times10^7$ kips

$GA_{v.eff} = 2.00\times10^7$ kips

$EI_{eff} = 1.20\times10^{12}$ kips-in.^{2}

Lateral displacement values are amplified by the C_d factor to obtain the amplified displacement. These amplified values are then used to calculate the ID, which is the difference in displacement between two floors normalized by the story height. The maximum design ID is limited according to ASCE/SEI 7, Section 12.12.1. In this case, the maximum design ID is 2%. The story displacement, amplified displacement, and ID are presented in Table 3-4.

The maximum ID ratio = 1.87% < 2% **o.k.**

Required Shear and Flexural Strength

The base shear given in Table 3-3 is amplified by a shear amplification factor of 4.0, following the recommendations of Section 3.3.2.1 of this Design Guide.

$$V_{amp} = 4.0V_{base} \quad (3\text{-}5)$$

$$= 4.0(2{,}940\text{ kips})$$

$$= 11{,}800\text{ kips}$$

$$V_{r.wall} = \frac{V_{amp}}{2}$$

$$= \frac{11{,}800\text{ kips}}{2}$$

$$= 5{,}900\text{ kips}$$

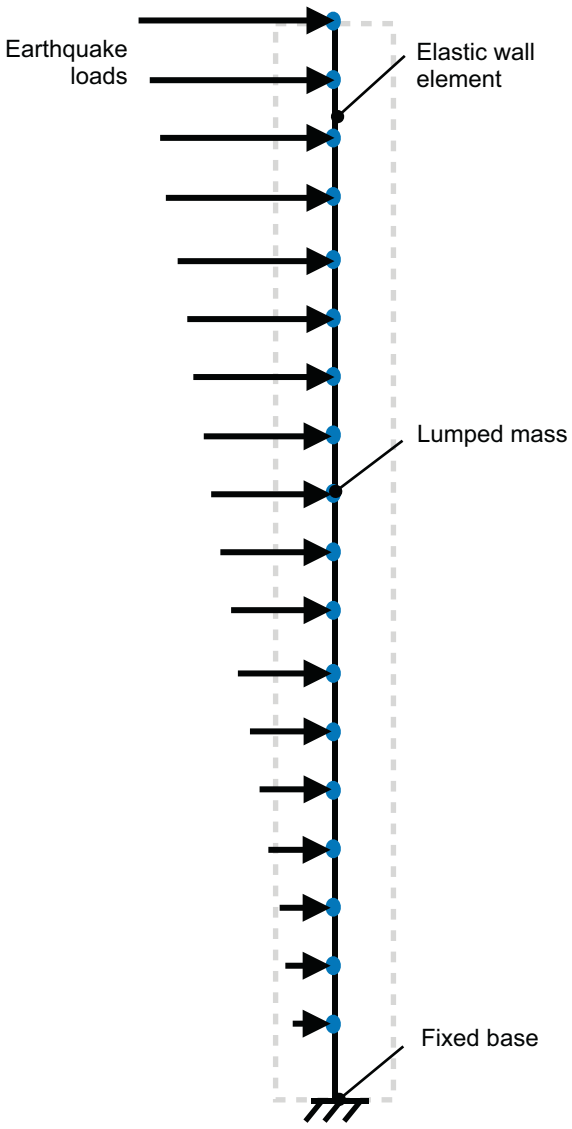


Fig. 3-16. Analysis model components.

Table 3-4. Lateral Displacement and ID Summary				
Story No.	Story Elevation (ft)	Displacement (in.)	Amplified Displacement (in.)	ID (%)
18	255	7.48	41.1	1.87
17	241	6.91	38.0	1.83
16	227	6.35	34.9	1.87
15	213	5.78	31.8	1.83
14	199	5.22	28.7	1.83
13	185	4.66	25.6	1.80
12	171	4.11	22.6	1.74
11	157	3.58	19.7	1.70
10	143	3.06	16.8	1.60
9	129	2.57	14.1	1.54
8	115	2.10	11.6	1.41
7	101	1.67	9.18	1.31
6	87	1.27	6.99	1.15
5	73	0.92	5.06	1.01
4	59	0.61	3.36	0.82
3	45	0.36	1.97	0.62
2	31	0.17	0.94	0.39
1	17	0.05	0.28	0.13
Base	0.00	0.00	0.00	0.00

The required flexural strength of each C-shaped wall is:

$$M_{r.wall} = \frac{M_{OT}}{2}$$
$$= \frac{6{,}720{,}000 \text{ kip-in.}}{2}$$
$$= 3{,}360{,}000 \text{ kip-in.}$$

Step 3. Design of Composite Walls

Step 3-1. Minimum and maximum area of steel

Gross area of the C-shaped wall:

$$A_g = L_{wall} t_{sc.w} + 2(W_{wall} - t_{sc.w}) t_{sc.f}$$
$$= (480 \text{ in.})(18 \text{ in.}) + 2(120 \text{ in.} - 18 \text{ in.})(18 \text{ in.})$$
$$= 12{,}300 \text{ in.}^2$$

The minimum steel required (AISC *Specification* Section I1.6) as given in Section 2.2.1:

$$A_{s.min} = 0.01A_g$$
$$= 0.01(12{,}300\ \text{in.}^2)$$
$$= 123\ \text{in.}^2$$

$$A_s = 735\ \text{in.}^2 > A_{s.min} = 123\ \text{in.}^2 \quad \textbf{o.k.}$$

The maximum steel required according to AISC *Specification* Section I1.6 is:

$$A_{s.max} = 0.1A_g$$
$$= 0.1(12{,}300\ \text{in.}^2)$$
$$= 1{,}230\ \text{in.}^2$$

$$A_s = 735\ \text{in.}^2 < A_{s.max} = 1{,}230\ \text{in.}^2 \quad \textbf{o.k.}$$

Step 3-2. Steel plate slenderness requirements for composite walls

The largest unsupported length between rows of steel anchors or tie bars is selected under the assumption that all support is through the tie bars and not the steel anchors; therefore, the plate slenderness and tie bar spacing requirements must be met. As discussed in Section 3.2.4.1 of the Design Guide, tie bars and steel anchors can be spaced farther apart outside of flexural yielding zones. The extent of the flexural yielding zones is addressed in the following.

A tie bar spacing of 12 in. is selected for the first story of the uncoupled SpeedCore wall, and a tie bar spacing of 14 in. is selected for the other stories.

$$S_{tie} = 12\ \text{in.}$$
$$S_{tie.top} = 14\ \text{in.}$$

From Section 3.2.4.1, the steel plate senderless ratio, b/t_p, is limited to:

$$\frac{b}{t_p} \le 1.05\sqrt{\frac{E_s}{R_yF_y}} \tag{3-1}$$

where b is the largest unsupported length of plate between rows of steel anchors or tie bars, which is this case is S_{tie}.

$$\frac{b}{t_p} = \frac{S_{tie}}{t_p}$$
$$= \frac{12\ \text{in.}}{½\ \text{in.}}$$
$$= 24.0$$

$$1.05\sqrt{\frac{E_s}{F_y}} = 1.05\sqrt{\frac{29{,}000\ \text{ksi}}{1.1(50\ \text{ksi})}}$$
$$= 24.1$$

$$\frac{b}{t_p} = 24.0 < 1.05\sqrt{\frac{E_s}{R_yF_y}} = 24.1 \quad \textbf{o.k.}$$

From Section 3.2.4.1, the steel plate senderless ratio, b/t_p, in the remainder of the stories is limited to:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{3-2}$$

where b is the largest unsupported length of plate between rows of steel anchors or tie bars, which in this case is $S_{tie.top}$.

$$\frac{b}{t_p} = \frac{S_{tie.top}}{t_p}$$
$$= \frac{14 \text{ in.}}{½ \text{ in.}}$$
$$= 28.0$$

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}}$$
$$= 28.9$$

$$\frac{b}{t_p} = 28.0 < 1.2\sqrt{\frac{E_s}{F_y}} = 28.9 \quad \textbf{o.k.}$$

The slenderness ratio requirements for the steel plates are met.

Flexural Yielding Zone

Tie bar and anchor spacing is intended to meet a more stringent requirement in zones of the wall where compression buckling is likely to occur or in the plastic hinge region.

Compression buckling of the faceplate is more likely to occur when the stress in the faceplate exceeds F_y. This load level will be called the yield moment in compression, M_{yc}. In addition to this limit, the plastic hinge region also requires more stringent tie bar and anchor spacing. This region is assumed to have a moment greater than $0.8M_p$. The lesser of these two limits, M_{yc} or $0.8M_p$, is used to determine where the tighter tie bar and anchor spacing can be relaxed in the SpeedCore walls.

The moments M_p, M_{yc}, and the yield moment in tension, M_{yt}, are calculated and compared to determine the height to which the more stringent tie bar spacing requirement should be applied. To calculate this height, the moment distribution is assumed to vary linearly with a magnitude of M_p at the base of the wall and 0 at the top of the structure.

The plastic neutral axis location of the uncoupled SpeedCore wall is:

$$C = \frac{2L_{wall}t_pF_y + 0.85f_c'(t_{c.w}t_{sc.f} - t_{c.f}W_c)}{4t_pF_y + 0.85f_c't_{c.w}}$$

The length of the flange wall without perpendicular steel plates is:

$$W_c = W_{wall} - 3t_p$$
$$= 120 \text{ in.} - 3(½ \text{ in.})$$
$$= 119 \text{ in.}$$

The width of concrete in the web wall is:

$$t_{c.w} = t_{sc.w} - 2t_p$$
$$= 18 \text{ in.} - 2(½ \text{ in.})$$
$$= 17.0 \text{ in.}$$

The width of concrete in the flange wall is:

$$t_{c.f} = t_{sc.f} - 2t_p$$
$$= 18 \text{ in.} - 2(½ \text{ in.})$$
$$= 17.0 \text{ in.}$$

And thus,

$$C = \frac{2(480 \text{ in.})(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})\left[(17.0 \text{ in.})(18 \text{ in.}) - (17.0 \text{ in.})(119 \text{ in.})\right]}{4(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(17.0 \text{ in.})}$$
$$= 81.6 \text{ in.}$$

The compression force in the exterior steel flange plate is:

$$C_1 = W_c t_p F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 2{,}980 \text{ kips}$$

The compression force in the interior steel flange plate is:

$$C_2 = W_c t_p F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 2{,}980 \text{ kips}$$

The compression force in the flange steel end plate is:

$$C_3 = t_{sc.f} t_p F_y$$
$$= (18 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 450 \text{ kips}$$

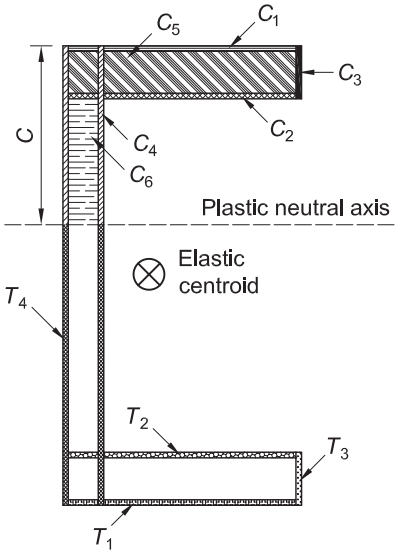


Fig. 3-17. Cross section with labeled regions for plastic moment calculation of tension wall.

The compression force in the steel plate in the web is:

$$C_4 = 2C\,t_p F_y$$
$$= 2(81.6\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 4{,}080\text{ kips}$$

The compression force in the concrete in the flange is:

$$C_5 = 0.85 f_c' t_{c.f} W_c$$
$$= 0.85(6\text{ ksi})(17\text{ in.})(119\text{ in.})$$
$$= 10{,}300\text{ kips}$$

The compression force carried by the concrete in the web is:

$$C_6 = 0.85 f_c' t_{c.w} (C - t_{sc.f})$$
$$= 0.85(6\text{ ksi})(17.0\text{ in.})(81.6\text{ in.} - 18\text{ in.})$$
$$= 5{,}510\text{ kips}$$

The tensile force in the exterior flange steel plate is:

$$T_1 = W_c\,t_p F_y$$
$$= (119\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 2{,}980\text{ kips}$$

The tension force in the interior flange steel plate is:

$$T_2 = W_c\,t_p F_y$$
$$= (119\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 2{,}980\text{ kips}$$

The tension force in the flange end steel plate:

$$T_3 = t_{sc.f}\,t_p F_y$$
$$= (18\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 450\text{ kips}$$

The tension force carried by the steel plate in the web is:

$$T_4 = 2(L_{wall} - C)\,t_p F_y$$
$$= 2(480\text{ in.} - 81.6\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 19{,}900\text{ kips}$$

The plastic flexural strength of the uncoupled SpeedCore wall is:

$$M_{P.wall} = C_1\left(C - \frac{t_p}{2}\right) + C_2\left(C - t_{sc.f} + \frac{t_p}{2}\right) + C_3\left(C - \frac{t_{sc.f}}{2}\right) + C_4\left(\frac{C}{2}\right) + C_5\left(C - \frac{t_{sc.f}}{2}\right) + C_6\left(\frac{C - t_{sc.f}}{2}\right)$$
$$+ T_1\left(L_{wall} - C - \frac{t_p}{2}\right) + T_2\left(L_{wall} - C - t_{sc.f} + \frac{t_p}{2}\right) + T_3\left(L_{wall} - C - \frac{t_{sc.f}}{2}\right) + T_4\left(\frac{L_{wall} - C}{2}\right)$$
$$= (2{,}980 \text{ kips})\left(81.6 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (2{,}980 \text{ kips})\left(81.6 \text{ in.} - 18 \text{ in.} + \frac{½ \text{ in.}}{2}\right) + (450 \text{ kips})\left(81.6 \text{ in.} - \frac{18 \text{ in.}}{2}\right)$$
$$+ (4{,}080 \text{ kips})\left(\frac{81.6 \text{ in.}}{2}\right) + (10{,}300 \text{ kips})\left(81.6 \text{ in.} - \frac{18 \text{ in.}}{2}\right) + (5{,}510 \text{ kips})\left(\frac{81.6 \text{ in.} - 18 \text{ in.}}{2}\right)$$
$$+ (2{,}980 \text{ kips})\left(480 \text{ in.} - 81.6 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (2{,}980 \text{ kips})\left(480 \text{ in.} - 81.6 \text{ in.} - 18 \text{ in.} + \frac{½ \text{ in.}}{2}\right)$$
$$+ (450 \text{ kips})\left(480 \text{ in.} - 81.6 \text{ in.} - \frac{18 \text{ in.}}{2}\right) + (19{,}900 \text{ kips})\left(\frac{480 \text{ in.} - 81.6 \text{ in.}}{2}\right)$$
$$= 8{,}200{,}000 \text{ kip-in.}$$

$$0.8M_{P.wall} = 0.8(8{,}200{,}000 \text{ kip-in.})$$
$$= 6{,}560{,}000 \text{ kip-in.}$$

Yield Moments, M_{yt} *and* M_{yc}

The yield moment is calculated from section moment curvature analysis considering nominal properties. The moment curvature plot is shown in Figure 3-18. This case did not consider axial load but considering the effects of axial loads in the analysis is also possible.

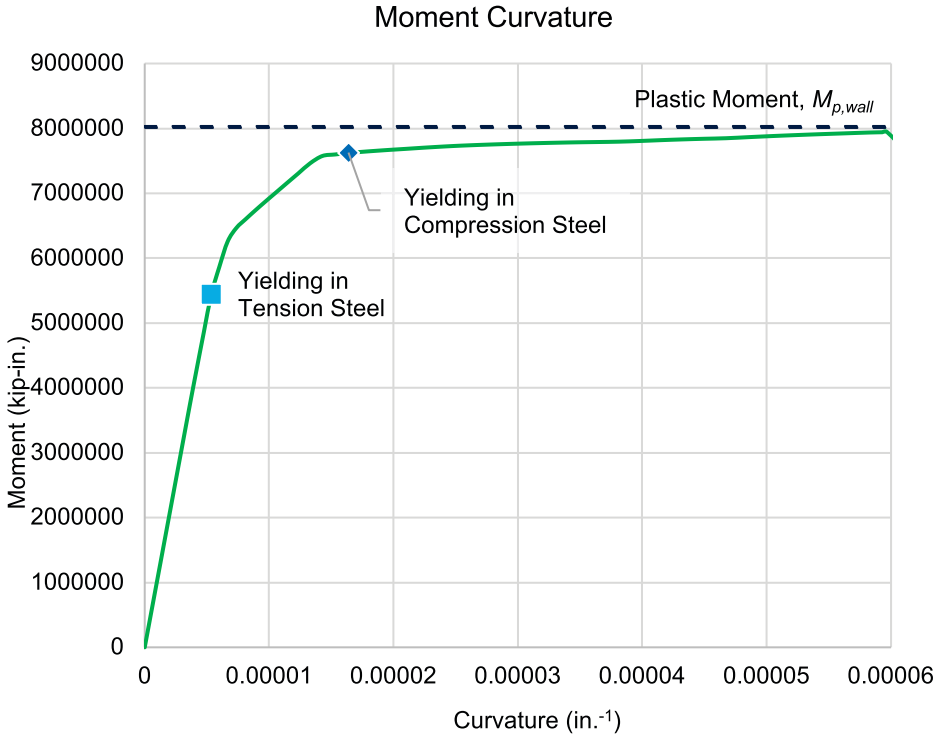


Fig. 3-18. Moment curvature plot.

From analysis:

$$M_{yt} = 5{,}450{,}000 \text{ kip-in.}$$

$$M_{yc} = 7{,}620{,}000 \text{ kip-in.}$$

$$\frac{M_{yt}}{M_{P.wall}} = \frac{5{,}450{,}000 \text{ kip-in.}}{8{,}200{,}000 \text{ kip-in.}} = 0.665$$

$$\frac{M_{yc}}{M_{P.wall}} = \frac{7{,}620{,}000 \text{ kip-in.}}{8{,}200{,}000 \text{ kip-in.}} = 0.929$$

The ratio of limiting moment to $M_{P.wall}$ is:

$$r_1 = \frac{\min(M_{yc}, 0.8M_{P.wall})}{M_{P.wall}} = \frac{\min(7{,}620{,}000 \text{ kip-in.}, 6{,}560{,}000 \text{ kip-in.})}{8{,}200{,}000 \text{ kip-in.}} = 0.800$$

The portion of the wall with demand higher than the moment limit is:

$$(1-r_1)\left[h_1 + (n-1)h_{typ}\right] = (1-0.800)\left[17 \text{ ft} + (18-1)(14 \text{ ft})\right] = 51.0 \text{ ft}$$

Based on these calculations, the steel anchor and tie bar spacing would be 12 in. o.c. through the fourth story (59 ft) and then could transition to 14 in. o.c. for the remainder of the height.

Step 3-3. Tie bar spacing requirements for composite walls

A tie bar diameter of 1 in. is selected for the uncoupled SpeedCore wall, and the tie bar requirement considering the stability of the empty steel module is checked as follows:

$$d_{tie} = 1 \text{ in.}$$

$$S_{tie.top} = 14 \text{ in.}$$

From Section 2.2.1.2, the tie bar spacing to plate thickness ratio, S/t_p, is limited as stated in Equation 2-4:

$$\frac{S_{tie.top}}{t_p} \le 1.0\sqrt{\frac{E_s}{2\alpha + 1}} \tag{2-4}$$

The value of α is calculated using Equation 2-5:

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$

$$= 1.7\left(\frac{18 \text{ in.}}{½ \text{ in.}} - 2\right)\left(\frac{½ \text{ in.}}{1 \text{ in.}}\right)^4 = 3.61$$

$$1.0\sqrt{\frac{E_s}{2\alpha+1}} = 1.0\sqrt{\frac{29{,}000 \text{ ksi}}{2(3.61)+1}}$$
$$= 59.4$$

$$\frac{S_{tie.top}}{t_p} = \frac{14 \text{ in.}}{½ \text{ in.}}$$
$$= 28.0$$

$$\frac{S_{tie.top}}{t_p} = 28.0 < 1.0\sqrt{\frac{E_s}{2\alpha+1}} = 59.4 \quad \textbf{o.k.}$$

The tie bar spacing requirements are met.

Step 3-4. Required wall shear strength

The required shear strength of each planar wall, including the amplification factor, was calculated in Step 2 as:

$V_{r.wall} = 5{,}900$ kips

Step 3-5. Required wall flexural strength

The required flexural strength of each planar wall, including the amplification factor, was calculated in Step 2 as:

$M_{r.wall} = 3{,}360{,}000$ kip-in.

Step 3-6. Composite wall resistance factor

The resistance factor for a composite wall in shear according to AISC *Specification* Section I4.4 is:

$\phi_v = 0.90$

The resistance factor for a composite wall in flexure according to AISC *Specification* Section I3.5 is:

$\phi_b = 0.90$

Step 3-7. Composite wall flexural strength check

The plastic flexural strength of wall as calculated in Step 3-2:

$M_{n.wall} = M_{P.wall} = 8{,}200{,}000$ kip-in.

$\phi_b M_{n.wall} = 0.90(8{,}200{,}000 \text{ kip-in.})$
$= 7{,}380{,}000 \text{ kip-in.} > M_{r.wall} = 3{,}360{,}000 \text{ kip-in.}$ **o.k.**

Step 3-8. Composite wall shear strength check

Design Shear Strength of the Uncoupled SpeedCore Wall

The steel in the direction of shear is:

$A_{sw} = 2L_{wall}t_p$
$= 2(480 \text{ in.})(½ \text{ in.})$
$= 480 \text{ in.}^2$

The area of concrete in the direction of shear is:

$A_{cw} = (L_{wall} - 4t_p)(t_{sc.w} - 2t_p)$
$= \left[480 \text{ in.} - 4(½ \text{ in.})\right]\left[18 \text{ in.} - 2(½ \text{ in.})\right]$
$= 8{,}130 \text{ in.}^2$

The K factors for the shear calculations are:

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200 \text{ ksi})(480 \text{ in.})$$
$$= 5{,}380{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_{cw})(E_s A_{sw})}{(4E_s A_{sw}) + (E_c A_{cw})} \quad \text{(from Eq. 2-16)}$$
$$= \frac{0.7(4{,}500 \text{ ksi})(8{,}130 \text{ in.}^2)(29{,}000 \text{ ksi})(480 \text{ in.}^2)}{4(29{,}000 \text{ ksi})(480 \text{ in.}^2) + (4{,}500 \text{ ksi})(8{,}130 \text{ in.}^2)}$$
$$= 3{,}860{,}000 \text{ kips}$$

The nominal shear strength of the wall is:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \tag{2-14}$$
$$= \frac{5{,}380{,}000 \text{ kips} + 3{,}860{,}000 \text{ kips}}{\sqrt{3(5{,}380{,}000 \text{ kips})^2 + (3{,}860{,}000 \text{ kips})^2}} (50 \text{ ksi})(480 \text{ in.}^2)$$
$$= 22{,}000 \text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the uncoupled SpeedCore wall is:

$$\phi_v V_{n.wall} = 0.90(22{,}000 \text{ kips})$$
$$= 19{,}800 \text{ kips} > V_{r.wall} = 5{,}900 \text{ kips} \quad \textbf{o.k.}$$

Step 4. Design of Connections

No connection design details are included in this design example. This section outlines the length of the protected zone and demands for the wall-to-foundation connection.

Protected zones extend from the base of the structure until no yielding in the cross section is expected; in other words, from the base to where the moment demand is yield moment in tension, M_{yt}. The moment at the base is considered to be the expected moment, $M_{p.exp}$, because this is the moment associated with the structure's plastic mechanism. This moment is calculated and compared to the yield moment from Step 3-2 to determine the height of the protected zone.

Figure 3-19 illustrates the plastic neutral axis location and the compression and tension force regions in the wall.

The expected plastic neutral axis location of the uncoupled SpeedCore wall is:

$$C_{exp} = \frac{2L_{wall} t_p R_y F_y + 0.85 R_c f_c' (t_{c.w} t_{sc.f} - t_{c.f} W_c)}{4 t_p R_y F_y + 0.85 R_c f_c' t_{c.w}}$$
$$= \frac{2(480 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})\left[(17.0 \text{ in.})(18 \text{ in.}) - (17.0 \text{ in.})(119 \text{ in.})\right]}{4(½ \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})(17.0 \text{ in.})}$$
$$= 67.4 \text{ in.}$$

The compression force in the exterior steel flange plate is:

$$C_{1.exp} = W_c t_p R_y F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 3{,}270 \text{ kips}$$

The compression force in the interior steel flange plate is:

$$C_{2.exp} = W_c t_p R_y F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 3{,}270 \text{ kips}$$

The compression force in the flange end steel plate is:

$$C_{3.exp} = t_{sc.f} t_p R_y F_y$$
$$= (18 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 495 \text{ kips}$$

The compression force in the steel plate in the web is:

$$C_{4.exp} = 2C_{exp} t_p R_y F_y$$
$$= 2(67.4 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 3{,}710 \text{ kips}$$

The compression force carried by the concrete in the flange is:

$$C_{5.exp} = 0.85 R_c f'_c t_{c.f} W_c$$
$$= 0.85(1.3)(6 \text{ ksi})(17 \text{ in.})(119 \text{ in.})$$
$$= 13{,}400 \text{ kips}$$

The compression force in the concrete in the web is:

$$C_{6.exp} = 0.85 R_c f'_c t_{c.w} (C_{exp} - t_{sc.f})$$
$$= 0.85(1.3)(6 \text{ ksi})(17.0 \text{ in.})(67.4 \text{ in.} - 18 \text{ in.})$$
$$= 5{,}570 \text{ kips}$$

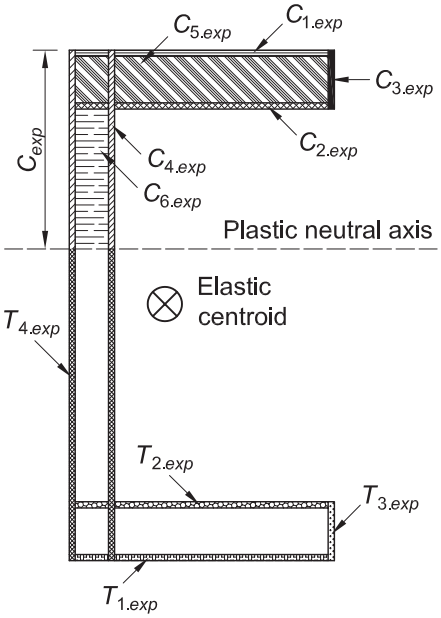


Fig. 3-19. Cross section with labeled regions for plastic moment calculation of tension wall.

The tension force in the exterior steel flange plate is:

$$T_{1.exp} = W_c t_p R_y F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 3{,}270 \text{ kips}$$

The tension force in the interior steel flange plate is:

$$T_{2.exp} = W_c t_p R_y F_y$$
$$= (119 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 3{,}270 \text{ kips}$$

The tensile force in the steel flange end plate is:

$$T_{3.exp} = t_{sc.f} t_p R_y F_y$$
$$= (18 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 495 \text{ kips}$$

The tensile force in the steel plate in the web is:

$$T_{4.exp} = 2(L_{wall} - C_{exp}) t_p R_y F_y$$
$$= 2(480 \text{ in.} - 67.4 \text{ in.})(½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 22{,}700 \text{ kips}$$

The plastic flexural strength of the uncoupled SpeedCore wall is calculated as:

$$M_{p.exp.wall} = C_{1.exp}\left(C_{exp} - \frac{t_p}{2}\right) + C_{2.exp}\left(C_{exp} - t_{sc.f} + \frac{t_p}{2}\right) + C_{3.exp}\left(C_{exp} - \frac{t_{sc.f}}{2}\right) + C_{4.exp}\left(\frac{C_{exp}}{2}\right)$$
$$+ C_{5.exp}\left(C_{exp} - \frac{t_{sc.f}}{2}\right) + C_{6.exp}\left(\frac{C_{exp} - t_{sc.f}}{2}\right) + T_{1.exp}\left(L_{wall} - C_{exp} - \frac{t_p}{2}\right)$$
$$+ T_{2.exp}\left(L_{wall} - C_{exp} - t_{sc.f} + \frac{t_p}{2}\right) + T_{3.exp}\left(L_{wall} - C_{exp} - \frac{t_{sc.f}}{2}\right) + T_{4.exp}\left(\frac{L_{wall} - C_{exp}}{2}\right)$$
$$= (3{,}270 \text{ kips})\left(67.4 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (3{,}270 \text{ kips})\left(67.4 \text{ in.} - 18 \text{ in.} + \frac{½ \text{ in.}}{2}\right)$$
$$+ (495 \text{ kips})\left(67.4 \text{ in.} - \frac{18 \text{ in.}}{2}\right) + (3{,}710 \text{ kips})\left(\frac{67.4 \text{ in.}}{2}\right)$$
$$+ (13{,}400 \text{ kips})\left(67.4 \text{ in.} - \frac{18 \text{ in.}}{2}\right) + (5{,}570 \text{ kips})\left(\frac{67.4 \text{ in.} - 18 \text{ in.}}{2}\right)$$
$$+ (3{,}270 \text{ kips})\left(480 \text{ in.} - 67.4 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (3{,}270 \text{ kips})\left(480 \text{ in.} - 67.4 \text{ in.} - 18 \text{ in.} + \frac{½ \text{ in.}}{2}\right)$$
$$+ (495 \text{ kips})\left(480 \text{ in.} - 67.4 \text{ in.} - \frac{18 \text{ in.}}{2}\right) + (22{,}700 \text{ kips})\left(\frac{480 \text{ in.} - 67.4 \text{ in.}}{2}\right)$$
$$= 8{,}980{,}000 \text{ kip-in.}$$

Ratio of limiting moment to $M_{p.exp.wall}$:

$$r_2 = \frac{M_{yt}}{M_{p.exp.wall}}$$
$$= \frac{5{,}450{,}000 \text{ kip-in.}}{8{,}980{,}000 \text{ kip-in.}}$$
$$= 0.607$$

Portion of wall with demand higher than moment limit:

$$(1-r_2)\left[h_1 + (n-1)h_{typ}\right] = (1-0.608)\left[17 \text{ ft} + (18-1)14 \text{ ft}\right]$$
$$= 100 \text{ ft}$$

Based on these calculations, the protected zone extends beyond the seventh floor (101 ft).

Wall-to-Foundation Connection Demands

The flexural demand for a connection at the base of the wall is based on the system's failure mechanism (hinging at the base of the walls):

$$1.1M_{p.exp.wall} = 1.1(8{,}980{,}000 \text{ kip-in.})$$
$$= 9{,}880{,}000 \text{ kip-in.}$$

The shear demand for a connection at the base of the wall is the amplified ELF shear as calculated previously in Step 2:

$$V_{r.wall} = 5{,}900 \text{ kips}$$

Chapter 4
Seismic Design of Coupled SpeedCore Walls

4.1 OVERVIEW

Coupled composite plate shear walls (CC-PSW/CF), also known as coupled SpeedCore walls, consist of two or more composite plate shear walls connected together using composite coupling beams. The composite walls may be planar, C-shaped, I-shaped, L-shaped, or closed cell cores where each linear wall segment consists of two steel web plates connected to flange (or closure) plates or other linear segments at the ends. The coupling beams are rectangular steel built-up box sections filled with concrete. There are no additional reinforcing bars in the system (in the walls or the coupling beams), thus making the system a complete modular steel-concrete composite structure.

Coupled SpeedCore walls are being considered as an alternative to coupled reinforced concrete shear wall systems. Significant research has been conducted to develop seismic design provisions for coupled SpeedCore systems. Experimental research has been conducted on the cyclic lateral loading behavior of planar (Shafaei et al., 2021b) and C-shaped composite plate shear walls (Kenarangi et al., 2021). Detailed FEMA P-695 studies have been conducted to justify seismic design factors R, Ω_0, and C_d for coupled composite plate shear walls (FEMA, 2009; Bruneau et al., 2019).

Capacity design principles were used to develop seismic design criteria for coupled SpeedCore systems. These principles are presented in detail along with extended commentary in the following section. As explained in the basis of design in Section 4.2.2, the system is designed to undergo significant inelastic deformation in large seismic events. The inelastic deformation has two sources: (1) flexural plastic hinges at the ends of the coupling beams and (2) flexural yielding in the walls. The preferred inelastic response consists of forming plastic hinges at both ends of the coupling beams and in the composite walls. The design implements a strong wall-weak coupling beam design approach for appropriate sizing of the composite members. This design approach helps achieve extensive plastic hinging in most of the coupling beams before significant yielding of the walls.

The detailed design criteria presented in the following section include: (1) system requirements and section detailing requirements for both composite walls and coupling beams; (2) stiffness, modeling, and analysis recommendations for calculating the design demands for the coupling beams and composite walls; (3) recommendations for calculating the strengths of composite walls and coupling beams; (4) requirements for the connections between the coupling beams and walls, and the connections between the walls and foundations; and (5) identification of protected zones and demand critical welds. Following the design criteria and extended commentary, a detailed seismic design procedure along with flowcharts is presented.

The seismic design criteria and the design procedure presented herein were used to develop archetype structures for a FEMA P-695 study of these walls. These structures included (1) 3-, 8-, and 12-story archetype structures with planar composite walls and (2) 18- and 22-story archetype structures with C-shaped composite walls and composite coupling beams. For each story height, three different structures were designed with coupling beam length-to-depth ratios of 3, 4, and 5. Nonlinear inelastic numerical models of the coupled SpeedCore archetype structures were developed and analyzed using OpenSEES and the modeling recommendations of Shafaei et al. (2021a) and Bruneau et al. (2019). The results from the numerical analyses confirmed that the archetype structures performed as designed, and significant inelastic deformations occurred through the formation of flexural plastic hinges at the ends of the coupling beams along the height of the structure and eventually by the formation of flexural plastic hinges at the base of the walls.

A brief summary of the seismic behavior and results from the FEMA P-695 studies of the system are included in Chapter 5. The results from the numerical analyses indicated that coupled SpeedCore systems have excellent seismic performance. Statistical analyses of the extensive numerical investigations indicated that the system designed in accordance with the seismic design criteria and procedure presented in the following sections have adjusted collapse margin ratios that are acceptable according to FEMA P-695 criteria (Bruneau et al., 2019). Consequently, coupled SpeedCore systems have been included in the 2020 edition of the *NEHRP Recommended Seismic Provisions for New Buildings and Other Structures* (FEMA, 2020), and in the 2022 ASCE/SEI 7, *Minimum Design Loads for Buildings and Other Structures* (ASCE, 2022). ASCE/SEI 7, Table 12.2.-1, includes the seismic response modification factors for coupled SpeedCore systems: $R = 8$, $\Omega_0 = 2.5$, and $C_d = 5.5$.

Two detailed design examples are presented at the end of this chapter: an 8-story structure with coupled, planar SpeedCore walls and a 22-story structure with coupled C-shaped walls. These examples include detailed calculations for the design of the composite walls, coupling beams, and coupling beam-to-wall connections. Results from nonlinear analysis of seismic behavior of these structures are discussed and included in Chapter 5.

4.2 DESIGN REQUIREMENTS

This section details the seismic design provisions for coupled SpeedCore systems. These provisions are covered in the 2022 AISC *Specification for Structural Steel Buildings* (AISC, 2022b) and the 2022 AISC *Seismic Provisions for Structural Steel Buildings* (AISC, 2022a).

4.2.1 Scope

Coupled SpeedCore systems are synonymous with the seismic system addressed in AISC *Seismic Provisions* Section H8.1: composite plate shear walls—concrete filled (CC-PSW/CF), consisting of concrete-filled composite plate shear walls and filled composite coupling beams.

The composite plate shear walls of coupled SpeedCore systems consist of planar, C-shaped, I-shaped, or L-shaped walls, where each wall element consists of two planar steel plates with concrete infill between them. Composite action between the plates and concrete infill is achieved using either tie bars or a combination of tie bars and steel headed stud anchors. In each wall element, the two steel plates have equal nominal thickness and are connected using tie bars. Examples of coupled SpeedCore systems are shown in Figure 4-1.

Coupled SpeedCore systems are an alternative to coupled reinforced concrete shear walls or steel plate shear walls (SPSW), especially when relatively large seismic demand on the walls leads to dense reinforcement and large thicknesses in conventional concrete shear walls and coupling beams, or to relatively large wall thicknesses of the web infill and boundary elements in coupled SPSW. The limits set forth in the scope are associated with the range of parameters considered in the investigations conducted on coupled SpeedCore systems (Bruneau et al., 2019; Agarwal et al., 2020).

A flange or closure plate is used at the open ends of the wall elements. No additional boundary elements besides the closure plate are required to be used with the composite walls. Walls without flange plates are not permitted. While their performance may be adequate, their construction can be difficult due to the absence of closure plates and need for additional formwork. This need for additional formwork defeats the whole purpose of the composite plate shear walls and compromises their construction advantage. Future provisions may allow other steel sections (I-shaped sections or C-shaped sections) as a closure element.

The height-to-length ratio, H_{wall}/L_{wall}, of the composite walls is greater than or equal to 4. This requirement ensures that the walls are flexure-critical (i.e., flexural yielding and failure govern behavior rather than shear failure.) Calculations can also be performed to show that the wall is flexure-critical—that is, plastic hinges with flexural strength equal to $1.2M_{p.exp}$ form at the base of the wall before shear failure occurs. The shortest archetype structure that was evaluated using the FEMA P-695 (FEMA, 2009) approach for this system was three stories with two 45-ft-tall composite walls of 10 ft length, corresponding to a height-to-length ratio equal to 4.5 for each wall that constituted the coupled wall.

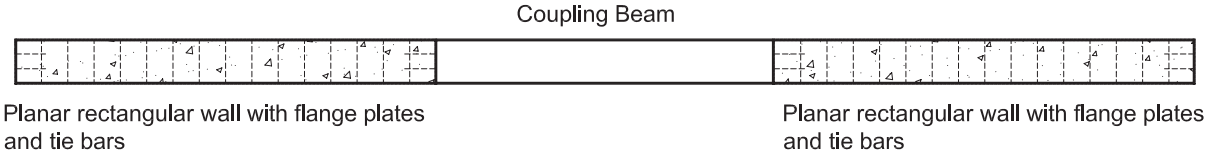
Coupling beams consist of concrete-filled built-up box sections of uniform cross section along their entire length and with a width equal to or greater than the wall thickness at the connection. For at least 90% of the stories of the building, the clear length-to-section depth ratios, L_{cb}/h_{CB}, of the coupling beams is greater than or equal to 3 and less than or equal to 5. This requirement for coupling beams to have length-to-depth ratios greater than or equal to 3 and less than or equal to 5 is based on (1) the range of parameters included in the FEMA P-695 studies conducted in order to establish the seismic response modification coefficient, R, for the system and (2) the fact that coupling beams with length-to-depth ratios less than 3 tend to be shear-critical, which is not recommended. AISC *Seismic Provisions* Section H8.5c explicitly requires coupling beams to be flexure-critical—that is, flexural yielding and failure governs their behavior rather than shear failure.

4.2.2 Basis of Design

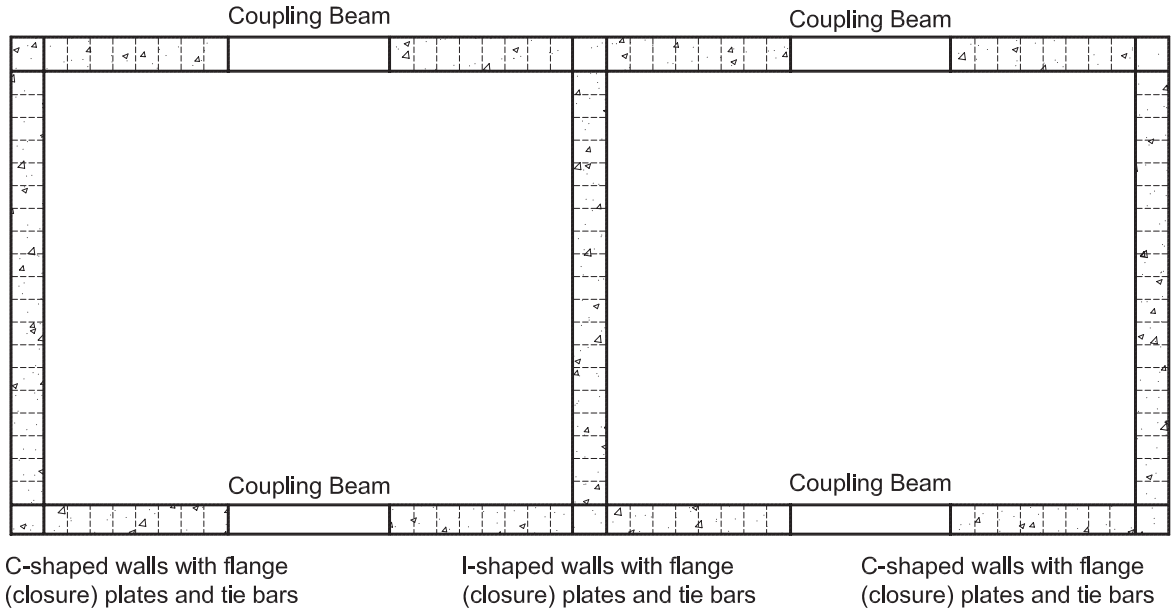
According to AISC *Seismic Provisions* Section H8.2, SpeedCore walls are expected to provide significant inelastic deformation capacity through developing the plastic flexural strength of the composite SpeedCore cross section by yielding of the steel plate and the concrete attaining its compressive strength. The cross section is detailed such that the section is able to attain its plastic flexural strength. Shear yielding of the steel web plates is not the governing mechanism.

The system uses coupled walls to resist lateral loads as shown in Figure 4-2. This system is expected to undergo significant inelastic deformation in large (design-basis and maximum considered) seismic events. The inelastic deformation has two sources: (1) flexural plastic hinges at the ends of coupling beams and (2) flexural yielding at the base of walls. The preferred inelastic failure mechanism consists of forming flexural plastic hinges at both ends of the coupling beams and at the base of the composite walls. It should be noted that flexural yielding in the walls is anticipated at the base of the wall element, but the location of yielding is dependent on the building geometry. For irregular buildings, the location of yielding in the walls should be confirmed by analysis.

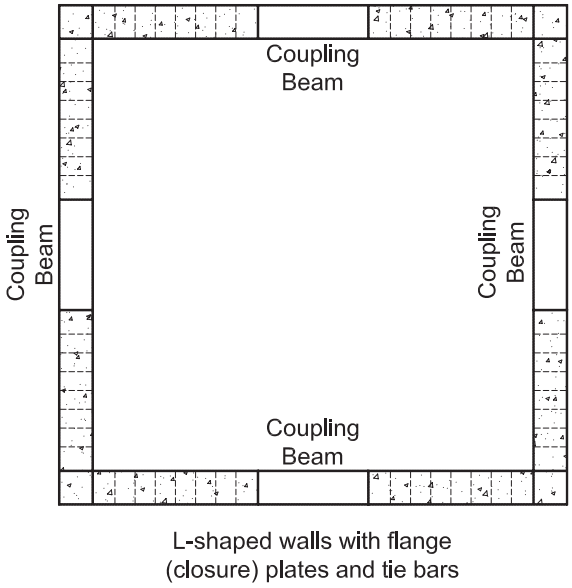
The design implements a strong wall/weak coupling beam approach for appropriately sizing the composite members. This design approach helps achieve development of extensive plastic hinging in most of the coupling beams before significant yielding of the walls.



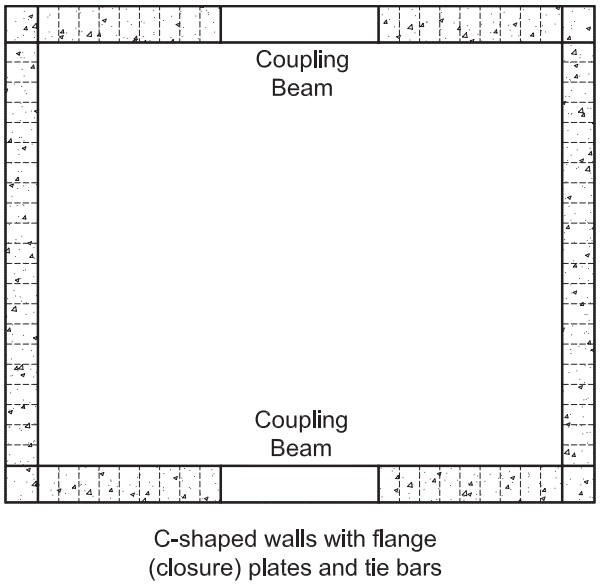
(a) CC-PSW/CF with planar rectangular C-PSW/CF



(b) CC-PSW/CF with C-shaped and I-shaped C-PSW/CF



(c) CC-PSW/CF with L-shaped C-PSW/CF



(d) CC-PSW/CF with C-shaped C-PSW/CF

Fig. 4-1. Examples of SpeedCore wall configurations.

4.2.3 Analysis

The design philosophy expressed in Section 4.2.2 leads to structures with the characteristic pushover behavior depicted in Figure 4-3. The initial branch represents the elastic behavior of the structure, and the slope of this branch represents the effective structural stiffness, which is approximated by elastic models. On the base shear-roof displacement curve, Point A represents the lateral load level corresponding to the design level seismic load distribution. The coupling beams are designed to reach their flexural capacity at this demand. As the lateral load (and base shear force) increases, the coupling beams along the height of the structure undergo flexural plastic hinging at both ends. The response reaches the next milestone, Point B, where all coupling beams have developed flexural hinges. The composite walls are designed to have a flexural capacity adequate to resist this demand level. The next milestone on the response, Point C, corresponds to the overall inelastic mechanism with flexural plastic hinging in all the coupling beams and the base of the composite walls. A final milestone, Point D, represents fracture of the composite walls. The overstrength factor for this system, defined as the ratio of ultimate load capacity-to-capacity, is approximately the ratio of base shear force at Point C to Point A.

The seismic response modification coefficient, *R*, given in ASCE/SEI 7, Table 12.2-1, for uncoupled composite plate shear walls is 6.5. A FEMA P-695 study was conducted to evaluate an appropriate seismic response modification coefficient, or *R* factor, for SpeedCore systems (Bruneau et al., 2019). This FEMA P-695 study demonstrated that coupled composite plate shear walls considered here can be designed with a greater *R* factor of 8. This increase in the value of *R* for coupled walls is due to the spread of plastic hinging and inelastic deformations (energy dissipation) in the coupling beams along the height of the structure. This lateral load behavior is illustrated in Figure 4-4 and Figure 4-5 using finite element analysis for an eight-story archetype structure having coupling beams with span-to-depth ratio of 5. The nonlinear static pushover behavior predicted by the finite element model in Figure 4-4 follows the expected behavior presented in Figure 4-3.

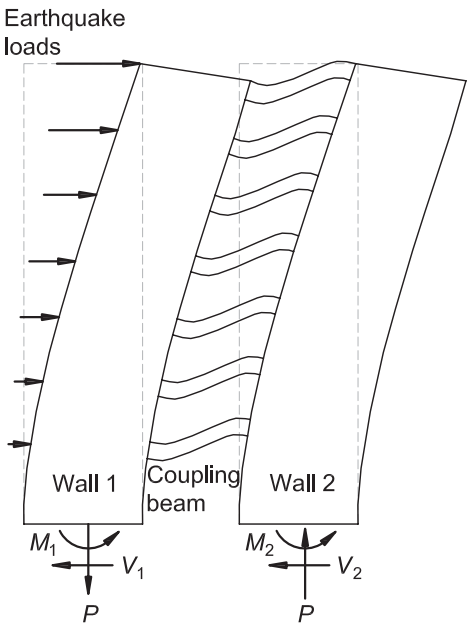


Fig. 4-2. Deformed shape of coupled SpeedCore walls under lateral loads.

In the FEMA P-695 study, archetype structures having 3, 8, 12, 18, and 22 stories and coupling beam span-to-depth ratios of 3, 4, and 5 were designed. The archetypes were designed using a seismic response modification coefficient, or *R* factor, of 8 and a C_d value of 5.5. The 3-, 8-, and 12-story archetype structures used planar composite walls, while the 18- and 22-story archetype structures used C-shaped walls. These archetype structures were doubly symmetric in plan and the wall thickness was uniform along the height of the structure. For the 18- and 22-story archetype structures, the thickness of the steel plates for the composite walls and the coupling beams was reduced in the top half of the structure. The 22-story archetype had an overall height of 311 ft. These structures were designed to meet the composite member and system requirements outlined in Sections 4.2.4 and 4.2.5. The coupling ratio for the archetype structures was about 50 to 80%, where the taller buildings had higher coupling ratios. In this context, coupling ratio is defined at Point B on the characteristic pushover curve as the proportion of the total overturning moment resisted by coupling action. It is important to note that there is no specific required or recommended coupling ratio for the coupled SpeedCore system.

Seismic demands followed standards set in ASCE/SEI 7 and the FEMA P-695 procedure. The numerical models for the structures accounted for the various complexities of flexural behavior of the coupling beams and composite walls, including the effects of concrete cracking, steel yielding, local buckling, concrete crushing, and steel inelastic behavior up to fracture due to cumulative plastic strains and low cycle fatigue. The numerical models were benchmarked using experimental data available in the literature.

Results from the FEMA P-695 analyses indicated that all archetypes reached collapse at drifts greater than 5%, but all collapse margin ratios established in this study were conservatively calculated based on results obtained at 5% drift (i.e., at less than actual collapse points). Results of the FEMA P-695 studies indicated that collapse margin ratios increased for the taller buildings, which is consistent with the fact that code-specified drift limits governed the design of the 18- and 22-story archetypes.

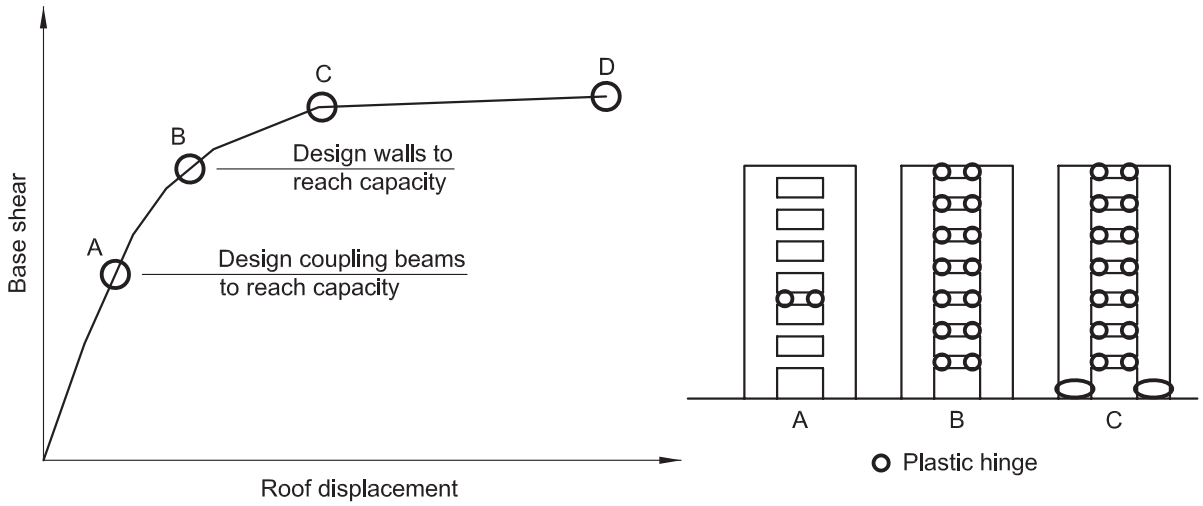


Fig. 4-3. Characteristic pushover (base shear-roof displacement) behavior for eight-story archetype structure.

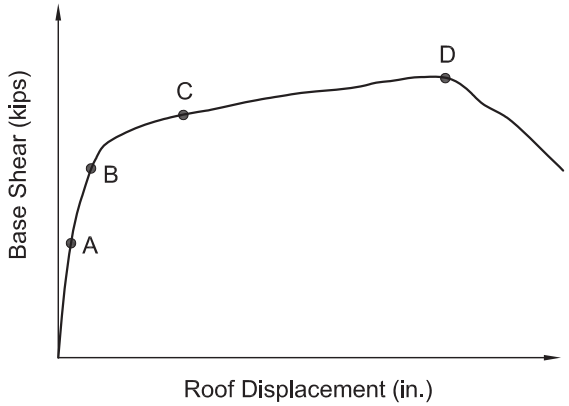


Fig. 4-4. Pushover behavior from 3D FEM analysis for eight-story archetype structure.

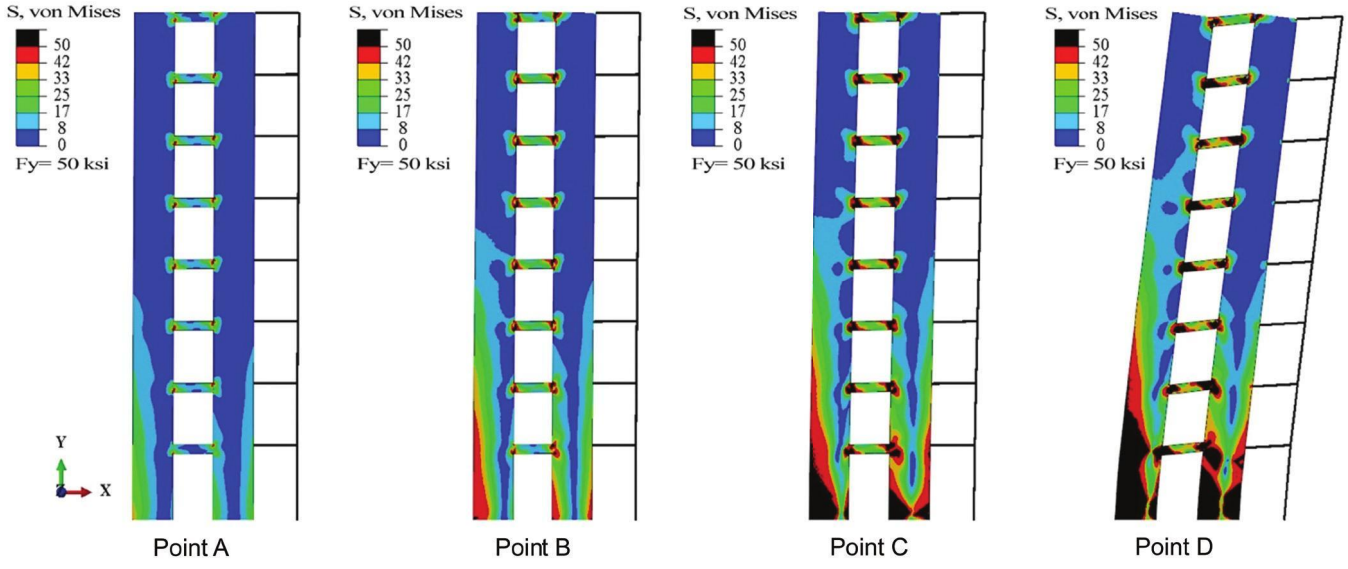


Fig. 4-5. Extent of steel yielding for points on pushover curve for eight-story archetype structure.

An elastic model of the structure is used to conduct structural analysis for design by the ELF procedure outlined in ASCE/SEI 7. The results of this analysis are used to determine the design demands for the coupling beams and the maximum elastic story drift ratio, which is amplified by C_d to estimate the inelastic story drift ratio for design. This analysis can be performed in accordance with AISC *Specification* Section I1.5, which is based on the direct analysis method, and includes recommendations for the flexural, shear, and axial effective stiffnesses of composite walls and filled composite members (i.e., composite coupling beams). These recommendations and stiffness values were discussed in detail in Section 2.2.2 of this Design Guide.

4.2.3.1 Stiffness

According to AISC *Seismic Provisions* Section H8.3a, the effective flexural, axial, and shear stiffnesses of composite walls and filled composite coupling beams are calculated in accordance with AISC *Specification* Section I1.5. These equations and stiffnesses are discussed in detail in Section 2.2.2 of this Design Guide.

4.2.3.2 Required Strengths for Coupling Beams

According to AISC *Seismic Provisions* Section H8.3b, analyses in conformance with the applicable building code are performed to calculate the required strengths for the coupling beams.

4.2.3.3 Required Strengths for Composite Walls

According to AISC *Seismic Provisions* Section H8.3c, the required strengths for the composite walls are determined using the capacity-limited seismic load effect. The capacity-limited seismic load effect is also presented in Section 4.2.3.4 in this Design Guide.

4.2.3.4 Capacity-Limited Seismic Load

According to AISC *Seismic Provisions* Section H8.3c, the capacity-limited horizontal seismic load effect, E_{cl}, is determined from an analysis in which all the coupling beams are assumed to develop plastic hinges at both ends with an expected plastic flexural strength of $1.2M_{p.exp.CB}$, and the maximum overturning moment is amplified to account for the increase in lateral loading from the formation of the earliest plastic hinges to the formation of plastic hinges in all coupling beams over the full wall height. This amplification factor, γ_1, can be calculated using Equation 4-1.

$$\gamma_1 = \frac{\sum_n 1.2M_{p.exp.CB}}{\sum_n M_{u.CB}} \quad (4\text{-}1)$$

where

$\sum 1.2M_{p.exp.CB}$ = sum of the expected flexural strength of coupling beams along structure height, kip-in.

$\sum M_{u.CB}$ = sum of the flexural design demands for the coupling beams along structure height, kip-in.

n = number of coupling beams along structure height

The earthquake-induced axial force in the walls used in determining the required wall strength is calculated as the sum of the capacity-limited coupling beam shear forces along the height of the structure. The portion of the total overturning moment resisted by the coupling action can be estimated as the equal and opposite axial forces at the base of the walls, P_w, multiplied by the distance between them. The remaining portion of the total overturning moment can be distributed to the individual walls based on their effective flexural stiffness, accounting for the effects of tensile or compressive axial force.

The required axial and flexural strengths for the composite walls are determined directly from this analysis, while the required wall shear strengths determined from this analysis are amplified by a factor of 4.0.

The shear force is amplified by a factor of 4.0 to conservatively account for (1) effects of higher modes and (2) the overstrength in the walls resulting from the difference between their expected flexural strength (at Point C in Figure 4-3) and design demand (Point B). For reinforced concrete walls, this amplification factor is approximately 2.0 to 3.0 (ACI, 2019). A conservative value of 4.0 was used for composite walls in the absence of better information, and in recognition of their inherent shear strength. The shear strength of these composite walls is very high due to the significant contribution of the steel plates and composite action.

4.2.4 System Requirements for Coupled SpeedCore Walls with Flange (Closure) Plates

System requirements for coupled SpeedCore components are the same as those described in Section 3.2.4 of this Design Guide for uncoupled walls.

4.2.5 System Requirements for Composite Coupling Beams

Composite coupling beams are designed in accordance with the requirements of Section 2.3 and additional requirements presented in this section.

4.2.5.1 Slenderness Requirement for Coupling Beams

According to AISC *Seismic Provisions* Section H8.5b, the slenderness ratios of the flanges and webs of the filled

composite coupling beam, b_c/t_f and h_c/t_w, are limited as follows:

$$\frac{b_{c.CB}}{t_{pf.CB}} \le 2.37\sqrt{\frac{E_s}{R_yF_y}} \quad (4\text{-}2)$$

$$\frac{h_{c.CB}}{t_{pw.CB}} \le 2.66\sqrt{\frac{E_s}{R_yF_y}} \quad (4\text{-}3)$$

where

$b_{c.CB}$ = clear width of the coupling beam flange plate, in.
$h_{c.CB}$ = clear width of the coupling beam web plate, in.
$t_{pf.CB}$ = thickness of the coupling beam flange plate, in.
$t_{pw.CB}$ = thickness of the coupling beam web plate, in.

The slenderness requirements are based on compact section requirements in the AISC *Specification* Section I1.4 for filled composite members. The web slenderness ratio requirement is based on developing the shear yielding strength of the web plates before shear buckling as per AISC *Specification* Section G4. Figure 4-6 shows a schematic of the coupling beam cross section along with the clear widths of the flange and web plates.

4.2.5.2 Flexure-Critical Coupling Beams

According to AISC *Seismic Provisions* Section H8.5c, composite coupling beams are proportioned to be flexure-critical with the expected shear strength, $V_{n.exp}$, calculated in accordance with AISC *Specification* Section I4.2 using the expected yield strength, R_yF_y, for steel and the expected compressive strength, $R_cf'_c$, for concrete. The equations for calculating the shear strength of composite filled sections were discussed in detail in Section 2.3.3 of this Design Guide. The expected shear strength is calculated as follows:

$$V_{n.exp} \ge \frac{2.4M_{p.exp}}{L_{cb}} \quad (4\text{-}4)$$

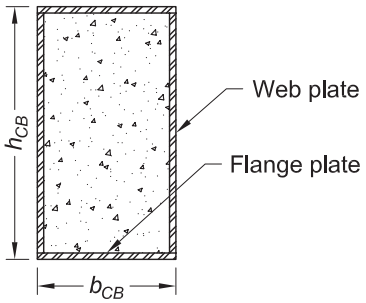


Fig. 4-6. Coupling beam cross section.

where

L_{cb} = clear span length of the coupling beam, in.
$M_{p.exp}$ = expected flexural strength of composite coupling beam calculated in accordance with Section 2.2.3 using the expected yield strength, R_yF_y, for steel and the expected compressive strength, $R_cf'_c$, for concrete, kip-in.

This requirement is based on achieving flexure-critical behavior in composite beams. The requirement increases the capacity-limited shear force capacity ($2M_{p.exp}/L_{cb}$) by a factor of 1.2 to account for the effects of steel inelastic hardening in tension, concrete confinement, and the biaxial (tensile) stress effect in the steel tension flange (Bruneau et al., 2019).

4.2.6 Composite Wall Strength

The nominal strengths of the composite walls are calculated in accordance with AISC *Seismic Provisions* Section H8.6. This section refers the user to AISC *Specification* Chapter I. The nominal compressive, tensile, flexural, shear, and combined flexure and axial strengths from the AISC *Specification* are detailed in Section 2.2.3 of this Design Guide.

4.2.7 Composite Coupling Beam Strength

The nominal strengths of the composite coupling beams are calculated in accordance with AISC *Seismic Provisions* Section H8.7. This section refers the user to AISC *Specification* Chapter I. The nominal flexural and shear strengths are covered in Section 2.3.3 of this Design Guide.

4.2.8 Coupling Beam-to-Wall Connections

According to AISC *Seismic Provisions* Section H8.8, the coupling beam-to-wall connections are designed to develop and transfer the expected flexural strength and corresponding capacity-limited shear force of the associated coupling beams. The required flexural and shear strengths are calculated based on the expected flexural strengths of the coupling beams. The nominal flexural strength, with or without concurrent axial force, can be calculated using nominal steel strength, F_y, and nominal concrete strength, f'_c. The expected flexural strength can be calculated using expected strengths, R_yF_y and $R_cf'_c$, for steel and concrete, respectively. The expected flexural strength, $M_{p.exp}$, of filled composite members is amplified by a factor of 1.2 to account for the effects of steel inelastic hardening in tension, concrete confinement, and the biaxial (tensile) stress effect in the steel tension flange. The coupling beam-to-wall connection is designed and detailed to resist this amplified ($1.2M_{p.exp}$) flexural strength of the beam and the associated capacity-limited shear ($2.4M_{p.exp}/L_{cb}$).

Figure 4-7 shows the envelope of the inelastic moment-rotation response assumed in the FEMA P-695 (analytical) studies for the flexural plastic hinges in the coupling beams. As shown, the plastic rotation before degradation of flexural strength due to fracture failure was assumed to be equal to 0.025 rad.

Coupling beam-to-wall connections have been tested in the past—for example, Nie et al. (2014)—and additional testing of coupling beam-to-wall connections is ongoing (Ahmad et al., 2021). Some details that have been demonstrated to be acceptable by testing (Ahmad et al., 2021) include those shown in Figure 4-8 and Figure 4-9.

Figure 4-8 shows a connection where (i) the web plates are continuous between the coupling beam and the composite walls, (ii) the coupling beam flange plates are extended into the wall and welded to the wall web plates to develop their expected tensile strength, and (iii) the wall closure plate is interrupted at the coupling beam.

Figure 4-9 shows a connection where (i) coupling beam web plates are lapped and welded to the composite wall web plates, (ii) the coupling beam flange plates are continued into the wall and welded to the wall web plates to develop their expected tensile strength, and (iii) the wall closure plate is not interrupted at the coupling beam.

4.2.8.1 Required Flexural Strength

According to AISC *Seismic Provisions* Section H8.8a, the required flexural strength, M_u, of the coupling beam-to-wall connection is 120% of the expected flexural strength of the coupling beam, $M_{p.exp.CB}$.

4.2.8.2 Required Shear Strength

According to AISC *Seismic Provisions* Section H8.8b, the required shear strength, V_u, for the coupling beam-to-wall connection is determined using the capacity-limited seismic load effect as follows:

$$V_u = \frac{2(1.2M_{p.exp.CB})}{L_{cb}} \tag{4-5}$$

4.2.8.3 Rotation Capacity

According to AISC *Seismic Provisions* Section H8.8c, the coupling beam-to-wall connection is detailed to develop a rotation capacity of 0.030 rad before flexural strength decreases to 80% of the flexural plastic strength of the beam. Connection details that have been previously demonstrated to have adequate rotation capacity—for example, Ahmad et al. (2021)—are approved for use. The available rotation capacity of the coupling beam using other connection details can be verified through testing, advanced analysis calibrated to physical testing, or a combination thereof. The acceptable envelope of coupling beam end moment-chord rotation behavior is shown in Figure 4-10.

To meet these requirements, designers select a connection that has either been verified through testing or advanced analysis calibrated to testing. In any nonlinear models, typical for performance-based design, the coupling beam-to-wall connections are assumed to degrade to 80% of M_p at a chord rotation of 0.030 rad beyond which complete or gradual loss of flexural strength can be assumed. In modeling approaches able to capture this loss, the analysis can continue beyond this point.

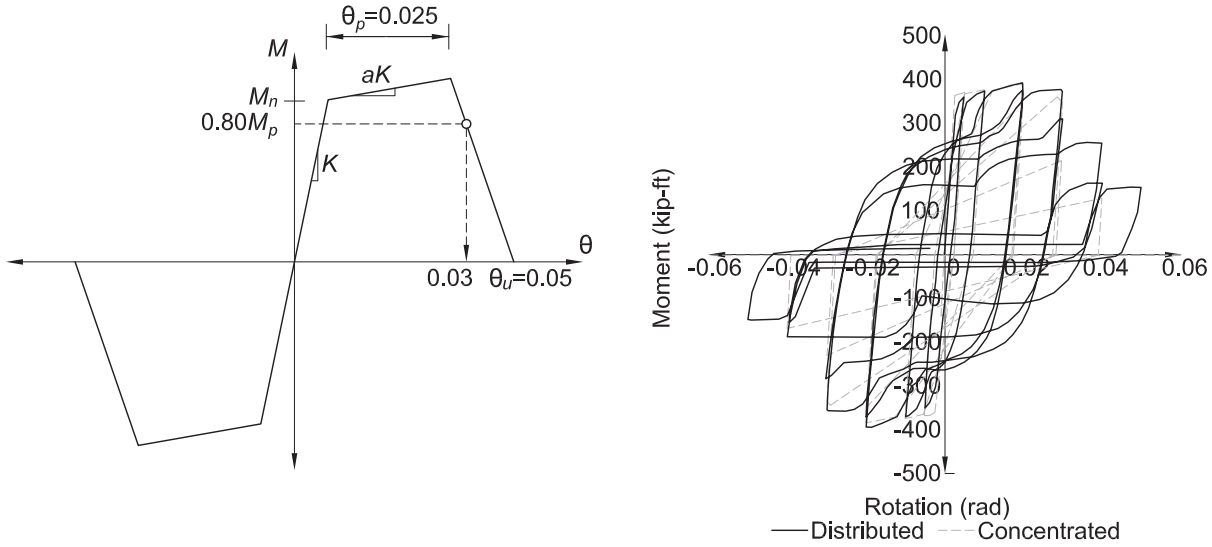
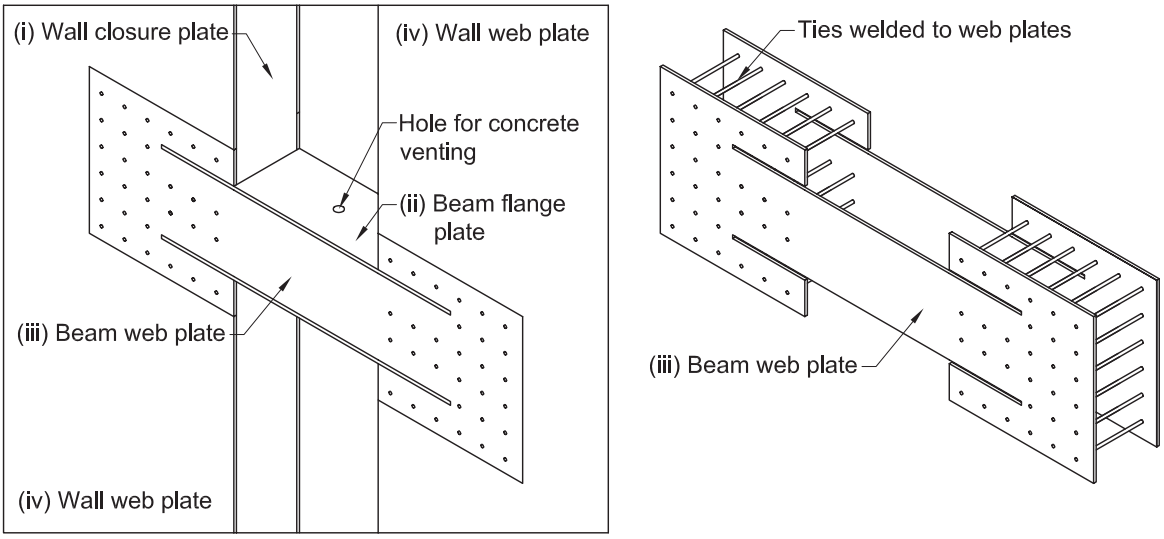
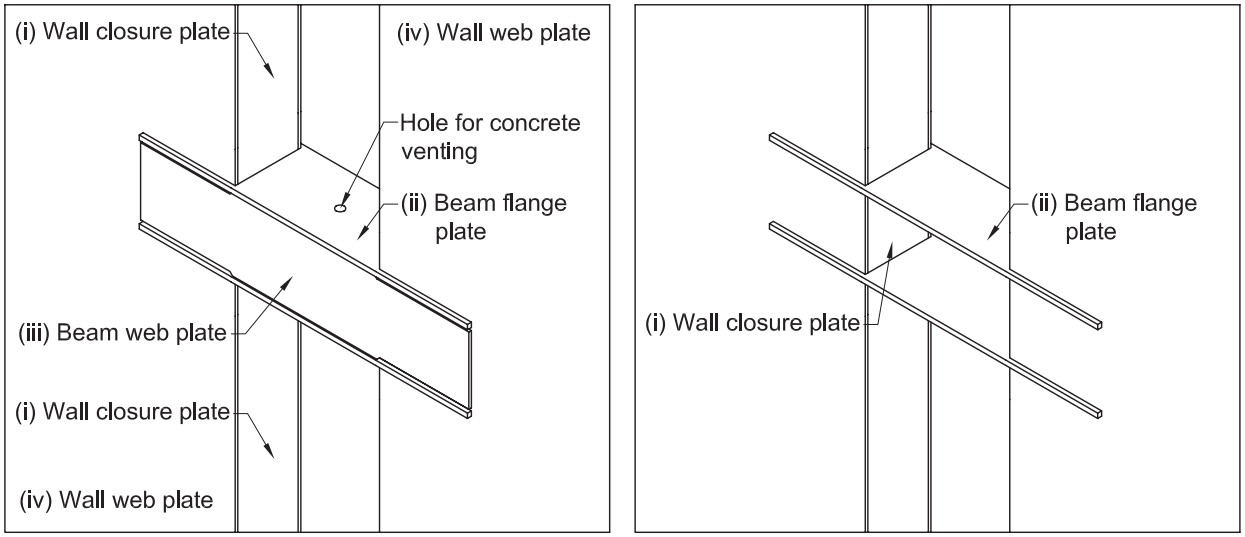


Fig. 4-7. Envelope of cyclic moment-chord rotation response and hysteretic behavior of plastic hinges in composite coupling beams (Broberg et al., 2022).



(i) Wall closure plate interrupted and welded to beam flange plate
(ii) Beam flange plate penetrates into wall and welded to wall web plate
(iii) Beam web plate continuous with wall web plate
(iv) Wall web plate welded to continuous beam web plate away from connection

Fig. 4-8. Coupling beam connection with continuous web plate and interrupted wall closure plate.



(i) Uninterrupted wall closure plate welded to beam flange plates
(ii) Beam flange plate penetrates into wall, and welded to wall web plate
(iii) Beam web plate lapped and weldedto wall web plate from outside

Fig. 4-9. Coupling beam connection with lapped web plate and continuous wall closure plate.

4.2.9 Composite Wall-to-Foundation Connections

AISC *Seismic Provisions* Section H8.9 applies to the design of composite wall-to-foundation connections, where the composite walls are connected directly to the foundation, at the point of maximum moment in the walls.

For structures with subgrade basement stories, the maximum shear force and overturning moment in the composite walls at the grade level can be transferred to the basement stories through a transverse diaphragm. The rate of force transfer depends on the diaphragm stiffness. In practice, upper and lower bound assumptions for stiffness of the primary transfer diaphragm are often considered as discussed in *Tall Buildings Initiative: Guidelines for Performance Based Seismic Design of Tall Buildings* (PEER, 2017).

For structures that are connected to the concrete basemat or foundation at the location of maximum shear force and overturning moment, the wall-to-basemat connections are designed for (1) the expected flexural strength of the composite walls, accounting for the effects of axial force; (2) the expected axial forces associated with capacity-limited shear forces in the coupling beams; and (3) and the amplified shear force demand, with an amplification factor of 4.0, used for the design of the composite walls. Some connection details that have been used in the past include those discussed in Section 3.2.6.3 with details such as welded base plates and rebar couplers or walls embedded in the concrete foundation as shown in Figure 3-3.

4.2.9.1 Required Strengths

According to AISC *Seismic Provisions* Section H8.9a, the required strength of the composite wall-to-foundation connection is determined using the capacity-limited seismic load effect. The coupling beams are assumed to have developed plastic hinges at both ends with the expected flexural strength of $1.2M_{p.exp}$. The composite walls also are assumed to have developed plastic hinges at the base with expected flexural strength of $1.2M_{p.exp}$, while accounting for the effects of simultaneous axial force. The required shear strength of the composite wall-to-foundation connections is equal to the required shear strength of the composite walls calculated in accordance with the AISC *Seismic Provisions* Section H8.3d (Section 2.2.3).

4.2.10 Other Connections

Other connections such as connection between tie bars and steel plates and connections between SpeedCore steel components are discussed in Section 2.4.

4.2.11 Protected Zones

According to AISC *Seismic Provisions* Section H8.10, the following regions are designated as protected zones and are designed to meet the requirements of AISC *Seismic Provisions* Section D1.3:

1. The regions at ends of the coupling beams subject to inelastic straining.
2. The regions at the base of the composite walls subject to inelastic straining.

Protected zones are defined in the AISC *Seismic Provisions* as regions of members or connections of members undergoing large inelastic strains or plastic hinging to provide significant inelastic deformation capacity and energy dissipation during design-basis or higher magnitude earthquakes. FEMA/SAC testing has demonstrated the sensitivity of these regions to discontinuities caused by fabrication or erection activities or from other attachments. For this reason, operations specified in AISC *Seismic Provisions* Section I2.1 are prohibited in the protected zones.

For the coupled SpeedCore system, the protected zones are designated as the regions at the ends of coupling beams

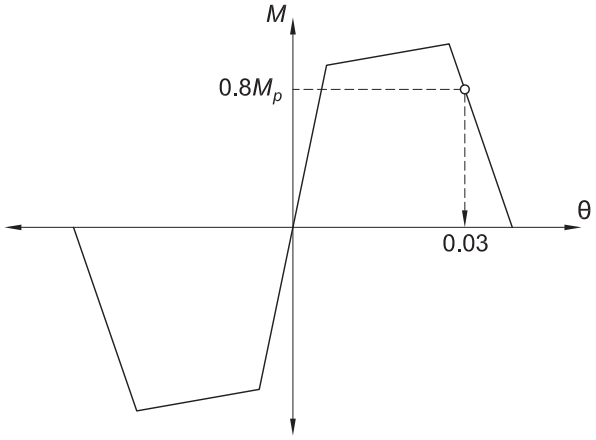


*Fig. 4-10. Envelope of coupling beam end moment-chord rotation (*M – θ*).*

that will undergo significant inelastic straining and plastic hinging, and portions of the adjacent wall (if any) undergoing yielding at the connection. The typical length of the plastic hinge region will extend from the face of the composite wall to a distance equal to the coupling beam depth. However, the extent of the plastic hinge (and the protected zone) can depend on the cross-section geometry, the flange and web plate thicknesses, and the length-to-depth ratio of the coupling beam. The extent of the protected zone can be determined from analysis.

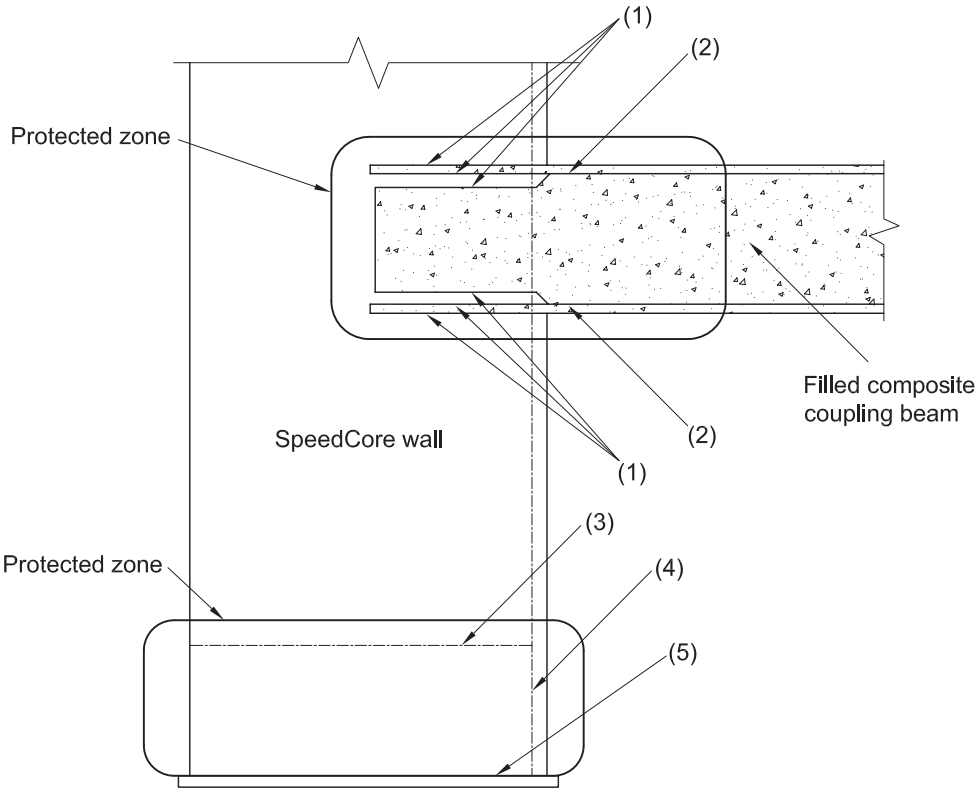
Additionally, the regions of the composite walls undergoing significant inelastic straining and plastic hinging are also designated as protected zones. The extent of the plastic hinge region undergoing significant inelastic strains (and the protected zone) can depend on wall cross-section geometry, web plate and flange (closure) plate thickness and length, and the height-to-length ratios of the walls. The extent of the protected zone can be determined from analysis.

4.2.12 Demand Critical Welds in Connections

According to AISC *Seismic Provisions* Section H8.11, where located within the protected zones identified in Section H8.10, the welds in composite wall steel plate splices are demand critical and must satisfy the applicable requirements of AISC *Seismic Provisions* Sections A3.4b and I2.3.

Coupled composite wall systems include connections with several welded details as shown in Figure 4-11. These include (1) welds connecting the coupling beam flanges and web plates to composite wall steel plates, (2) welds connecting the coupling beam web plates to flange plates in built-up box sections, (3) welds in the composite wall steel plate splices, (4) welds connecting the composite wall flange (closure) plates to the web plates, and (5) welds at the composite wall steel plate-to-base plate connections.

Most of these welds can be designed and detailed appropriately using AISC *Specification* Chapter J for the available



(1) Welds connecting the coupling beam flanges and web plates to composite wall steel plates
(2) Welds connecting the coupling beam web plates to flange plates in built-up box sections
(3) Welds in the composite wall steel plate splices
(4) Welds connecting the composite wall flange (closure) plates to the web plates
(5) Welds at the composite wall plate-to-base plate connections

Fig. 4-11. Critical welds in coupling beam-to-wall connection.

strengths, such that the weld stresses remain in the elastic range. Consequently, only the welds in the composite wall steel plate splices, when located in the protected zones, are designated as demand critical.

4.3 GENERAL DESIGN PROCEDURE FOR COUPLED WALLS

4.3.1 Overview

This section describes the overall design process for coupled SpeedCore walls. This design process heavily references the design requirements from Section 4.2. This process is summarized in Figure 4-12.

4.3.2 Step 1. Design Inputs

At the start of the design process, a series of parameters are considered as inputs. These parameters include (1) requirements of the governing building code; (2) floor loading; (3) location considerations (site class and soil conditions); (4) building importance factor (I_e); (5) material properties (f_c' and F_y); and (6) architectural considerations like typical bay widths, story heights, coupling beam length and depth, and core wall configurations.

4.3.3 Step 2. Analysis for Design

The seismic forces on the coupled SpeedCore walls are calculated following ASCE/SEI 7. It is important to note that ASCE/SEI 7 offers multiple options for determining the seismic force demand, including the equivalent lateral force method and response spectra analysis. From this analysis, the overturning moment, base shear, story forces, and coupling beam shear are calculated. Using these outputs, preliminary sizes for wall and coupling beam elements are selected. These components are then analyzed to determine the capacity-limited seismic loads. The wall and coupling dimensions may then be adjusted to meet these demands and prescribed limits. An overview of this process is presented in Figure 4-13.

4.3.3.1 Base Shear Force and Amplified Base Shear

The base shear, V, is amplified by a factor of 4.0. This base shear amplification factor is used to account for various effects discussed earlier. This amplification factor is used only to calculate shear demand and is not used for overturning moment calculation.

$$V_{amp} = 4.0V_{base} \quad (4\text{-}6)$$

4.3.3.2 Wall and Coupling Beam Stiffnesses

Wall and coupling beam stiffnesses are determined based on AISC *Specification* Section I1.5. These requirements are

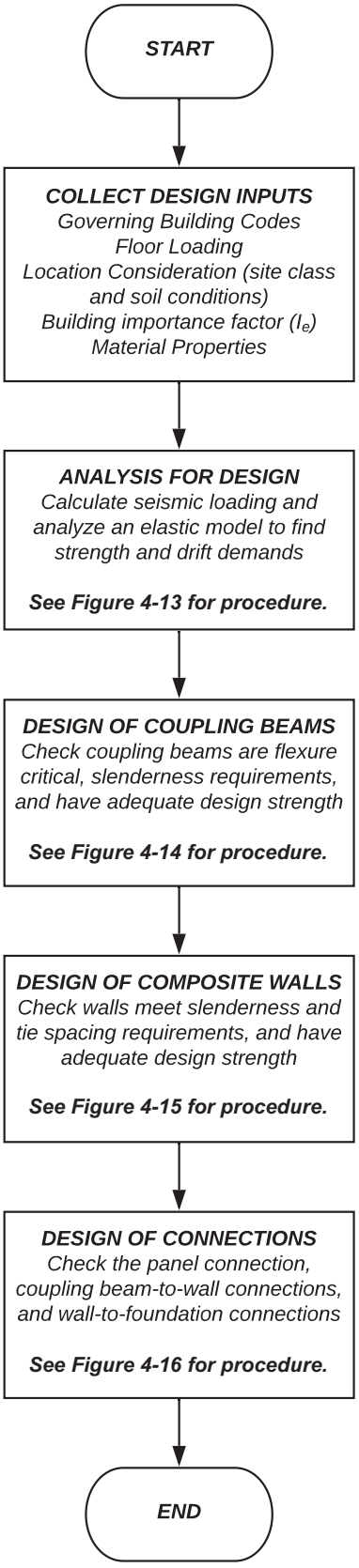


Fig. 4-12. Overview of coupled SpeedCore design procedure.

discussed in Section 2.2.2 and Section 2.3.2 of this Design Guide for SpeedCore walls and composite coupling beams, respectively.

4.3.3.3 Required Strength for Coupling Beams

Analyses in conformance with the applicable building code are performed to calculate the required strengths for the coupling beams.

These analyses involve modeling the seismic force-resisting system, namely the SpeedCore walls and composite coupling beams. Walls and coupling beams can be modeled as general frame sections with effective stiffness properties (per Section 4.3.3.2). Wall frame elements are located at the elastic centroid of the wall section. Coupling beam elements span from one wall node to the next and are assigned rigid properties from the elastic centroid of the wall to the start of the coupling beam. Design forces are applied at each story level. The analysis results can then be used to determine the shear and moment demand in the coupling beams. The required coupling beam shear strength can be calculated from analysis. The required strength is the average shear demand of the coupling beams along the height of the structure. Alternatively, the designer can choose to use the maximum shear demand.

4.3.3.4 Required Strength of Composite Walls

The required strength of the composite wall is determined using the capacity-limited seismic load effect in accordance with Sections 4.2.3.3. and 4.2.3.4.

4.3.3.5 Interstory Drift

Analysis is performed in conformance with the applicable building code to calculate the ID and compare against the applicable limit.

This analysis follows the modeling approach and loads described in Section 4.3.3.3. To calculate the ID, the story displacement results determined from this analysis are amplified by C_d (C_d = 5.5 for this system) and normalized by the story height to determine the ID. This drift is then compared to the applicable limit. Per ASCE/SEI 7, Table 12.12-1, this limit is 2% for a Risk Category II building with greater than four stories, where Risk Category is determined by code requirements.

4.3.4 Step 3. Design of Coupling Beams

After initial analysis, the coupling beams are designed to meet the average demands calculated in the analysis as

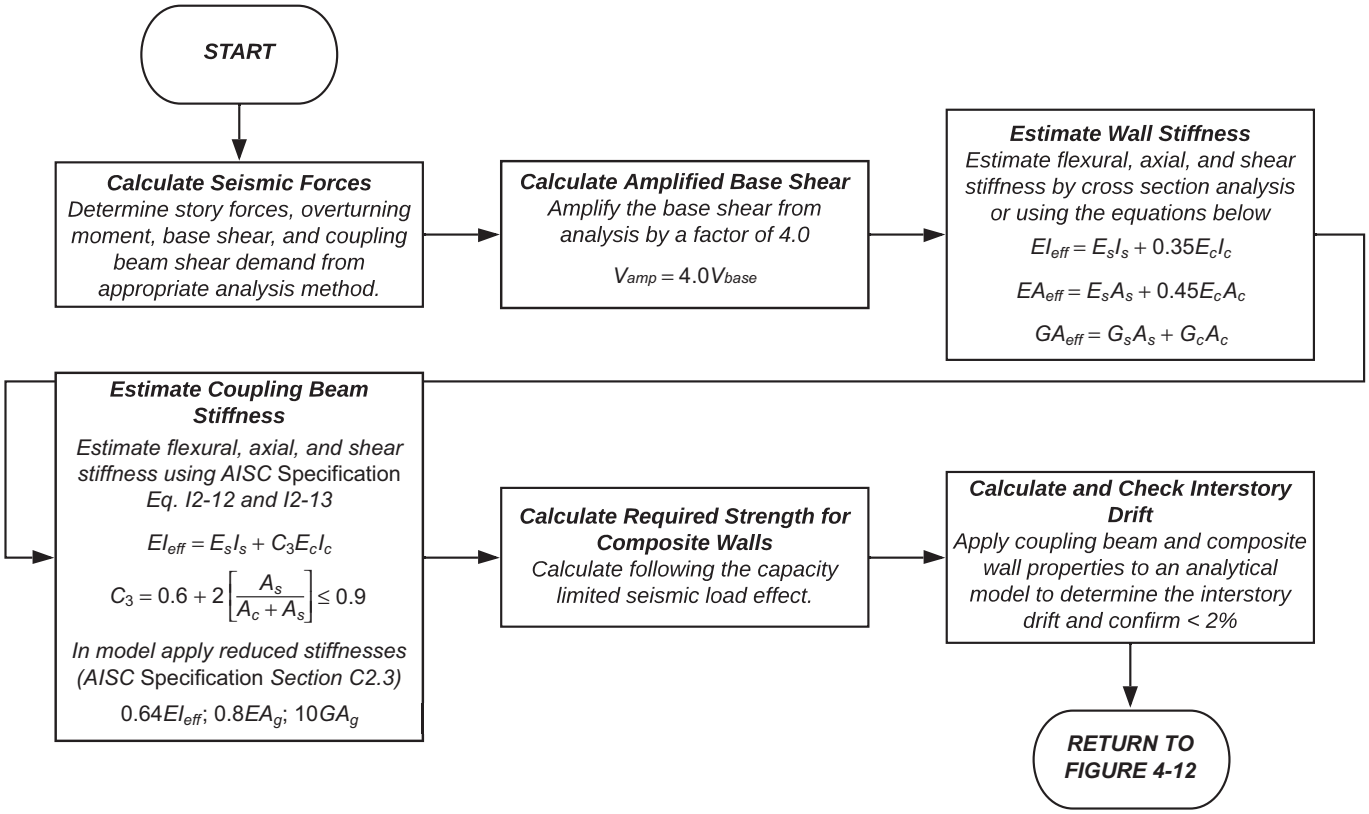


Fig. 4-13. Analysis for design flowchart.

discussed in Section 4.3.3.3. An overview of the process is shown in Figure 4-14.

4.3.4.1 Expected Flexural Strength, $M_{p.exp.CB}$

The expected flexural strength of coupling beams is the moment corresponding to a plastic stress distribution over the composite cross section. Steel components are assumed to have reached a yield stress of R_yF_y in either tension or compression, and concrete components in compression due to axial force and/or flexure are assumed to have reached a stress of $R_cf'_c$.

4.3.4.2 Flexure-Critical Coupling Beams

Coupling beams are proportioned to meet the minimum area requirements of Section 2.3.

4.3.4.3 Minimum Area of Steel

The steel plates comprise at least 1% of the total composite cross-sectional area.

4.3.4.4 Slenderness Requirement

The slenderness ratios of the flange and web components, $b_{c.CB}/t_{pf.CB}$ and $h_{c.CB}/t_{pw.CB}$, are limited as detailed in Section 4.2.5.1.

4.3.4.5 Coupling Beam Flexural Strength, M_n

The coupling beam flexural strength is calculated according to Section 4.2.7.

4.3.4.6 Coupling Beam Nominal Shear Strength, V_n

The nominal shear strength is calculated as the summation of the nominal shear strengths of concrete infill, and nominal shear strength of the webs of the coupling beam. This expression is presented in Section 2.3.3.

4.3.5 Step 4. Design of Composite Walls

This step discusses the calculation of the required strengths for the composite walls. These force demands are associated

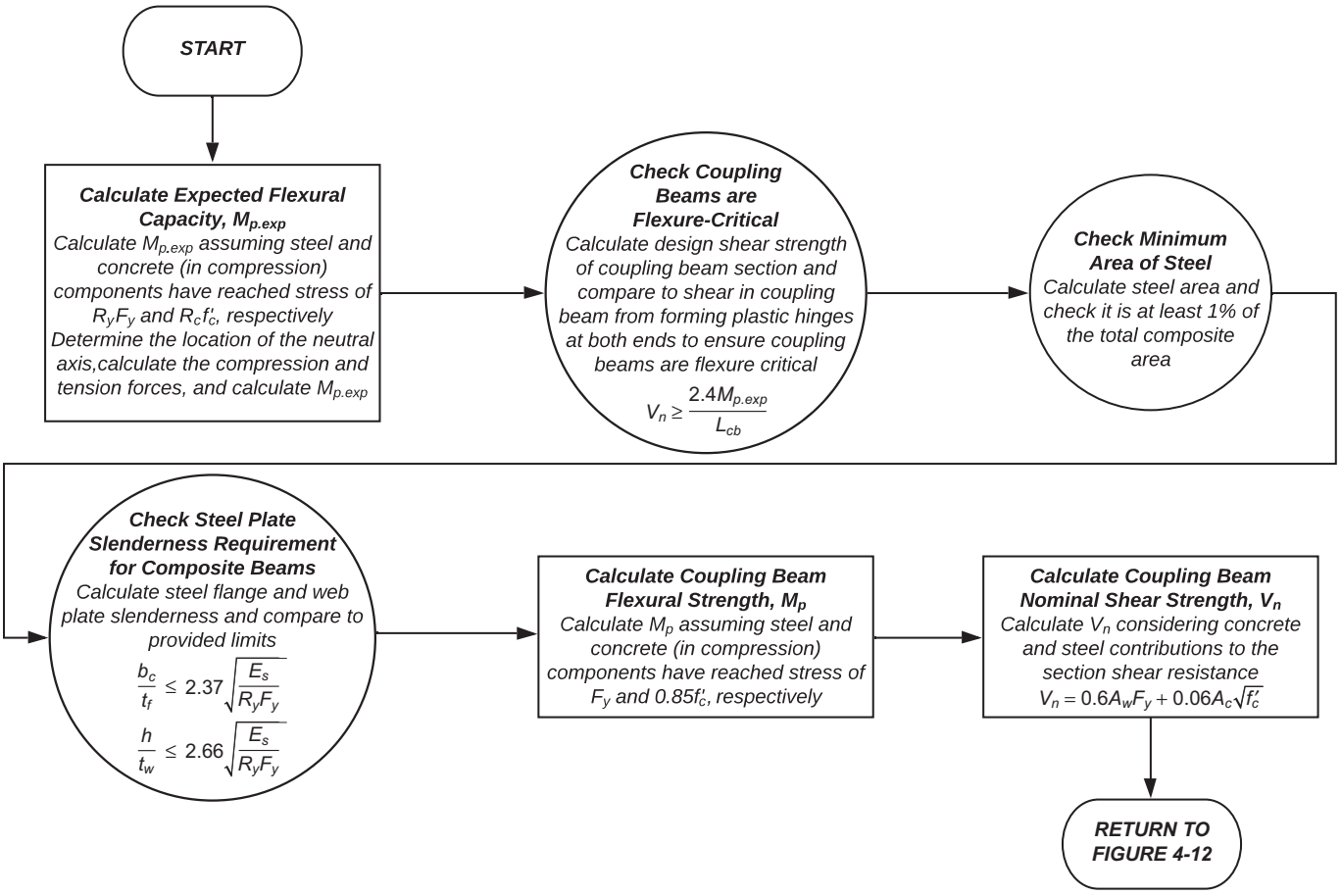


Fig. 4-14. Design of coupling beams flowchart.

with the formation of the plastic hinge mechanisms at the ends of the coupling beams and are calculated using the coupling beams designed in Step 3. An overview of this process is shown in Figure 4-15.

4.3.5.1 Minimum and Maximum Area of Steel

Walls are proportioned to meet the minimum area requirements discussed in Section 2.2.1.

4.3.5.2 Steel Plate Slenderness Requirement for Composite Walls

Wall slenderness ratios meet the requirements of Section 4.2.4.

4.3.5.3 Tie Bar Spacing Requirement for Composite Walls

The tie bar spacing to plate thickness ratio, S_{tie}/t_p, is limited according to Section 4.2.4.

4.3.5.4 Required Wall Shear Strength

The required in-plane shear strength is V_{amp} per Section 4.3.3.1.

4.3.5.5 Required Wall Flexural Strength

The required flexural strength is calculated per Section 4.2.3.3.

4.3.5.6 Wall Axial Tensile Strength

The nominal tensile strength, P_n, is determined using Section 4.2.6.

4.3.5.7 Wall Axial Compressive Strength

The nominal compressive strength is determined using Section 4.2.6. The flexural stiffness calculated per Section 2.2.3 should be calculated about the minor axis.

4.3.5.8 Wall Flexural Strength

The nominal flexural strength is determined using Section 4.2.6.

4.3.5.9 Combined Axial Force and Flexure for Wall

The nominal strength of composite walls subjected to combined axial force and flexure is determined using Section 4.2.6.

4.3.5.10 Wall Shear Strength

The nominal in-plane shear strength, $V_{n.wall}$, is determined using Section 2.2.3.

4.3.6 Step 5. Design of Connections

This step focuses on the design of various connections in the coupled SpeedCore system. The connection elements can be designed in accordance with AISC *Specification* Chapter J, as applicable. As discussed in Section 2.1.5, these include connections at the section, member, and structural level. For example, consider Figure 4-11, which identifies various connections at the section, member, and structural levels. An overview of this process is shown in Figure 4-16.

At the section level, these include the tie bar-to-steel plate connections, composite coupling beam flange-to-web connections (e.g., weld 2 in Figure 4-11), SpeedCore flange-to-web plate connections (e.g., weld 4 in Figure 4-11), and welds in SpeedCore splices (e.g., weld 3 in Figure 4-11). All these section level connections are designed to achieve tensile yielding of the gross section as the governing limit state and, thus, achieve local ductility. For example, the tie bar-to-steel plate connections are designed to develop the yield strength of the tie bar in axial tension as mentioned earlier. Similarly, the flange-to-web plate connections can be designed, if needed, to develop the expected yield strength of the weaker of the two connected plate elements.

Connections at the member level include the coupling beam-to-composite wall connection (e.g., weld 1 in Figure 4-11), and connections at the structural level include the composite wall-to-foundation connection (e.g., weld 5 in Figure 4-11 is a part of this connection). These connections at the member and structural levels are designed in accordance with the calculated required strengths, namely Sections 4.2.8 and 4.2.9, and the applicable provisions of AISC *Specification* Chapter J for calculating available strengths.

4.3.6.1 Coupling Beam-to-Wall Connections—Required Strengths

The composite coupling beam is comprised of flange plates, web plates, and concrete infill. The forces in each of these elements is transferred to the composite wall at the coupling beam-to-wall connection. It is important to note that the composite wall is comprised of its web plates, flange/closure plate, and concrete infill. Thus, the forces from the coupling beam elements are transferred to the composite wall elements through identifiable, robust force transfer mechanisms.

As explained earlier, the composite coupling beam develops its expected flexural strength ($1.2M_{p.exp}$) and shear force ($2.4M_{p.exp}/L_b$) that must be transferred to the composite wall. These section level strengths can be used to calculate the corresponding forces in the coupling beam elements, namely, the flange plates, web plates, and concrete infill.

For example, the corresponding force in the coupling beam flange plates can be estimated as the lesser of 120% of the expected tensile yield capacity or 100% of the expected

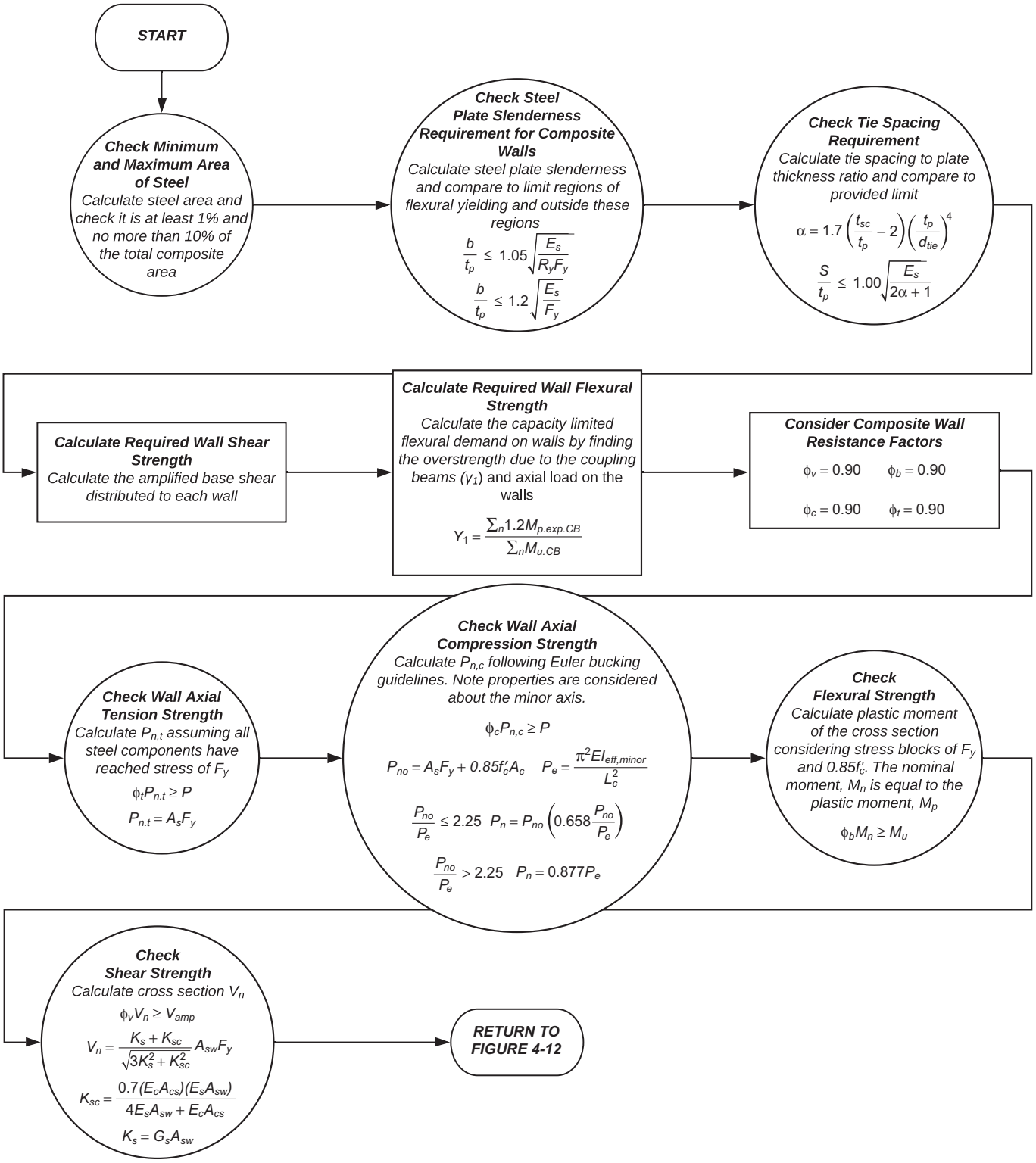


Fig. 4-15. Design of composite walls flowchart.

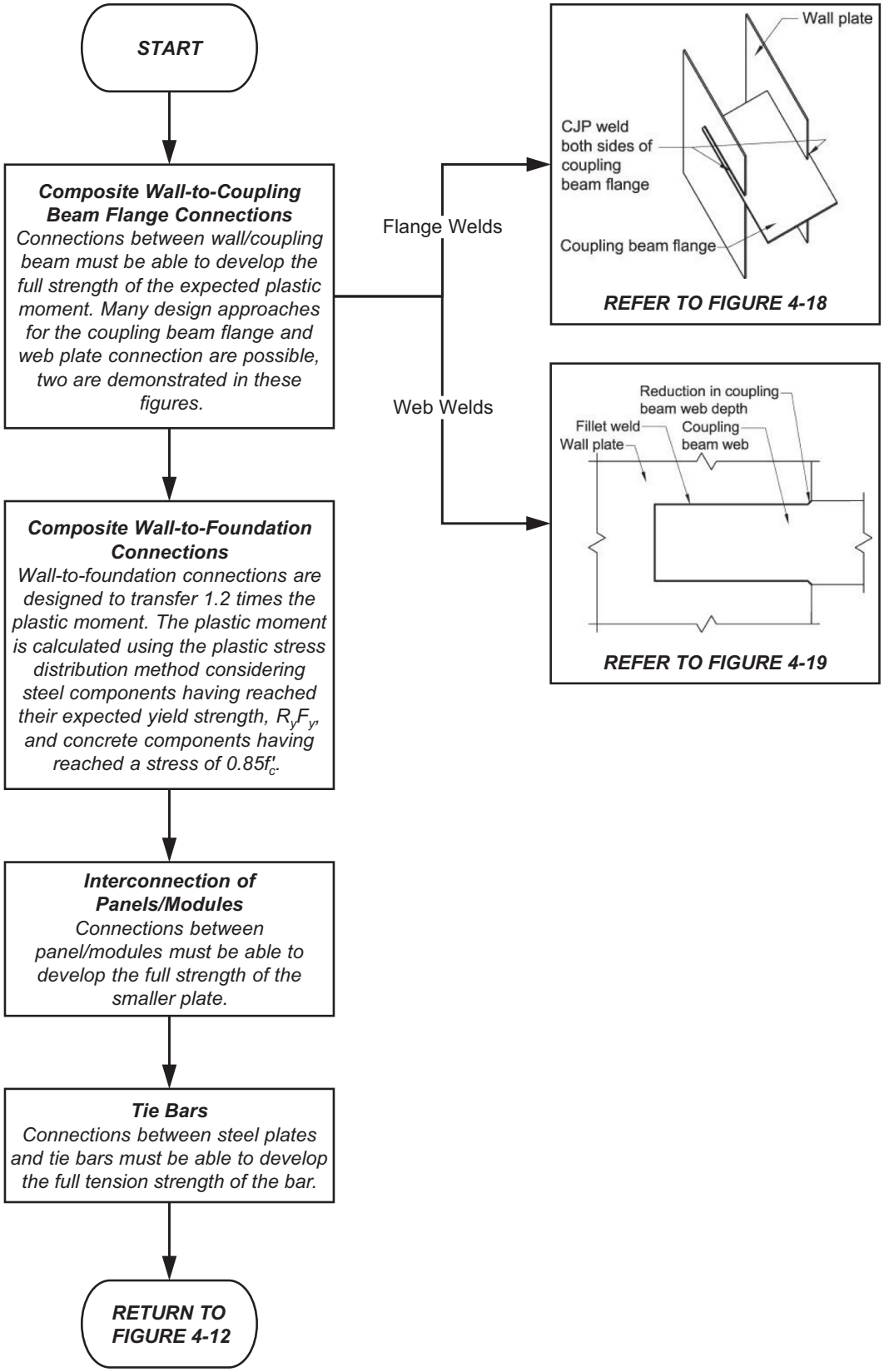


Fig. 4-16. Design of connections flowchart.

tensile rupture capacity of the flange plate (Shafaei and Varma, 2021). This becomes the required strength for the coupling beam flange plate-to-composite wall web plate welded connection (shown as weld 1 in Figure 4-11 and also in Figures 4-8 and 4-9).

Similarly, the corresponding force in the coupling beam web plates can be estimated as a combination of axial tension, flexure, and shear force by considering the stress blocks in the composite coupling beam section as shown in Figure 4-17. As shown in Figure 4-17, the plastic neutral axis for the composite section does not coincide with the centroid. Consequently, the web plate is subjected to net tension and flexure associated with the stress blocks, which are assumed to be at a stress level of the lesser of $1.2R_yF_y$ and R_tF_t. Additionally, the shear force ($V_u = 2.4M_{p.exp}/L_{cb}$) can conservatively be assumed to be transferred from the coupling beam web plates to the composite wall. Thus, the required strengths for the coupling beam web plate-to-composite wall web plate welded connection (shown as weld 1 in Figure 4-11 and also in Figure 4-9) become a combination of net tension, flexure, and shear.

The axial compression in the concrete infill of the coupling beam is transferred through direct bearing to the concrete infill of the composite wall. This assumption is reasonable whether the composite wall flange/closure plate is interrupted (Figure 4-9 and Figure 4-11) or uninterrupted (Figure 4-8).

4.3.6.2 Coupling Beam-to-Wall Connections—Available and Design Strengths

In this Design Guide, welded connections are used to transfer forces from the coupling beam flanges and webs to the composite wall web plates. Alternative connection types are possible and would require a similar design procedure and calculations.

The design of the welded connection between the coupling beam flange and the composite wall web plates (weld 1 in Figure 4-11) considers multiple limit states: (1) weld metal and base metal strength following AISC *Specification* Section J2.4 and (2) shear yielding and shear rupture of the coupling beam flange plates and wall web plates following AISC *Specification* Section J4.2. Additionally, the coupling beam flange is detailed such that its net section expected tension rupture strength is greater than its gross section expected yield strength. This is done in order to achieve local ductility and prevent premature fracture of the coupling beam flange plate before gross yielding. An overview of this procedure is shown in Figure 4-18.

The design of the welded connection between the coupling beam web plate and the composite wall web plates (weld 1 in Figure 4-11) can be performed in accordance with AISC *Specification* Section J2. The weld metal design strength for simultaneous tension, flexure, and shear can be calculated directly by following the instantaneous center of rotation method for eccentrically loaded weld groups in AISC *Manual* Part 8; however, this may become complicated, requiring the development of a spreadsheet program. Alternately, the weld metal design strength can be checked using the following method, which is generally conservative. An overview of this procedure is shown in Figure 4-19.

The weld metal strength, $P_{weld.V}$, to resist simultaneous flexure and shear can be calculated following AISC *Manual* Table 8-8, which considers an eccentric shear force, $V_{C.weld}$, creating an equivalent combination of simultaneous flexure and shear. The weld metal strength, $P_{weld.T}$, to resist the simultaneous tension force, $T_{C.weld}$, can be calculated considering only the horizontal portion of the weld to resist tension following AISC *Specification* Section J2.4. Then the utilization ratio for the weld group for simultaneous tension, flexure, and shear can be calculated using Equation 4-7.

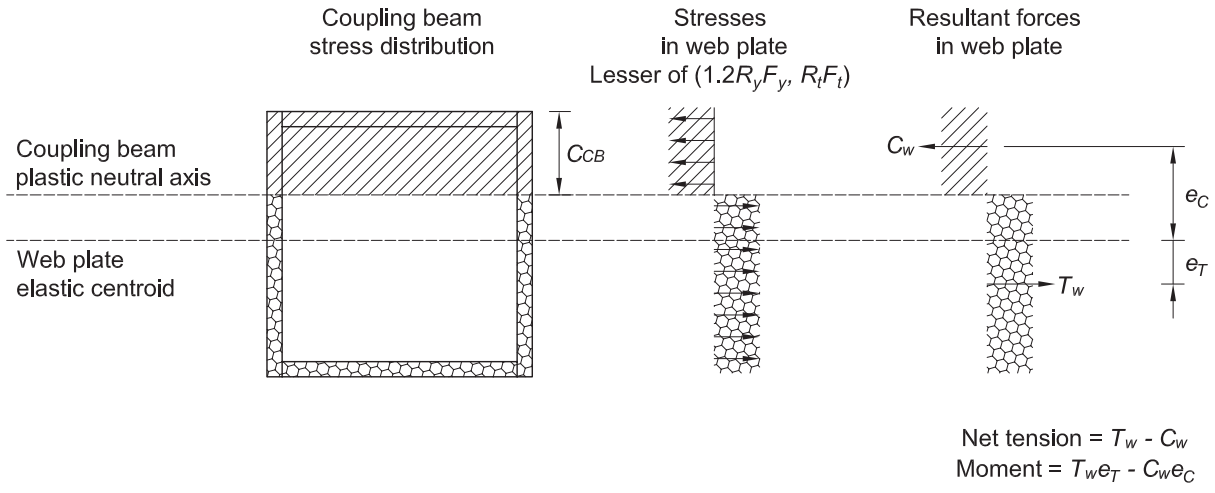


Fig. 4-17. Tension and moment in coupling beam web plates.

This utilization ratio uses the square root of the sum of the squared terms (SRSS) method because the weld group has to resist the vector shear resulting from all the demands. The utilization ratio must be less than 1.0 for design.

$$Utilization = \sqrt{\left(\frac{V_{C.weld}}{P_{weld.V}}\right)^2 + \left(\frac{T_{C.weld}}{P_{weld.T}}\right)^2} \qquad (4\text{-}7)$$

4.3.6.3 Coupling Beam-to-Wall Connections—Rotation Capacity

Connections should meet the requirements of Section 4.2.8.3. Example connections that satisfy the rotation requirements are included in Ahmad et al. (2021) and are similar to those shown in Figures 4-9 and 4-11.

4.3.6.4 Composite Wall-to-Foundation Connections

Composite wall-to-foundation connections demands can be difficult to conceptualize as the wall-to-foundation connection may not occur at grade, where the moment and shear demands are generally calculated. When the wall-to-foundation connection is located at grade level, the force demands for shear, moment, and axial load are discussed in Section 4.2.9. Some examples of composite wall-to-foundation connections are shown in Section 3.2.6.3. For example, if the base-plate connection design shown in Figure 3-3(a) is used, several force transfers and associated limit states will have to be considered. Some of these are listed in the following. Please note that this is only an indicative list, not an exhaustive one:

1. The welded connection between the composite wall plates and the base plate will have to be designed to develop the expected tensile strength of the web plate.
2. The rebar below the base plate will have to be designed to be stronger than the composite wall plates—that is, the expected yield strength of the rebar will have to be greater than or equal to the tensile strength of the wall plates.
3. The base plate will have to be designed to transfer the tensile strength of the composite wall plates to the welded rebar anchors, while considering the limit state of base plate bending and prying action.

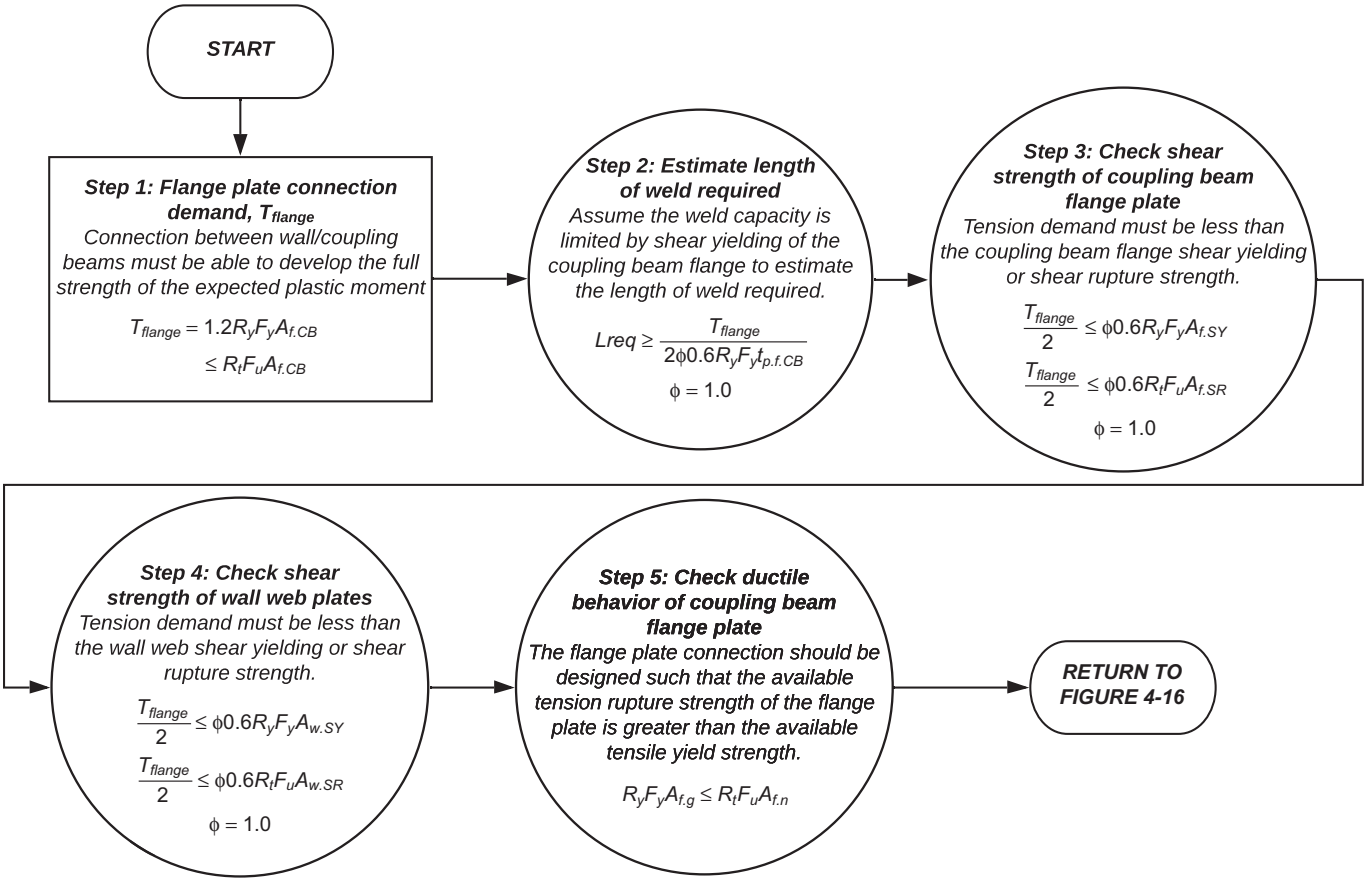


Fig. 4-18. Coupling beam-to-wall connection design: flange weld design flowchart.

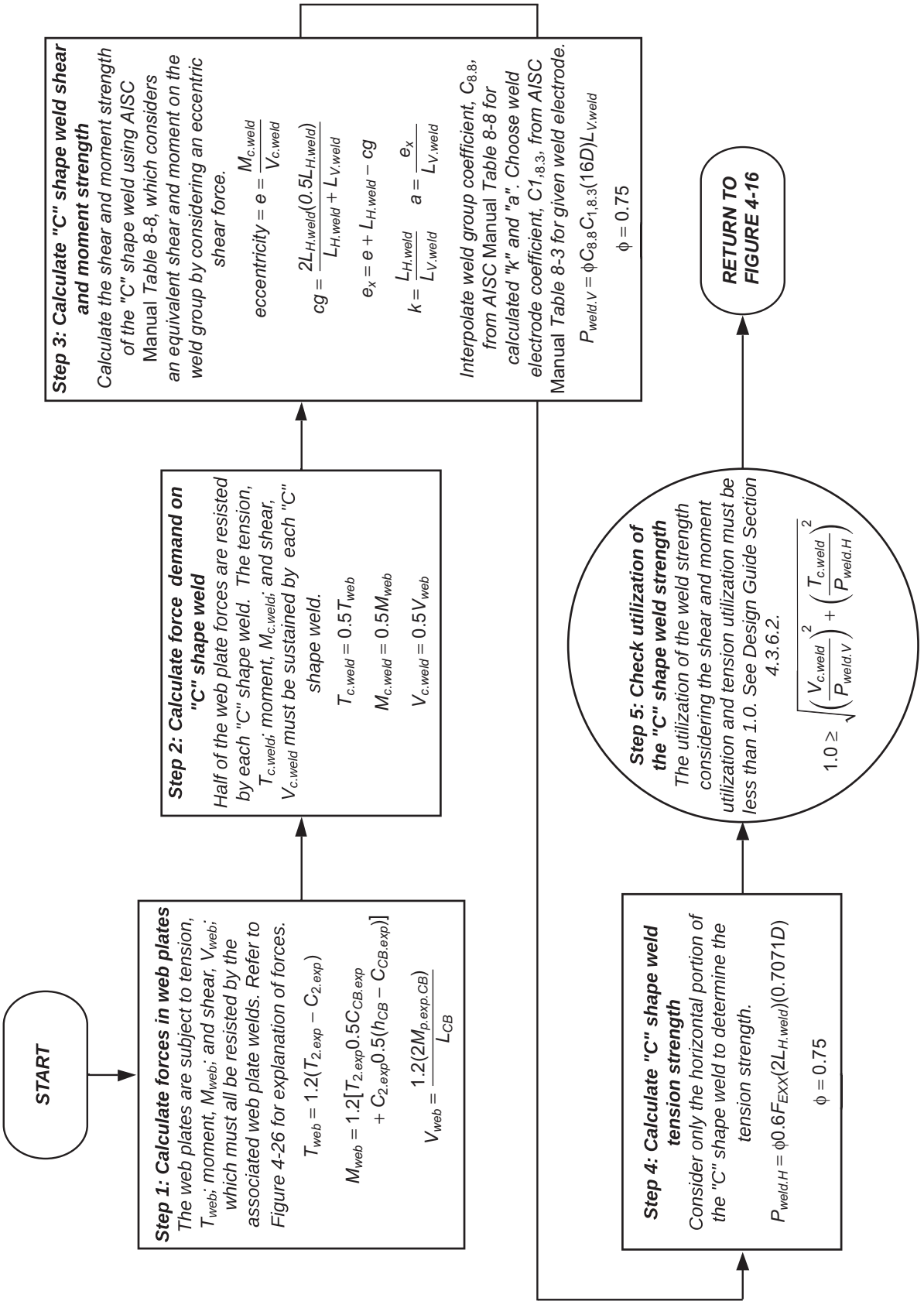


Fig. 4-19. Coupling beam-to-wall connection design: web weld design flowchart.

4. The connection between the rebar and the base plate will have to be designed to develop the tensile strength of the rebar. This can be achieved using commercial weldable couplers or other options.
5. The base plate connection detail will also have to be checked for transferring the shear force in the walls. This can be done using shear friction on the rebar or direct bearing on the rebar and associated couplers.

Some of these force transfer mechanisms and associated limit states are discussed in detail along with example calculations in AISC Design Guide 32, *Design of Modular Steel Plate Composite (SC) Walls for Safety-Related Nuclear Facilities* (Bhardwaj and Varma, 2017) (refer to Chapter 11.5 and Appendix A, Step 12). However, it is important to note that there are fundamental differences between the wall-to-foundation connections discussed in Design Guide 32 and in this Design Guide. Not all of the limit states and calculations from Design Guide 32 are applicable; it should be used only as a reference.

4.4 DESIGN EXAMPLES

EXAMPLE 4.1—Seismic Design of Eight-Story Structure Using Coupled, Planar SpeedCore System

This example details the design of an eight-story office building with typical design loads, floor geometry, and high seismic design loads. The steps followed in this design mirror the design procedure presented in Section 4.2. For simplicity, this design example does not consider accidental eccentricity and assumes a seismic redundancy factor of 1.0.

Design the coupled, planar SpeedCore walls as shown in Figure 4-20 using the given geometry, material properties, and loads. The effects of gravity loads, which can slightly increase the axial compression in the composite walls, have been conservatively ignored. However, design calculations including the effects of axial compression from gravity loads are included in the Mathcad file that can be downloaded from the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

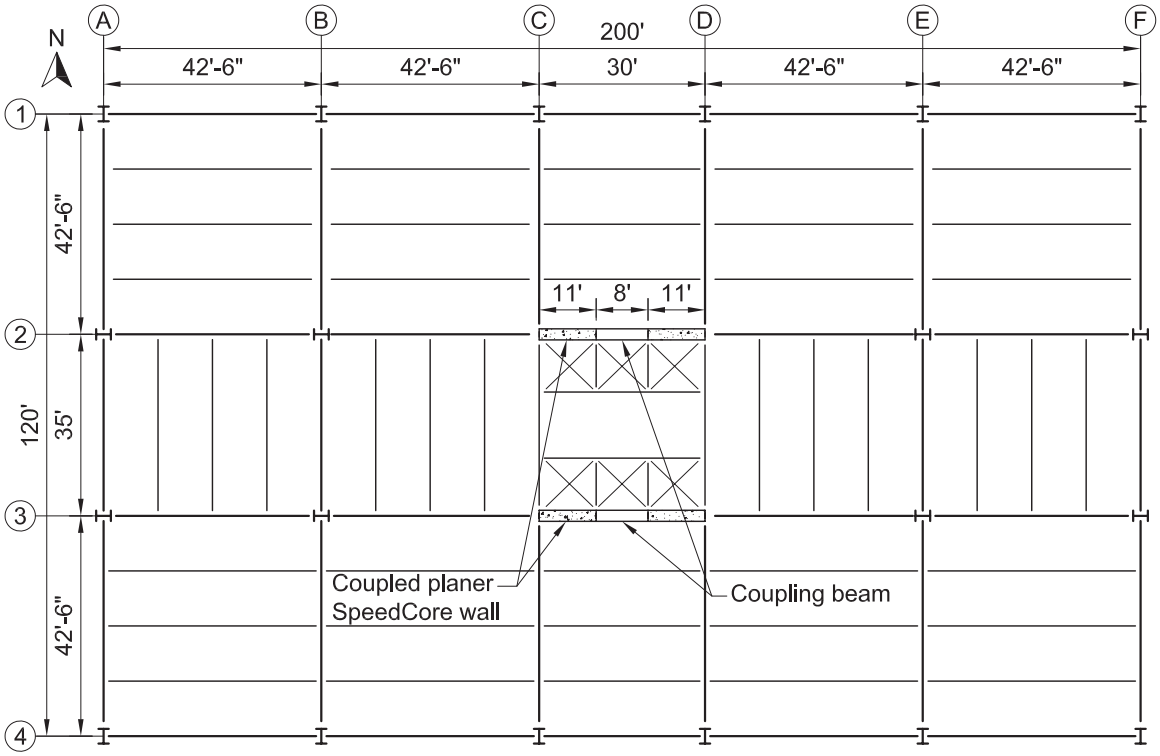


Fig. 4-20. Building floor plan for Example 4.1.

Table 4-1. Seismic Design Loads Calculated from ASCE/SEI 7		
Story No.	**Story Elevation (ft)**	**Story Force (kips)**
1	17	19.6
2	31	41.4
3	45	65.8
4	59	92.1
5	73	120
6	87	149
7	101	180
8	115	211
Total base shear, V_{base}		879
Overturning moment, OTM		890,000 kip-in.

Given:

Building geometry:

L_f = building length
= 200 ft
W_f = building width
= 120 ft
h_{typ} = typical story height
= 14 ft
h_1 = first story height
= 17 ft
n = number of stories
= 8

Floor loads:

DL = floor dead load
= 0.12 ksf

Seismic design loads:

Step 1. General Information of the Considered Building

The steel and concrete material properties are as follows:

ASTM A572/A572M Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi
R_c = 1.3

Weld metal (E70XX):

F_{EXX} = 70 ksi

Seismic design coefficients:

C_d = deflection amplification factor (ASCE/SEI 7, Table 12.2-1)
= 5.5
I_e = importance factor (ASCE/SEI 7, Section 11.5.1 for an office building)
= 1
R = seismic response modification coefficient (from ASCE/SEI 7, Table 12.2-1)
= 8

Risk Category = II (ASCE/SEI 7, Table 1.5-1)

Ω_0 = overstrength factor (from ASCE/SEI 7, Table 12.2-1)
= 2.5
ρ = seismic redundancy factor (ASCE/SEI 7, Section 12.3.4)
= 1

Solution:

For ASTM A572/A572M Grade 50 steel, from AISC *Manual* Table 2-4 and AISC *Seismic Provisions* Table A3.2, the material properties are as follows:

F_y = 50 ksi
F_u = 65 ksi
R_y = 1.1

SpeedCore walls and coupling beam components are sized based on initial estimates of loads; these sizes are refined through iteration. Acceptable dimensions are presented here so that this example can focus on the appropriate limit states to check.

The SpeedCore wall dimensions are shown in Figure 4-21.

L_{wall} = wall length
= 132 in.
t_p = wall plate thickness
= 9/16 in.
t_{sc} = wall thickness
= 24 in.

Coupling beam dimensions as shown in Figure 4-22:

L_{cb} = coupling beam length
= 96 in.
b_{CB} = coupling beam width
= 24 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf.CB}$ = coupling beam flange plate thickness
= ½ in.
$t_{pw.CB}$ = coupling beam web plate thickness
= ⅜ in.

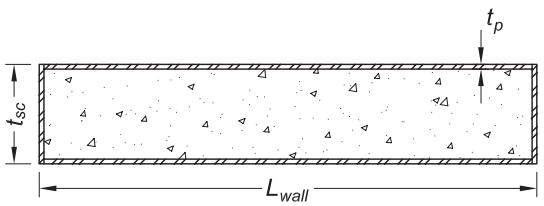


Fig. 4-21. Wall cross-section dimensions.

Step 2. Analysis for Design

In this step, a computer model is built and analyzed using an elastic method to determine the force on system members and overall system deflection.

The effective distance between wall centroids as shown in Figure 4-23 is:

$$L_{eff} = L_{cb} + L_{wall}$$
$$= \frac{(96 \text{ in.} + 132 \text{ in.})}{12 \text{ in./ft}}$$
$$= 19.0 \text{ ft}$$

Planar SpeedCore Wall Properties

The effective axial, shear, and flexural stiffnesses are calculated for SpeedCore elements for elastic analysis.

The area of steel in the wall is:

$$A_s = t_p\left[2(L_{wall} - 2t_p) + 2t_{sc}\right]$$
$$= (9/16 \text{ in.})\left\{2\left[132 \text{ in.} - 2(9/16 \text{ in.})\right] + 2(24 \text{ in.})\right\}$$
$$= 174 \text{ in.}^2$$

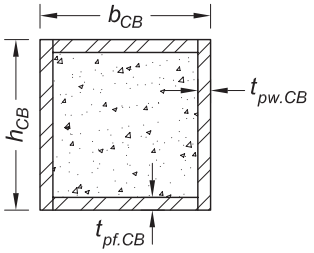


Fig. 4-22. Coupling beam cross-section dimensions.

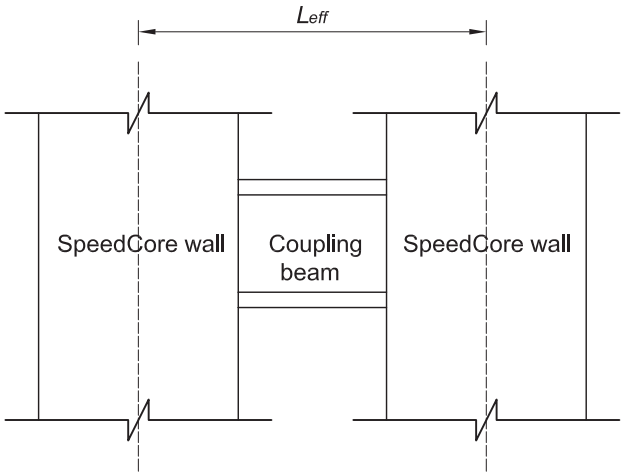


Fig. 4-23. Effective length.

The area of concrete in the wall is:

$$A_c = L_{wall} t_{sc} - A_s$$
$$= (132\text{ in.})(24\text{ in.}) - 174\text{ in.}^2$$
$$= 2{,}990\text{ in.}^2$$

The effective stiffness of the wall is calculated from Equation 2-10 as:

$$EA_{eff} = E_s A_s + 0.45 E_c A_c \quad (2\text{-}10)$$
$$= (29{,}000\text{ ksi})(174\text{ in.}^2) + 0.45(4{,}500\text{ ksi})(2{,}990\text{ in.}^2)$$
$$= 1.11 \times 10^7\text{ kips}$$

The steel area in the direction of shear is:

$$A_{sw} = 2 L_{wall}\, t_p$$
$$= 2(132\text{ in.})(9/16\text{ in.})$$
$$= 149\text{ in.}^2$$

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_s A_{sw} + G_c A_c \quad (2\text{-}11)$$
$$= (11{,}200\text{ ksi})(149\text{ in.}^2) + (1{,}800\text{ ksi})(2{,}990\text{ in.}^2)$$
$$= 7.05 \times 10^6\text{ kips}$$

The moment of inertia of steel in the wall is:

$$I_s = 2\left[\frac{t_p (L_{wall} - 2t_p)^3}{12}\right] + 2\left[\left(\frac{t_{sc} t_p^3}{12}\right) + t_{sc} t_p \left(\frac{L_{wall}}{2} - \frac{t_p}{2}\right)^2\right]$$
$$= 2\left\{\frac{(9/16\text{ in.})\left[132\text{ in.} - 2(9/16\text{ in.})\right]^3}{12}\right\} + 2\left\{\frac{(24\text{ in.})(9/16\text{ in.})}{12} + (24\text{ in.})(9/16\text{ in.})\left[\frac{132\text{ in.}}{2} - \frac{(9/16\text{ in.})}{2}\right]^2\right\}$$
$$= 327{,}000\text{ in.}^4$$

The moment of inertia of concrete in the wall is:

$$I_c = \frac{(L_{wall} - 2t_p)^3 (t_{sc} - 2t_p)}{12}$$
$$= \frac{\left[132\text{ in.} - 2(9/16\text{ in.})\right]^3 \left[24\text{ in.} - 2(9/16\text{ in.})\right]}{12}$$
$$= 4{,}270{,}000\text{ in.}^4$$

The effective flexural stiffness of the wall is:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \quad (2\text{-}9)$$
$$= (29{,}000\text{ ksi})(327{,}000\text{ in.}^4) + 0.35(4{,}500\text{ ksi})(4{,}270{,}000\text{ in.}^4)$$
$$= 1.62 \times 10^{10}\text{ kip-in.}^2$$

Coupling Beam Geometric Properties

The effective axial, shear, and flexural stiffnesses are calculated for the coupling beam elements for elastic analysis.

The area of steel in the coupling beam is:

$$\begin{aligned} A_{s.CB} &= 2t_{pw.CB}h_{CB} + 2(b_{CB} - 2t_{pw.CB})t_{pf.CB} \\ &= 2(⅜\text{ in.})(24\text{ in.}) + 2\left[24\text{ in.} - 2(⅜\text{ in.})\right](½\text{ in.}) \\ &= 41.3\text{ in.}^2 \end{aligned}$$

The area of concrete in the coupling beam is:

$$\begin{aligned} A_{c.CB} &= (b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB}) \\ &= \left[24\text{ in.} - 2(⅜\text{ in.})\right]\left[24\text{ in.} - 2(½\text{ in.})\right] \\ &= 535\text{ in.}^2 \end{aligned}$$

The axial stiffness of the uncracked section is:

$$\begin{aligned} EA_{uncr.CB} &= E_s A_{s.CB} + E_c A_{c.CB} \\ &= (29{,}000\text{ ksi})(41.3\text{ in.}^2) + (4{,}500\text{ ksi})(535\text{ in.}^2) \\ &= 3{,}610{,}000\text{ kips} \end{aligned}$$

The moment of inertia of steel in the coupling beam is:

$$\begin{aligned} I_{s.CB} &= 2\left[\frac{t_{pw.CB}h_{CB}^3}{12} + \frac{(b_{CB} - 2t_{pw.CB})t_{pf.CB}^3}{12} + (b_{CB} - 2t_{pw.CB})t_{pf.CB}\left(\frac{h_{CB}}{2} - \frac{t_{pf.CB}}{2}\right)^2\right] \\ &= 2\left\{\frac{(⅜\text{ in.})(24\text{ in.})^3}{12} + \frac{\left[24\text{ in.} - 2(⅜\text{ in.})\right](½\text{ in.})^3}{12} + \left[24\text{ in.} - 2(⅜\text{ in.})\right](½\text{ in.})\left(\frac{24\text{ in.}}{2} - \frac{½\text{ in.}}{2}\right)^2\right\} \\ &= 4{,}070\text{ in.}^4 \end{aligned}$$

The moment of inertia of the concrete in the coupling beam is:

$$\begin{aligned} I_{c.CB} &= \frac{(b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB})^3}{12} \\ &= \frac{\left[24\text{ in.} - 2(⅜\text{ in.})\right]\left[24\text{ in.} - 2(½\text{ in.})\right]^3}{12} \\ &= 23{,}600\text{ in.}^4 \end{aligned}$$

From AISC *Specification* Section I2.2b, Equation I2-13, the effective rigidity coefficient is:

$$\begin{aligned} C_{3.CB} &= 0.45 + 3\left(\frac{A_{s.CB}}{b_{CB}h_{CB}}\right) \le 0.9 \qquad \text{(from Eq. 2-18)} \\ &= 0.45 + 3\left[\frac{41.3\text{ in.}^2}{(24\text{ in.})(24\text{ in.})}\right] \\ &= 0.665 \end{aligned}$$

The effective stiffness of the coupling beam is:

$$EI_{eff.CB} = E_s I_{s.CB} + C_3 E_c I_{c.CB} \quad \text{(from Eq. 2-17)}$$
$$= (29{,}000 \text{ ksi})(4{,}070 \text{ in.}^4) + 0.665(4{,}500 \text{ ksi})(23{,}600 \text{ in.}^4)$$
$$= 1.89 \times 10^8 \text{ kip-in.}^2$$

The area of steel in the direction of shear is:

$$A_{sw.CB} = 2h_{CB} t_{pw.CB}$$
$$= 2(24 \text{ in.})(\tfrac{3}{8} \text{ in.})$$
$$= 18.0 \text{ in.}^2$$

The effective shear stiffness of the coupling beam is:

$$GA_{v.CB} = G_s A_{sw.CB} + G_c A_{c.CB} \quad \text{(from Eq. 2-11)}$$
$$= (11{,}200 \text{ ksi})(18.0 \text{ in.}^2) + (1{,}800 \text{ ksi})(535 \text{ in.}^2)$$
$$= 1{,}160{,}000 \text{ kips}$$

From Section 2.3.2:

$$0.64EI_{eff.CB} = 0.64(1.89 \times 10^8 \text{ kips})$$
$$= 1.21 \times 10^8 \text{ kip-in.}^2$$

$$0.8EA_{uncr.CB} = 0.8(3{,}610{,}000 \text{ kips})$$
$$= 2{,}890{,}000 \text{ kips}$$

Numerical Model

To determine the ID ratio and shear demand on coupling beams, an analysis model was built using commercial software. First-order, linear elastic analysis was performed on a model consisting of frame elements for only the SpeedCore walls and composite coupling beams. This model was subjected to the earthquake loading previously defined. It employed rigid offsets at the end of the coupling beams to account for wall length. All mass was applied at the story level and distributed to the two joints. A pictorial representation of the model is shown in Figure 4-24.

Frame elements were assigned the axial, flexural, and shear stiffnesses calculated in the preceding calculations.

From this analysis model, the required coupling beam shear strengths, as shown in Table 4-2, and story displacement values, as shown in Table 4-3, were determined. This model also confirms the base shear and overturning moment values presented in the Given section.

Lateral displacement values are amplified by the C_d factor to obtain the amplified displacement. These amplified values are then used to calculate the ID, which is the difference in displacement between two floors normalized by the story height. The maximum design ID is limited according to ASCE/SEI 7, Section 12.12.1. In this case, the maximum design ID is 2% for Risk Category II. The story displacement, amplified displacement, and ID are presented in Table 4-3. This distribution corresponds to the deflected shape shown in Figure 4-25.

The maximum ID = 1.34% < 2%.

Table 4-2. Coupling Beam Required Shear Strengths		
Story No.	**Story Elevation (ft)**	**Coupling Beam Required Shear Strength (kips)**
1	17	297
2	31	393
3	45	420
4	59	404
5	73	359
6	87	299
7	101	234
8	115	185
Average		324

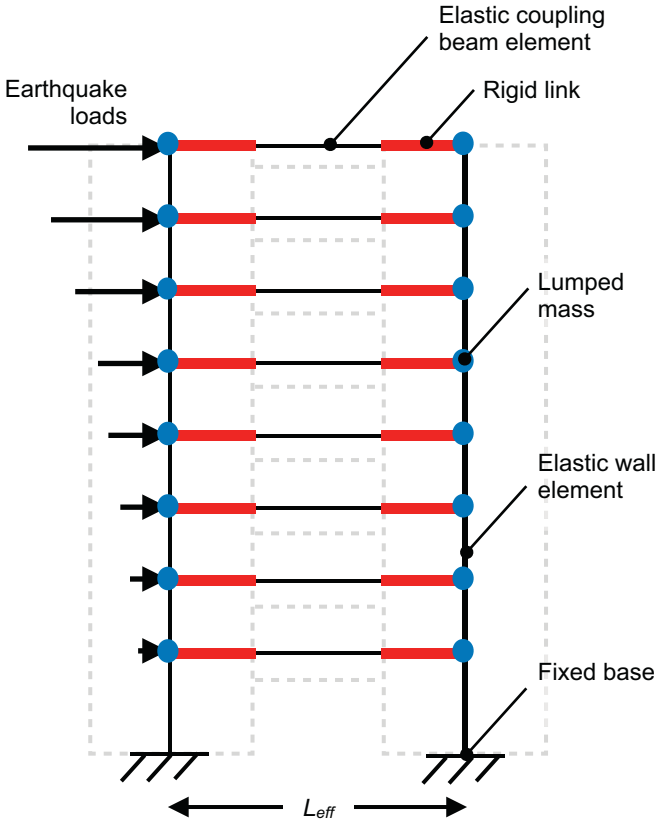


Fig. 4-24. Pictorial representation of analysis model.

Table 4-3. Lateral Displacement and ID Summary				
Story No.	Story Height (ft)	Displacement (in.)	Amplified Displacement (in.)	ID (%)
8	14	2.74	15.1	1.14
7	14	2.39	13.1	1.24
6	14	2.01	11.1	1.31
5	14	1.61	8.86	1.34
4	14	1.20	6.60	1.28
3	14	0.81	4.46	1.18
2	14	0.45	2.48	0.95
1	17	0.16	0.88	0.43

Required Shear and Flexural Strength

The required coupling beam shear strength is calculated as the average of all coupling beam shear demands. Shear demand differs along the height of the structure due to the seismic load distribution and the nature of linear elastic modeling. Alternatively, the designer can choose to use the maximum shear demand. Because a portion of the overturning moment will be resisted by the coupling action and the remainder by the individual walls, the result of this choice is the relative proportioning of wall and coupling beam elements. The system is designed to ensure plasticity spreads along the height of the structure, and therefore, either method is acceptable.

From Table 4-2, the required coupling beam shear strength is:

$$V_{r.CB} = 324 \text{ kips}$$

The coupling beam required flexural strength is:

$$M_{r.CB} = \frac{V_{r.CB} L_{cb}}{2}$$
$$= \frac{(324 \text{ kips})(96 \text{ in.})}{2}$$
$$= 15{,}600 \text{ kip-in.}$$

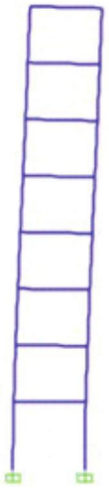


Fig. 4-25. Analysis model deflected shape.

The base shear for each planar wall is amplified by a shear amplification factor of 4.0, following the recommendations of Section 4.3.3.1. The shear associated with the earthquake loads is V_{base}, as provided in the Given section.

$$V_{amp} = 4.0V_{base} \tag{4-6}$$
$$= 4.0(879 \text{ kips})$$
$$= 3{,}520 \text{ kips}$$

$$V_{r.wall} = \frac{V_{amp}}{2}$$
$$= \frac{3{,}520 \text{ kips}}{2}$$
$$= 1{,}760 \text{ kips}$$

Step 3. Design of Coupling Beams

Step 3-1. Flexure-critical coupling beams

Composite coupling beams are proportioned to be flexure critical with design shear strength:

$$V_{n.exp} \geq \frac{2.4M_{p.exp}}{L_{cb}} \tag{4-4}$$

Because this equation requires knowing the capacity of the coupling beam, this check will be completed at the end of the section.

*Step 3-2. Expected flexural strength (*Mp.exp.CB*)*

The expected flexural strength of the composite coupling beams is calculated using the plastic stress distribution method with the assumed stress blocks noted in Figure 4-26. This is used in Step 3-7 as part of the shear strength check.

The width of concrete in the coupling beam is:

$$t_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 24 \text{ in.} - 2(⅜ \text{ in.})$$
$$= 23.3 \text{ in.}$$

The plastic neutral axis location of the coupling beam from the top flange is:

$$C_{CB.exp} = \frac{2t_{pw.CB}\,h_{CB}\,R_yF_y + 0.85R_cf'_c\,t_{c.CB}\,t_{pf.CB}}{4t_{pw.CB}R_yF_y + 0.85R_cf'_c\,t_{c.CB}}$$
$$= \frac{2(⅜ \text{ in.})(24 \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})(23.3 \text{ in.})(½ \text{ in.})}{4(⅜ \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})(23.3 \text{ in.})}$$
$$= 4.50 \text{ in.}$$

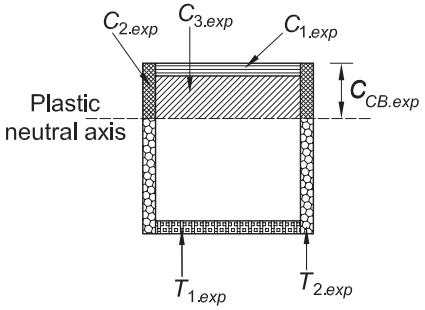


Fig. 4-26. Cross section with labeled regions for plastic moment calculation of coupling beam.

The expected compression force in the flange is:

$$C_{1.exp} = (b_{CB} - 2t_{pw.CB})t_{pf.CB} R_y F_y$$
$$= \left[24 \text{ in.} - 2(⅜ \text{ in.})\right](½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 639 \text{ kips}$$

The compression force in the web is:

$$C_{2.exp} = 2t_{pw.CB} C_{CB.exp} R_y F_y$$
$$= 2(⅜ \text{ in.})(4.50 \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 186 \text{ kips}$$

The compression force in the concrete is:

$$C_{3.exp} = 0.85 R_c f'_c t_{c.CB}(C_{CB.exp} - t_{pf.CB})$$
$$= 0.85(1.3)(6 \text{ ksi})(23.3 \text{ in.})(4.50 \text{ in.} - ½ \text{ in.})$$
$$= 618 \text{ kips}$$

The tension force in the flange is:

$$T_{1.exp} = (b_{CB} - 2t_{pw.CB})t_{pf.CB} R_y F_y$$
$$= \left[24 \text{ in.} - 2(⅜ \text{ in.})\right](½ \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 639 \text{ kips}$$

The tension force in the web is:

$$T_{2.exp} = 2t_{pw.CB}(h_{CB} - C_{CB.exp}) R_y F_y$$
$$= 2(⅜ \text{ in.})(24 \text{ in.} - 4.50 \text{ in.})(1.1)(50 \text{ ksi})$$
$$= 804 \text{ kips}$$

The expected plastic flexural strength of the coupling beam is:

$$M_{p.exp.CB} = C_{1.exp}\left(C_{CB.exp} - \frac{t_{pf.CB}}{2}\right) + C_{2.exp}\left(\frac{C_{CB.exp}}{2}\right) + C_{3.exp}\left(\frac{C_{CB.exp} - t_{pf.CB}}{2}\right) + T_{1.exp}\left(h_{CB} - C_{CB.exp} - \frac{t_{pf.CB}}{2}\right)$$
$$+ T_{2.exp}\left(\frac{h_{CB} - C_{CB.exp}}{2}\right)$$
$$= (639 \text{ kips})\left(4.50 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (186 \text{ kips})\left(\frac{4.50 \text{ in.}}{2}\right) + (618 \text{ kips})\left(\frac{4.50 \text{ in.} - ½ \text{ in.}}{2}\right)$$
$$+ (639 \text{ kips})\left(24 \text{ in.} - 4.50 \text{ in.} - \frac{½ \text{ in.}}{2}\right) + (804 \text{ kips})\left(\frac{24 \text{ in.} - 4.50 \text{ in.}}{2}\right)$$
$$= 24{,}500 \text{ kip-in.}$$

Step 3-3. Minimum area of steel

The minimum area of steel required according to AISC *Specification* Section I2.2a is:

$$A_{s.CB.min} = 0.01 b_{CB} h_{CB}$$
$$= 0.01(24 \text{ in.})(24 \text{ in.})$$
$$= 5.76 \text{ in.}^2$$

$A_{s.CB} = 41.3 \text{ in.}^2 > A_{s.CB\ min} = 5.76 \text{ in.}^2$ **o.k.**

Step 3-4. Steel plate slenderness requirement for coupling beams

The clear width of the coupling beam flange plate is:

$$b_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 24 \text{ in.} - 2(⅜ \text{ in.})$$
$$= 23.3 \text{ in.}$$

The clear width of the coupling beam web plate is:

$$h_{c.CB} = h_{CB} - 2t_{pf.CB}$$
$$= 24 \text{ in.} - 2(½ \text{ in.})$$
$$= 23.0 \text{ in.}$$

The slenderness requirements for coupling beams are checked using AISC *Seismic Provisions* Section H8.5b.

Slenderness check for coupling beam flange plates:

$$\frac{b_{c.CB}}{t_{pf.CB}} = \frac{23.3 \text{ in.}}{½ \text{ in.}}$$
$$= 46.6$$

$$2.37\sqrt{\frac{E_s}{R_y F_y}} = 2.37\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}}$$
$$= 54.4$$

$$\frac{b_{c.CB}}{t_{pf.CB}} = 46.6 < 2.37\sqrt{\frac{E_s}{R_y F_y}} = 54.4 \quad \textbf{o.k.}$$

Slenderness check for coupling beam web plates:

$$\frac{h_{c.CB}}{t_{pw.CB}} = \frac{23.0 \text{ in.}}{⅜ \text{ in.}}$$
$$= 61.3$$

$$2.66\sqrt{\frac{E_s}{R_y F_y}} = 2.66\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}}$$
$$= 61.1$$

$$\frac{h_{c.CB}}{t_{pw.CB}} = 61.3 > 2.66\sqrt{\frac{E_s}{R_y F_y}} = 61.1$$

The coupling beam design would be adjusted so that the beam's web plates are compact, i.e., they satisfy the limit given in Section H8.5b. Increasing the coupling beam flange plate thickness, $t_{pw.CB}$, to 7⁄16 in. would satisfy that requirement. The iterative design process is not fully shown in this example.

Step 3-5. Flexural strength, $M_{p.CB}$

The flexural strength of the composite coupling beams is calculated using the plastic stress distribution method with the assumed stress blocks noted in Figure 4-26.

The plastic neutral axis location of the coupling beam from the top flange is:

$$C_{CB} = \frac{2t_{pw.CB}h_{CB}F_y + 0.85f_c't_{c.CB}t_{pf.CB}}{4t_{pw.CB}F_y + 0.85f_c't_{c.CB}}$$

$$= \frac{2(⅜\text{ in.})(24\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})(23.3\text{ in.})(½\text{ in.})}{4(⅜\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})(23.3\text{ in.})}$$

$$= 4.95\text{ in.}$$

Note: The plastic neutral axis location, C_{CB}; compression forces, C_1, C_2, C_3; and tension forces, T_1, T_2, for a rectangular coupling beam are depicted in Appendix A, Figure A-13.

The compression force in the top flange is:

$$C_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$

$$= \left[24\text{ in.} - 2(⅜\text{ in.})\right](½\text{ in.})(50\text{ ksi})$$

$$= 581\text{ kips}$$

The compression force in the web is:

$$C_2 = 2t_{pw.CB}C_{CB}F_y$$

$$= 2(⅜\text{ in.})(4.95\text{ in.})(50\text{ ksi})$$

$$= 186\text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85f_c't_{c.CB}(C_{CB} - t_{pf.CB})$$

$$= 0.85(6\text{ ksi})(23.3\text{ in.})(4.95\text{ in.} - ½\text{ in.})$$

$$= 529\text{ kips}$$

The tension force in the flange is:

$$T_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$

$$= \left[24\text{ in.} - 2(⅜\text{ in.})\right](½\text{ in.})(50\text{ ksi})$$

$$= 581\text{ kips}$$

The tension force in the web is:

$$T_2 = 2t_{pw.CB}(h_{CB} - C_{CB})F_y$$

$$= 2(⅜\text{ in.})(24\text{ in.} - 4.95\text{ in.})(50\text{ ksi})$$

$$= 714\text{ kips}$$

The plastic flexural strength of the coupling beam is:

$$M_{Pn.CB} = C_1\left(C_{CB} - \frac{t_{pf.CB}}{2}\right) + C_2\left(\frac{C_{CB}}{2}\right) + C_3\left(\frac{C_{CB} - t_{pf.CB}}{2}\right) + T_1\left(h_{CB} - C_{CB} - \frac{t_{pf.CB}}{2}\right) + T_2\left(\frac{h_{CB} - C_{CB}}{2}\right)$$

$$= (581\text{ kips})\left(4.95\text{ in.} - \frac{½\text{ in.}}{2}\right) + (186\text{ kips})\left(\frac{4.95\text{ in.}}{2}\right) + (529\text{ kips})\left(\frac{4.95\text{ in.} - ½\text{ in.}}{2}\right)$$

$$+ (581\text{ kips})\left(24\text{ in.} - 4.95\text{ in.} - \frac{½\text{ in.}}{2}\right) + (714\text{ kips})\left(\frac{24\text{ in.} - 4.95\text{ in.}}{2}\right)$$

$$= 22{,}100\text{ kip-in.}$$

Using AISC *Specification* Section I3.4b with $\phi_b = 0.90$, the design flexural strength is:

$$M_{n.CB} = M_{Pn.CB} \quad \text{(AISC } \textit{Spec.} \text{ Eq. I3-5a)}$$

$$\phi_b M_{n.CB} = 0.90(22{,}100 \text{ kip-in.})$$
$$= 19{,}900 \text{ kip-in.} > M_{r.CB} = 15{,}600 \text{ kip-in.} \quad \textbf{o.k.}$$

Step 3-6. Nominal shear strength

The area of steel in the webs was calculated in Step 2:

$$A_{sw.CB} = 18.0 \text{ in.}^2$$

The nominal shear strength of the coupling beam is:

$$V_{n.CB} = 0.6F_y A_{sw.CB} + 0.06K_c A_{c.CB}\sqrt{f'_c} \quad \text{(from Eq. 2-19)}$$

From Section 2.3.3, $K_c = 1.0$:

$$V_{n.CB} = 0.6(50 \text{ ksi})(18.0 \text{ in.}^2) + 0.06(1.0)(535 \text{ in.}^2)\sqrt{6 \text{ ksi}}$$
$$= 619 \text{ kips}$$

Using $\phi_v = 0.90$, the design shear strength of the coupling beam is:

$$\phi_v V_{n.CB} = 0.90(619 \text{ kips})$$
$$= 557 \text{ kips} > V_{r.CB} = 324 \text{ kips} \quad \textbf{o.k.}$$

Step 3-7. Flexure-critical coupling beams (revisited)

The expected shear strength for the coupling beam is:

$$V_{n.exp} = 0.6R_y F_y A_{sw.CB} + 0.06A_{c.CB}\sqrt{R_c f'_c} \quad \text{(from Eq. 2-19)}$$
$$= 0.6(1.1)(50 \text{ ksi})(18.0 \text{ in.}^2) + 0.06(535 \text{ in.}^2)\sqrt{1.3(6 \text{ ksi})}$$
$$= 684 \text{ kips}$$

$$V_u = \frac{2.4M_{p.exp.CB}}{L_{cb}} \quad \text{(from Eq. 4-5)}$$
$$= \frac{2.4(24{,}500 \text{ kip-in.})}{96 \text{ in.}}$$
$$= 613 \text{ kips}$$

$$V_{n.exp} = 684 \text{ kips} > \frac{2.4M_{p.exp.CB}}{L_{cb}} = 613 \text{ kips} \quad \textbf{o.k.}$$

Step 4. Design of Composite Walls

Step 4-1. Minimum and maximum area of steel

The minimum and maximum areas of steel are calculated according to AISC *Seismic Provisions* Section H8.4a.

The minimum area of steel required is:

$$A_{s.min} = 0.01L_{wall}\, t_{sc}$$
$$= 0.01(132 \text{ in.})(24 \text{ in.})$$
$$= 31.7 \text{ in.}^2$$

$$A_s = 174 \text{ in.}^2 > A_{s.min} = 31.7 \text{ in.}^2 \quad \textbf{o.k.}$$

The maximum area of steel allowed is:

$$A_{s.max} = 0.1 L_{wall}\, t_{sc}$$
$$= 0.1(132 \text{ in.})(24 \text{ in.})$$
$$= 317 \text{ in.}^2$$

$$A_s = 174 \text{ in.}^2 < A_{s.max} = 317 \text{ in.}^2 \quad \textbf{o.k.}$$

Step 4-2. Steel plate slenderness requirements for composite walls

The largest unsupported length between rows of steel anchors or tie bars for this example is selected under the assumption that all support is through tie bars (no steel anchors). Plate slenderness and tie bar spacing requirements must be met with this choice. As discussed in Section 3.2.4.1, tie bars and steel anchors can be spaced farther apart outside of flexural yielding zones. Calculating the extent of the flexural yielding zone is discussed in Examples 3.1 and 3.2 but omitted here for simplicity.

Tie bar spacing is selected to meet the limits shown in the following checks. In this case, a uniform tie bar spacing of 12 in. was acceptable for the entire structure.

Tie bar spacing in the flexural yielding zone is checked as follows:

$$S_{tie} = 12 \text{ in.}$$

$$\frac{S_{tie}}{t_p} = \frac{12 \text{ in.}}{9/16 \text{ in.}}$$
$$= 21.3$$

$$1.05\sqrt{\frac{E_s}{R_y F_y}} = 1.05\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}} \quad \text{(from Eq. 3-1)}$$
$$= 24.1$$

$$\frac{S_{tie}}{t_p} = 21.3 < 1.05\sqrt{\frac{E_s}{R_y F_y}} = 24.1 \quad \textbf{o.k.}$$

The tie bar spacing above the flexural yielding zone is:

$$S_{tie.top} = 12 \text{ in.}$$

$$\frac{S_{tie.top}}{t_p} = \frac{12 \text{ in.}}{9/16 \text{ in.}}$$
$$= 21.3$$

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}} \quad \text{(from Eq. 3-2)}$$
$$= 28.9$$

$$\frac{S_{tie.top}}{t_p} = 21.3 < 1.2\sqrt{\frac{E_s}{F_y}} = 28.9 \quad \textbf{o.k.}$$

Step 4-3. Tie bar spacing requirements for composite walls

A tie bar diameter of 1 in. is selected for the SpeedCore wall. The bar requirement considering the stability of the empty steel module is checked as follows:

$$d_{tie} = 1 \text{ in.}$$

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$

$$= 1.7\left(\frac{24.0 \text{ in.}}{9/16 \text{ in.}} - 2\right)\left(\frac{9/16 \text{ in.}}{1 \text{ in.}}\right)^4$$

$$= 6.92$$

$$\frac{S_{tie}}{t_p} = \frac{12 \text{ in.}}{9/16 \text{ in.}}$$

$$= 21.3$$

$$1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 1.0\sqrt{\frac{29{,}000 \text{ ksi}}{2(6.92) + 1}} \quad \text{(from Eq. 2-4)}$$

$$= 44.2$$

$$\frac{S_{tie}}{t_p} = 21.3 < 1.0\sqrt{\frac{E_s}{2\alpha + 1}} = 44.2 \quad \textbf{o.k.}$$

Step 4-4. Required wall shear strength

The amplified shear force to be resisted by the wall as calculated in Step 1:

$$V_{r.wall} = 1{,}760 \text{ kips}$$

Step 4-5. Required wall flexural strength

The required wall flexural strength is determined based on the capacity-limited forces in the coupling beam elements. Walls are sized to resist the axial load associated with the formation of plastic hinges in every coupling beam and the flexural load associated with the amplified overturning moment less the contribution from the axial force couple in the adjacent walls (see Figure 4-27).

The expected plastic flexural strength of the coupling beams as calculated in Step 3-2 is:

$$M_{p.exp.CB} = 24{,}500 \text{ kip-in.}$$

The coupling beam shear strength corresponding to the expected moment is:

$$V_{n.Mp,exp} = \frac{2(1.2M_{p.exp.CB})}{L_{cb}} \tag{4-5}$$

$$= \frac{2(1.2)(24{,}500 \text{ kip-in.})}{96 \text{ in.}}$$

$$= 613 \text{ kips}$$

The amplification factor discussed in Section 4.2.3.4, and considering $M_{r.CB} = M_{u.CB}$ calculated in Step 2, is:

$$\gamma_1 = \frac{\sum_n 1.2M_{p.exp.CB}}{\sum_n M_{u.CB}} \tag{4-1}$$

$$= \frac{8(1.2)(24{,}500 \text{ kip-in.})}{8(15{,}600 \text{ kip-in.})}$$

$$= 1.88$$

The axial force applied to the walls is:

$$P_r = \sum_n V_{n.Mp,exp}$$
$$= 8(613 \text{ kips})$$
$$= 4{,}900 \text{ kips}$$

The total factored moment in the walls, calculated using the overturning moment from the Given section and the effective length of the wall as calculated in Step 2, is:

$$\text{OTM} = 890{,}000 \text{ kip-in.}$$

$$M_{walls} = \gamma_1 \text{OTM} - P_r L_{eff}$$
$$= 1.88(890{,}000 \text{ kip-in.}) - (4{,}900 \text{ kips})(19.0 \text{ ft})(12 \text{ in./ft})$$
$$= 556{,}000 \text{ kip-in.}$$

A cross-sectional analysis was performed using commercial structural analysis software. The results of this analysis are presented in Figure 4-28. The assumed material behavior for this analysis is shown in Figure 4-29.

$$EI_{T.walls} = 7.70 \times 10^9 \text{ kip-in.}^2$$

$$EI_{C.walls} = 1.81 \times 10^{10} \text{ kip-in.}^2$$

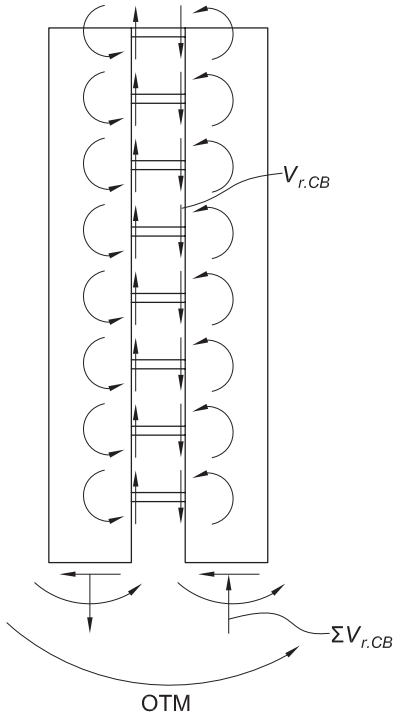


Fig. 4-27. Forces acting within the coupling beams.

The portion of the overturning moment resisted by the individual wall is:

$$M_{UT.wall} = \left(\frac{EI_{T.wall}}{EI_{T.wall} + EI_{C.wall}}\right) M_{walls}$$

$$= \left(\frac{7.70 \times 10^9 \text{ kip-in.}^2}{7.70 \times 10^9 \text{ kip-in.}^2 + 1.81 \times 10^{10} \text{ kip-in.}^2}\right) (556{,}000 \text{ kip-in.})$$

$$= 166{,}000 \text{ kip-in.}$$

$$M_{UC.wall} = \left(\frac{EI_{C.wall}}{EI_{T.wall} + EI_{C.wall}}\right) M_{walls}$$

$$= \left(\frac{1.81 \times 10^{10} \text{ kip-in.}^2}{7.70 \times 10^9 \text{ kip-in.}^2 + 1.81 \times 10^{10} \text{ kip-in.}^2}\right) (556{,}000 \text{ kip-in.})$$

$$= 390{,}000 \text{ kip-in.}$$

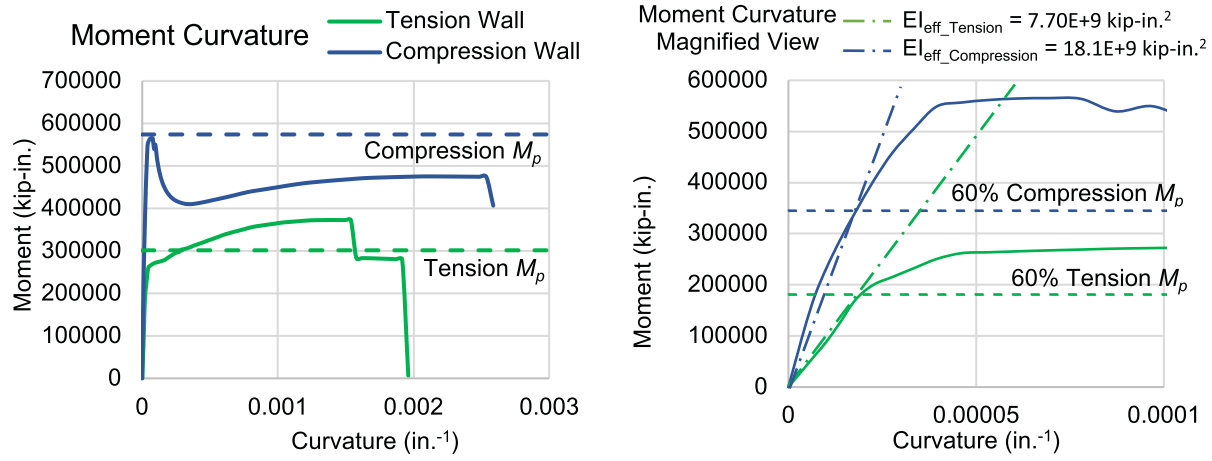


Fig. 4-28. Moment curvature plot from commercial structural analysis software.

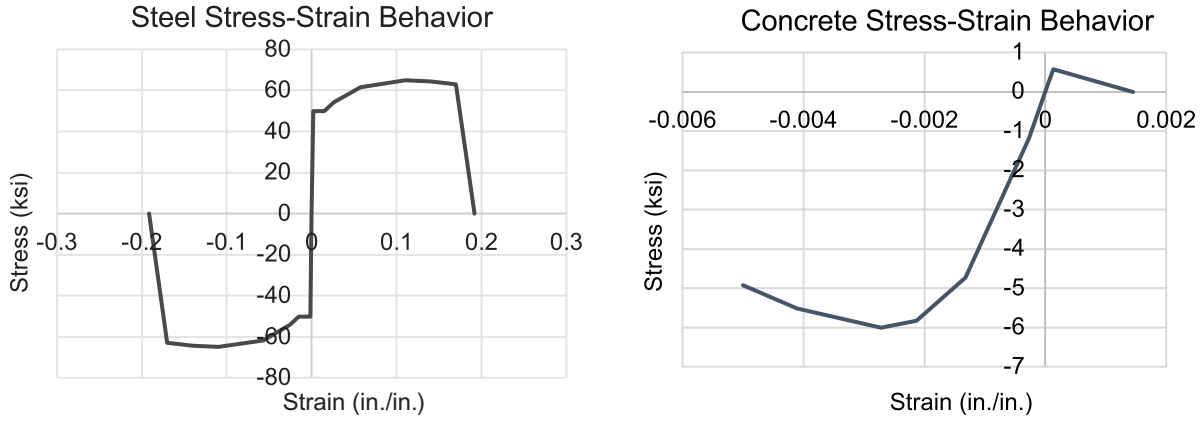


Fig. 4-29. Stress-strain behavior of steel and concrete from commercial structural analysis software.

Step 4-6. Composite wall resistance factor

The resistance factor for the composite wall in shear is given in AISC *Specification* Section I4.4:

$\phi_v = 0.90$

The resistance factor for the composite wall in flexure, from AISC *Specification* Section I3.5, is:

$\phi_b = 0.90$

The resistance factor for the composite wall in compression from Section 2.2.3 of this Design Guide is:

$\phi_c = 0.90$

The resistance factor for the composite wall in tension from Section 2.2.3 of this Design Guide is:

$\phi_t = 0.90$

Step 4-7. Wall tensile strength

The nominal tensile strength is:

$$P_{n.T} = F_y A_s \tag{2-13}$$
$$= (50 \text{ ksi})(174 \text{ in.}^2)$$
$$= 8{,}700 \text{ kips}$$

The design tensile strength is:

$$\phi_t P_{n.T} = 0.90(8{,}700 \text{ kips})$$
$$= 7{,}830 \text{ kips} > P_r = 4{,}900 \text{ kips} \quad \textbf{o.k.}$$

Step 4-8. Wall compression strength

The nominal compressive strength of the SpeedCore wall without consideration of length effects is:

$$P_{no} = F_y A_s + 0.85 f'_c A_c \tag{2-12}$$
$$= 8{,}700 \text{ kips} + 0.85(6 \text{ ksi})(2{,}990 \text{ in.}^2)$$
$$= 23{,}900 \text{ kips}$$

The moment of inertia of the steel in the wall about the minor axis is:

$$I_{s.min} = 2\left[\frac{(L_{wall} - 2t_p)t_p^3}{12} + (L_{wall} - 2t_p)t_p\left(\frac{t_{sc} - t_p}{2}\right)^2\right] + 2\left(\frac{t_p t_{sc}^3}{12}\right)$$
$$= 2\left\{\frac{\left[132 \text{ in.} - 2(9/16 \text{ in.})\right](9/16 \text{ in.})^3}{12} + \left[132 \text{ in.} - 2(9/16 \text{ in.})\right](9/16 \text{ in.})\left(\frac{24 \text{ in.} - 9/16 \text{ in.}}{2}\right)^2\right\} + 2\left[\frac{(9/16 \text{ in.})(24 \text{ in.})^3}{12}\right]$$
$$= 21{,}500 \text{ in.}^4$$

About the minor axis, the moment of inertia of the concrete in the wall is:

$$I_{c.min} = \frac{(L_{wall} - 2t_p)(t_{sc} - 2t_p)^3}{12}$$

$$= \frac{\left[132 \text{ in.} - 2(\frac{9}{16} \text{ in.})\right]\left[24 \text{ in.} - 2(\frac{9}{16} \text{ in.})\right]^3}{12}$$

$$= 131{,}000 \text{ in.}^4$$

The effective flexural stiffness of the wall about the minor axis is:

$$EI_{eff.min} = E_s I_{s.min} + 0.35 E_c I_{c.min} \quad \text{(from Eq. 2-9)}$$

$$= (29{,}000 \text{ ksi})(21{,}500 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(131{,}000 \text{ in.}^4)$$

$$= 8.30 \times 10^8 \text{ kip-in.}^2$$

The critical unsupported height for buckling of the wall is:

$$L_c = \max(h_1, h_{typ})$$

$$= \max(17 \text{ ft}, 14 \text{ ft})$$

$$= 17 \text{ ft}$$

The elastic critical buckling load is:

$$P_e = \frac{\pi^2 EI_{eff.min}}{L_c^2} \quad \text{(from AISC } \textit{Spec.} \text{ Eq. I2-4)}$$

$$= \frac{\pi^2 (8.30 \times 10^8 \text{ kip-in.}^2)}{\left[(17 \text{ ft})(12 \text{ in./ft})\right]^2}$$

$$= 197{,}000 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{23{,}900 \text{ kips}}{197{,}000 \text{ kips}}$$

$$= 0.121$$

Because $P_{no}/P_e < 2.25$, AISC *Specification* Equation I2-2 applies, and the nominal compressive strength is:

$$P_n = P_{no}\left(0.658^{\frac{P_{no}}{P_e}}\right) \quad \text{(AISC } \textit{Spec.} \text{ Eq. I2-2)}$$

$$= (23{,}900 \text{ kips})\left(0.658^{\frac{23{,}900 \text{ kips}}{197{,}000 \text{ kips}}}\right)$$

$$= 22{,}700 \text{ kips}$$

The design compression strength is:

$$\phi_c P_n = 0.90(22{,}700 \text{ kips})$$

$$= 20{,}400 \text{ kips} > P_r = 4{,}900 \text{ kips} \quad \textbf{o.k.}$$

Step 4-9. Wall flexural strength

The flexural strength of the SpeedCore wall is calculated using the plastic stress distribution method with the assumed stress blocks noted in Figure 4-30 and Figure 4-31 for tension and compression walls, respectively. The axial force is assumed to act at the elastic centroid of the walls.

As calculated previously, the axial forces in the walls are:

$P_{r.wall} = 4{,}900$ kips
$T_{r.wall} = 4{,}900$ kips

Flexural Strength of Tension SpeedCore Wall

The location of the plastic neutral axis of the wall is:

$$C_T = \frac{-T_{r.wall} + 2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$

$$= \frac{-4{,}900\,\text{kips} + 2(9/16\text{ in.})(132\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[24\text{ in.} - 2(9/16\text{ in.})\right](9/16\text{ in.})}{4(9/16\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[24\text{ in.} - 2(9/16\text{ in.})\right]}$$

$= 11.3$ in.

The compression force in the flange is:

$$C_{1.T} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[24\text{ in.} - 2(9/16\text{ in.})\right](9/16\text{ in.})(50\text{ ksi})$$
$= 643$ kips

The compression force in the webs is:

$$C_{2.T} = 2t_p C_T F_y$$
$$= 2(9/16\text{ in.})(11.3\text{ in.})(50\text{ ksi})$$
$= 636$ kips

The compression force in the concrete is:

$$C_{3.T} = 0.85 f_c'(t_{sc} - 2t_p)(C_T - t_p)$$
$$= 0.85(6\text{ ksi})\left[24\text{ in.} - 2(9/16\text{ in.})\right](11.3\text{ in.} - 9/16\text{ in.})$$
$= 1{,}250$ kips

The tension force in the flange is:

$$T_{1.T} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[24\text{ in.} - 2(9/16\text{ in.})\right](9/16\text{ in.})(50\text{ ksi})$$
$= 643$ kips

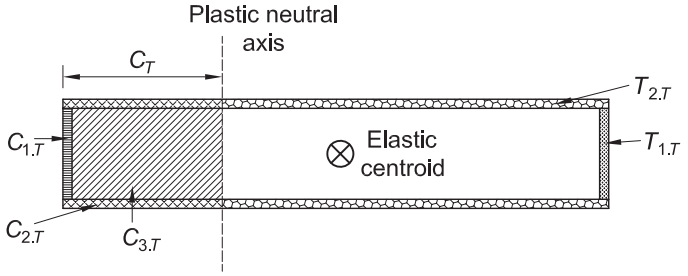


Fig. 4-30. Cross section with labeled regions for plastic moment calculation of tension wall.

The tension force in the web is:

$$T_{2.T} = 2t_p(L_{wall} - C_T)F_y$$
$$= 2(9/16\text{ in.})(132\text{ in.} - 11.3\text{ in.})(50\text{ ksi})$$
$$= 6{,}790\text{ kips}$$

The plastic flexural strength of the composite wall in tension is:

$$M_{PT.wall} = C_{1.T}\left(C_T - \frac{t_p}{2}\right) + C_{2.T}\left(\frac{C_T}{2}\right) + C_{3.T}\left(\frac{C_T - t_p}{2}\right) + T_{1.T}\left(L_{wall} - C_T - \frac{t_p}{2}\right) + T_{2.T}\left(\frac{L_{wall} - C_T}{2}\right) + T_{r.wall}\left(C_T - \frac{L_{wall}}{2}\right)$$
$$= (643\text{ kips})\left(11.3\text{ in.} - \frac{9/16\text{ in.}}{2}\right) + (636\text{ kips})\left(\frac{11.3\text{ in.}}{2}\right) + (1{,}250\text{ kips})\left(\frac{11.3\text{ in.} - 9/16\text{ in.}}{2}\right)$$
$$+ (643\text{ kips})\left(132\text{ in.} - 11.3\text{ in.} - \frac{9/16\text{ in.}}{2}\right) + (6{,}790\text{ kips})\left(\frac{132\text{ in.} - 11.3\text{ in.}}{2}\right)$$
$$+ (4{,}900\text{ kips})\left(11.3\text{ in.} - \frac{132\text{ in.}}{2}\right)$$
$$= 237{,}000\text{ kip-in.}$$

$$M_{nT.wall} = M_{PT.wall} = 237{,}000\text{ kip-in.}$$

Flexural Strength of Compression SpeedCore Wall

The plastic neutral axis location in the compression wall is:

$$C_C = \frac{P_{r.wall} + (2t_p L_{wall} F_y) + 0.85f'_c(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85f'_c(t_{sc} - 2t_p)}$$
$$= \frac{4{,}900\text{ kips} + 2(9/16\text{ in.})(132\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[24\text{ in.} - 2(9/16\text{ in.})\right](9/16\text{ in.})}{4(9/16\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})\left[24\text{ in.} - 2(9/16\text{ in.})\right]}$$
$$= 54.1\text{ in.}$$

The compression force in the flange is:

$$C_{1.C} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[24\text{ in.} - 2(9/16\text{ in.})\right](9/16\text{ in.})(50\text{ ksi})$$
$$= 643\text{ kips}$$

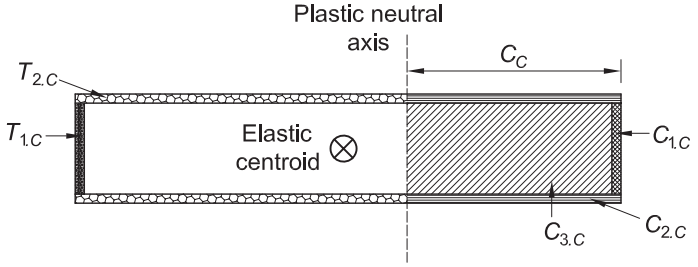


Fig. 4-31. Cross section with labeled regions for plastic moment calculation of compression wall.

The compression force in the webs is:

$$C_{2.C} = 2t_p C_C F_y$$
$$= 2(9/16 \text{ in.})(54.1 \text{ in.})(50 \text{ ksi})$$
$$= 3{,}040 \text{ kips}$$

The compression force in the concrete is:

$$C_{3.C} = 0.85 f_c'(t_{sc} - 2t_p)(C_C - t_p)$$
$$= 0.85(6 \text{ ksi})\left[24 \text{ in.} - 2(9/16 \text{ in.})\right](54.1 \text{ in.} - 9/16 \text{ in.})$$
$$= 6{,}250 \text{ kips}$$

The tension force in the flange is:

$$T_{1.C} = (t_{sc} - 2t_p)t_p F_y$$
$$= \left[24 \text{ in.} - 2(9/16 \text{ in.})\right](9/16 \text{ in.})(50 \text{ ksi})$$
$$= 643 \text{ kips}$$

The tension force in the webs is:

$$T_{2.C} = 2t_p(L_{wall} - C_C)F_y$$
$$= 2(9/16 \text{ in.})(132 \text{ in.} - 54.1 \text{ in.})(50 \text{ ksi})$$
$$= 4{,}380 \text{ kips}$$

The plastic flexural strength of the SpeedCore wall in compression is:

$$M_{PC.wall} = C_{1.C}\left(C_C - \frac{t_p}{2}\right) + C_{2.C}\left(\frac{C_C}{2}\right) + C_{3.C}\left(\frac{C_C - t_p}{2}\right) + T_{1.C}\left(L_{wall} - C_C - \frac{t_p}{2}\right) + T_{2.C}\left(\frac{L_{wall} - C_C}{2}\right) + P_{r.wall}\left(\frac{L_{wall}}{2} - C_C\right)$$
$$= (643 \text{ kips})\left(54.1 \text{ in.} - \frac{9/16 \text{ in.}}{2}\right) + (3{,}040 \text{ kips})\left(\frac{54.1 \text{ in.}}{2}\right) + (6{,}250 \text{ kips})\left(\frac{54.1 \text{ in.} - 9/16 \text{ in.}}{2}\right)$$
$$+ (643 \text{ kips})\left(132 \text{ in.} - 54.1 \text{ in.} - \frac{9/16 \text{ in.}}{2}\right) + (4{,}380 \text{ kips})\left(\frac{132 \text{ in.} - 54.1 \text{ in.}}{2}\right) + (4{,}900 \text{ kips})\left(\frac{132 \text{ in.}}{2} - 54.1 \text{ in.}\right)$$
$$= 563{,}000 \text{ kip-in.}$$

$$M_{nC.wall} = M_{PC.wall} = 563{,}000 \text{ kip-in.}$$

The available flexural strength of the tension wall is:

$$\phi_b M_{nT.wall} = 0.90(237{,}000 \text{ kip-in.})$$
$$= 213{,}000 \text{ kip-in.} > M_{UT.wall} = 166{,}000 \text{ kip-in.} \quad \textbf{o.k.}$$

The available flexural strength of the compression wall is:

$$\phi_b M_{nC.wall} = 0.90(563{,}000 \text{ kip-in.})$$
$$= 507{,}000 \text{ kip-in.} > M_{UC.wall} = 390{,}000 \text{ kip-in.} \quad \textbf{o.k.}$$

Step 4-10. Wall shear strength

The K factors for the shear calculations are as follows:

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200 \text{ ksi})(149 \text{ in.}^2)$$
$$= 1{,}670{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_c)(E_s A_{sw})}{4E_s A_{sw} + E_c A_c} \tag{2-16}$$
$$= \frac{0.7(4{,}500 \text{ ksi})(2{,}990 \text{ in.}^2)(29{,}000 \text{ ksi})(149 \text{ in.}^2)}{4(29{,}000 \text{ ksi})(149 \text{ in.}^2) + (4{,}500 \text{ ksi})(2{,}990 \text{ in.}^2)}$$
$$= 1{,}320{,}000 \text{ kips}$$

The nominal shear strength of the wall is:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \tag{2-14}$$
$$= \frac{1{,}670{,}000 \text{ kips} + 1{,}320{,}000 \text{ kips}}{\sqrt{3(1{,}670{,}000 \text{ kips})^2 + (1{,}320{,}000 \text{ kips})^2}} (50 \text{ ksi})(149 \text{ in.}^2)$$
$$= 7{,}010 \text{ kips}$$

The design shear strength of the uncoupled SpeedCore wall is:

$$\phi_v V_{n.wall} = 0.90(7{,}010 \text{ kips})$$
$$= 6{,}310 \text{ kips} > V_{r.wall} = 1{,}760 \text{ kips} \quad \textbf{o.k.}$$

EXAMPLE 4.2—Seismic Design of 22-Story Structure Using Coupled, C-Shaped SpeedCore Wall

This example details the design of a 22-story office building with typical design loads, floor geometry, and high seismic design loads. The steps followed in this design mirror the design procedure laid out in Section 4.2. For simplicity, this design example does not consider accidental eccentricity and assumes a seismic redundancy factor, ρ, of 1.0.

Given:

Design the coupled, C-shaped SpeedCore walls shown in Figure 4-32 using the given geometry, material properties, and loads. The effects of gravity loads, which can slightly increase the axial compression in the composite walls, have been conservatively ignored. However, design calculations, including the effects of axial compression from gravity loads, are included in the Mathcad file that can be downloaded from the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

Building geometry:

L_f = building length
= 200 ft
W_f = building width
= 120 ft
h_{typ} = typical story height
= 14 ft
h_1 = first story height
= 17 ft
n = number of stories
= 22

Floor loads:

DL = floor dead load
= 0.10 ksf

Seismic design loads in the east-west direction are listed in Table 4-4.

Step 1. General Information of the Considered Building

The steel and concrete material properties are as follows:

ASTM A572/A572M, Grade 50 steel:

E_s = 29,000 ksi
G_s = 11,200 ksi

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi
R_c = 1.3

Weld metal (E70XX):

F_{EXX} = 70 ksi

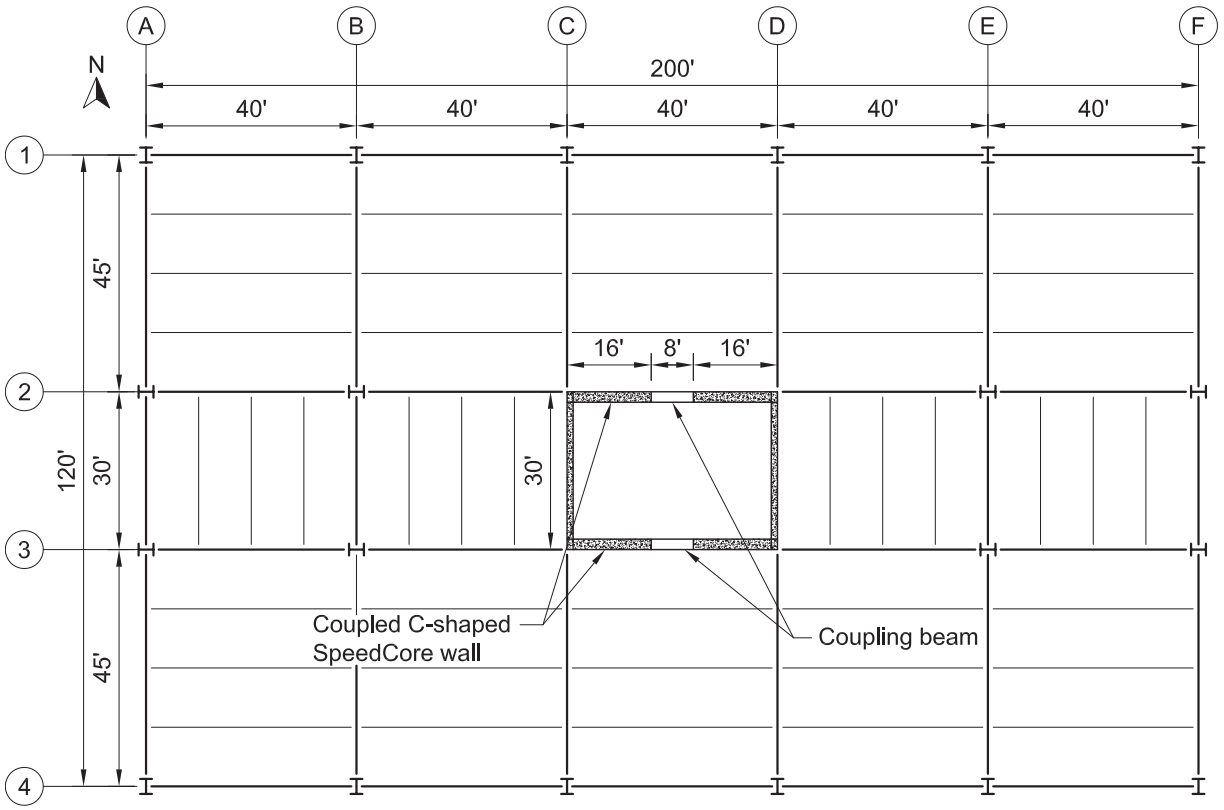


Fig. 4-32. Building floor plan for Example 4.2.

Table 4-4. Seismic Design Loads Calculated from ASCE/SEI 7		
Story No.	**Story Elevation (ft)**	**Story Force (kips)**
1	17	1.52
2	31	4.45
3	45	8.66
4	59	14.1
5	73	20.6
6	87	28.1
7	101	36.7
8	115	46.3
9	129	56.9
10	143	68.4
11	157	80.8
12	171	94.1
13	185	108
14	199	123
15	213	139
16	227	156
17	241	174
18	255	192
19	269	211
20	283	232
21	297	252
22	311	274
Total base shear, V_{base}		2320
Overturning moment, OTM		6,520,000 kip-in.

Seismic design coefficients:

C_d = deflection amplification factor (ASCE/SEI 7, Table 12.2-1)
= 5.5

I_e = importance factor (ASCE/SEI 7, Section 11.5.1, for an office building)
= 1

R = seismic response modification coefficient (from ASCE/SEI 7, Table 12.2-1)
= 8

Risk Category = II (ASCE/SEI 7, Table 1.5-1)

Ω_0 = overstrength factor (from ASCE/SEI 7, Table 12.2-1)
= 2.5

ρ = seismic redundancy factor (ASCE/SEI 7, Section 12.3.4)
= 1

Solution:

From AISC *Manual* Table 2-4 and AISC *Seismic Provisions* Table A3.2, the material properties are as follows:

F_u = 65 ksi
F_y = 50 ksi
R_y = 1.1

SpeedCore wall and coupling beam components are sized based on initial estimates of loads and these sizes are refined through iteration. Acceptable dimensions are presented here so that this example can focus on the appropriate limit states to check.

SpeedCore wall dimensions as shown in Figure 4-33:

L_{wall} = wall web length
= (30 ft)(12 in./ft)
= 360 in.
W_{wall} = wall flange length
= (16 ft)(12 in./ft)
= 192 in.
t_p = wall plate thickness
= ½ in.
$t_{sc.f}$ = wall flange thickness
= 24 in.
$t_{sc.w}$ = wall web thickness
= 14 in.

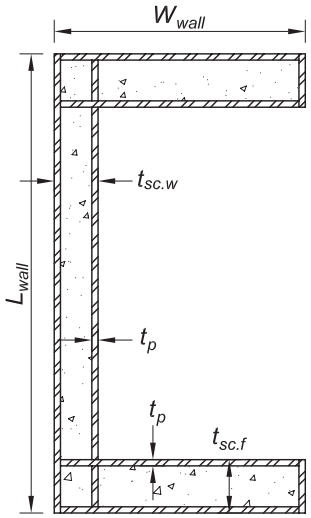


Fig. 4-33. Wall cross-section dimensions.

As shown in Figure 4-34, the coupling beam dimensions are:

L_{cb} = coupling beam length
= (8 ft)(12 in./ft)
= 96.0 in.
b_{CB} = coupling beam width
= 24 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf.CB}$ = coupling beam flange plate thickness
= 9/16 in.
$t_{pw.CB}$ = coupling beam web plate thickness
= ½ in.

Step 2. Analysis for Design

In this step, an elastic computer model is built and analyzed to determine the force on system members and overall system deflection.

C-Shaped SpeedCore Wall Properties

The effective axial, shear, and flexural stiffnesses are calculated for SpeedCore elements for elastic analysis.

The area of steel in the C-shaped wall is:

$$A_s = t_p\left[4W_{wall} + 2L_{wall} + 2t_{sc.f} - (12t_p)\right]$$
$$= (½\text{ in.})\left[4(192\text{ in.}) + 2(360\text{ in.}) + 2(24\text{ in.}) - 12(½\text{ in.})\right]$$
$$= 765\text{ in.}^2$$

The area of concrete in the wall is:

$$A_c = 2W_{wall}t_{sc.f} + (L_{wall} - 2t_{sc.f})t_{sc.w} - A_s$$
$$= 2(192\text{ in.})(24\text{ in.}) + \left[360\text{ in.} - 2(24\text{ in.})\right](14\text{ in.}) - 765\text{ in.}^2$$
$$= 12{,}800\text{ in.}^2$$

The effective axial stiffness of the wall is:

$$EA_{eff} = E_sA_s + 0.45E_cA_c \quad (2\text{-}10)$$
$$= (29{,}000\text{ ksi})(765\text{ in.}^2) + 0.45(4{,}500\text{ ksi})(12{,}800\text{ in.}^2)$$
$$= 4.81\times10^7\text{ kips}$$

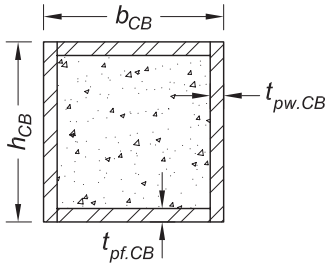


Fig. 4-34. Coupling beam cross-section dimensions.

The area of steel in the direction of shear is:

$$\begin{aligned} A_{sw} &= 4W_{wall}t_p \\ &= 4(192\text{ in.})(½\text{ in.}) \\ &= 384\text{ in.}^2 \end{aligned}$$

The area of concrete in the direction of shear is:

$$\begin{aligned} A_{cw} &= 2(W_{wall} - 3t_p)(t_{sc.f} - 2t_p) \\ &= 2[192\text{ in.} - 3(½\text{ in.})][24\text{ in.} - 2(½\text{ in.})] \\ &= 8{,}760\text{ in.}^2 \end{aligned}$$

The effective shear stiffness of the wall is:

$$GA_{v.eff} = G_sA_{sw} + G_cA_c \tag{2-11}$$

$$\begin{aligned} &= (11{,}200\text{ ksi})(384\text{ in.}^2) + (1{,}800\text{ ksi})(8{,}760\text{ in.}^2) \\ &= 2.01\times10^7\text{ kips} \end{aligned}$$

The center of area in the *y*-direction of the C-shaped SpeedCore wall is:

$$\bar{y} = \frac{\sum Ay}{A_s + \frac{E_c}{E_s}A_c}$$

$$\begin{aligned} \sum Ay &= 4W_{wall}t_p\frac{W_{wall}}{2} + (L_{wall} - 4t_p)t_p\left(t_{sc.w} - \frac{t_p}{2}\right) + (L_{wall} - 4t_p)t_p\left(\frac{t_p}{2}\right) + 2(t_{sc.f} - 2t_p)t_p\left(W_{wall} - \frac{t_p}{2}\right) \\ &\quad + \frac{E_c}{E_s}\left[2(W_{wall} - t_{sc.w} - t_p)(t_{sc.f} - 2t_p)\left(\frac{W_{wall} - t_{sc.w} - t_p}{2} + t_{sc.w}\right) + (L_{wall} - 4t_p)(t_{sc.w} - 2t_p)\left(\frac{t_{sc.w}}{2}\right)\right] \\ &= 4(192\text{ in.})(½\text{ in.})\left(\frac{192\text{ in.}}{2}\right) + [360\text{ in.} - 4(½\text{ in.})](½\text{ in.})\left(14\text{ in.} - \frac{½\text{ in.}}{2}\right) + [360\text{ in.} - 4(½\text{ in.})](½\text{ in.})\left(\frac{½\text{ in.}}{2}\right) \\ &\quad + 2[24\text{ in.} - 2(½\text{ in.})](½\text{ in.})\left(192\text{ in.} - \frac{½\text{ in.}}{2}\right) \\ &\quad + \left(\frac{4{,}500\text{ ksi}}{29{,}000\text{ ksi}}\right)\left\{\begin{array}{l} 2(192\text{ in.} - 14\text{ in.} - ½\text{ in.})[24\text{ in.} - 2(½\text{ in.})]\left(\frac{192\text{ in.} - 14\text{ in.} - ½\text{ in.}}{2} + 14\text{ in.}\right) \\ + [360\text{ in.} - 4(½\text{ in.})][14\text{ in.} - 2(½\text{ in.})]\left(\frac{14\text{ in.}}{2}\right) \end{array}\right\} \\ &= 179{,}000\text{ in.}^3 \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{179{,}000\text{ in.}^3}{765\text{ in.}^2 + \left(\frac{4{,}500\text{ ksi}}{29{,}000\text{ ksi}}\right)(12{,}800\text{ in.}^2)} \\ &= 65.1\text{ in.} \end{aligned}$$

The effective distance between the centroids of the two coupled walls is:

$$L_{eff} = L_{cb} + 2(W_{wall} - \overline{y})$$
$$= 96.0 \text{ in.} + 2(192 \text{ in.} - 65.1 \text{ in.})$$
$$= 350 \text{ in.}$$

The moment of inertia of steel in the wall is:

$$I_s = 4\left[\frac{t_p W_{wall}^3}{12} + (W_{wall}\, t_p)\left(\frac{W_{wall}}{2} - \overline{y}\right)^2\right] + 2\left[\frac{(t_{sc.f} - 2t_p)t_p^3}{12} + (t_{sc.f} - 2t_p)\, t_p \left(W_{wall} - \frac{t_p}{2} - \overline{y}\right)^2\right]$$
$$+ \left[\frac{(L_{wall} - 4t_p)\, t_p^3}{12}\right] + (L_{wall} - 4t_p)\, t_p \left(\overline{y} - \frac{t_p}{2}\right)^2 + \left[\frac{(L_{wall} - 4t_p)\, t_p^3}{12}\right] + (L_{wall} - 4t_p)\, t_p \left[\overline{y} - \left(t_{sc.w} - \frac{t_p}{2}\right)\right]^2$$
$$= 4\left\{\frac{(½ \text{ in.})(192 \text{ in.})^3}{12} + \left[(192 \text{ in.})(½ \text{ in.})\right]\left(\frac{192 \text{ in.}}{2} - 65.1 \text{ in.}\right)^2\right\}$$
$$+ 2\left\{\frac{\left[24 \text{ in.} - 2(½ \text{ in.})\right](½ \text{ in.})^3}{12} + \left[24 \text{ in.} - 2(½ \text{ in.})\right](½ \text{ in.})\left(192 \text{ in.} - \frac{½ \text{ in.}}{2} - 65.1 \text{ in.}\right)^2\right\}$$
$$+ \frac{\left[360 \text{ in.} - 4(½ \text{ in.})\right](½ \text{ in.})^3}{12} + \left[360 \text{ in.} - 4(½ \text{ in.})\right](½ \text{ in.})\left(65.1 \text{ in.} - \frac{½ \text{ in.}}{2}\right)^2$$
$$+ \frac{\left[360 \text{ in.} - 4(½ \text{ in.})\right](½ \text{ in.})^3}{12} + \left[360 \text{ in.} - 4(½ \text{ in.})\right](½ \text{ in.})\left[65.1 \text{ in.} - \left(14.0 \text{ in.} - \frac{½ \text{ in.}}{2}\right)\right]^2$$
$$= 3{,}140{,}000 \text{ in.}^4$$

The moment of inertia of concrete in the wall is:

$$I_c = 2\left[\frac{(t_{sc.f} - 2t_p)(W_{wall} - t_{sc.w} - t_p)^3}{12}\right] + 2(t_{sc.f} - 2t_p)\,(W_{wall} - t_{sc.w} - t_p)\left(\frac{W_{wall} - t_{sc.w} - t_p}{2} + t_{sc.w} - \overline{y}\right)^2$$
$$+ \frac{(L_{wall} - 4t_p)(t_{sc.w} - 2t_p)^3}{12} + (t_{sc.w} - 2t_p)(L_{wall} - 4t_p)\left(\overline{y} - \frac{t_{sc.w}}{2}\right)^2$$
$$= 2\left\{\frac{\left[24 \text{ in.} - 2(½ \text{ in.})\right](192 \text{ in.} - 14 \text{ in.} - ½ \text{ in.})^3}{12}\right\}$$
$$+ 2\left[24 \text{ in.} - 2(½ \text{ in.})\right](192 \text{ in.} - 14 \text{ in.} - ½ \text{ in.})\left(\frac{192 \text{ in.} - 14 \text{ in.} - ½ \text{ in.}}{2} + 14 \text{ in.} - 65.1 \text{ in.}\right)^2$$
$$+ \frac{\left[360 \text{ in.} - 4(½ \text{ in.})\right]\left[14 \text{ in.} - 2(½ \text{ in.})\right]^3}{12} + \left[14 \text{ in.} - 2(½ \text{ in.})\right]\left[360 \text{ in.} - 4(½ \text{ in.})\right]\left(65.1 \text{ in.} - \frac{14 \text{ in.}}{2}\right)^2$$
$$= 4.88 \times 10^7 \text{ in.}^4$$

The effective flexural stiffness of the wall is:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \qquad (2\text{-}9)$$
$$= (29{,}000 \text{ ksi})(3.14 \times 10^6 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(4.88 \times 10^7 \text{ in.}^4)$$
$$= 1.68 \times 10^{11} \text{ kip-in.}^2$$

Coupling Beam Geometric Properties

The effective axial, shear, and flexural stiffnesses are calculated for the coupling beam.

The area of steel in the coupling beam is:

$$\begin{aligned} A_{s.CB} &= 2t_{pw.CB}h_{CB} + 2(b_{CB} - 2t_{pw.CB})t_{pf.CB} \\ &= 2(½\text{ in.})(24\text{ in.}) + 2\left[24\text{ in.} - 2(½\text{ in.})\right](9⁄16\text{ in.}) \\ &= 49.9\text{ in.}^2 \end{aligned}$$

The area of concrete in the coupling beam is:

$$\begin{aligned} A_{c.CB} &= (b_{CB} - 2t_{pw.CB})(h_{CB} - 2t_{pf.CB}) \\ &= \left[24\text{ in.} - 2(½\text{ in.})\right]\left[24\text{ in.} - 2(9⁄16\text{ in.})\right] \\ &= 526\text{ in.}^2 \end{aligned}$$

The axial stiffness of the uncracked section is:

$$\begin{aligned} EA_{uncr.CB} &= E_sA_{s.CB} + E_cA_{c.CB} \\ &= (29{,}000\text{ ksi})(49.9\text{ in.}^2) + (4{,}500\text{ ksi})(526\text{ in.}^2) \\ &= 3{,}810{,}000\text{ kips} \end{aligned}$$

The moment of inertia of steel in the coupling beam is:

$$\begin{aligned} I_{s.CB} &= 2\left[\frac{t_{pw.CB}h_{CB}^3}{12} + \frac{(b_{CB} - 2t_{pw.CB})t_{pf.CB}^3}{12} + (b_{CB} - 2t_{pw.CB})t_{pf.CB}\left(\frac{h_{CB}}{2} - \frac{t_{pf.CB}}{2}\right)^2\right] \\ &= 2\left\{\frac{(½\text{ in.})(24\text{ in.})^3}{12} + \frac{\left[24\text{ in.} - 2(½\text{ in.})\right](9⁄16\text{ in.})^3}{12} + \left[24\text{ in.} - 2(½\text{ in.})\right](9⁄16\text{ in.})\left(\frac{24\text{ in.}}{2} - \frac{9⁄16\text{ in.}}{2}\right)^2\right\} \\ &= 4{,}710\text{ in.}^4 \end{aligned}$$

The moment of inertia of concrete in the coupling beam is:

$$\begin{aligned} I_{c.CB} &= \frac{(b_{CB} - 2t_{pw.CB})(b_{CB} - 2t_{pf.CB})^3}{12} \\ &= \frac{\left[24\text{ in.} - 2(½\text{ in.})\right]\left[24\text{ in.} - 2(9⁄16\text{ in.})\right]^3}{12} \\ &= 22{,}900\text{ in.}^4 \end{aligned}$$

From AISC *Specification* Section I2.2b, Equation I2-13, the effective rigidity coefficient is:

$$\begin{aligned} C_{3.CB} &= 0.45 + 3\left(\frac{A_{s.CB}}{b_{CB}h_{CB}}\right) \le 0.9 \qquad \text{(from Eq. 2-18)} \\ &= 0.45 + 3\left[\frac{49.9\text{ in.}^2}{(24\text{ in.})(24\text{ in.})}\right] \\ &= 0.710 \end{aligned}$$

The effective flexural stiffness of the coupling beam is:

$$EI_{eff.CB} = E_s I_{s.CB} + C_3 E_c I_{c.CB} \quad \text{(from Eq. 2-17)}$$
$$= (29{,}000 \text{ ksi})(4{,}710 \text{ in.}^4) + 0.710(4{,}500 \text{ ksi})(22{,}900 \text{ in.}^4)$$
$$= 2.10 \times 10^8 \text{ kip-in.}^2$$

The area of steel in the direction of shear is:

$$A_{sw.CB} = 2h_{CB} t_{pw.CB}$$
$$= 2(24 \text{ in.})(½ \text{ in.})$$
$$= 24.0 \text{ in.}^2$$

The effective shear stiffness of the coupling beam is:

$$GA_{v.CB} = G_s A_{sw.CB} + G_c A_{c.CB} \quad \text{(from Eq. 2-11)}$$
$$= (11{,}200 \text{ ksi})(24 \text{ in.}^2) + (1{,}800 \text{ ksi})(526 \text{ in.}^2)$$
$$= 1{,}220{,}000 \text{ kips}$$

From Section 2.3.2:

$$0.64EI_{eff.CB} = 0.64(2.10 \times 10^8 \text{ kip-in.}^2)$$
$$= 1.34 \times 10^8 \text{ kip-in.}^2$$

$$0.8EA_{uncr.CB} = 0.8(3.81 \times 10^6 \text{ kips})$$
$$= 3.05 \times 10^6 \text{ kips}$$

Numerical Model

To determine the ID ratio and shear demand on coupling beams, an analysis model was built using commercial software. First-order, linear elastic analysis was performed on a model consisting of frame elements for only the SpeedCore walls and composite coupling beams. This model was subjected to the earthquake loading previously defined. It employed rigid offsets at the end of the coupling beams to account for wall length. All mass was applied at the story level and distributed to the two joints. A pictorial representation of the model is shown in Figure 4-35. To model the structure as a 2D frame, the area and moment of inertia properties assigned to the coupling beams are doubled to simulate the two coupling beams as depicted in Figure 4-36.

The frame elements were assigned section properties calculated previously.

From this analysis model, the required coupling beam shear strengths and story displacements are determined and shown in Tables 4-5 and 4-6. This model also confirms the base shear and overturning moment values presented in the Given section.

Lateral displacement values are amplified by the C_d factor to obtain the amplified displacement. These amplified values are then used to calculate the ID, which is the difference in displacement between two floors divided by the story height. The maximum design ID is limited according to ASCE/SEI 7, Section 12.12.1. In this case, the maximum design ID limit is 2% for Risk Category II. The story displacement, amplified displacement, and ID are presented in Table 4-6. This distribution corresponds to the deflected shape shown in Figure 4-37.

The maximum ID ratio = 1.95%, which is less than 2%.

Required Shear and Flexural Strength

The required coupling beam shear strength, determined from analysis, considers two coupling beams. Demand on individual coupling beams is found by dividing the shear by 2.

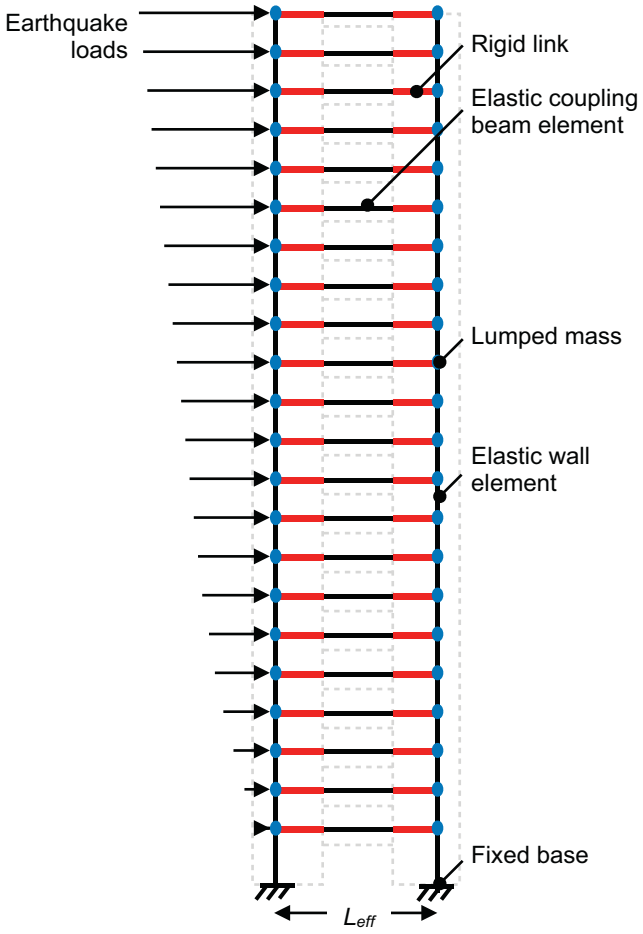


Fig. 4-35. Pictorial representation of analysis model.

Table 4-5. Coupling Beam Required Shear Strengths		
Story No.	**Story Elevation (ft)**	**Required Coupling Beam Shear Strength (kips)**
1	17	438
2	31	648
3	45	777
4	59	855
5	73	899
6	87	920
7	101	925
8	115	918
9	129	902
10	143	878
11	157	847
12	171	810
13	185	766
14	199	716
15	213	661
16	227	599
17	241	533
18	255	464
19	269	394
20	283	327
21	297	269
22	311	232
Average		672

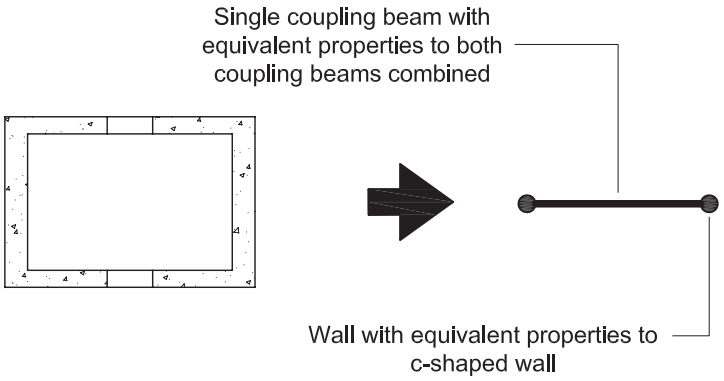


Fig. 4-36. Analysis model equivalencies.

Table 4-6. Lateral Displacement and ID Summary				
Story No.	**Story Height (ft)**	**Displacement (in.)**	**Amplified Displacement (in.)**	**ID (%)**
22	14	10.6	58.3	1.82
21	14	10.1	55.3	1.84
20	14	9.48	52.2	1.87
19	14	8.91	49.0	1.90
18	14	8.33	45.8	1.93
17	14	7.74	42.6	1.94
16	14	7.15	39.3	1.95
15	14	6.55	36.0	1.95
14	14	5.96	32.8	1.94
13	14	5.37	29.5	1.91
12	14	4.78	26.3	1.87
11	14	4.21	23.2	1.82
10	14	3.65	20.1	1.76
9	14	3.12	17.1	1.69
8	14	2.60	14.3	1.59
7	14	2.11	11.6	1.49
6	14	1.66	9.13	1.36
5	14	1.24	6.84	1.22
4	14	0.87	4.79	1.05
3	14	0.55	3.02	0.85
2	14	0.29	1.59	0.61
1	14	0.10	0.56	0.34
Base	17	0.00	0.00	0.00

The required coupling beam shear strength is calculated as the average of the coupling beam shear demand at each story from analysis. Shear demand differs along the height of the structure due to the applied seismic force distribution using the ASCE/SEI 7 equivalent lateral force procedure, and due to the nature of linear elastic modeling. Alternatively, the designer can choose to use the maximum shear demand. Because a portion of the overturning moment will be resisted by the coupling action and the remainder by the individual walls, the result of this choice is the relative proportioning of wall and coupling beam elements. Because the system is designed to ensure that plasticity spreads along the height of the structure, either method is acceptable.

$$V_{r.CB} = \frac{672 \text{ kips}}{2}$$
$$= 336 \text{ kips}$$

The coupling beam required flexural strength is:

$$M_{r.CB} = \frac{V_{r.CB} L_{cb}}{2}$$
$$= \frac{(336 \text{ kips})(96 \text{ in.})}{2}$$
$$= 16{,}100 \text{ kip-in.}$$

The base shear is amplified by a shear amplification factor of 4.0, following the recommendations of Section 4.3.3.1.

$$V_{amp} = 4.0 V_{base} \quad (4\text{-}6)$$
$$= 4.0(2{,}320 \text{ kips})$$
$$= 9{,}280 \text{ kips}$$

$$V_{r.wall} = \frac{V_{amp}}{2}$$
$$= \frac{9{,}280 \text{ kips}}{2}$$
$$= 4{,}640 \text{ kips}$$

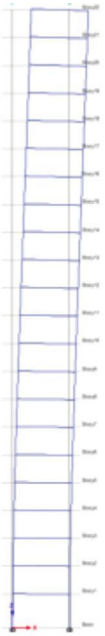


Fig. 4-37. Deflected shape.

Because the required flexural strength of the SpeedCore wall is determined using a capacity-based seismic load effect, these calculations are presented later.

Step 3. Design of Coupling Beams

Step 3-1. Flexure-critical coupling beams

Composite coupling beams are proportioned to be flexure-critical with design shear strength calculated from:

$$V_{n.exp} \geq \frac{2.4M_{p.exp.CB}}{L_{cb}} \qquad \text{(from Eq. 4-4)}$$

Because this equation is based on the strength of the coupling beam, this check will be completed at the end of the section.

Step 3-2. Expected flexural strength

Figure 4-38 illustrates the plastic neutral axis location and the compression and tension regions in the coupling beam.

The width of concrete in the coupling beam is:

$$\begin{aligned} t_{c.CB} &= b_{CB} - 2t_{p.w.CB} \\ &= 24 \text{ in.} - 2(\tfrac{1}{2} \text{ in.}) \\ &= 23.0 \text{ in.} \end{aligned}$$

The expected plastic neutral axis location of the coupling beam is:

$$\begin{aligned} C_{CB.exp} &= \frac{2t_{pw.CB}h_{CB}R_yF_y + 0.85R_cf_c't_{c.CB}t_{pf.CB}}{4t_{pw.CB}R_yF_y + 0.85R_cf_c't_{c.CB}} \\ &= \frac{2(\tfrac{1}{2} \text{ in.})(24 \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})(23.0 \text{ in.})(\tfrac{9}{16} \text{ in.})}{4(\tfrac{1}{2} \text{ in.})(1.1)(50 \text{ ksi}) + 0.85(1.3)(6 \text{ ksi})(23.0 \text{ in.})} \\ &= 5.36 \text{ in.} \end{aligned}$$

The compression force in the flange is:

$$\begin{aligned} C_{1.exp} &= (b_{CB} - 2t_{pw.CB})t_{pf.CB}R_yF_y \\ &= \left[24 \text{ in.} - 2(\tfrac{1}{2} \text{ in.})\right](\tfrac{9}{16} \text{ in.})(1.1)(50 \text{ ksi}) \\ &= 712 \text{ kips} \end{aligned}$$

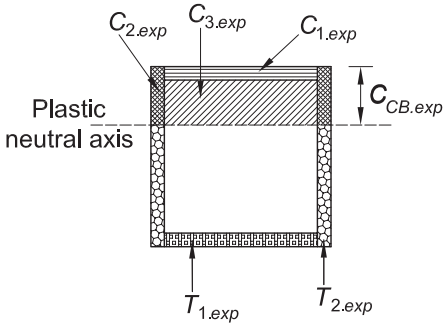


Fig. 4-38. Cross section with labeled regions for plastic moment calculation of coupling beam.

The compression force in each web is:

$$C_{2.exp} = 2t_{pw.CB}C_{CB.exp}R_yF_y$$
$$= 2(½\text{ in.})(5.36\text{ in.})(1.1)(50\text{ ksi})$$
$$= 295\text{ kips}$$

The compression force in the concrete is:

$$C_{3.exp} = 0.85R_cf_c't_{c.CB}(C_{CB.exp} - t_{pf.CB})$$
$$= 0.85(1.3)(6\text{ ksi})(23.0\text{ in.})(5.36\text{ in.} - 9/16\text{ in.})$$
$$= 732\text{ kips}$$

The tension force in the flange is:

$$T_{1.exp} = (b_{CB} - 2t_{pw.CB})t_{pf.CB}R_yF_y$$
$$= \left[24\text{ in.} - 2(½\text{ in.})\right](9/16\text{ in.})(1.1)(50\text{ ksi})$$
$$= 712\text{ kips}$$

The tension force in the web is:

$$T_{2.exp} = 2t_{pw.CB}(h_{CB} - C_{CB.exp})R_yF_y$$
$$= 2(½\text{ in.})(24.0\text{ in.} - 5.36\text{ in.})(1.1)(50\text{ ksi})$$
$$= 1{,}030\text{ kips}$$

The expected plastic flexural strength of the coupling beam is:

$$M_{p.exp.CB} = C_{1.exp}\left(C_{CB.exp} - \frac{t_{pf.CB}}{2}\right) + C_{2.exp}\left(\frac{C_{CB.exp}}{2}\right) + C_{3.exp}\left(\frac{C_{CB.exp} - t_{pf.CB}}{2}\right)$$
$$+ T_{1.exp}\left(h_{CB} - C_{CB.exp} - \frac{t_{pf.CB}}{2}\right) + T_{2.exp}\left(\frac{h_{CB} - C_{CB.exp}}{2}\right)$$
$$= (712\text{ kips})\left(5.36\text{ in.} - \frac{9/16\text{ in.}}{2}\right) + (295\text{ kips})\left(\frac{5.36\text{ in.}}{2}\right) + (732\text{ kips})\left(\frac{5.36\text{ in.} - 9/16\text{ in.}}{2}\right)$$
$$+ (712\text{ kips})\left(24\text{ in.} - 5.36\text{ in.} - \frac{9/16\text{ in.}}{2}\right) + (1{,}030\text{ kips})\left(\frac{24\text{ in.} - 5.36\text{ in.}}{2}\right)$$
$$= 28{,}800\text{ kip-in.}$$

Step 3-3. Minimum area of steel

The minimum area of steel required according to AISC *Specification* Section I2.2a is:

$$A_{s.CB.min} = 0.01b_{CB}h_{CB}$$
$$= 0.01(24\text{ in.})(24\text{ in.})$$
$$= 5.76\text{ in.}^2$$

$A_{s.CB} = 49.9\text{ in.}^2 > A_{s.CB.min} = 5.76\text{ in.}^2$ **o.k.**

Step 3-4. Steel plate slenderness requirement for coupling beams

The clear width of the coupling beam flange plate is:

$$b_{c.CB} = b_{CB} - 2t_{pw.CB}$$
$$= 24 \text{ in.} - 2(½ \text{ in.})$$
$$= 23.0 \text{ in.}$$

The clear width of the coupling beam web plate is:

$$h_{c.CB} = h_{CB} - 2t_{pf.CB}$$
$$= 24 \text{ in.} - 2(9/16 \text{ in.})$$
$$= 22.9 \text{ in.}$$

Slenderness check for coupling beam flange plates

According to AISC *Seismic Provisions* Section H8.5b, the maximum slenderness ratio for the coupling beam flanges is:

$$2.37\sqrt{\frac{E_s}{R_y F_y}} = 2.37\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}}$$
$$= 54.4$$

The width-to-thickness ratio of the coupling beam flange is:

$$\frac{b_{c.CB}}{t_{pf.CB}} = \frac{23.0 \text{ in.}}{9/16 \text{ in.}}$$
$$= 40.9$$

$$\frac{b_{c.CB}}{t_{pf.CB}} = 40.9 < 2.37\sqrt{\frac{E_s}{R_y F_y}} = 54.4 \quad \textbf{o.k.}$$

Slenderness Check for Coupling Beam Web Plates

According to AISC *Seismic Provisions* Section H8.5b, the maximum slenderness ratio for coupling beam webs is:

$$2.66\sqrt{\frac{E_s}{R_y F_y}} = 2.66\sqrt{\frac{29{,}000 \text{ ksi}}{1.1(50 \text{ ksi})}}$$
$$= 61.1$$

The width-to-thickness ratio of the coupling beam web is:

$$\frac{h_{c.CB}}{t_{pw.CB}} = \frac{22.9 \text{ in.}}{½ \text{ in.}}$$
$$= 45.8$$

$$\frac{h_{c.CB}}{t_{pw.CB}} = 45.8 < 2.66\sqrt{\frac{E_s}{R_y F_y}} = 61.1 \quad \textbf{o.k.}$$

The slenderness ratio requirements for the steel plates are met, and therefore the coupling beam section is considered compact.

Step 3-5. Flexural strength

The plastic neutral axis location in the coupling beam is:

$$C_{CB} = \frac{2t_{pw.CB}h_{CB}F_y + 0.85f'_c t_{c.CB}t_{pf.CB}}{4t_{pw.CB}F_y + 0.85f'_c t_{c.CB}}$$

$$= \frac{2(½\text{ in.})(24\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})(23.0\text{ in.})(9/16\text{ in.})}{4(½\text{ in.})(50\text{ ksi}) + 0.85(6\text{ ksi})(23.0\text{ in.})}$$

$$= 5.83\text{ in.}$$

Note: The plastic neutral axis location, C_{CB}; compression forces, C_1, C_2, C_3; and tension forces, T_1, T_2, for a rectangular coupling beam are depicted in Appendix A, Figure A-13.

The compression force in the flange is:

$$C_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$

$$= \left[24\text{ in.} - 2(½\text{ in.})\right](9/16\text{ in.})(50\text{ ksi})$$

$$= 647\text{ kips}$$

The compression force in the webs is:

$$C_2 = 2t_{pw.CB}C_{CB}F_y$$

$$= 2(½\text{ in.})(5.83\text{ in.})(50\text{ ksi})$$

$$= 292\text{ kips}$$

The compression force in the concrete is:

$$C_3 = 0.85f'_c t_{c.CB}(C_{CB} - t_{pf.CB})$$

$$= 0.85(6\text{ ksi})(23.0\text{ in.})(5.83\text{ in.} - 9/16\text{ in.})$$

$$= 618\text{ kips}$$

The tension force in the flange is:

$$T_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$

$$= \left[24\text{ in.} - 2(½\text{ in.})\right](9/16\text{ in.})(50\text{ ksi})$$

$$= 647\text{ kips}$$

The tension force in the webs is:

$$T_2 = 2t_{pw.CB}(h_{CB} - C_{CB})F_y$$

$$= 2(½\text{ in.})(24\text{ in.} - 5.83\text{ in.})(50\text{ ksi})$$

$$= 909\text{ kips}$$

The plastic flexural strength of the coupling beam is:

$$M_{pn.CB} = C_1\left(C_{CB} - \frac{t_{pf.CB}}{2}\right) + C_2\left(\frac{C_{CB}}{2}\right) + C_3\left(\frac{C_{CB} - t_{pf.CB}}{2}\right) + T_1\left(h_{CB} - C_{CB} - \frac{t_{pf.CB}}{2}\right) + T_2\left(\frac{h_{CB} - C_{CB}}{2}\right)$$

$$= (647 \text{ kips})\left(5.83 \text{ in.} - \frac{9/16 \text{ in.}}{2}\right) + (292 \text{ kips})\left(\frac{5.83 \text{ in.}}{2}\right) + (618 \text{ kips})\left(\frac{5.83 \text{ in.} - 9/16 \text{ in.}}{2}\right)$$

$$+ (647 \text{ kips})\left(24 \text{ in.} - 5.83 \text{ in.} - \frac{9/16 \text{ in.}}{2}\right) + (909 \text{ kips})\left(\frac{24 \text{ in.} - 5.83 \text{ in.}}{2}\right)$$

$$= 25{,}900 \text{ kip-in.}$$

The design flexural strength according to AISC *Specification* Section I3.4b is:

$\phi_b = 0.90$

$M_{n.CB} = M_{Pn.CB}$

$\phi_b M_{n.CB} = 0.90(25{,}900 \text{ kip-in.})$

$= 23{,}300 \text{ kip-in.} > M_{r.CB} = 16{,}100 \text{ kip-in.}$ **o.k.**

Step 3-6. Nominal shear strength

The area of the steel web in the direction of shear for the coupling beam is:

$A_{sw.CB} = 2h_{CB}t_{pw.CB}$

$= 2(24 \text{ in.})(½ \text{ in.})$

$= 24.0 \text{ in.}^2$

According to Equation 2-19 and AISC *Specification* Section I4.2, the nominal shear strength of the coupling beam is:

$$V_{n.CB} = 0.6F_y A_{sw.CB} + 0.06K_c A_{c.CB}\sqrt{f'_c} \quad \text{(from Eq. 2-19)}$$

$$= 0.6(50 \text{ ksi})(24.0 \text{ in.}^2) + 0.06(1)(526 \text{ in.}^2)\sqrt{6 \text{ ksi}}$$

$$= 797 \text{ kips}$$

The design shear strength is:

$\phi_v = 0.90$

$\phi_v V_{n.CB} = 0.90(797 \text{ kips})$

$= 717 \text{ kips} > V_{r.CB} = 336 \text{ kips}$ **o.k.**

Step 3-7. Flexure-critical coupling beams (revisited)

The expected shear strength is calculated using Equation 2-19, using the expected material strengths:

$$V_{n.exp.CB} = 0.6R_y F_y A_{sw.CB} + 0.06A_{c.CB}\sqrt{R_c f'_c} \quad \text{(from Eq. 2-19)}$$

$$= 0.6(1.1)(50 \text{ ksi})(24.0 \text{ in.}^2) + 0.06(526 \text{ in.}^2)\sqrt{1.3(6 \text{ ksi})}$$

$$= 880 \text{ kips}$$

The required shear strength of the coupling beam is determined according to AISC *Seismic Provisions* Section H8.5c:

$$\frac{2.4M_{p.exp.CB}}{L_{cb}} = \frac{2.4(28{,}800 \text{ kip-in.})}{96 \text{ in.}} \quad \text{(from Eq. 4-5)}$$
$$= 720 \text{ kips}$$

$V_{n.exp.CB} = 880 \text{ kips} > 720 \text{ kips}$ **o.k.**

Step 4. Design of Composite Walls

Step 4-1. Minimum and maximum area of steel

The gross area of the C-shaped SpeedCore wall is:

$$A_g = L_{wall}t_{sc.w} + 2(W_{wall} - t_{sc.w})t_{sc.f}$$
$$= (360 \text{ in.})(14 \text{ in.}) + 2(192 \text{ in.} - 14 \text{ in.})(24 \text{ in.})$$
$$= 13{,}600 \text{ in.}^2$$

The minimum area of steel required according to AISC *Seismic Provisions* Section H8.4a is:

$$A_{s.min} = 0.01A_g$$
$$= 0.01(13{,}600 \text{ in.}^2)$$
$$= 136 \text{ in.}^2$$

$A_s = 765 \text{ in.}^2 > A_{s.min} = 136 \text{ in.}^2$ **o.k.**

The maximum area of steel required according to AISC *Seismic Provisions* Section H8.4a is:

$$A_{s.max} = 0.1A_g$$
$$= 0.1(13{,}600 \text{ in.}^2)$$
$$= 1{,}360 \text{ in.}^2$$

$A_s = 765 \text{ in.}^2 < A_{s.max} = 1{,}360 \text{ in.}^2$ **o.k.**

Step 4-2. Steel plate slenderness requirements for composite walls

The largest unsupported length between rows of steel anchors or tie bars for this example is selected under the assumption that all support is through tie bars (no steel anchors). Therefore, the anchor and tie bar spacing are different. As discussed in Section 3.2.4.1, tie bars and steel anchors can be spaced farther apart outside of flexural yielding zones. Calculating the extent of the flexural yielding zone is discussed in Example 3.1 and 3.2 and omitted here for simplicity.

A tie bar spacing of 12 in. is selected for all stories.

$S_{tie} = 12 \text{ in.}$

$S_{tie.top} = 12 \text{ in.}$

From Section 3.2.4.1, the steel plate slenderness ratio, b/t_p, is limited to:

$$\frac{b}{t_p} \le 1.05\sqrt{\frac{E_s}{R_yF_y}} \quad (3\text{-}1)$$

where b is the largest unsupported length of the plate between rows of steel anchors or tie bars, which in this case is S_{tie}.

$$\frac{b}{t_p} = \frac{12\text{ in.}}{½\text{ in.}}$$
$$= 24.0$$

$$1.05\sqrt{\frac{E_s}{R_yF_y}} = 1.05\sqrt{\frac{29{,}000\text{ ksi}}{1.1(50\text{ ksi})}}$$
$$= 24.1$$

$$\frac{b}{t_p} = 24.0 < 1.05\sqrt{\frac{E_s}{R_yF_y}} = 24.1 \quad \textbf{o.k.}$$

From Section 2.2.1.1, the steel plate slenderness ratio, b/t_p, in the remainder of the stories is limited to:

$$\frac{b}{t_p} \le 1.2\sqrt{\frac{E_s}{F_y}} \tag{3-2}$$

where b is the largest unsupported length of the plate between rows of steel anchors or tie bars, which in this case is $S_{tie.top}$.

$$\frac{b}{t_p} = \frac{S_{tie.top}}{t_p}$$
$$= \frac{12\text{ in.}}{½\text{ in.}}$$
$$= 24.0$$

$$1.2\sqrt{\frac{E_s}{F_y}} = 1.2\sqrt{\frac{29{,}000\text{ ksi}}{50\text{ ksi}}}$$
$$= 28.9$$

$$\frac{b}{t_p} = 24.0 < 1.2\sqrt{\frac{E_s}{F_y}} = 28.9 \quad \textbf{o.k.}$$

The slenderness ratio requirements for the steel plates are met.

Step 4-3. Tie bar spacing requirements for composite walls

A tie bar diameter of 1 in. is selected for the SpeedCore wall, and the bar requirement considering the stability of the empty steel module is checked as follows, according to AISC *Seismic Provisions* Section H8.4c:

$S_{tie} = 12$ in.

$d_{tie} = 1$ in.

$$\alpha = 1.7\left(\frac{t_{sc}}{t_p} - 2\right)\left(\frac{t_p}{d_{tie}}\right)^4 \tag{2-5}$$
$$= 1.7\left(\frac{24\text{ in.}}{½\text{ in.}} - 2\right)\left(\frac{½\text{ in.}}{1\text{ in.}}\right)^4$$
$$= 4.89$$

$$\frac{S_{tie}}{t_p} = \frac{12\text{ in.}}{½\text{ in.}}$$
$$= 24.0$$

The slenderness limit is given in AISC *Seismic Provisions* Equation H8-2:

$$1.00\sqrt{\frac{E_s}{2\alpha+1}} = 1.00\sqrt{\frac{29{,}000 \text{ ksi}}{2(4.89)+1}} \quad \text{(from Eq. 2-4)}$$
$$= 51.9$$

$$\frac{S_{tie}}{t_p} = 24.0 < 1.00\sqrt{\frac{E_s}{2\alpha+1}} = 51.9 \quad \textbf{o.k.}$$

Step 4-4. Required wall shear strength

The amplified shear force to be resisted by the wall, as calculated in Step 2, is:

$V_{r.wall} = 4{,}640$ kips

Step 4-5. Required wall flexural strength

The required wall flexural strength is determined based on the capacity-limited forces in the coupling beam elements. Walls are sized to resist the axial load associated with the formation of plastic hinges in every coupling beam and the flexural load associated with the amplified overturning moment, less the contribution from the axial force couple in the adjacent walls.

The expected plastic flexural strength of the coupling beams, as calculated in Step 3-2, is:

$M_{p.exp.CB} = 28{,}800$ kip-in.

The coupling beam shear corresponding to the expected flexural strength is calculated from Equation 4-5:

$$V_{n.Mp.exp} = \frac{2(1.2M_{p.exp.CB})}{L_{cb}} \quad \text{(from Eq. 4-5)}$$
$$= \frac{2(1.2)(28{,}800 \text{ kip-in.})}{96 \text{ in.}}$$
$$= 720 \text{ kips}$$

The overstrength factor, as discussed in Section 4.2.3.4 of this Design Guide, is calculated as follows using Equation 4-1, with $M_{r.CB}$ calculated in Step 2:

$$\gamma_1 = \frac{\sum_n 1.2M_{p.exp.CB}}{\sum_n M_{r.CB}} \quad (4\text{-}1)$$
$$= \frac{22(1.2)(28{,}800 \text{ kip-in.})}{22(16{,}100 \text{ kip-in.})}$$
$$= 2.15$$

The walls are subjected to the following axial force:

$$P_{r.wall} = 2\sum_n V_{n.Mp.exp}$$
$$= 2(22)(720 \text{ kips})$$
$$= 31{,}700 \text{ kips}$$

The overturning moment in the walls is:

$$\text{OTM} = 6{,}520{,}000 \text{ kip-in.}$$

$$\begin{aligned} M_{wall} &= \gamma_1 \text{OTM} - P_{r.wall} L_{eff} \\ &= 2.15(6{,}520{,}000 \text{ kip-in.}) - (31{,}700 \text{ kips})(350 \text{ in.}) \\ &= 2{,}920{,}000 \text{ kip-in.} \end{aligned}$$

Cross-section analysis was performed using commercial structural analysis software. The results of this analysis are presented in Figure 4-39. The secant stiffness method used herein often yields effective stiffness values for the tension and compression walls close to E_sI_s and $E_sI_s + 0.7E_cI_c$, respectively. The agreement between these values depends on the axial load level. The assumed material behavior for this analysis is shown in Figure 4-40.

$$EI_{T.walls} = 8.67 \times 10^{10} \text{ kip-in.}^2$$

$$EI_{C.walls} = 2.06 \times 10^{11} \text{ kip-in.}^2$$

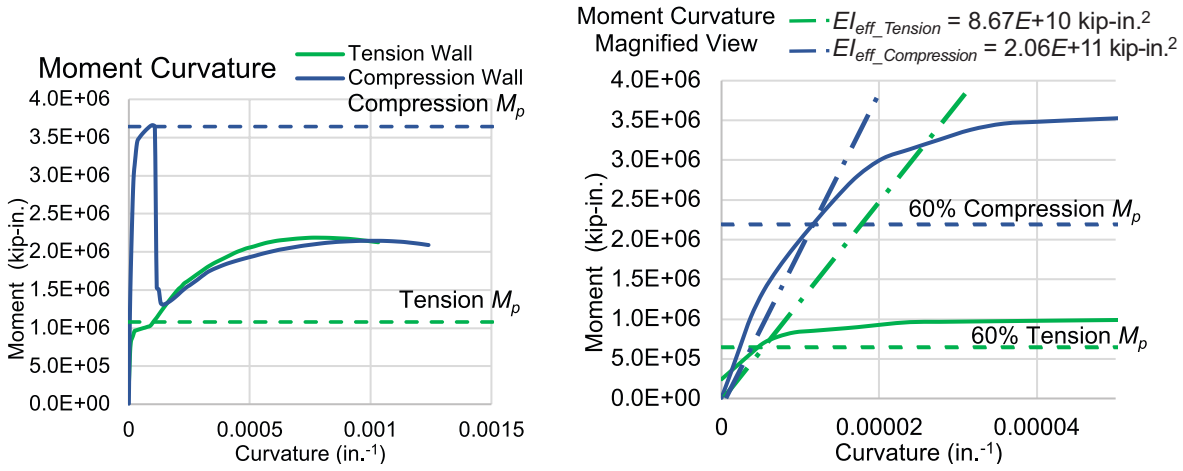


Fig. 4-39. Moment curvature plot from computer model.

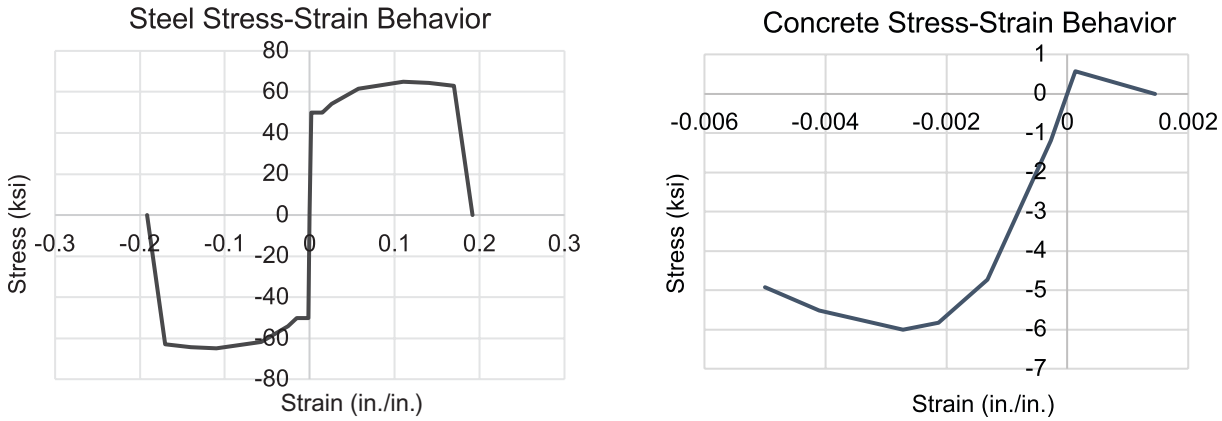


Fig. 4-40. Stress strain behavior of steel and concrete from computer model.

The portion of the OTM resisted by the individual walls is:

$$M_{UT.wall} = \left(\frac{EI_{T.wall}}{EI_{T.wall} + EI_{C.wall}}\right) M_{wall}$$

$$= \left(\frac{8.67 \times 10^{10} \text{ kip-in.}^2}{8.67 \times 10^{10} \text{ kip-in.}^2 + 2.06 \times 10^{11} \text{ kip-in.}^2}\right) (2{,}920{,}000 \text{ kip-in.})$$

$$= 865{,}000 \text{ kip-in.}$$

$$M_{UC.wall} = \left(\frac{EI_{C.wall}}{EI_{T.wall} + EI_{C.wall}}\right) M_{wall}$$

$$= \left(\frac{2.06 \times 10^{11} \text{ kip-in.}^2}{8.67 \times 10^{10} \text{ kip-in.}^2 + 2.06 \times 10^{11} \text{ kip-in.}^2}\right) (2{,}920{,}000 \text{ kip-in.})$$

$$= 2{,}060{,}000 \text{ kip-in.}$$

Step 4-6. Composite wall resistance factor

The resistance factor for the composite wall in shear is given in AISC *Specification* Section I4.4:

$\phi_v = 0.90$

The resistance factor for the composite wall in flexure is given in AISC *Specification* Section I3.5:

$\phi_b = 0.90$

The resistance factor for a composite wall in compression is from Section 2.2.3 of this Design Guide:

$\phi_c = 0.90$

From Section 2.2.3, the resistance factor for a composite wall in tension is:

$\phi_t = 0.90$

Step 4-7. Wall tensile strength

Equation 2-13 is used to calculate the nominal tensile strength:

$$P_{n.T} = F_y A_s \qquad \text{(from Eq. 2-13)}$$

$$= (50 \text{ ksi})(765 \text{ in.}^2)$$

$$= 38{,}300 \text{ kips}$$

The design tensile strength is:

$$\phi_t P_{n.T} = 0.90(38{,}300 \text{ kips})$$

$$= 34{,}500 \text{ kips} > T_{r.wall} = 31{,}700 \text{ kips} \quad \textbf{o.k.}$$

Step 4-8. Wall compression strength

The compression strength of the SpeedCore wall is given by Equation 2-12:

$$P_{no} = F_y A_s + 0.85 f'_c A_c \qquad (2\text{-}12)$$

$$= (50 \text{ ksi})(765 \text{ in.}^2) + 0.85(6 \text{ ksi})(12{,}800 \text{ in.}^2)$$

$$= 104{,}000 \text{ kips}$$

The moment of inertia of steel about the minor axis as calculated in Step 2 is:

$I_{s.min} = 3{,}140{,}000 \text{ in.}^4$

The moment of inertia of concrete about the minor axis as calculated in Step 2 is:

$$I_{c.min} = 4.88 \times 10^7 \text{ in.}^4$$

The effective stiffness of the wall about the minor axis:

$$EI_{eff.min} = E_s I_{s.min} + 0.35 E_c I_{c.min} \quad \text{(from Eq. 2-9)}$$
$$= (29{,}000 \text{ ksi})(3.14 \times 10^6 \text{ in.}^4) + 0.35(4{,}500 \text{ ksi})(4.88 \times 10^7 \text{ in.}^4)$$
$$= 1.68 \times 10^{11} \text{ kip-in.}^2$$

The critical unsupported length for buckling of the wall:

$$L_c = \max(h_1, h_{typ})$$
$$= \max(17 \text{ ft}, 14 \text{ ft})$$
$$= 17 \text{ ft}$$

The elastic critical buckling load from AISC *Specification* Equation I2-5 is:

$$P_e = \frac{\pi^2 EI_{eff.min}}{L_c^2}$$
$$= \frac{\pi^2 (1.68 \times 10^{11} \text{ kip-in.}^2)}{\left[(17 \text{ ft})(12 \text{ in./ft})\right]^2}$$
$$= 3.98 \times 10^7 \text{ kips}$$

$$\frac{P_{no}}{P_e} = \frac{104{,}000 \text{ kips}}{3.98 \times 10^7 \text{ kips}}$$
$$= 2.61 \times 10^{-3} < 2.25$$

The Euler buckling load is calculated from AISC *Specification* Equation I2-2:

$$P_{n.C} = P_{no} 0.658^{\frac{P_{no}}{P_e}} \quad \text{(from AISC } \textit{Spec.} \text{ Eq. I2-2)}$$
$$= (104{,}000 \text{ kips}) \left(0.658^{\frac{104{,}000 \text{ kips}}{3.98 \times 10^7 \text{ kips}}} \right)$$
$$= 104{,}000 \text{ kips}$$

The design compression strength is:

$$\phi_c P_{n.C} = 0.90(104{,}000 \text{ kips})$$
$$= 93{,}600 \text{ kips} > P_{r.wall} = 31{,}700 \text{ kips} \quad \textbf{o.k.}$$

Step 4-9. Wall flexural strength

The flexural strength of the SpeedCore wall is calculated using the plastic stress distribution method with the assumed stress blocks noted in Figures 4-41 and 4-42 for tension and compression walls, respectively. The axial force is assumed to act at the elastic centroid of the walls.

The axial forces in the walls are:

$$P_{r.wall} = 31{,}700 \text{ kips}$$

$$T_{r.wall} = 31{,}700 \text{ kips}$$

Flexural Strength of Tension Wall

The width of concrete in the flange is:

$$t_{c.f} = t_{sc.f} - 2t_p$$
$$= 24 \text{ in.} - 2(½ \text{ in.})$$
$$= 23.0 \text{ in.}$$

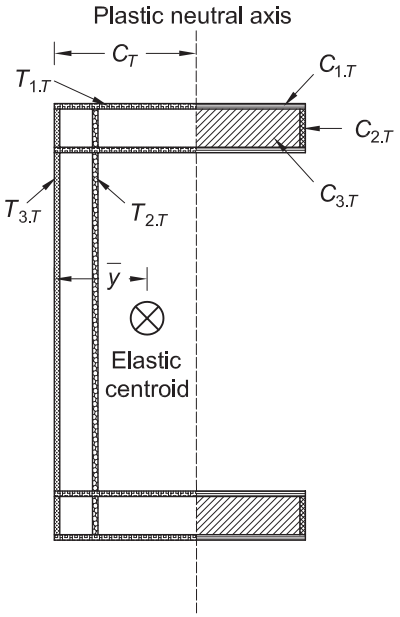


Fig. 4-41. Cross section with labeled regions for plastic moment calculation of tension wall.

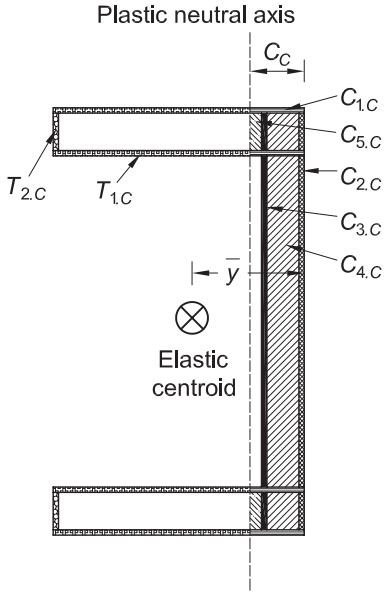


Fig. 4-42. Cross section with labeled regions for plastic moment calculation of compression wall.

The width of concrete in the web is:

$$t_{c.w} = t_{sc.w} - 2t_p$$
$$= 14 \text{ in.} - 2(½ \text{ in.})$$
$$= 13.0 \text{ in.}$$

The location of the plastic neutral axis of the tension wall is calculated assuming that the plastic neutral axis is in the concrete flange:

$$C_T = \frac{T_{r.wall} + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y + 0.85 f_c' 2t_{c.f} (W_{wall} - t_p) - 2t_p (L_{wall} - 4t_p) F_y}{8t_p F_y + 0.85 f_c' 2t_{c.f}}$$
$$= \frac{31{,}700 \text{ kips} + 4(½ \text{ in.})(192 \text{ in.})(50 \text{ ksi}) + 2(½ \text{ in.})(23.0 \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(2)(23.0 \text{ in.})(192 \text{ in.} - ½ \text{ in.}) - 2(½ \text{ in.})\left[360 \text{ in.} - 4(½ \text{ in.})\right](50 \text{ ksi})}{8(½ \text{ in.})(50 \text{ ksi}) + 0.85(6 \text{ ksi})(2)(23.0 \text{ in.})}$$
$$= 182 \text{ in.}$$

The plastic neutral axis location, $C_T = 182$ in., was correctly assumed to be in the concrete flange.

The compression force in the flange edge plates is:

$$C_{1.T} = 4(W_{wall} - C_T) t_p F_y$$
$$= 4(192 \text{ in.} - 182 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 1{,}000 \text{ kips}$$

The compression force in the flange end plates is:

$$C_{2.T} = 2t_{c.f} t_p F_y$$
$$= 2(23.0 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 1{,}150 \text{ kips}$$

The compression force in the concrete is:

$$C_{3.T} = 0.85 f_c' 2t_{c.f} (W_{wall} - C_T - t_p)$$
$$= 0.85(6 \text{ ksi})(2)(23.0 \text{ in.})(192 \text{ in.} - 182 \text{ in.} - ½ \text{ in.})$$
$$= 2{,}230 \text{ kips}$$

The tensile force in the steel in the wall perpendicular to the web is:

$$T_{1.T} = 4C_T t_p F_y$$
$$= 4(182 \text{ in.})(½ \text{ in.})(50 \text{ ksi})$$
$$= 18{,}200 \text{ kips}$$

The tensile force in the inside steel plate in the web is:

$$T_{2.T} = (L_{wall} - 4t_p) t_p F_y$$
$$= \left[360 \text{ in.} - 4(½ \text{ in.})\right](½ \text{ in.})(50 \text{ ksi})$$
$$= 8{,}950 \text{ kips}$$

The tensile force in the web is:

$$T_{3.T} = (L_{wall} - 4t_p)t_p F_y$$
$$= [360\text{ in.} - 4(½\text{ in.})](½\text{ in.})(50\text{ ksi})$$
$$= 8{,}950\text{ kips}$$

The plastic moment of the composite wall in tension is:

$$M_{PT.wall} = C_{1.T}\left(\frac{W_{wall} - C_T}{2}\right) + C_{2.T}\left(W_{wall} - C_T - \frac{t_p}{2}\right) + C_{3.T}\left(\frac{W_{wall} - C_T - t_p}{2}\right)$$
$$+ T_{1.T}\left(\frac{C_T}{2}\right) + T_{2.T}\left(C_T - t_{sc.w} + \frac{t_p}{2}\right) + T_{3.T}\left(C_T - \frac{t_p}{2}\right) + P(\bar{y} - C_T)$$
$$= (1{,}000\text{ kips})\left(\frac{192\text{ in.} - 182\text{ in.}}{2}\right) + (1{,}150\text{ kips})\left(192\text{ in.} - 182\text{ in.} - \frac{½\text{ in.}}{2}\right)$$
$$+ (2{,}230\text{ kips})\left(\frac{192\text{ in.} - 182\text{ in.} - ½\text{ in.}}{2}\right) + (18{,}200\text{ kips})\left(\frac{182\text{ in.}}{2}\right)$$
$$+ (8{,}950\text{ kips})\left(182\text{ in.} - 14\text{ in.} + \frac{½\text{ in.}}{2}\right) + (8{,}950\text{ kips})\left(182\text{ in.} - \frac{½\text{ in.}}{2}\right) + (31{,}700\text{ kips})(65.1\text{ in.} - 182\text{ in.})$$
$$= 1{,}110{,}000\text{ kip-in.}$$

$$M_{n.T.wall} = M_{PT.wall} = 1{,}110{,}000\text{ kip-in.}$$

Flexural strength of compression wall

The location of the plastic neutral axis of the compression wall is calculated with the assumption that the plastic neutral axis is in the concrete flange:

$$C_C = \frac{P_{r.wall} - 2t_p(L_{wall} - 4t_p)F_y - t_{c.w}(L_{wall} - 4t_p)\,0.85f_c' + 2t_{c.f}t_{sc.w}\,0.85f_c' + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y}{8t_p F_y + 2t_{c.f}\,0.85f_c'}$$

$$= \frac{\begin{array}{l}31{,}700\text{ kips} - 2(½\text{ in.})[360\text{ in.} - 4(½\text{ in.})](50\text{ ksi}) - (13.0\text{ in.})[360\text{ in.} - 4(½\text{ in.})](0.85)(6\text{ ksi}) \\ + 2(23.0\text{ in.})(14\text{ in.})(0.85)(6\text{ ksi}) + 4(½\text{ in.})(192\text{ in.})(50\text{ ksi}) + 2(½\text{ in.})(23.0\text{ in.})(50\text{ ksi})\end{array}}{8(½\text{ in.})(50\text{ ksi}) + 2(23.0\text{ in.})0.85(6\text{ ksi})}$$

$$= 31.5\text{ in.}$$

The compression force in the steel plates perpendicular to the web is:

$$C_{1.C} = 4C_C t_p F_y$$
$$= 4(31.5\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 3{,}150\text{ kips}$$

The compression force in the outside steel plate of the web is:

$$C_{2.C} = t_p(L_{wall} - 4t_p)F_y$$
$$= (½\text{ in.})[360\text{ in.} - 4(½\text{ in.})](50\text{ ksi})$$
$$= 8{,}950\text{ kips}$$

The compression force in the inside steel plate of the web is:

$$C_{3.C} = t_p(L_{wall} - 4t_p)F_y$$
$$= (½\text{ in.})\left[360\text{ in.} - 4(½\text{ in.})\right](50\text{ ksi})$$
$$= 8{,}950\text{ kips}$$

The compression force in the concrete in the web is:

$$C_{4.C} = 0.85f_c't_{c.w}(L_{wall} - 4t_p)$$
$$= 0.85(6\text{ ksi})(13.0\text{ in.})\left[360\text{ in.} - 4(½\text{ in.})\right]$$
$$= 23{,}700\text{ kips}$$

The compression force in the concrete in the flange is:

$$C_{5.C} = 0.85f_c'2t_{c.f}(C_C - t_{sc.w})$$
$$= 0.85(6\text{ ksi})(2)(23.0\text{ in.})(31.5\text{ in.} - 14\text{ in.})$$
$$= 4{,}110\text{ kips}$$

The tensile force in the steel in the wall perpendicular to the web is:

$$T_{1.C} = 4(W_{wall} - C_C)t_pF_y$$
$$= 4(192\text{ in.} - 31.5\text{ in.})(½\text{ in.})(50\text{ ksi})$$
$$= 16{,}100\text{ kips}$$

The tension force in the flange end plate is:

$$T_{2.C} = 2t_pt_{c.f}F_y$$
$$= 2(½\text{ in.})(23.0\text{ in.})(50\text{ ksi})$$
$$= 1{,}150\text{ kips}$$

The plastic flexural strength of the composite wall in compression is:

$$M_{PC.wall} = C_{1.C}\left(\frac{C_C}{2}\right) + C_{2.C}\left(C_C - \frac{t_p}{2}\right) + C_{3.C}\left(C_C - t_{sc.w} + \frac{t_p}{2}\right) + C_{4.C}\left(C_C - \frac{t_{sc.w}}{2}\right) + C_{5.C}\left(\frac{C_C - t_{sc.w}}{2}\right)$$
$$+ T_{1.C}\left(\frac{W_{wall} - C_C}{2}\right) + T_{2.C}\left(W_{wall} - C_C - \frac{t_p}{2}\right) + P(\bar{y} - C_C)$$
$$= (3{,}150\text{ kips})\left(\frac{31.5\text{ in.}}{2}\right) + (8{,}950\text{ kips})\left(31.5\text{ in.} - \frac{½\text{ in.}}{2}\right) + (8{,}950\text{ kips})\left(31.5\text{ in.} - 14\text{ in.} + \frac{½\text{ in.}}{2}\right)$$
$$+ (23{,}700\text{ kips})\left(31.5\text{ in.} - \frac{14\text{ in.}}{2}\right) + (4{,}110\text{ kips})\left(\frac{31.5\text{ in.} - 14\text{ in.}}{2}\right) + (16{,}100\text{ kips})\left(\frac{192\text{ in.} - 31.5\text{ in.}}{2}\right)$$
$$+ (1{,}150\text{ kips})\left(192\text{ in.} - 31.5\text{ in.} - \frac{½\text{ in.}}{2}\right) + (31{,}700\text{ kips})(65.1\text{ in.} - 31.5\text{ in.})$$
$$= 3{,}650{,}000\text{ kip-in.}$$

$$M_{nC.wall} = M_{PC.wall} = 3{,}650{,}000\text{ kip-in.}$$

The available flexural strength in the tension wall is:

$$\phi_bM_{nT.wall} = 0.90(1{,}110{,}000\text{ kip-in.})$$
$$= 999{,}000\text{ kip-in.} > M_{r.wall} = 865{,}000\text{ kip-in.} \quad \textbf{o.k.}$$

The available flexural strength in the compression wall is:

$$\phi_b M_{nC.wall} = 0.90(3{,}650{,}000)$$
$$= 3{,}290{,}000 \text{ kip-in.} > M_{r.wall} = 2{,}060{,}000 \text{ kip-in.} \quad \textbf{o.k.}$$

Step 4-10. Wall shear strength

The area of steel in the direction of shear is:

$$A_{sw} = 4W_{wall}t_p$$
$$= 4(192 \text{ in.})(½ \text{ in.})$$
$$= 384 \text{ in.}^2$$

The area of concrete in the direction of shear is:

$$A_{cw} = 2(W_{wall} - 3t_p)(t_{sc.f} - 2t_p)$$
$$= 2\left[192 \text{ in.} - 3(½ \text{ in.})\right]\left[24 \text{ in.} - 2(½ \text{ in.})\right]$$
$$= 8{,}760 \text{ in.}^2$$

The K factors for the shear calculations are calculated according to Equations 2-15 and 2-16 of this Design Guide:

$$K_s = G_s A_{sw} \tag{2-15}$$
$$= (11{,}200 \text{ ksi})(384 \text{ in.}^2)$$
$$= 4{,}300{,}000 \text{ kips}$$

$$K_{sc} = \frac{0.7(E_c A_{cw})(E_s A_{sw})}{4E_s A_{sw} + E_c A_{cw}} \quad \text{(from Eq. 2-16)}$$
$$= \frac{0.7(4{,}500 \text{ ksi})(8{,}760 \text{ in.}^2)(29{,}000 \text{ ksi})(384 \text{ in.}^2)}{4(29{,}000 \text{ ksi})(384 \text{ in.}^2) + (4{,}500 \text{ ksi})(8{,}760 \text{ in.}^2)}$$
$$= 3{,}660{,}000 \text{ kips}$$

The nominal shear strength of the wall is calculated from Equation 2-14:

$$V_{n.wall} = \frac{K_s + K_{sc}}{\sqrt{3K_s^2 + K_{sc}^2}} F_y A_{sw} \quad \text{(from Eq. 2-14)}$$
$$= \frac{4{,}300{,}000 \text{ kips} + 3{,}660{,}000 \text{ kips}}{\sqrt{3(4{,}300{,}000 \text{ kips})^2 + (3{,}660{,}000 \text{ kips})^2}} (50 \text{ ksi})(384 \text{ in.}^2)$$
$$= 18{,}400 \text{ kips}$$

The design shear strength of the uncoupled wall:

$$\phi_v V_{n.wall} = 0.90(18{,}400 \text{ kips})$$
$$= 16{,}600 \text{ kips} > V_{r.wall} = 4{,}640 \text{ kips} \quad \textbf{o.k.}$$

EXAMPLE 4.3—Continuous Web Plate Connection—Seismic Design of Coupling Beam-to-SpeedCore Wall Connection

This example details the design of a composite coupling beam-to-SpeedCore wall connection. This design example can be downloaded as a Mathcad file from the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

Design a welded connection for the composite coupling beam-to-planar wall connection using the given coupling beam and wall geometry, material properties, and loads.

Given:

For the coupling beam connection to the wall, the coupling beam flange plates are CJP groove welded to the inside of the wall web plates. Coupling beam web plates are continuous with the wall web plate (i.e., the coupling beam web plate and wall plate thickness must be the same). See Figure 4-43 for further details.

The steel and concrete material properties are as follows:

For ASTM A572/A572M, Grade 50 steel, from AISC *Manual* Table 2-4:

$F_u = 65$ ksi
$F_y = 50$ ksi

From AISC *Seismic Provisions* Table A3.2:

$R_t = 1.2$
$R_y = 1.1$

Concrete:

$f'_c = 6$ ksi
$E_c = 4{,}500$ ksi
$G_c = 1{,}800$ ksi
$R_c = 1.3$

Weld metal (E70XX):

$F_{EXX} = 70$ ksi

SpeedCore wall dimensions (see Figure 4-44):

t_p = wall plate thickness
= 9/16 in.
t_{sc} = wall thickness
= 24 in.

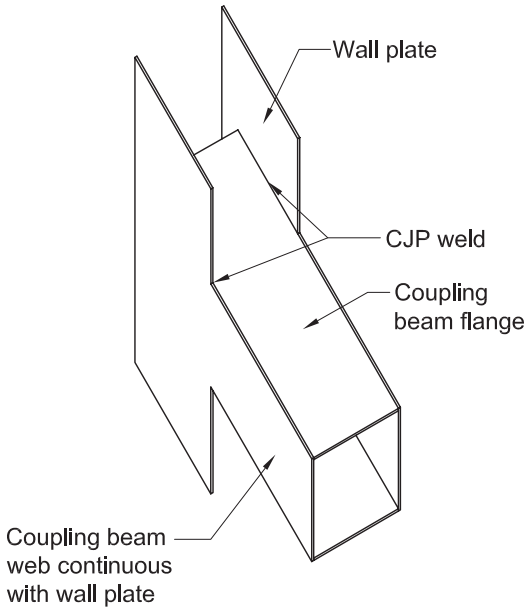


Fig. 4-43. Coupling beam connection for Example 4.3.

Coupling beam dimensions (see Figure 4-45):

b_{CB} = coupling beam width
= 24 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf.CB}$ = coupling beam flange plate thickness
= ½ in.
$t_{pw.CB}$ = coupling beam web plate thickness
= 9⁄16 in.

Solution:

This connection is designed for the expected connection strength using the capacity-limited design philosophy implemented for coupled SpeedCore system design.

Step 1. Flange Plate Connection Demand

The required strength of the coupling beam flange plate connection is the minimum of 120% of the tensile yield strength and 100% of the tensile rupture strength of the flange plate, accounting for the expected material strength.

The area of the coupling beam flange plate is:

$$A_{f.CB} = (b_{CB} - 2t_{pw.CB})t_{pf.CB}$$
$$= [24 \text{ in.} - 2(9/16 \text{ in.})](1/2 \text{ in.})$$
$$= 11.4 \text{ in.}^2$$

The required strength of the flange plate connection is:

$$T_{flange} = \min(1.2R_yF_yA_{f.CB}, R_tF_uA_{f.CB})$$
$$= \min[(1.2)(1.1)(50 \text{ ksi})(11.4 \text{ in.}^2), (1.2)(65 \text{ ksi})(11.4 \text{ in.}^2)]$$
$$= \min(752 \text{ kips}, 889 \text{ kips})$$
$$= 752 \text{ kips}$$

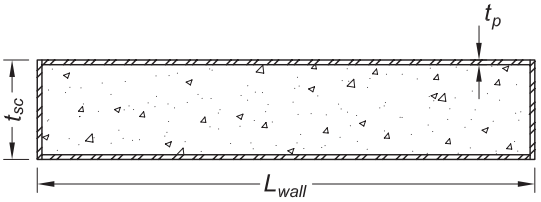


Fig. 4-44. Wall cross-section dimensions.

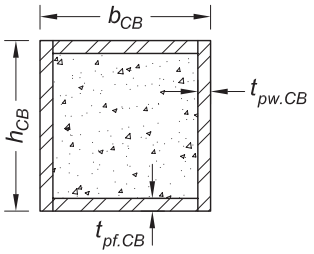


Fig. 4-45. Coupling beam cross-section dimensions.

Step 2. Estimate Length of Weld Required

From AISC *Specification* Section J4.2, determine the length of the flange plate CJP groove weld by assuming that shear yielding of the flange plate will govern the weld strength. The welds of interest are depicted in Figure 4-46.

The design strength of the CJP flange plate weld on one edge of the flange plate must be greater than half the tensile demand on the flange plate.

$$\frac{T_{flange}}{2} \le \phi 0.60 F_y t_{pf.CB} L_{req}$$

Therefore, the required weld length, L_{req}, is:

$$L_{req} \ge \frac{T_{flange}}{2\phi 0.60 F_y t_{pf.CB}}$$
$$= \frac{752 \text{ kips}}{2(1.00)(0.60)(50 \text{ ksi})(½ \text{ in.})}$$
$$= 25.1 \text{ in.}$$

Try a flange weld length of $L_{f.w} = 26$ in.

Step 3. Check Shear Strength of Coupling Beam Flange Plate

The shear yielding and rupture plane of interest for the flange plate is shown in Figure 4-47. Multiple shear rupture planes can be checked depending upon the weld type, but only one plane will be checked in this example.

The gross shear area of the coupling beam flange plate for shear yielding is:

$$A_{f.SY} = t_{pf.CB} L_{f.w}$$
$$= (½ \text{ in.})(26 \text{ in.})$$
$$= 13.0 \text{ in.}^2$$

For shear yielding, the design strength of the coupling beam flange plate from AISC *Specification* Equation J4-3 is:

$$\phi 0.60 F_y A_{f.SY} = (1.00)(0.60)(50 \text{ ksi})(13.0 \text{ in.}^2) \quad \text{(from AISC } Spec. \text{ Eq. J4-3)}$$
$$= 390 \text{ kips}$$

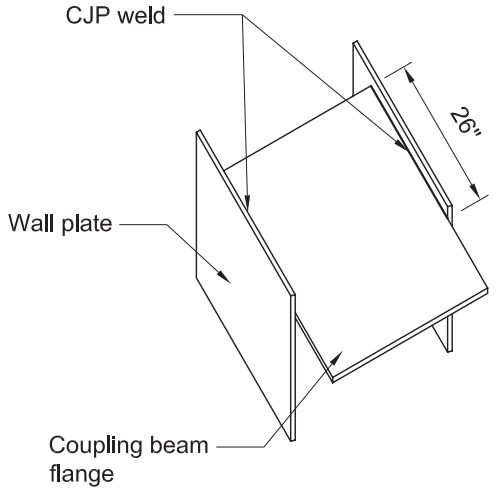


Fig. 4-46. Flange plate connection detail.

The net shear area of the coupling beam flange plate for shear rupture is equal to the gross area:

$$A_{f.SR} = A_{f.SY}$$
$$= 13.0 \text{ in.}^2$$

For shear rupture, the design strength of the coupling beam flange plate from AISC *Specification* Equation J4-4 is:

$$\phi 0.6 F_u A_{f.SR} = (0.75)(0.60)(65 \text{ ksi})(13.0 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-4)}$$
$$= 380 \text{ kips}$$

Shear rupture controls, and the shear strength of the coupling beam flange plate is adequate:

$$\phi 0.6 F_u A_{f.SR} = 380 \text{ kips} > T_{flange}/2 = 376 \text{ kips}$$

Step 4. Check Shear Strength of Wall Web Plates

The shear yielding and rupture planes of interest for the web plates are shown in Figure 4-48. Multiple shear rupture planes can be checked depending upon the weld type, but only one plane is checked in this example.

The gross shear area of the wall web plate for shear yielding is:

$$A_{w.SY} = 2 t_p L_{f.w}$$
$$= 2(\tfrac{9}{16} \text{ in.})(26 \text{ in.})$$
$$= 29.3 \text{ in.}^2$$

For shear yielding, the design strength of the wall web plate from AISC *Specification* Equation J4-3 is:

$$\phi 0.60 F_y A_{w.SY} = 1.00(0.60)(50 \text{ ksi})(29.3 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-3)}$$
$$= 879 \text{ kips}$$

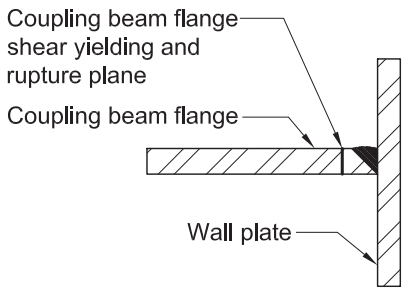


Fig. 4-47. Coupling beam flange plate shear yielding and shear rupture plane of interest.

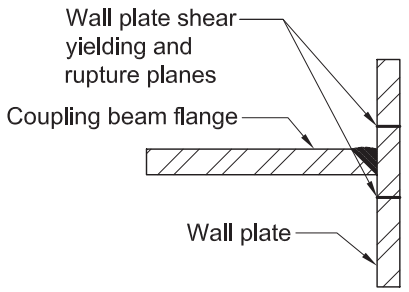


Fig. 4-48. Wall web plate shear yielding and shear rupture planes of interest.

The net shear area of the wall web plate for shear rupture is equal to the gross area:

$$A_{w.SR} = A_{w.SY}$$
$$= 29.3 \text{ in.}^2$$

For shear rupture, the design strength of the wall web plate from AISC *Specification* Equation J4-4 is:

$$\phi 0.60 F_u A_{w.SR} = 0.75(0.60)(65 \text{ ksi})(29.3 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-4)}$$
$$= 857 \text{ kips}$$

Shear rupture controls, and the shear strength of the wall plate is adequate:

$$\phi 0.60 F_u A_{w.SR} = 857 \text{ kips} > T_{flange}/2 = 376 \text{ kips}$$

Step 5. Check Ductile Behavior of Flange Plates

The flange plate connection is designed such that the available tension rupture strength of the flange plate is greater than the available tensile yield strength.

The gross area of the flange plate in tension is:

$$A_{f.g} = (t_{sc} - 2t_p)t_{pf.CB}$$
$$= [24.0 \text{ in.} - 2(9/16 \text{ in.})](1/2 \text{ in.})$$
$$= 11.4 \text{ in.}^2$$

The net area of the flange plate in tension is equal to the gross area:

$$A_{f.n} = A_{f.g}$$
$$= 11.4 \text{ in.}^2$$

The shear lag factor from AISC *Specification* Table D3.1 is:

$$U = \frac{3L_{f.w}^2}{3L_{f.w}^2 + (t_{sc} - 2t_p)^2}\left(1 - \frac{\bar{x}}{L_{f.w}}\right)$$
$$= \frac{3(26 \text{ in.})^2}{3(26 \text{ in.})^2 + [24 \text{ in.} - 2(9/16 \text{ in.})]^2}\left(1 - \frac{0}{26 \text{ in.}}\right)$$
$$= 0.795$$

The effective area of the flange plate in tension from AISC *Specification* Equation D3-1 is:

$$A_{f.e} = UA_{f.n}$$
$$= (0.795)(11.4 \text{ in.}^2)$$
$$= 9.06 \text{ in.}^2$$

The available tensile yield strength from AISC *Specification* Equation J4-1 is:

$$R_y F_y A_{f.g} = (1.1)(50 \text{ ksi})(11.4 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-1)}$$
$$= 627 \text{ kips}$$

The available tensile rupture strength from AISC *Specification* Equation J4-2 is:

$$R_t F_u A_{f.e} = (1.2)(65 \text{ ksi})(9.06 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-2)}$$
$$= 707 \text{ kips}$$

The available tension rupture strength of the flange plate is greater than the available tensile yield strength, and the design intent is met.

This check completes the weld design for the composite coupling beam-to-SpeedCore wall connection. The coupling beam flange plate is CJP groove welded to the SpeedCore wall web plate.

EXAMPLE 4.4—Lapped Web Plate Connection—Seismic Design of Coupling Beam-to-Wall Connection

This example details the design of a composite coupling beam-to-SpeedCore wall connection. This design example can be downloaded as a Mathcad file from the link given on the AISC Design Guide 38 webpage (www.aisc.org/dg).

Given:

Design a welded connection for the composite coupling beam-to-planar wall using the lapped web plate connection detail show in Figure 4-49 using the given coupling beam and wall geometry, material properties, and loads.

For the coupling beam connection to the wall, the coupling beam flange plate is inserted into slots in the wall web plates. The edge of the flange plate extends 1 in. out from the wall section on both sides of the connection. The wall web plates are beveled on two sides of the slot and CJP groove welded to the coupling beam flange plate. Coupling beam web plates are lapped over the exterior of the wall web plates. The depth of the coupling beam web plates is reduced to fit between the coupling beam flange plates. Coupling beam web plates are fillet welded to the wall web plate on three sides to form a C-shaped weld.

The steel and concrete material properties are as follows:

For ASTM A572/A572M, Grade 50 steel, from AISC *Manual* Table 2-4:

$F_u = 65$ ksi
$F_y = 50$ ksi

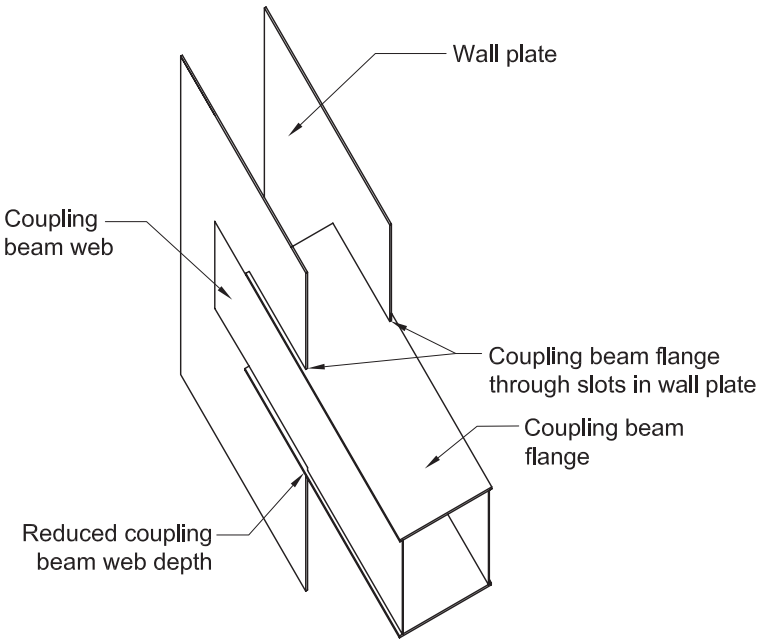


Fig. 4-49. Coupling beam connection for Example 4.4.

From AISC *Seismic Provisions* Table A3.2:

R_t = 1.2
R_y = 1.1

Concrete:

f'_c = 6 ksi
E_c = 4,500 ksi
G_c = 1,800 ksi
R_c = 1.3

Weld metal (E70XX):

F_{EXX} = 70 ksi

SpeedCore wall dimensions (see Figure 4-50):

t_p = wall plate thickness
= ½ in.
t_{sc} = wall thickness
= 24 in.

Coupling beam dimensions (see Figure 4-51):

L_{CB} = coupling beam length
= 96 in.
b_{CB} = coupling beam width
= 26 in.
h_{CB} = coupling beam depth
= 24 in.
$t_{pf\cdot CB}$ = coupling beam flange plate thickness
= 9/16 in.
$t_{pw\cdot CB}$ = coupling beam web plate thickness
= ½ in.

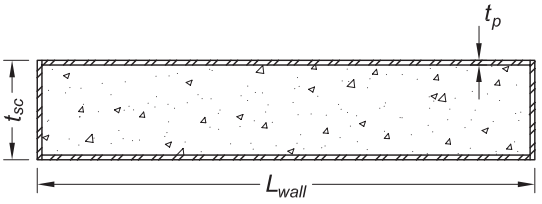


Fig. 4-50. Wall cross-section dimensions.

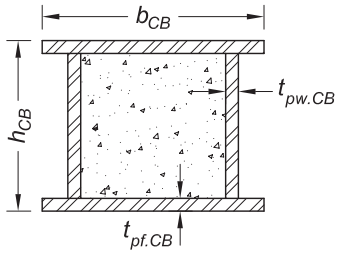


Fig. 4-51. Coupling beam cross-section dimensions.

Expected Coupling Beam Forces

These forces are typically determined in the design process of the coupled wall system and considered as givens for the sake of the coupling beam-to-wall connection design. Refer to Examples 4.1 and 4.2 for similar calculations.

The expected plastic flexural strength of the coupling beam is:

$M_{p.exp.CB} = 28{,}800$ kip-in.

The expected tensile force in the web is:

$T_{2.exp} = 1{,}030$ kips

The expected compression force in the web is:

$C_{2.exp} = 295$ kips

The location of the coupling beam plastic neutral axis considering expected strength is:

$C_{CB.exp} = 5.36$ in.

Solution:

The connection was designed for the expected connection strength using the capacity-limited design philosophy implemented for coupled SpeedCore wall design.

Step 1. Flange Plate Connection Demand

The required strength of the coupling beam flange plate connection is the minimum of 120% of the tensile yield strength and 100% of the tensile rupture strength of the flange plate, accounting for the expected material strength.

The area of the coupling beam flange plate is:

$$\begin{aligned} A_{f.CB} &= b_{CB} t_{pf.CB} \\ &= (26 \text{ in.})(9/16 \text{ in.}) \\ &= 14.6 \text{ in.}^2 \end{aligned}$$

The coupling beam flange plate connection required strength is:

$$\begin{aligned} T_{flange} &= \min\left(1.2R_y F_y A_{f.CB}, R_t F_u A_{f.CB}\right) \\ &= \min\left[(1.2)(1.1)(50 \text{ ksi})(14.6 \text{ in.}^2), (1.2)(65 \text{ ksi})(14.6 \text{ in.}^2)\right] \\ &= \min(964 \text{ kips}, 1{,}140 \text{ kips}) \\ &= 964 \text{ kips} \end{aligned}$$

Step 2. Estimate Length of Weld Required

From AISC *Specification* Section J4.2, determine the length of the flange plate CJP groove weld by assuming that shear yielding of the flange plate will govern the weld strength. The welds of interest are shown in Figure 4-52.

The design strength of the CJP flange plate weld on one edge of the flange plate must be greater than half the tensile demand on the flange plate.

$$\frac{T_{flange}}{2} \le \phi 0.60 F_y t_{pf.CB} L_{req}$$

Therefore, the estimated required weld length, L_{req}, is:

$$L_{req} \geq \frac{T_{flange}}{2\phi 0.60 F_y t_{pf.CB}}$$
$$\geq \frac{964 \text{ kips}}{2(1.00)(0.60)(50 \text{ ksi})(9/16 \text{ in.})}$$
$$\geq 28.6 \text{ in.}$$

Try a flange weld length of $L_{f.w} = 30$ in.

Step 3. Check Shear Strength of Coupling Beam Flange Plate

The shear yielding and rupture plane of interest for the flange plate is shown in Figure 4-53. Multiple shear rupture planes can be checked, depending upon the weld type but only one plane will be checked in this example.

The gross shear area of the coupling beam flange plate for shear yielding is:

$$A_{f.SY} = t_{pf.CB} L_{f.w}$$
$$= (9/16 \text{ in.})(30 \text{ in.})$$
$$= 16.9 \text{ in.}^2$$

For shear yielding, the design strength of the coupling beam flange plate from AISC *Specification* Equation J4-3 is:

$$\phi 0.60 F_y A_{f.SY} = 1.00(0.60)(50 \text{ ksi})(16.9 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-3)}$$
$$= 507 \text{ kips}$$

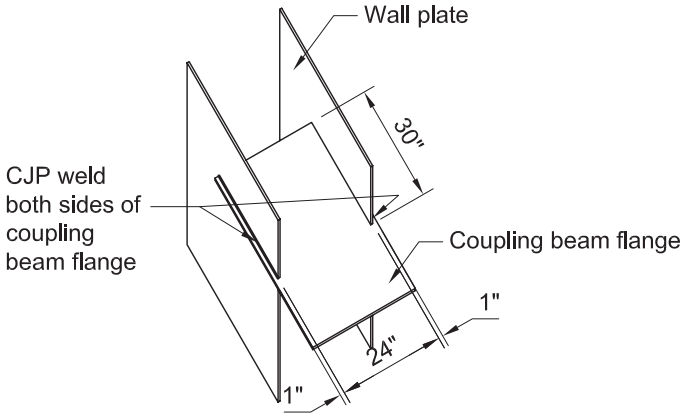


Fig. 4-52. Flange plate connection detail for Example 4.4.

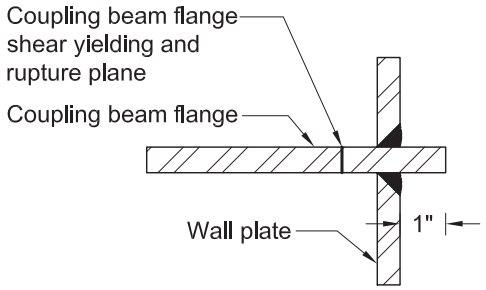


Fig. 4-53. Coupling beam flange plate shear yielding and shear rupture planes of interest.

The net shear area of the coupling beam flange plate for shear rupture is equal to the gross area:

$$A_{f.SR} = A_{f.SY}$$
$$= 16.9 \text{ in.}^2$$

For shear rupture, the design strength of the coupling beam flange plate from AISC *Specification* Equation J4-4 is:

$$\phi 0.60 F_u A_{f.SR} = 0.75(0.60)(65 \text{ ksi})(16.9 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-4)}$$
$$= 494 \text{ kips}$$

Shear rupture controls, and the shear strength of the coupling beam flange plate is adequate:

$$\phi 0.60 F_u A_{f.SR} = 494 \text{ kips} > T_{flange}/2 = 482 \text{ kips}$$

Step 4. Check Shear Strength of Wall Web Plates

The shear yielding and rupture planes of interest for the web plates are shown in Figure 4-54. Depending upon the weld type, shear rupture may also need to be checked.

The gross shear area of the wall web plate for shear yielding is:

$$A_{w.SY} = 2 t_p L_{f.w}$$
$$= 2(½ \text{ in.})(30 \text{ in.})$$
$$= 30.0 \text{ in.}^2$$

For shear yielding, the design strength of the wall web plate from AISC *Specification* Equation J4-3 is:

$$\phi 0.60 F_y A_{w.SY} = 1.00(0.60)(50 \text{ ksi})(30.0 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-3)}$$
$$= 900 \text{ kips} > 483 \text{ kips}$$

The net shear area of the wall web plate for shear rupture is equal to the gross area:

$$A_{w.SR} = A_{w.SY}$$
$$= 30.0 \text{ in.}^2$$

For shear rupture, the design strength of the wall web plate from AISC *Specification* Equation J4-4 is:

$$\phi 0.6 F_u A_{w.SR} = 0.75(0.60)(65 \text{ ksi})(30.0 \text{ in.}^2) \quad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-4)}$$
$$= 878 \text{ kips} > 446 \text{ kips}$$

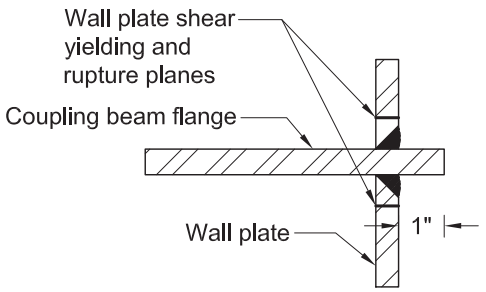


Fig. 4-54. Wall web plate shear yielding and shear rupture planes of interest.

Shear rupture controls, and the shear strength of the wall web plate is adequate:

$$\phi 0.6F_uA_{w.SR} = 878 \text{ kips} > T_{flange}/2 = 446 \text{ kips}$$

Step 5. Check Ductile Behavior of Flange Plates

The flange plate connection is designed such that the available tensile rupture strength of the flange plate is greater than the available tensile yield strength.

The gross area of the flange plate in tension is:

$$\begin{aligned} A_{f.g} &= b_{CB}t_{pf.CB} \\ &= (26 \text{ in.})(9/16 \text{ in.}) \\ &= 14.6 \text{ in.}^2 \end{aligned}$$

The net area of the flange plate in tension is:

$$\begin{aligned} A_{f.n} &= (b_{CB} - 2 \text{ in.})t_{pf.CB} \\ &= (26 \text{ in.} - 2 \text{ in.})(9/16 \text{ in.}) \\ &= 13.5 \text{ in.}^2 \end{aligned}$$

The shear lag factor from AISC *Specification* Table D3.1 is:

$$\begin{aligned} U &= \frac{3L_{f.w}^2}{3L_{f.w}^2 + (b_{CB} - 2 \text{ in.})^2}\left(1 - \frac{\bar{x}}{L_{f.w}}\right) \\ &= \frac{3(30 \text{ in.})^2}{3(30 \text{ in.})^2 + (26 \text{ in.} - 2 \text{ in.})^2}\left(1 - \frac{0 \text{ in.}}{30 \text{ in.}}\right) \\ &= 0.824 \end{aligned}$$

The effective area of the flange plate in tension from AISC *Specification* Equation D3-1 is:

$$\begin{aligned} A_{f.e} &= UA_{f.n} \\ &= (0.824)(13.5 \text{ in.}^2) \\ &= 11.1 \text{ in.}^2 \end{aligned}$$

The available tensile yield strength from AISC *Specification* Equation J4-1 is:

$$\begin{aligned} R_yF_yA_{f.g} &= 1.1(50 \text{ ksi})(14.6 \text{ in.}^2) \qquad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-1)} \\ &= 803 \text{ kips} \end{aligned}$$

The available tensile rupture strength from AISC *Specification* Equation J4-2 is:

$$\begin{aligned} R_tF_uA_{f.n} &= 1.2(65 \text{ ksi})(11.1 \text{ in.}^2) \qquad \text{(from AISC } \textit{Spec.} \text{ Eq. J4-2)} \\ &= 866 \text{ kips} \end{aligned}$$

The available tensile rupture strength of the flange plate is greater than the available tensile yield strength, and the design intent is met.

Step 6. Calculate Forces in Web Plates

Design the coupling beam web-to-wall web C-shaped fillet weld, as shown in Figure 4-55, for 120% of the ratio of the moment contribution of the coupling beam web plate to the plastic flexural strength of the coupling beam, accounting for the expected material strength. Also design the weld for 120% of the resulting tensile force in the coupling beam web due to flexure about

the plastic neutral axis of the coupling beam and 120% of the shear in the coupling beam corresponding to the plastic flexural strength of the coupling beam. Figure 4-56 shows the forces and moments for the C-shaped fillet weld.

The tensile force in the webs of the coupling beam is:

$$T_{web} = 1.2(T_{2.exp} - C_{2.exp})$$
$$= 1.2(1{,}030 \text{ kips} - 295 \text{ kips})$$
$$= 882 \text{ kips}$$

The moment in the web of the coupling beam relative to the centroid of the web is:

$$M_{web} = 1.2\left[T_{2.exp}\frac{C_{CB.exp}}{2} + C_{2.exp}\left(\frac{h_{CB} - C_{CB.exp}}{2}\right)\right]$$
$$= 1.2\left[(1{,}030 \text{ kips})\left(\frac{5.36 \text{ in.}}{2}\right) + (295 \text{ kips})\left(\frac{24 \text{ in.} - 5.36 \text{ in.}}{2}\right)\right]$$
$$= 6{,}610 \text{ kip-in.}$$

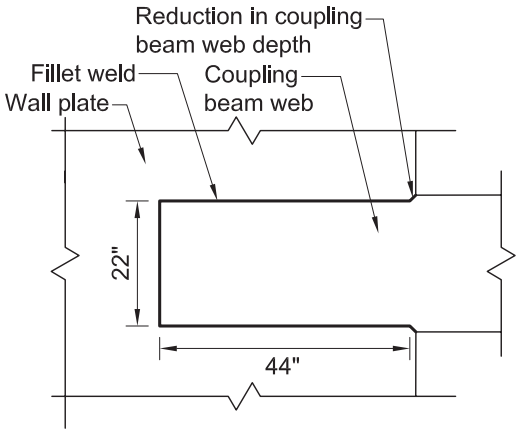


Fig. 4-55. Web plate connection detail.

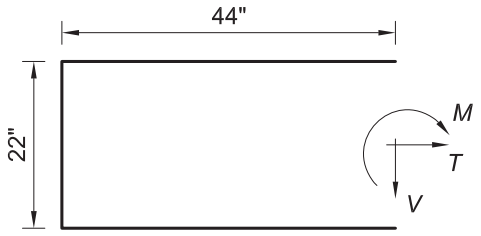


Fig. 4-56. Forces on web plate connection.

Shear in the web of the coupling beam corresponding to the plastic flexural strength is:

$$V_{web} = \frac{2(1.2M_{p.exp.CB})}{L_{cb}} \quad \text{(from Eq. 4-5)}$$
$$= \frac{2\left[1.2(28{,}800 \text{ kip-in.})\right]}{96 \text{ in.}}$$
$$= 720 \text{ kips}$$

Note that this is conservative because the concrete contribution is not considered.

Step 7. Calculate Force Demand on C-Shaped Weld

Assign half the forces in the coupling beam web to each C-shaped weld.

The tensile demand on the C-shaped weld is:

$$T_{C.weld} = \frac{T_{web}}{2}$$
$$= \frac{882 \text{ kips}}{2}$$
$$= 441 \text{ kips}$$

The moment demand on the C-shaped weld is:

$$M_{C.weld} = \frac{M_{web}}{2}$$
$$= \frac{6{,}610 \text{ kip-in.}}{2}$$
$$= 3{,}310 \text{ kip-in.}$$

The shear demand on the C-shaped weld is:

$$V_{C.weld} = \frac{V_{web}}{2}$$
$$= \frac{720 \text{ kips}}{2}$$
$$= 360 \text{ kips}$$

Step 8. Select Weld Geometry

The vertical length of the weld is:

$$L_{V.weld} = h_{CB} - 2 \text{ in.}$$
$$= 24 \text{ in.} - 2 \text{ in.}$$
$$= 22.0 \text{ in.}$$

An initial assumption for the horizontal length of the weld is:

$$L_{H.weld} = 44 \text{ in.}$$

The minimum fillet weld size is 3⁄16 in. according to AISC *Specification* Table J2.4, and according to AISC *Specification* Section J2.2b, the maximum fillet weld size is 7⁄16 in. A 3⁄8-in. weld diameter is selected for the weld size.

Step 9. Calculate C-Shaped Weld Shear and Flexural Strength

The moment and shear demand on the C-shaped welds, as shown in Figure 4-57(a), can be designed for an eccentric shear force, as shown in Figure 4-57(b).

Use AISC *Manual* Table 8-8 to calculate the available strength of the C-shaped weld group for shear and flexure. AISC *Manual* Table 8-8 uses an eccentric shear force to represent an equivalent shear and moment acting on a C-shaped weld group.

The eccentricity of shear force to generate equivalent moment is:

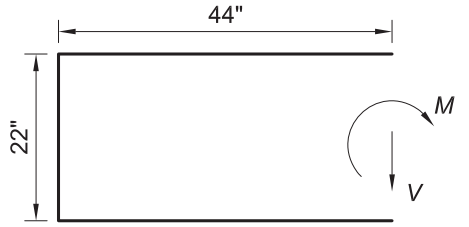
$$e = \frac{M_{C.weld}}{V_{C.weld}}$$
$$= \frac{3{,}310 \text{ kip-in.}}{360 \text{ kips}}$$
$$= 9.19 \text{ in.}$$

The horizontal centroid of the weld group is:

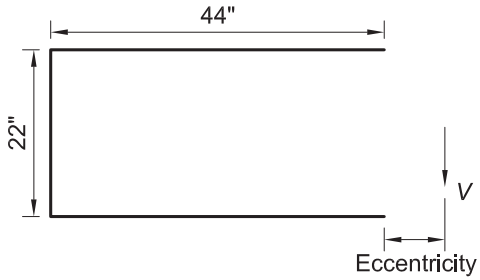
$$cg = \frac{L_{H.weld}^2}{2L_{H.weld} + L_{V.weld}}$$
$$= \frac{(44.0 \text{ in.})^2}{2(44.0 \text{ in.}) + 22.0 \text{ in.}}$$
$$= 17.6 \text{ in.}$$

The horizontal eccentricity of shear from the vertical weld is:

$$e_x = e + L_{H.weld} - cg$$
$$= 9.19 \text{ in.} + 44 \text{ in.} - 17.6 \text{ in.}$$
$$= 35.6 \text{ in.}$$



(a) Moment and shear demand



(b) Equivalent eccentric shear force

Fig. 4-57. Weld strength for eccentric shear and moment.

The ratio of horizontal weld length to vertical weld length is:

$$k = \frac{L_{H.weld}}{L_{V.weld}} = \frac{44 \text{ in.}}{22.0 \text{ in.}} = 2$$

The ratio of force eccentricity to vertical weld length is:

$$a = \frac{e_x}{L_{V.weld}} = \frac{35.6 \text{ in.}}{22.0 \text{ in.}} = 1.62$$

The coefficient for eccentrically loaded weld groups, linearly interpolated from AISC *Manual* Table 8-8 for calculated k and a values is:

$$C_{8.8} = 4.60$$

From AISC Manual Table 8-3, the electrode strength coefficient for an E70 weld electrode is:

$$C_{1-8.3} = 1.00$$

The design weld strength to resist eccentric shear is:

$$P_{weld.V} = \phi C_{8.8} C_{1-8.3} (16D) L_{V.weld}$$
$$= 0.75(4.60)(1.00)[16(\tfrac{3}{8} \text{ in.})](22.0 \text{ in.})$$
$$= 455 \text{ kips}$$

Step 10. Calculate C-Shaped Weld Tensile Strength

Consider the horizontal portion of the weld to calculate the tensile strength, as shown in Figure 4-58.

The design weld strength to resist tension from AISC *Manual* Equation 8-1 is:

$$P_{weld.T} = \phi 0.60 F_{EXX} 2 L_{H.weld} 0.707 D \quad \text{(from AISC } \textit{Manual}\text{. Eq. 8-1)}$$
$$= 0.75(0.60)(70 \text{ ksi})(2)(44 \text{ in.})(0.707)(\tfrac{3}{8} \text{ in.})$$
$$= 735 \text{ kips}$$

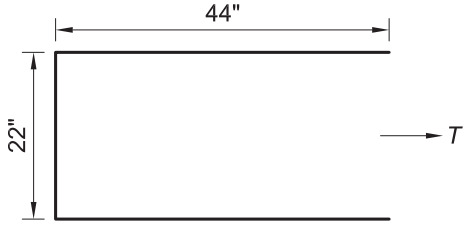


Fig. 4-58. Weld strength in tension.

Step 11. Calculate the Utilization of C-Shaped Weld Strength

Calculate the overall utilization of weld strength for shear, moment, and tension by taking the square root of the sum of the squared utilization of the eccentric shear strength and tensile strength. Refer to Section 4.3.6.2.

$$Utilization = \sqrt{\left(\frac{V_{C.weld}}{P_{weld.V}}\right)^2 + \left(\frac{T_{C.weld}}{P_{weld.T}}\right)^2} \tag{4-7}$$

$$= \sqrt{\left(\frac{360 \text{ kips}}{455 \text{ kips}}\right)^2 + \left(\frac{441 \text{ kips}}{735 \text{ kips}}\right)^2}$$

$$= 0.993 < 1.0 \quad \textbf{o.k.}$$

This check completes the weld design for the composite coupling beam-to-SpeedCore wall connection. The coupling beam flange plate is CJP groove welded to the wall web plate. The coupling beam web plate is welded to the wall web plate using a C-shaped ⅜-in. fillet weld.

Chapter 5
Seismic Performance Evaluation

The discussion presented in this chapter is based on the research results presented in Bruneau et al. (2019) and Agarwal et al. (2020). Seismic performance evaluation is performed on SpeedCore system designs to better understand the behavior of the system. This performance evaluation can range from nonlinear static to linear time history to nonlinear time history to incremental dynamic analysis. This chapter briefly discusses modeling SpeedCore systems for seismic performance evaluation, nonlinear static and dynamic analysis, and incremental dynamic analysis using procedures defined in FEMA P-695, *Quantification of Building Seismic Performance* (FEMA, 2009). The 8-story planar coupled example structure detailed in Chapter 4 is used for nonlinear static and dynamic analyses in Section 5.2. Incremental dynamic results for all four example structures from Chapters 3 and 4 are discussed in Section 5.3. The focus of this chapter is largely on coupled SpeedCore systems as the behavior of uncoupled SpeedCore systems is straightforward. The modeling and analysis techniques discussed herein would also be appropriate for both uncoupled and coupled systems.

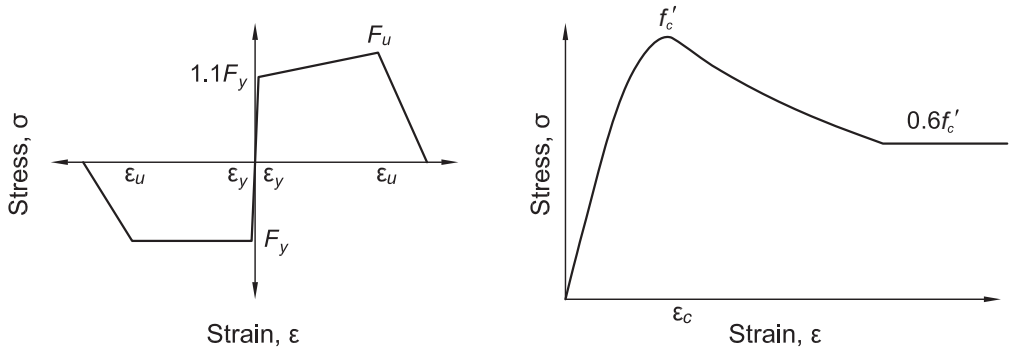
5.1 MODELING APPROACH

As discussed in Bruneau et al. (2019), SpeedCore walls and composite coupling beams can be modeled using different approaches. Three common approaches are 3D finite element analysis, 2D finite element analysis, and fiber analysis. While 3D finite element methods can directly capture complexities like local buckling, multiaxial stress states, and concrete confinement, this level of analysis is computationally expensive and often not needed to understand system level behavior. This section will focus on modeling both SpeedCore and coupled SpeedCore systems using 2D finite element and fiber analysis.

5.1.1 Material Models

A variety of nonlinear material models for steel and concrete components have been proposed for various applications. Any reasonable material models that capture existing experimental results for SpeedCore and composite coupling beam materials could be used. The material models used herein were derived from 3D finite element models and analyses (Shafaei et al., 2021a) of the experimental tests presented in Shafaei et al. (2021b) and Nie et al. (2014) on planar SpeedCore systems and composite coupling beams, respectively. To develop these material models, the SpeedCore walls and composite coupling beams were modeled in commercially available software and subjected to the loading protocol of the appropriate test. Effective stress-strain curves appropriate for 2D and finite element analysis were extracted from these models.

This enabled the 2D and fiber-based models to indirectly account for 3D phenomena like steel yielding, local buckling, biaxial stress states, steel fracture, concrete confinement, and composite action. The SpeedCore and composite coupling beam material models are shown in Figure 5-1 and Figure 5-2, respectively. The steel behavior is elastic-plastic in compression and elastic-plastic with strain hardening in tension. The model includes an increase in the tensile capacity of steel due to the biaxial stress state of the steel (Shafaei et al., 2021a). The concrete model follows the confined concrete model proposed by Tao et al. (2013). The unloading



(a) Steel *(b) Concrete*

Fig. 5-1. Material models for SpeedCore walls.

branch of this model is modified to account for the residual capacity of the concrete due to confinement after crushing. This confinement has a greater impact on the coupling beam behavior because the coupling beam concrete is confined to a higher degree. For further discussion on the development of these effective stress-strain curves, refer to Shafaei et al. (2021a).

In addition to nonlinear materials, fiber-based models can also use nonlinear hinge elements. The hinge serves as a representation of the behavior of the coupling beam plastic hinge without requiring discretization into individual fibers. Many concentrated hinges have been recommended in literature and implemented in finite element software. The hinge used herein was able to capture the backbone curve, account for cyclic deterioration, and be computationally stable. The behavior of the concentrated plasticity hinge was calibrated to fit the hysteretic behavior seen from trial coupling beam fiber models. A comparison between a distributed plasticity coupling beam model and a concentrated plasticity model is shown in Figure 5-3. After looking at 12 example structures, a generalized behavior for these hinges was extracted. This generalized hinge backbone curve is replicated in Figure 5-4.

When applicable, linear elements with effective properties can be assigned to elements not expected to undergo yielding. These elements include upper stories of the building and middle sections of coupling beams. Effective properties are assigned to these elements following the axial, shear, and flexural stiffness recommendations presented in Chapter 2 of this Design Guide.

5.1.2 2D Finite Element Model

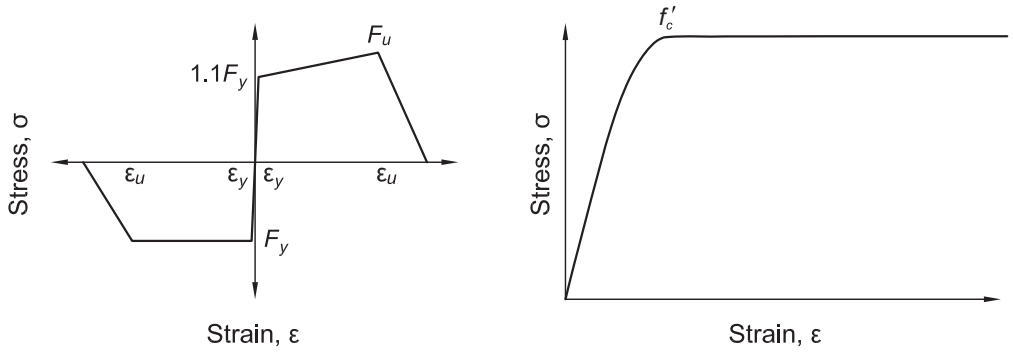
A 2D multi-story finite element model of coupled SpeedCore walls was modeled using the effective stress-strain curves presented in Section 5.1.1. Shear walls and coupling beams were modeled using four-node composite shell elements with reduced integration (S4R). Steel flange plates of shear walls and coupling beams were modeled using two-node truss elements (T3D2). Additionally, to consider P-Δ effects, the gravity frames are modeled with truss elements (T3D2). A typical 2D multi-story finite element model of coupled SpeedCore walls is shown in Figure 5-5.

5.1.3 Fiber-Based Model

For fiber-based models, additional simplifications were made. Elements outside of the expected plastic hinge zone in wall elements were assigned effective elastic properties. Coupling beam elements were assigned concentrated plasticity elements to capture all nonlinear behavior, and these elements were connected to elements with effective stiffness properties. This modeling approach was chosen because the concentrated plasticity elements are more computationally efficient than the distributed plasticity elements. A representation of this model is presented in Figure 5-6. The model also included an elastic P-delta column connected to the walls via rigid links (not depicted in the figure).

5.1.4 Analysis

Nonlinear pushover analysis was performed by applying monotonically increasing lateral loads until failure. This analysis used the load distribution established for equivalent lateral force analysis as presented in ASCE/SEI 7. Uncoupled systems are expected to form a plastic hinge at the base of each wall; coupled systems are expected to have the characteristic pushover curve presented in Chapter 4 of this Design Guide. Time-history analysis was performed for design basis, maximum considered, and failure level earthquakes. Earthquake records were scaled to these hazard levels. The results from these nonlinear pushover and time-history analyses for an 8-story structure are presented in the following section.



(a) Steel *(b) Concrete*

Fig. 5-2 Material models for composite coupling beams.

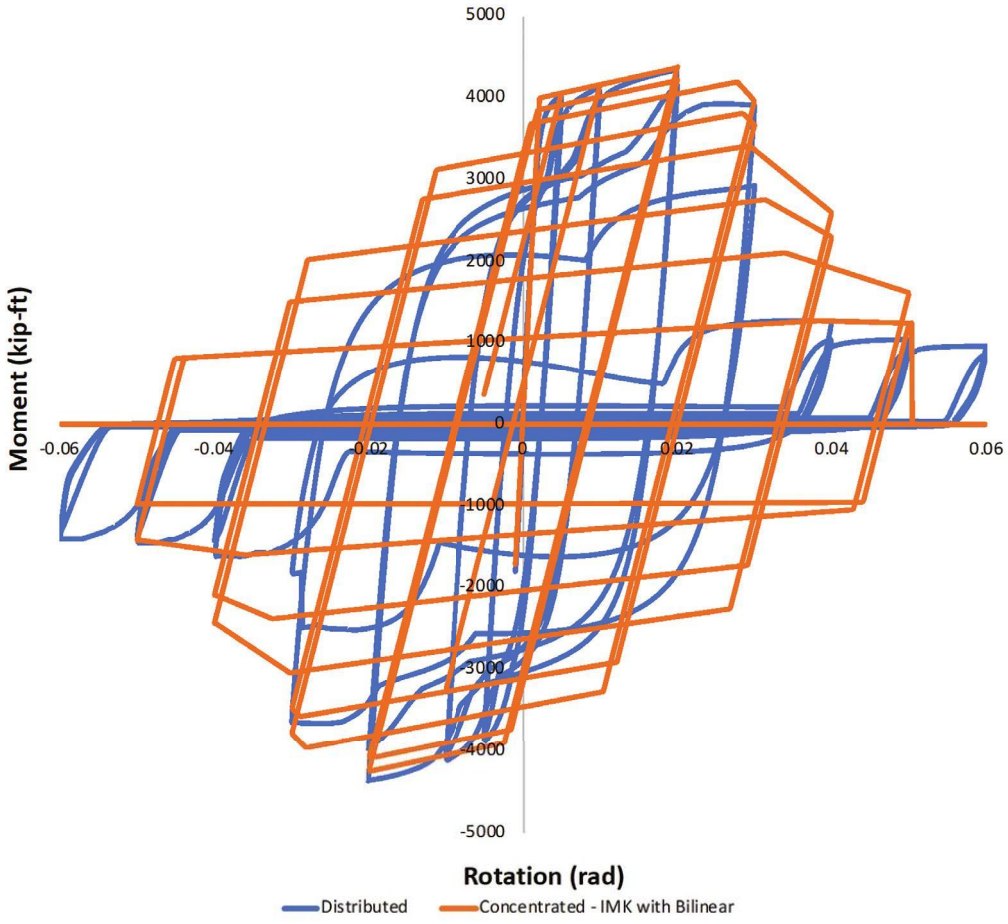


Fig. 5-3. Moment versus rotation behavior for coupling beams.

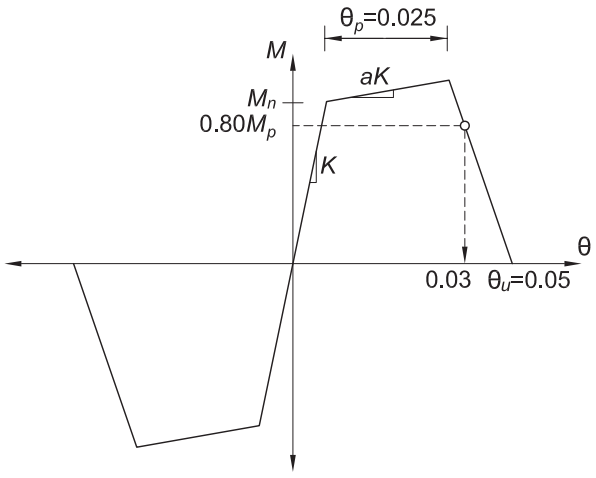


Fig. 5-4. Concentrated plasticity generalized hinge backbone curve.

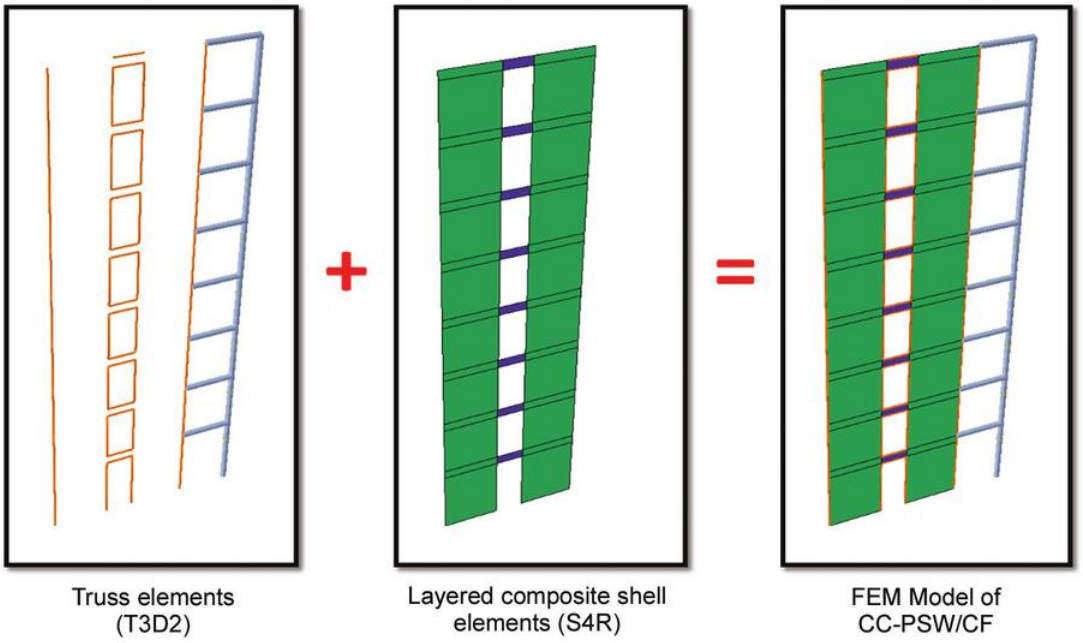


Fig. 5-5. 2D multi-story finite element model of coupled SpeedCore walls.

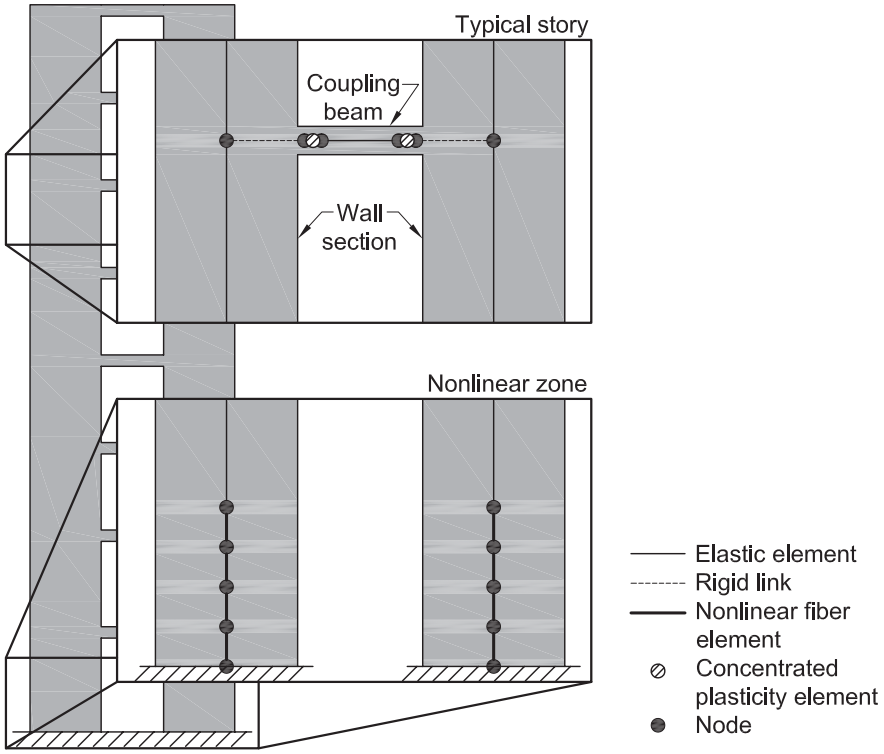


Fig. 5-6. Depiction of element and node distribution in fiber models.

Incremental dynamic analyses were performed following the recommendations of FEMA P-695, which outlines a suite of 22 ground motions (44 components). This set includes diverse earthquakes from across the world with differing intensity, peak ground velocity, and peak ground acceleration. To normalize these ground motions, their peak ground velocity is used. These normalized ground motions can then be scaled until the model structure reaches failure. Please refer to FEMA P-695 for a full list of ground motions and further details of ground motion scaling (FEMA, 2009).

5.2 SEISMIC PERFORMANCE OF COUPLED SPEEDCORE SYSTEMS

This section describes the seismic performance of the 8-story design example from Section 4.3. The geometry of these coupled walls is depicted in Figure 5-7.

5.2.1 Nonlinear Pushover Analysis

The coupled SpeedCore system is proportioned such that the general order of events is yielding of coupling beams, followed by the formation of plastic hinges in the coupling beams, yielding of the walls, and formation of a plastic hinge at the base of the walls. Using the material models discussed previously, a fiber-based model was developed to analyze the nonlinear pushover behavior of the 8-story coupled wall system and track the occurrence of events or milestones. This model was subjected to a loading profile proportional to the equivalent lateral force analysis loads prescribed in ASCE/SEI 7. This load was increased monotonically resulting in the pushover base shear-roof displacement curve shown in Figure 5-8(a). The occurrence of various events was identified by post-processing the analysis results and marked on the response.

Following the progression of events shown in Figure 5-8(a), the structure reached equivalent lateral force (ELF) level loads, initiation of yielding in the coupling beam, yielding of all coupling beams, yielding at the base of the walls, and coupling beam fracture. From this analysis, an analytical γ_1 factor can be derived as the ratio of the base shear corresponding to the initiation of wall yielding to the initiation of coupling beam yielding. This ratio is 1.6. The γ_1 value calculated using Equation 4-1 is 1.9, which is conservative for design.

Figure 5-8(b) compares the moment contributions from the tension and compression walls and the axial couple moments with the total overturning moment. The axial couple moment initially contributes over 60% of the total

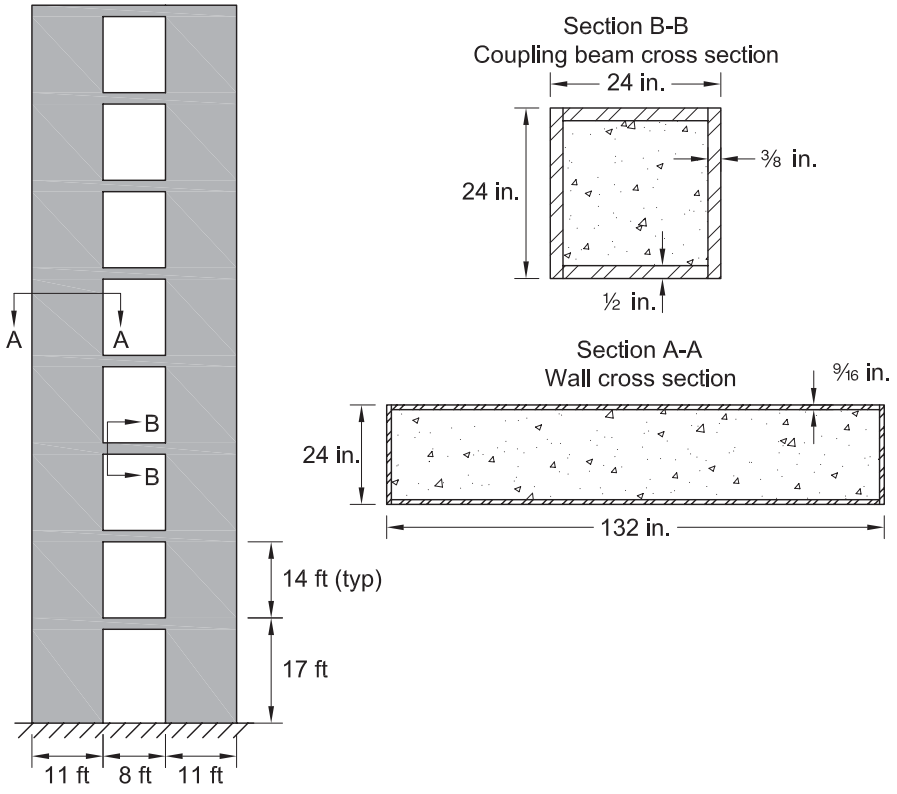
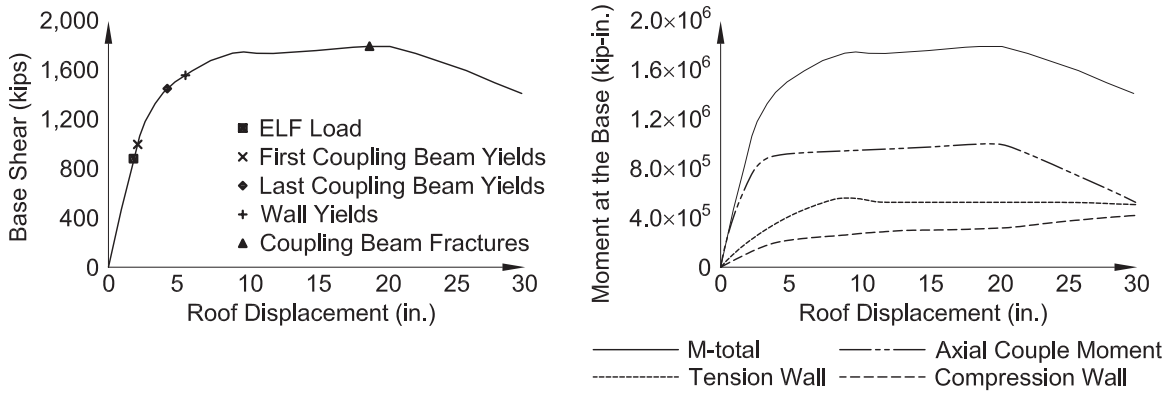
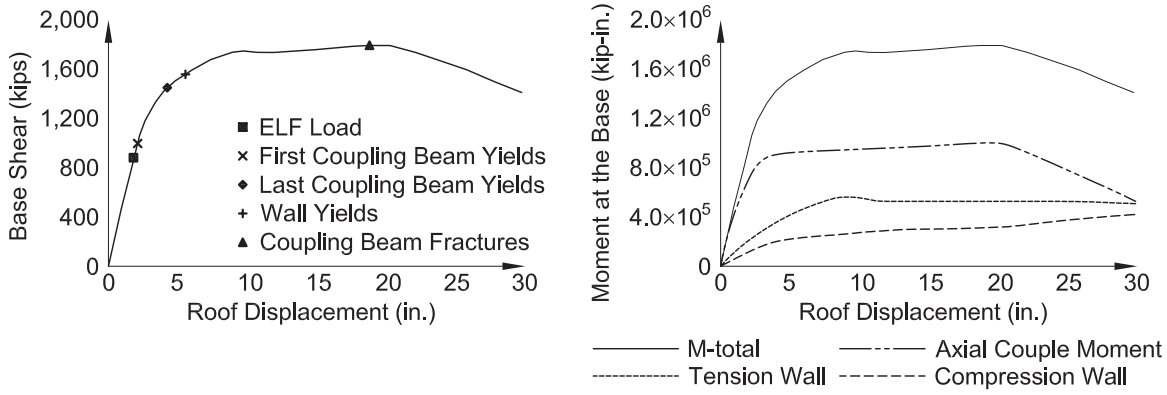


Fig. 5-7. Structure geometry.



(a) Base shear versus roof displacement



(b) Moment versus roof displacement

Fig. 5-8. Pushover results for the 8-story structure.

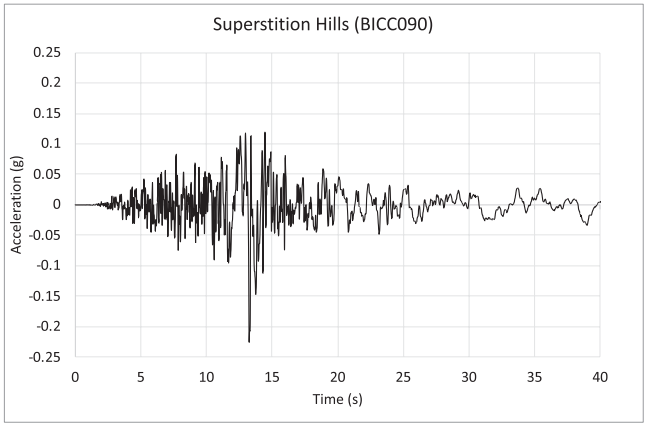
moment, but levels out at approximately 50% of the total moment contribution, then drops as the coupling beams begin to fracture. As expected, the compression wall carries a higher portion of the moment in the individual walls due to its higher effective stiffness. The compression and tension wall contributions approach each other as the coupling beams fracture and the walls begin to act as two independent systems.

5.2.2 Nonlinear Time-History Analysis

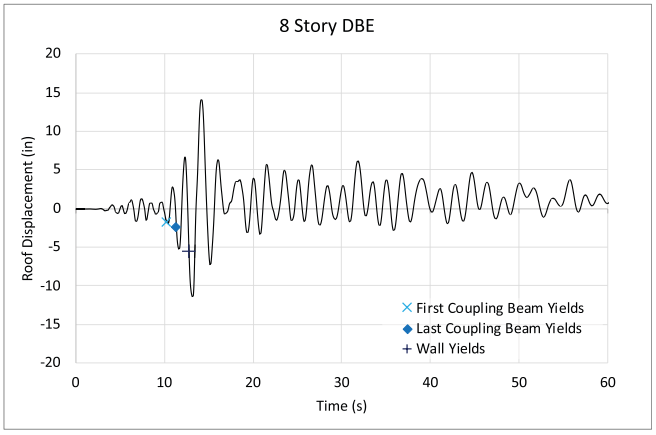
Nonlinear time-history analysis was performed using the fiber model described previously. Analyses were conducted for design basis, maximum considered, and failure level (5% interstory drift) earthquakes. The 1987 Superstition Hills ground motion (PEER NGA SUPERST/B-ICC090) was amplified to these hazard levels after normalizing the records by their peak ground velocity. The performance milestones expected in the nonlinear time-history analysis match those observed in the nonlinear pushover analysis.

Figure 5-9 shows the ground motion record and the time-history responses of the roof displacement for design basis, maximum considered, and failure level (5% interstory drift) earthquakes. As shown in the figure, the general order of events—namely, coupling beam yielding, wall yielding, and coupling beam fracture—remained relatively unchanged for the design basis, maximum considered, and failure level (5% interstory) earthquakes. For the design basis earthquake, only yielding occurred in the coupling beam and wall elements. The maximum considered earthquake resulted in fracturing of a coupling beam. For the failure level (5% interstory drift) earthquake, all coupling beams failed. For this hazard level, a distinct change (elongation) in the structural period is observed after fracturing all coupling beams.

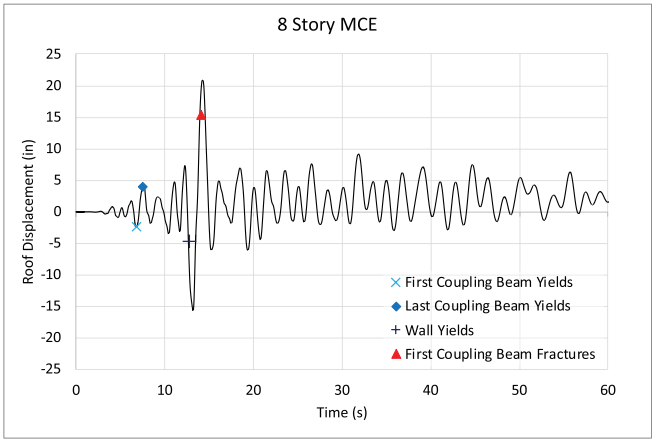
To better illustrate these events, the 2D finite element model mentioned earlier was analyzed and post-processed.



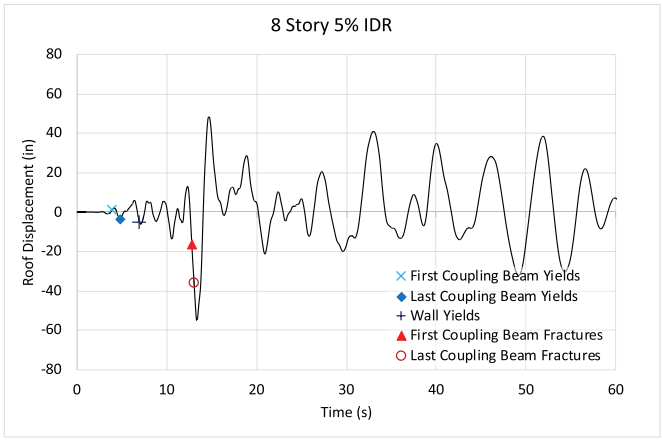
(a) Superstition Hills acceleration versus time ground motion



(b) Design basis earthquake response



(c) Maximum considered earthquake response



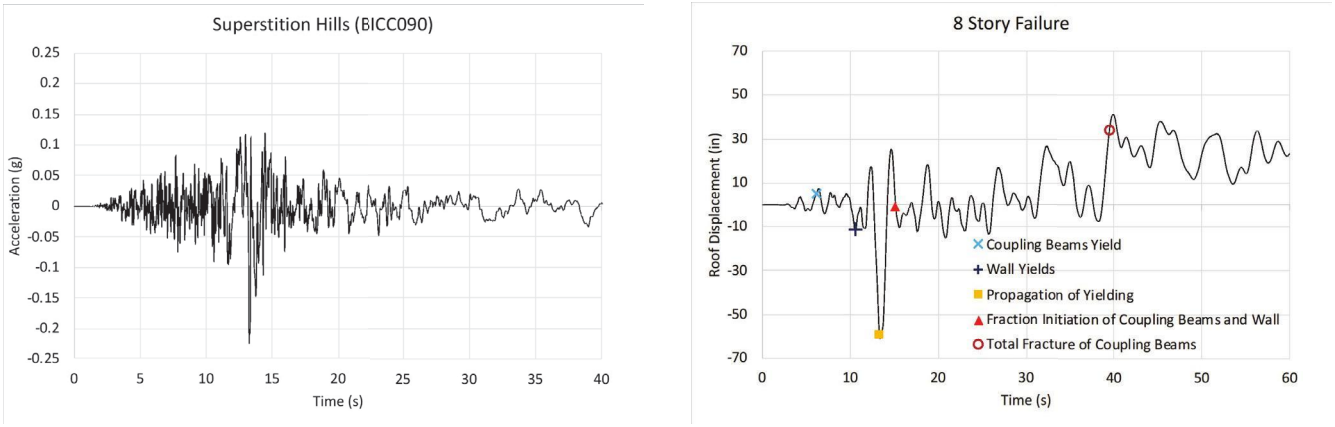
(d) Failure level earthquakes response

Fig. 5-9. Fiber analysis results for the 8-story structure.

The response of this structure to the failure level (5% inter-story drift) earthquake is shown in Figure 5-10. These results are further post-processed to highlight the events as shown in Figure 5-11. These illustrations represent the equivalent plastic strain (PEEQ) in the steel at the points indicated in Figure 5-10(b). The red shading indicates the strain exceeds the yield strain and the black shading indicates the strain exceeds the fracture criteria.

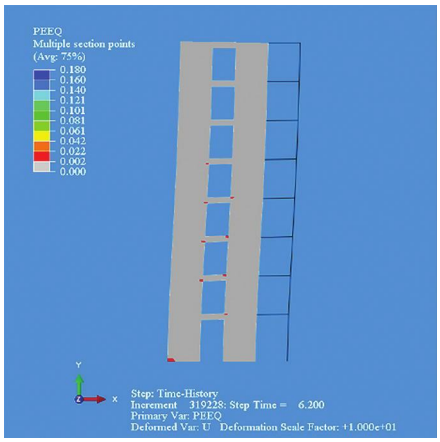
5.2.3 Summary

This example looked at the nonlinear static pushover and nonlinear time history response of an 8-story coupled SpeedCore wall structure. The structure was designed following the recommendation of Chapter 4 and the resulting behavior followed anticipated events of coupling beam yielding followed by wall yielding.

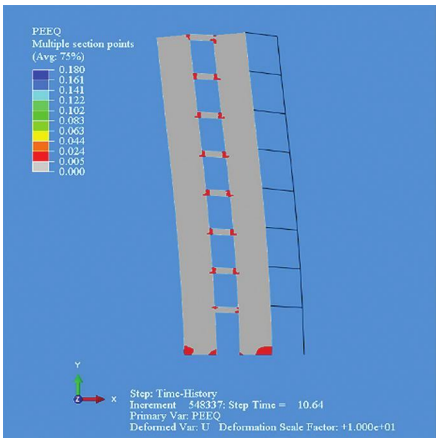


(a) Acceleration vs. time *(b) Roof displacement vs. time*

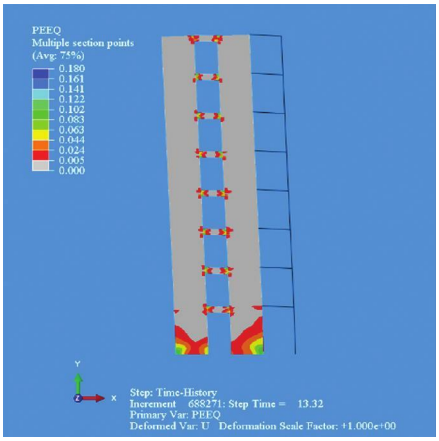
Fig. 5-10. 2D finite element results for the 8-story structure subjected to the failure level earthquake.



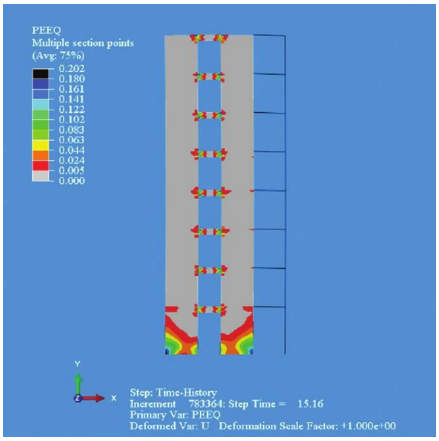
(a) Coupling beams yield



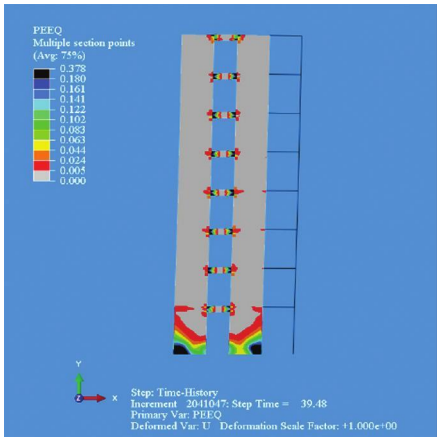
(b) Wall yields



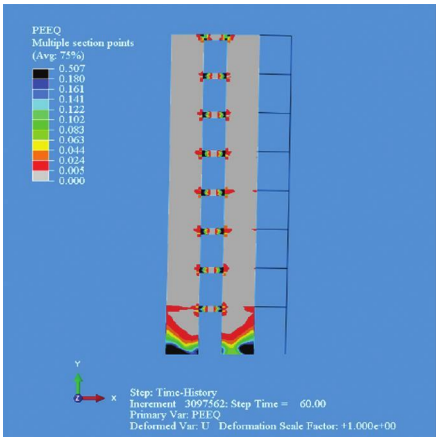
(c) Propagation of yielding



(d) Fracture initiation of coupling beams and wall



(e) Total fracture of coupling beams



(f) End of ground motion

Fig. 5-11. Milestones observed in failure level finite element analysis of the 8-story structure.

5.3 SEISMIC PERFORMANCE EVALUATION—FEMA P-695 ANALYSIS

FEMA P-695 specifies the methodology for performing nonlinear analyses to quantify seismic performance of lateral force-resisting systems. This document outlines the procedure to verify or confirm the seismic performance of systems designed with the assumed response modification, R, displacement amplification, C_d, and overstrength, Ω_0, factors. These parameters are of interest because they are part of the ASCE/SEI 7 approach for estimating equivalent linear elastic loads for expected seismic loading scenarios. Design-level earthquakes can cause buildings to undergo nonlinear deformations, but using R, C_d, and Ω_0 factors allows engineers to consider the inherent system ductility and design for less severe loads without having to develop or benchmark nonlinear models. Instead, these models are developed and verified for several scenarios to ensure that the resulting design factors are appropriate for general use.

This process includes (1) developing a concept for the system; (2) collecting design inputs such as design criteria, seismic coefficient, and system nonlinear behavior; (3) designing archetype structures covering the design space of interest; (4) developing and benchmarking numerical models; (5) conducting incremental dynamic analysis for a set of ground motions; and (6) analyzing the results. FEMA P-695 goes into extensive detail for the requirements of each of these components.

Design examples introduced in Chapter 3 (uncoupled SpeedCore systems) and Chapter 4 (coupled SpeedCore systems) of this Design Guide are evaluated according to the FEMA P-695 analysis procedure. For further information on the seismic performance evaluation of these and other structures refer to Bruneau et al. (2019) and Agrawal et al. (2020).

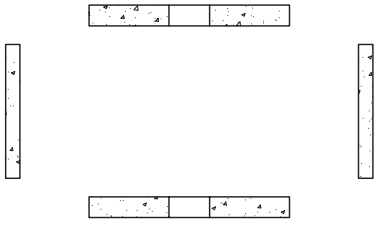
5.3.1 Structure Designs

Coupled and uncoupled structures were designed according to the provisions of Chapters 3 and 4. These structures were intended to represent typical configurations and loadings of mid- to high-rise buildings. Two wall configurations were used—planar and C-shaped—as shown in Figure 5-12. Depending on the direction of consideration, these types are either coupled or uncoupled (uncoupled in the north-south direction, coupled in the east-west direction).

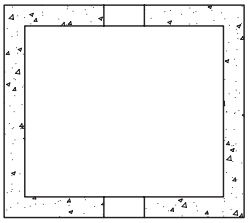
The member sizes of the example structures designed in Chapters 3 and 4 are replicated in Tables 5-1 and 5-2. These dimensions are depicted on planar and C-shaped walls in Figure 5-13.

5.3.2 Incremental Dynamic Analysis

Nonlinear analysis for 44 records (22 ground motions with two components each) was performed on each example structure. This analysis is performed so that earthquake intensity—in this case, spectral acceleration at collapse for each ground motion—can be recorded. From this set of failure spectral acceleration levels, the value corresponding to the median collapse is identified and normalized by the design spectral acceleration for the structure. This value is called the collapse margin ratio, CMR. This value is then adjusted based on confidence in the modeling approach, design parameters, and analysis methods and aptly renamed



(a) Type I Coupled and uncoupled walls

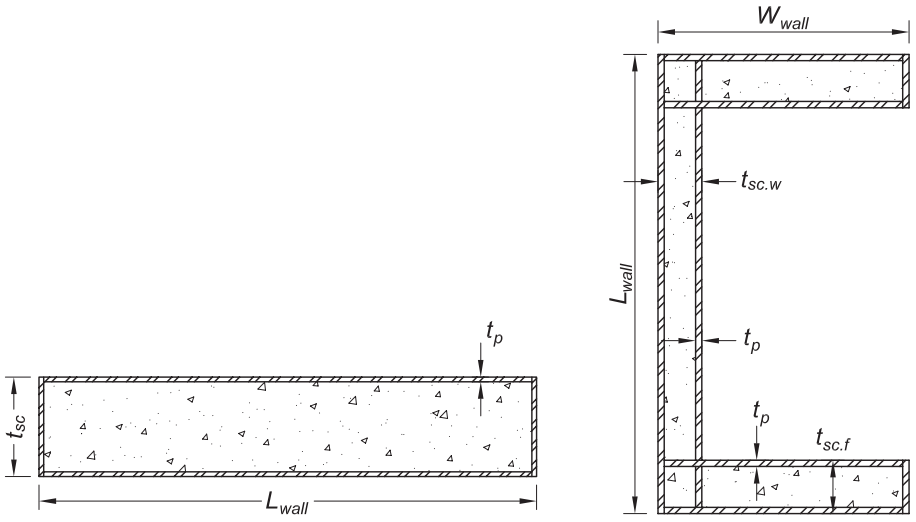


(b) Type II C-shaped coupled walls

Fig. 5-12. Basic configuration Type I and II.

Table 5-1. Planar Wall Geometry						
Structure	***R* Factor**	**Coupled Wall Length, L_{wall} (in.)**	**Wall Thickness, t_{sc} (in.)**	**Plate Thickness, t_p (in.)**	**Coupling Beam Length (in.)**	**Coupling Beam Section**
6-story, uncoupled	6.5	300	16	5/16	N/A	N/A
8-story, coupled	8	132	24	9/16	96	24×24×1/2(f), 3/8(w)

Table 5-2. C-Shaped Wall Geometry								
Structure	***R* Factor**	**Wall Depth, L_{wall} (in.)**	**Flange Length, h_w (in.)**	**$t_{sc.f}$ (in.)**	**$t_{sc.w}$ (in.)**	**t_p (in.)**	**Coupling Beam Length (in.)**	**Coupling Beam Section (in.)**
18-story, uncoupled	6.5	480	120	18	18	1/2	N/A	N/A
22-story, coupled	8	360	192	24	14	1/2	96	24×24×7/16(f), 3/8(w)



(a) Planar wall *(b) C-shaped walls*

Fig. 5-13. Labeled dimensions for planar and C-shaped walls.

Table 5-3. CMR and ACMR Values					
Structure	Period, s	Overstrength Factor, Ω_0	CMR	ACMR	FEMA ACMR Threshold Criteria
6-story, uncoupled	1.03	1.69	2.12	3.08	1.56
8-story, coupled	1.08	2.00	2.45	3.64	
18-story, uncoupled	2.49	2.66	2.82	4.53	
22-story, coupled	3.31	2.00	1.98	2.84	

the adjusted collapse margin ratio, ACMR. This ACMR value is then compared to limits prescribed in FEMA P-695. For further details on this process, please refer to FEMA P-695 (2009).

This process is shown for the 8-story coupled structure in Figure 5-14. Hundreds of analyses were performed to develop this plot and ultimately extract one number—the ACMR. The design level earthquake spectral acceleration for this structure was 0.84g, while the median collapse spectral acceleration was 2.05g. This resulted in a CMR of 2.45. This number is then adjusted by a factor accounting for the level of uncertainty in the process; in this case, the factor was 1.3. This adjustment leads to an ACMR of 3.10. The ACMR is then compared to the limit defined by FEMA, 1.56. Because the ACMR from the analysis is greater than this limit, the assumptions made to design the structure—namely, R, C_d, and Ω_0—are adequate for this design. This process is repeated for other structures and the limits are similarly checked.

5.3.3 Summary of Results

A summary of important results for the four example structures in Chapters 3 and 4 is shown in Table 5-3. All structures pass the required ACMR threshold, as seen by comparing the ACMR and the FEMA ACMR Threshold Criteria columns. Passing this criterion indicates that the seismic design values used were appropriate for the nonlinear behavior. This analysis is extensively detailed in Bruneau et al. (2019) and Agrawal et al. (2020) for several more coupled and uncoupled SpeedCore systems, respectively.

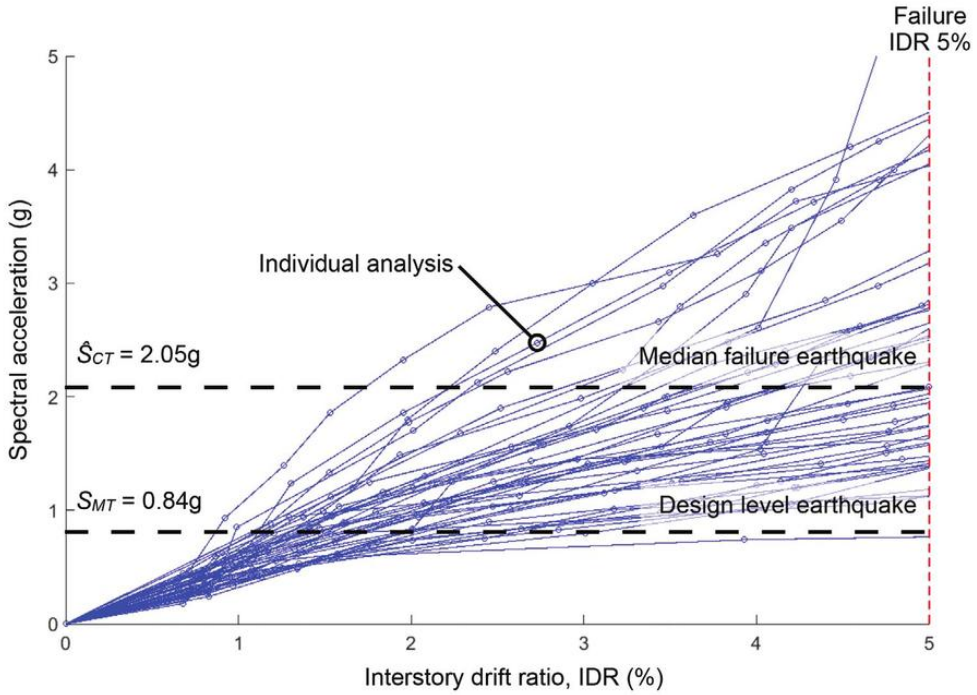


Fig. 5-14. Annotated incremental dynamic analysis results.

Chapter 6
Fire Design of SpeedCore Systems

6.1 PERFORMANCE OF SPEEDCORE SYSTEMS UNDER FIRE LOADING

6.1.1 General

An important aspect of SpeedCore design for commercial construction is performance under fire loading. SpeedCore systems, like other structural systems, must have the capability to endure fire loading during a fire incident. The system should have sufficient fire endurance to resist the applied mechanical and thermal loads to provide sufficient time for the occupants to evacuate the building and extinguish the fire before failure or collapse. Because steel plates (faceplates) are on the exterior surface of the walls, they are directly exposed to elevated temperatures from fire in the absence of fire protection. Fire loading results in elevated temperatures in the steel and concrete components and causes nonlinear thermal gradients through the cross section of the wall. Elevated temperatures result in the degradation of the mechanical properties of steel and concrete that may cause local or global instability. Elevated temperatures in SpeedCore walls can result in the collapse of walls at gravity load magnitudes lower than the axial capacity of the walls at ambient temperatures. The walls can be designed to provide the required fire resistance rating under fire loading based on the guidelines provided in this chapter.

In a research study performed by Anvari et al. (2020a), the results from prior experimental investigations were compiled, and five additional fire tests were conducted to address knowledge gaps in the performance of SpeedCore walls under fire loading. The tests were conducted on laboratory-scale specimens subjected to axial loading and simulated standard fire loading (heating). During the fire tests, the applied axial load to the wall specimens was kept constant until the tests were terminated. The parameters considered in the fire tests were axial loading magnitude (20–28% of section concrete strength, $A_c f_c'$); steel plate slenderness (tie spacing/plate thickness in the range of 24–48); maximum steel surface temperature (1,427–1,909°F); and fire scenario (uniform and nonuniform heating), where A_c is the area of concrete in the composite cross section, A_s is the area of steel in the composite cross section, and f_c' is the specified compressive strength of concrete.

Anvari et al. (2020a) simulated SpeedCore walls under fire loading using two independent numerical methods—namely, finite element analysis (FEA) and fiber-based analysis. The numerical models were benchmarked using test results, and the benchmarked models were used to conduct analytical parametric studies to expand the database. The temperature-dependent thermal and mechanical properties recommended by Eurocode standards were using in the analysis. The parameters considered were the wall thickness (8–24 in.), wall slenderness (story height/thickness ratio, L/t_{sc}, in the range of 5–20), axial loading ($P_u \leq 30\%$ of section concrete strength, $A_c f_c'$), fire scenario (uniform and nonuniform heating), boundary conditions (pinned and fixed), steel plate reinforcement ratio (area of steel plates/wall cross-section area, A_s/A_g in the range of 1.3–5.3%), steel plate slenderness (tie or stud spacing/plate thickness ratio, S/t_p, varying from 20 to 75), tie bar spacing to wall thickness ratio ($S_{tie}/t_{sc} = 0.5$–1.0), and concrete compressive strength ($f_c' = 5.8$–8.0 ksi).

The research results reported by Anvari et al. (2020a) were used to develop the SpeedCore fire resistance design provisions and methods presented in this chapter. These provisions and methods conform to the requirements of *International Building Code*, Section 703.3 (ICC, 2018), with particular reference to method 5—alternative protection methods allowed by Section 104.11 of the code. This chapter is focused on determining the load bearing capacity under fire loading and the fire resistance rating of SpeedCore systems. The post-fire residual capacity of the system to lateral loading was not evaluated. This post-fire residual capacity depends on the duration of the fire incident, permanent deformations and damage after exposure, etc. Additional research may be needed to evaluate the post-fire performance of the SpeedCore system under lateral loading.

6.1.2 Standard Time-Temperature (Fire Loading)

During fire tests and numerical studies (Anvari et al., 2020a), the temperature of the gas around the specimens (or models) followed the ISO 834 standard time-temperature curve. Due to losses associated with heat conduction, convection, and radiation, the surface temperatures of members are much lower than the gas temperature. In Figure 6-1, a comparison of measured surface temperature for a SpeedCore wall with the applied time-temperature fire curve is shown.

6.1.3 Failure Criteria

ISO 834 (ISO, 1999) thermal and structural failure criteria were investigated in both experimental and analytical studies. Walls were considered failed when the axial shortening of walls exceeded $L/100$, where L is the height of walls. For single-sided fire scenarios, walls were considered failed when the average temperature increase on the unexposed

surface of the wall exceeded 252°F above the initial temperature or the temperature increase at any point on the unexposed surface exceeded 324°F above the initial temperature.

6.1.4 Thermal Response of SpeedCore Systems under Fire Loading

The analytical parametric studies showed that the faceplate thickness variation leads to minor differences in the surface temperature. A comparison of surface temperatures with various faceplate thicknesses is illustrated in Figure 6-2. The surface temperature of walls with thicker faceplates was slightly lower than the thinner faceplates. Surface temperatures diverged at the early stages of heating. The surface temperature difference for 3⁄16 in. and ½ in. thick faceplates reached about 338°F at 35 minutes. Because the rate of temperature rise decreases in the standard fire time-temperature curve, surface temperatures converged after about 80 minutes of heating.

The temperature profiles through the thickness of walls with a thickness of 12 in. and 24 in. at different time points are plotted in Figure 6-3. The walls experienced a nonlinear thermal gradient through the wall thickness. Low thermal conductivity of concrete resulted in lower temperatures in the middle of the wall. The temperature profile through wall thickness is symmetric about the walls' mid-thickness. The comparison of the temperature profiles (Figure 6-3)

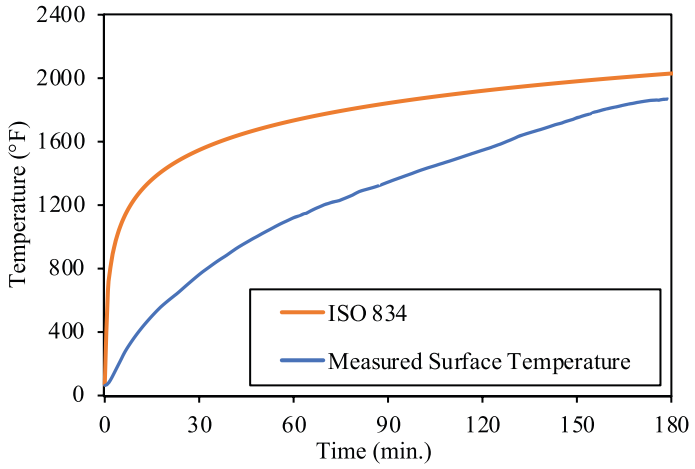


Fig. 6-1. Comparison of surface temperature and ISO 834 time-temperature curve.

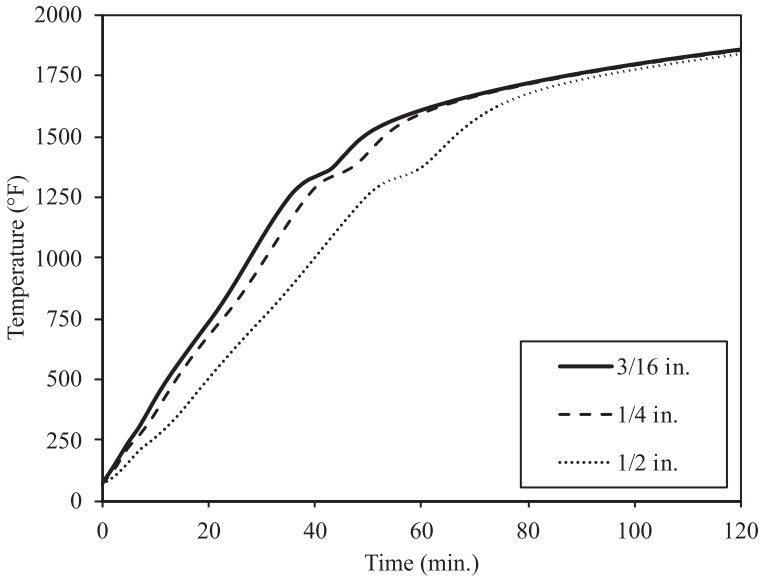


Fig. 6-2. Comparison of the surface temperature of SpeedCore walls with different faceplate thicknesses.

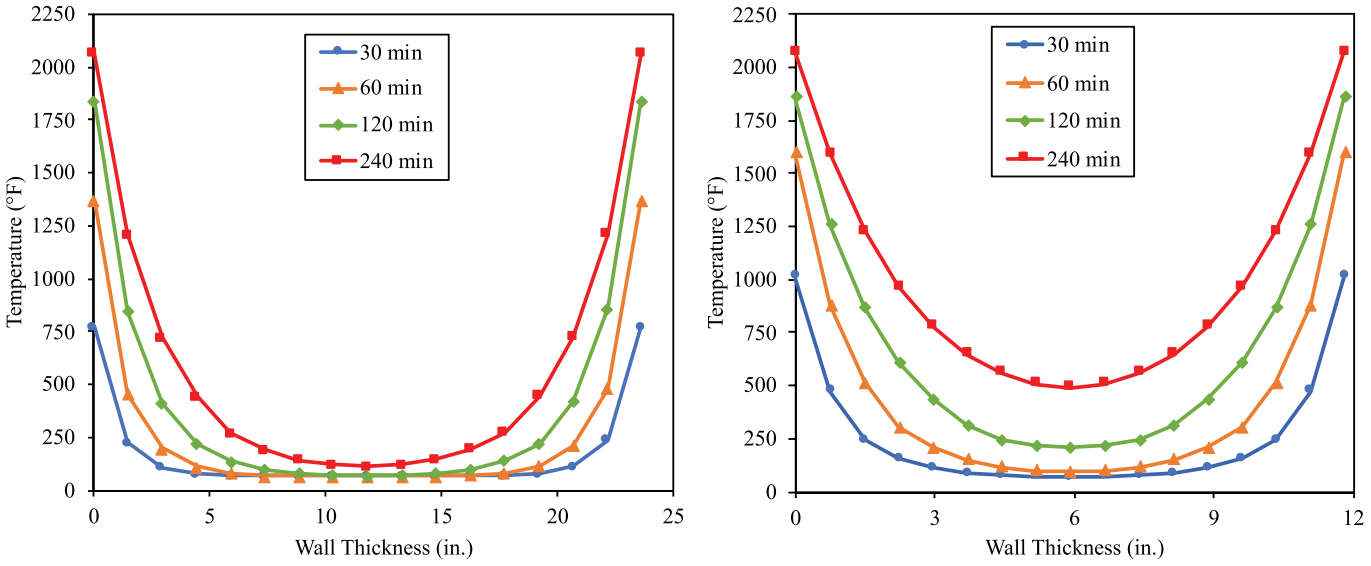
shows that temperature in the middle of the wall is a function of the wall thickness. After 4 hr of heating, the mid-thickness temperature of the 12 in. and 24 in. thick walls achieved a temperature of 492°F and 111°F with similar surface temperatures, respectively. The cooler parts of concrete in the middle of the walls improve the axial compressive capacity of SpeedCore walls during a fire incident. Due to the heat sink effect of concrete, the thermal bridging effect of the tie bars was negligible. No major temperature increase was observed at the center of the wall thickness due to the thermal bridging effect of the tie bars.

6.1.5 Structural Response of SpeedCore Walls under Fire Loading

The axial displacement of SpeedCore walls followed a general trend under fire loading in the experimental and numerical studies, as shown in Figure 6-4. In Figure 6-4, point A represents the beginning of the heating. At early steps of fire exposure (AB), the walls experienced a thermal expansion. Faceplates were exposed to fire, and they expanded due to the temperature increase. The thermal expansion length depended on the wall height; taller walls expanded more than shorter walls. The load-bearing capacity of faceplates was reduced due to the local buckling in faceplates and degradation of material strength and stiffness at elevated temperatures. Thus, a large portion of the axial load was sustained by the concrete infill. Axial shortening due to degradation of concrete mechanical properties overcame the thermal expansion (BC). The shortening of the walls continued at a slower rate (CD) due to the temperature increase of the concrete infill. Due to temperature increase, the shortening happened at a quicker rate (DE). Finally, a rapid wall shortening occurred that resulted in the wall failure (EF).

The results from the analytical parametric studies (Anvari et al., 2020a, 2020b) indicated that wall slenderness ratio (story height/wall thickness), wall thickness, applied load ratio, and end conditions have a significant influence on the fire resistance of the walls. Local buckling of faceplates between tie bars was observed, but local buckling did not result in significant degradation of the structural performance of the walls. Higher wall slenderness ratios and load ratios had a detrimental effect on the fire resistance of SpeedCore walls. Walls with higher wall slenderness ratios failed due to global buckling. In thicker walls, the lower temperatures in the middle regions of the concrete core helped to maintain the axial load-bearing capacity of walls at elevated temperatures. Limiting the plate slenderness ratio can slightly improve the fire resistance of unprotected walls.

The results from the analytical parametric studies were simplified into Equation 6-1 for the design of SpeedCore walls under fire loading. The load-bearing capacity of the walls can be estimated at elevated temperatures by using Equation 6-1. This equation is appropriate for unprotected walls with the range of parameters described in the parametric studies. The wall story height-to-wall thickness ratio, L/t_{sc}, has a limiting value of 20, which is quite large and generally beyond the practical range. Walls with slenderness ratios greater than 20 should be fire protected, at least on the side exposed to fire. The expansion of the exposed face of SpeedCore walls imposes moments on the wall cross section in nonuniform fire scenarios. This causes early failure



(a) 24-in.-thick wall

(b) 12-in.-thick wall

Fig. 6-3. Comparison of the through-wall thickness temperature.

of walls with wall slenderness ratios greater than 20 (Anvari et al., 2021).

The total (linear) width of a SpeedCore wall can be discretized into unit width columns, where each column's width is equal to the tie bar spacing. S_{tie}. Thus, each unit width is like a column with steel plates on the surfaces, concrete infill, and tie bars distributed uniformly along the height. The temperature profile through the wall thickness can be calculated by discretizing the section into fibers (or elements). Since the temperature of the elements is uniform along the height and width of walls, 1D thermal analysis (through wall thickness) can be performed using heat transfer equations or the fiber tool developed in the study (presented later in Equations 6-1 to 6-4) can be utilized to calculate the axial load capacity of a unit width column of the wall at elevated temperature (Anvari et al., 2021). The axial load capacity of the wall can be estimated conservatively as the axial load capacity of the unit width wall multiplied by the linear width of the wall divided by unit width (tie bar spacing). Equation 6-5 can be used to conservatively estimate the fire resistance rating (in hours) of unprotected SpeedCore walls exposed to the ISO-834 standard fire time-temperature curve.

6.1.6 Vent Holes

Anvari et al. (2020a) recommended the use of vent holes to relieve the buildup of vapor (or steam) pressure between the steel plates and concrete due to the evaporation of water from concrete drying at elevated temperatures. They developed a rational method to design vent holes for SpeedCore walls, which depends on the allowable pressure, concrete moisture content, vent hole spacing, and thermal gradient through the wall thickness. The vent hole size should be at least 1 in. diameter and the vent hole spacing should not be larger than the story height or 12 ft in the horizontal and vertical directions. The water-to-cement ratio of the concrete mix design has a minor influence on the pressure buildup but can be controlled to reduce the buildup pressure due to concrete water evaporation.

6.2 DESIGN REQUIREMENTS FOR SPEEDCORE WALLS UNDER FIRE LOADING

6.2.1 Temperatures under Fire Conditions

Temperatures within structural members, components, and frames due to the heating conditions posed by the design-basis fire can be determined by a heat transfer analysis. In the case of SpeedCore walls with uniform heating along the width, a 1D analysis can be conducted per unit width of the wall. A 2D or 3D analysis may be required for composite plate shear wall systems with nonuniform heating or with special configurations or boundary conditions.

6.2.2 Design for Compression

According to AISC *Specification* Appendix 4, Section 4.2.4d, the nominal compression strength for concrete-filled composite plate shear walls can be determined using the provisions of AISC *Specification* Section I2.3 with steel and concrete properties as stipulated in AISC *Specification* Appendix 4.2.3b and Equation 6-1 to calculate the nominal compressive strength for flexural buckling at elevated temperatures:

$$P_n(T) = \left\{ 0.32^{\left[\frac{P_{no}(T)}{P_e(T)}\right]^{0.3}} \right\} P_{no}(T) \qquad (6\text{-}1)$$

where $P_{no}(T)$ is calculated at elevated temperature using Equation 6-2, and $P_e(T)$ is calculated at elevated temperature using Equation 6-4. $F_y(T)$, $f'_c(T_c)$, $E_s(T_c)$, and $E_c(T_c)$

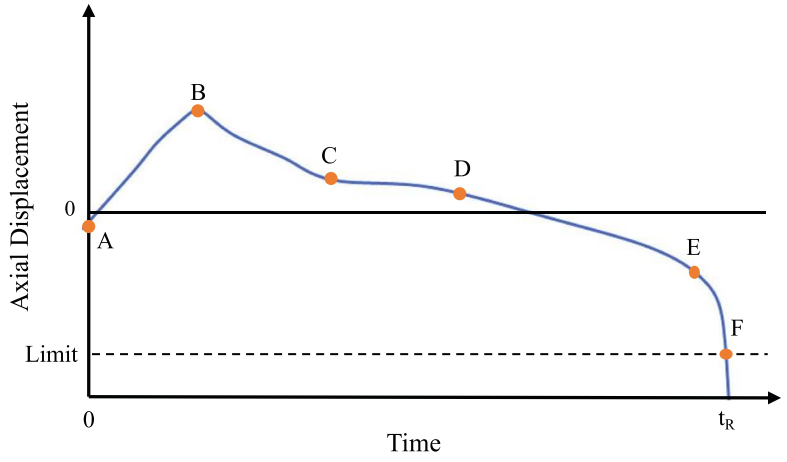


Fig. 6-4. Overall axial displacement of SpeedCore wall under fire loading.

are obtained using coefficients from AISC *Specification* Tables A-4.2.1 and A-4.2.2.

$$P_{no}(T) = A_s F_y(T) + \sum_{i=elements} 0.85 f_c'(T_i) A_{ci} \qquad (6\text{-}2)$$

$$EI_{eff}(T) = E_s(T) I_s + 0.35 \sum_{i=elements} E_c(T_i) I_{ci} \qquad (6\text{-}3)$$

$$P_e(T) = \frac{\pi^2 EI_{eff}(T)}{L_c^2} \qquad (6\text{-}4)$$

where

A_{ci} = area of a concrete element, in.^{2}

A_s = area of a steel element, in.^{2}

$E_c(T)$ = modulus of elasticity of concrete at elevated temperatures, ksi

$E_s(T)$ = modulus of elasticity of steel at elevated temperatures, ksi

$F_y(T)$ = specified minimum yield stress of steel at elevated temperatures, ksi

I_{ci} = moment of inertia of a concrete element about the elastic neutral axis of the composite section, in.^{4}

I_s = moment of inertia of steel section about the elastic neutral axis of the composite section, in.^{4}

L_c = effective length of the member, in.

$f_c'(T)$ = specified compressive strength of concrete at elevated temperatures, ksi

For composite members, the steel temperature is determined using heat transfer equations with heat input corresponding to the design-basis fire. The temperature distribution in concrete infill can be calculated using 1D or 2D heat transfer equations. The regions of concrete infill will have varying temperatures and mechanical properties. The concrete contribution to axial strength and effective stiffness can, therefore, be calculated by discretizing the cross section into smaller elements, with each concrete element considered to have a uniform temperature and summing up the contribution of individual elements.

Simple methods may suffice when a structural member or component can be assumed to be subjected to uniform heat flux on all sides and the assumption of a uniform temperature is reasonable as, for example, in a free-standing column surrounded by fire. For composite shear walls, the simplified analyses can be conducted per unit width of the wall and 1D heat-transfer equations can be used to model the thermal response. The unit width method for shear walls is conservative and can be used for different configurations of the walls, including planar walls and C-shaped walls (Anvari et al., 2020b).

The equations for compression strength of composite plate shear walls at elevated temperatures have been developed based on parametric studies conducted by Anvari et al. (2020b) using 3D finite element models validated against experimental data. Figure 6-5 shows the comparison of Equation 6-1 with results from finite element analyses. The data plotted includes both complete planar walls and their corresponding unit width columns (with no flange plates). Equation 6-1 provides a lower-bound estimate of composite shear wall compression strength at elevated temperatures.

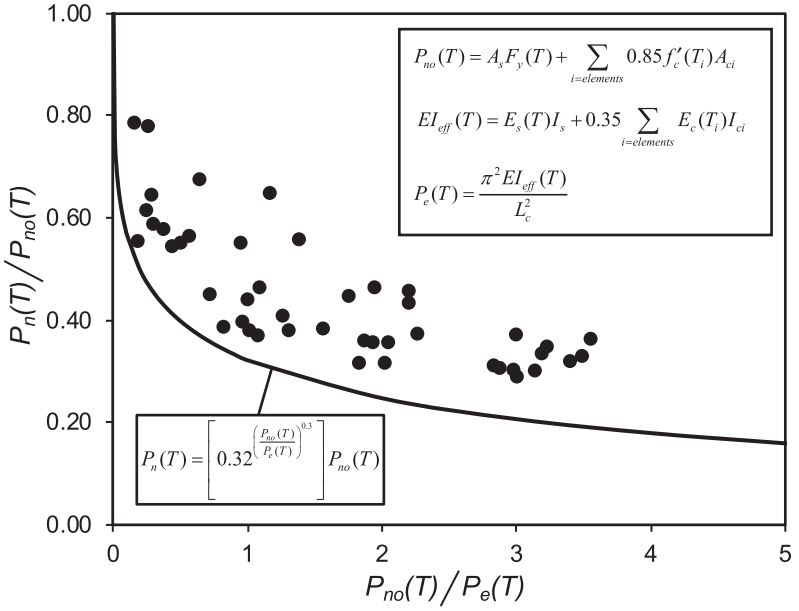


Fig. 6-5. Comparison of compression strength of SpeedCore walls with Equation 6-1.

6.2.3 Fire Resistance Rating

For unprotected composite plate shear walls meeting the requirements of AISC *Specification* Chapter I and Appendix 4, the fire resistance rating is permitted to be determined in accordance with Equation 6-5. To calculate P_n in Equation 6-5, P_{no} is calculated at ambient temperature using Equation 6-6b, P_e is calculated at ambient temperature using Equation 6-6c, and EI_{eff} is calculated at ambient temperature using Equation 6-6d. These equations are based on compression design of SpeedCore walls discussed in Sections 2.2.2 and 2.2.3 of this Design Guide.

The use of Equation 6-5 is limited to unprotected walls satisfying all the following conditions:

(1) Wall slenderness ratio (L/t_{sc}) is less than or equal to 20.

(2) Axial load ratio (P_u/P_n) is less than or equal to 0.2.

$$R = \left[-18.5\left(\frac{P_u}{P_n}\right)^{\left(0.24-\frac{L/t_{sc}}{230}\right)}+15\right]\left(\frac{1.9t_{sc}}{8}-1\right) \le 8 \text{ hr} \quad (6\text{-}5)$$

where

L = length of member, in.

P_n = compressive strength at ambient temperature, kips

P_u = ultimate axial load, kips

R = fire resistance rating, hr

t_{sc} = wall thickness, in.

$$P_n = \left(0.658^{\frac{P_{no}}{P_e}}\right)P_{no} \quad (6\text{-}6a)$$

$$P_{no} = F_y A_s + 0.85 f'_c A_c \quad (6\text{-}6b)$$

$$P_e = \frac{\pi^2 EI_{eff}}{L_c^2} \quad (6\text{-}6c)$$

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \quad (6\text{-}6d)$$

Equation 6-5 for determining the fire resistance rating of composite plate shear walls is based on research conducted by Anvari et al. (2020a). Equation 6-5 provides conservative failure times for walls subjected to standard ISO or ASTM fire scenarios. This equation is based on data obtained from experiments and benchmarked numerical models. The equation can be used for composite plate shear walls that meet the detailing and design requirements of AISC *Specification* Chapter I—namely, the steel plate slenderness and tie spacing requirements identified in Sections 2.2.1.1 and 2.2.1.2, respectively, of this Design Guide. The limits of applicability of Equation 6-5 are based on the range of parameters considered in the study. For walls with slenderness greater than 20, nonuniform fire scenarios may start controlling the failure of the wall, and additional fire protection may need to be provided. Typical axial load ratios for composite plate shear walls are in the range of 10–20%.

6.2.4 General Structural Integrity

The size and spacing of steam vent holes in concrete-filled columns and concrete-filled composite plate shear walls can be evaluated using any rational method that considers heat transfer through the cross section, water content in concrete, fire protection, and the allowable pressure in the member.

For example, Anvari et al. (2020a) have developed a rational method that considers these factors and can be used to calculate the size and spacing of vent holes required to limit the internal pressure build-up to a chosen design (allowable) value.

6.3 GENERAL DESIGN PROCEDURE FOR SPEEDCORE WALLS UNDER FIRE LOADING

This section describes the design approach for SpeedCore walls at elevated temperatures. This design approach considers the provisions discussed in Section 6.2 and the applicable building code. This design procedure has five main steps: (1) collecting design input, (2) calculating the fire resistance rating, (3) heat transfer analysis, (4) calculation of compressive strength at elevated temperatures, and (5) design of vent holes. A summary of this design approach is presented in Figure 6-6.

6.3.1 Design Inputs

The design inputs include the governing building code, applied axial load, geometric properties of the wall such as wall thickness, wall height, steel and concrete material properties at ambient temperature, fire scenario (uniform or nonuniform heating), and target fire resistance rating.

6.3.2 Fire Resistance Rating

The fire resistance rating can be calculated using Equation 6-5. This equation is a function of the ratio of the required strength to the nominal compressive strength at ambient temperature, P_u/P_n, wall slenderness ratio, L/t_{sc}, and wall thickness, t_{sc}.

6.3.3 Heat Transfer Analysis

The axial strength of a wall depends on the strength of the mechanical material properties of the wall's component at elevated temperatures. Thus, transient heat transfer analysis is required to obtain the temperature profile through the wall thickness. The wall thickness can be divided into several elements and the material properties of each element can be calculated based on the temperature of each element. The temperature profile through the wall thickness can be obtained using the program by Varma et al. (2020). A preview of this tool is provided in Section 6.5.

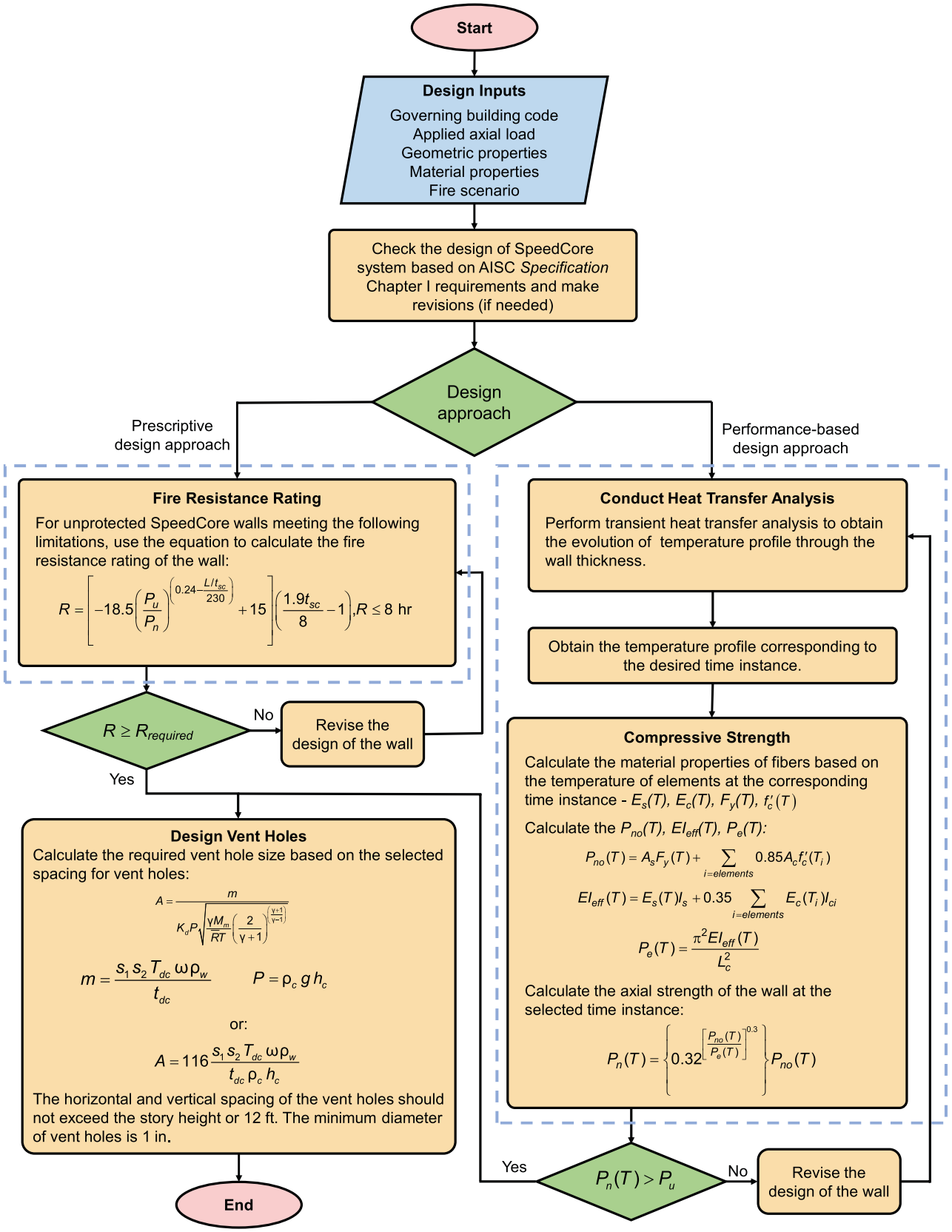


Fig. 6-6. Overview of fire design of SpeedCore systems under fire loading.

6.3.4 Compressive Strength of SpeedCore Walls at Elevated Temperatures

The nominal compressive strength of SpeedCore walls at elevated temperatures can be calculated using Equations 6-1 to 6-4. The compressive strength of walls can be calculated per unit width of the wall. A unit width column of the wall has a width equal to the tie bar spacing. This method uses the temperature-dependent material properties of concrete and steel. One-dimensional (for unit width columns) or 2D (for whole wall section) heat transfer analysis is conducted to calculate the temperature distribution through the wall thickness. The variation of the temperature distribution through the width and height of the wall can be neglected conservatively in the heat transfer analysis. The mechanical properties of each element can be calculated based on the calculated element temperature.

6.3.5 Design of Vent Holes

The existing water in the concrete infill of SpeedCore walls evaporates at elevated temperatures. The faceplate surrounding the concrete infill can trap vapor between faceplates and concrete, and the wall can act like an enclosed vessel during a fire event. The temperature rise builds up pressure between faceplates and concrete. This pressure can build up to large values, resulting in yielding of the steel plates. Providing vent holes can help to release the built-up pressure. Vent holes should not be blocked by any structural or nonstructural elements. A rational method has been developed by Anvari et al. (2020a) based on using spring-operated relief pressure device design methods (Crowl and Tipler, 2013).

Equation 6-7 can be used to calculate the required vent hole size for a designated effective area. It is assumed that vent holes are located in the middle of the effective area. The maximum allowable vapor pressure can be equated to the maximum allowable hydrostatic pressure on the steel plates during concrete casting. This allowable hydrostatic pressure was calculated by Bhardwaj et al. (2018) on the basis of not significantly altering the local buckling behavior and compressive strength of SpeedCore walls in the composite phase after concrete casting.

The vapor generation rate, m, can be determined based on the thickness of the dried concrete. This can be calculated by dividing the amount of evaporated water content from the dry concrete thickness, t_{dc}, with the time duration in seconds associated with drying. The discharge rate of every vent hole can be conservatively taken equal to the vapor generation rate, m. The vapor generation rate and the allowable pressure based on the concrete pouring height can be calculated using Equations 6-8 and 6-9, respectively. At least one vent hole should be provided at the top and bottom of the wall at every floor. Based on the height of the wall, additional vent holes should be provided between the vent holes at the top and bottom of a wall. The horizontal and vertical spacing of the vent holes should not exceed the story height or 12 ft. The minimum diameter of vent holes is 1 in.

In this method, the following assumptions were made to design vent holes:

- Temperature of vapor does not exceed (392°F) while traveling inside the wall until it reaches a vent hole (T = 392°F, $\gamma = 1.315$).
- Water content of concrete evaporates when the temperature exceeds 212°F.
- The flow of vapor through vent holes is reversible and isentropic.
- Vent holes have a square edge for calculation purposes.
- The generated vapor rate is equal to the vapor discharge.

Equation 6-10 was obtained after simplifying Equation 6-7 and taking into account all the listed assumptions.

$$A = \frac{m}{K_d P \sqrt{\frac{\gamma M_m}{\overline{R} T}\left(\frac{2}{\gamma+1}\right)^{\left(\frac{\gamma+1}{\gamma-1}\right)}}} \tag{6-7}$$

$$m = \frac{s_1 s_2 T_{dc} \omega \rho_w}{t_{dc}} \tag{6-8}$$

$$P = \rho_c g h_c \tag{6-9}$$

$$A = 116 \frac{s_1 s_2 T_{dc} \omega \rho_w}{t_{dc} \rho_c h_c} \tag{6-10}$$

where

A = vent hole area, ft^2
K_d = discharge coefficient (a square edge hole = 0.62)
M_m = molar weight of water, lb/mol
P = allowable pressure, lb/ft-s^2
$\overline{R}$ = ideal gas constant, lb-ft^2/s^2-K-mol
T = maximum vapor temperature, K
T_{dc} = dry concrete thickness, ft
g = gravitational acceleration, ft/s^2
h_c = concrete pouring height, ft
m = vapor generation rate, lb/s
s_1 = horizontal spacing of vent holes, ft
s_2 = vertical spacing of vent holes, ft
t_{dc} = heating duration associated with the selected dry concrete thickness, s
γ = specific heat ratio
ρ_c = concrete density, lb/ft^3
ρ_w = water density, lb/ft^3
ω = concrete water content, % by volume

6.4 DESIGN EXAMPLE

EXAMPLE 6.1—Fire Design of SpeedCore Walls

This example presents the fire design of a SpeedCore wall, which includes calculating the fire resistance rating and the axial strength of the wall. It is assumed that a fire resistance rating of 3 hr is required for the wall. It is also assumed that the wall is exposed to uniform fire loading and both sides of the wall are heated. A simplified design method using unit width columns of the wall can be used to calculate the compressive axial strength of the wall at elevated temperatures. This is a conservative method that can be used for walls with different configurations such as planar and C-shaped walls. Tie bars are located in the middle of the unit width columns, which have width equal to the tie bar spacing as shown in Figure 6-7.

Given:

ASTM A572/A572M, Grade 50 steel:

E_s = 29,000 ksi

Concrete (self-consolidating):

f'_c = 6 ksi
E_c = 4,280 ksi
K_d = discharge coefficient (square edge)
= 0.62
L_{wall} = wall length
= 11 ft
M_m = molar weight of water
= 0.04 lb/mol
$P_{u,wall}$ = applied axial load to the wall
= 3,600 kips
$\overline{R}$ = ideal gas constant
= 199.9 lb-ft^2/(s^2-K-mol)
S_{tie} = tie bar spacing
= 12 in.
T = assumed maximum vapor temperature
= 473 K
h = story height
= 17 ft
h_c = assumed concrete pouring height
= 14 ft
s_1 = horizontal spacing of vent holes
= 10 ft
s_2 = vertical spacing of vent holes
= 10 ft
t_p = wall plate thickness
= b in.
t_{sc} = wall thickness
= 24 in.
γ = specific heat ratio at 392°F
= 1.315
ρ_c = concrete density
= 145 lb/ft^3
ρ_w = water density
= 62.5 lb/ft^3
ω = concrete water content
= 15%

Solution:

Step 1. Calculation of the Fire Resistance Rating

To use Equation 6-5 for calculating the fire resistance rating of the wall, the properties of the wall or the unit width column need to satisfy the limitations of Equation 6-5.

Check that the wall slenderness ratio limit, L/t_{sc}, is less than or equal to 20.

Member length:

$$L = h = 17 \text{ ft}$$

$$\frac{L}{t_{sc}} = \frac{17 \text{ ft}(12 \text{ in./ft})}{24 \text{ in.}}$$
$$= 8.50 < 20$$

Check that the applied axial load ratio limit, P_u/P_n, is less than or equal to 0.2.

The area of steel in the cross section of the unit width column is:

$$A_s = 2 S_{tie} t_p$$
$$= 2(12 \text{ in.})(9/16 \text{ in.})$$
$$= 13.5 \text{ in.}^2$$

The area of concrete in the cross section is:

$$A_c = (t_{sc} - 2t_p) S_{tie}$$
$$= \left[24 \text{ in.} - 2(9/16 \text{ in.})\right](12 \text{ in.})$$
$$= 275 \text{ in.}^2$$

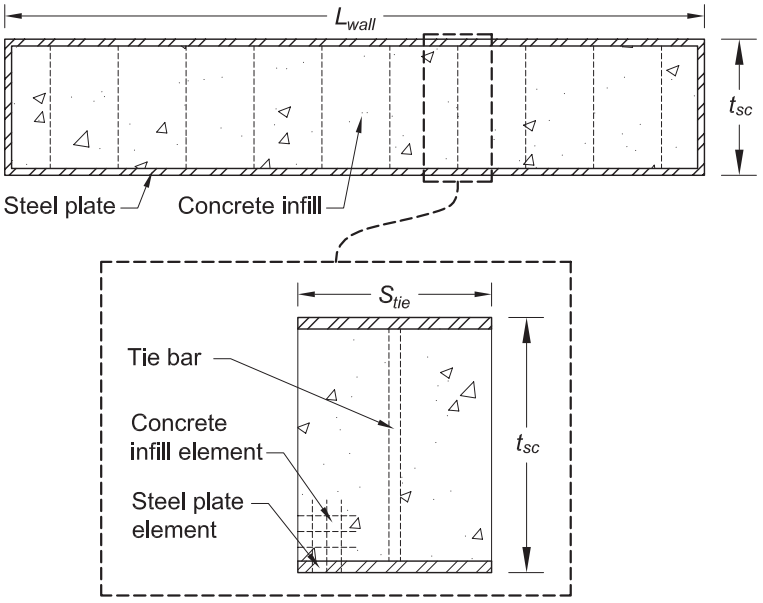


Fig. 6-7. SpeedCore cross section and unit width column for fire design.

The nominal axial compressive strength of a zero length (stub) column at ambient temperature is calculated from Equation 6-6b:

$$P_{no} = A_s F_y + 0.85 f'_c A_c \quad \text{(6-6b)}$$
$$= (13.5 \text{ in.}^2)(50 \text{ ksi}) + 0.85 (6 \text{ ksi})(275 \text{ in.}^2)$$
$$= 2{,}080 \text{ kips}$$

The moment of inertia of the concrete section about the elastic neutral axis of the composite section for a unit width is:

$$I_c = \frac{S_{tie} (t_{sc} - 2t_p)^3}{12}$$
$$= \frac{(12 \text{ in.})\left[24 \text{ in.} - 2(\tfrac{5}{16} \text{ in.})\right]^3}{12}$$
$$= 12{,}000 \text{ in.}^4$$

The moment of inertia of steel about the elastic neutral axis of the composite section is:

$$I_s = \frac{S_{tie} t_{sc}^3}{12} - I_c$$
$$= \frac{(12 \text{ in.})(24 \text{ in.})^3}{12} - 12{,}000 \text{ in.}^4$$
$$= 1{,}820 \text{ in.}^4$$

The effective moment of inertia of a unit width column at ambient temperature is calculated from Equation 6-6d:

$$EI_{eff} = E_s I_s + 0.35 E_c I_c \quad \text{(6-6d)}$$
$$= (29{,}000 \text{ ksi})(1{,}820 \text{ in.}^4) + 0.35(4{,}280 \text{ ksi})(12{,}000 \text{ in.}^4)$$
$$= 7.08 \times 10^7 \text{ kip-in.}^2$$

The elastic critical buckling load at ambient temperature is:

$$P_e = \frac{\pi^2 EI_{eff}}{L_c^2} \quad \text{(6-6c)}$$

Using K= 1.0 for a pinned-end member:

$$P_e = \frac{\pi^2 (7.08 \times 10^7 \text{ kip-in.}^2)}{\left[(1.0)(17 \text{ ft})(12 \text{ in./ft})\right]^2}$$
$$= 16{,}800 \text{ kips}$$

The axial compressive strength of the unit width column at ambient temperature is calculated using Equation 6-6a:

$$P_n = \left(0.658^{\frac{P_{no}}{P_e}}\right) P_{no} \quad \text{(6-6a)}$$
$$= \left(0.658^{\frac{2{,}080 \text{ kips}}{16{,}800 \text{ kips}}}\right)(2{,}080 \text{ kips})$$
$$= 1{,}970 \text{ kips}$$

The applied axial load to a unit width column of the wall is:

$$P_u = P_{u,wall}\left(\frac{S_{tie}}{L_{wall}}\right)$$
$$= (3{,}600 \text{ kips})\left[\frac{12 \text{ in.}}{(11 \text{ ft})(12 \text{ in./ft})}\right]$$
$$= 327 \text{ kips}$$

Check the load ratio against the limitation:

$$\frac{P_u}{P_n} = \frac{327 \text{ kips}}{1{,}970 \text{ kips}}$$
$$= 0.166 < 0.20 \quad \textbf{o.k.}$$

The limitations of Equation 6-5 are satisfied, and the fire-resistance rating is:

$$R = \left[-18.5\left(\frac{P_u}{P_n}\right)^{\left(0.24-\frac{L/t_{sc}}{230}\right)} + 15\right]\left(\frac{1.9\,t_{sc}}{8} - 1.0\right) \tag{6-5}$$
$$= \left\{-18.5\left(\frac{327 \text{ kips}}{1{,}970 \text{ kips}}\right)^{\left[0.24-\frac{(17 \text{ ft})(12 \text{ in./ft})/(24 \text{ in.})}{230}\right]} + 15\right\}\left[\frac{1.9\,(24 \text{ in.})}{8} - 1.0\right]$$
$$= 10.1 \text{ hr} > 8 \text{ hr}$$
$$= 8 \text{ hr}$$

Step 2. Calculation of the Thermal Profile Through Wall Thickness

Heat transfer analysis was conducted to calculate the temperature profile through the wall thickness after 3 hr (the target fire rating) of heating following ISO 834 gas time-temperature. This analysis was performed using the program developed by Varma et al. (2020).

The concrete infill was divided into several elements through the width and thickness of the unit width column. The thermal and mechanical material properties of each element were calculated based on the element's temperature. The calculated temperature profile through the wall thickness is shown in Figure 6-8.

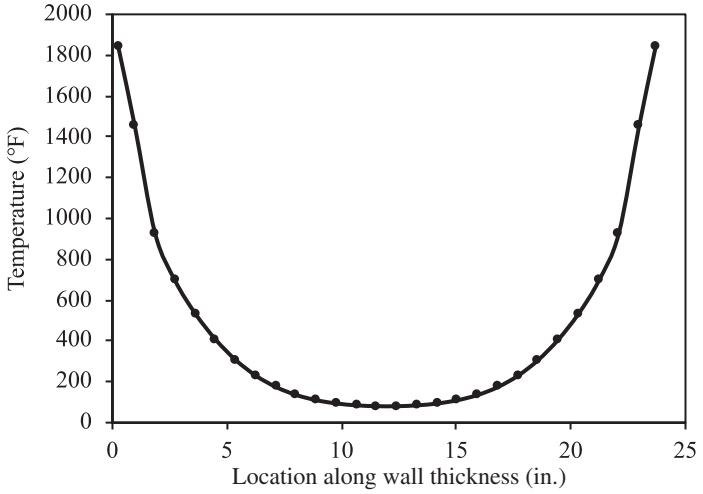


Fig. 6-8. Temperature profile through the wall thickness after 3 hr of heating.

Step 3. Calculation of the Material Properties at Elevated Temperatures

Table 6-1 shows a summary of the thermal analysis and the mechanical properties of each element at elevated temperatures. The size of elements through the wall thickness is shown in Table 6-1. The cross section of the concrete infill was discretized into 26 elements (13 elements for half of the wall thickness), and the steel plate was discretized into one element. The required fire resistance rating for this example was defined as 3 hr (180 min). The temperature of each element was obtained from the thermal analysis results using the computer program developed by Varma et al. (2020). Because the temperature profile through the wall thickness is symmetric, the material properties of the elements $[F_y(T),\ f_c'(T),\ E_s(T),\ E_c(T)]$ for half of the wall thickness are calculated after 180 min in Table 6-1. The modulus of elasticity and material strength for each element at elevated temperatures were calculated using AISC *Specification* Tables A-4.2.1 and A-4.2.2.

The area of the elements, $A_{element}$, was calculated by multiplying the thickness of each element by the tie bar spacing (unit width column's length as shown in Figure 6-7). The distance of each element, $d_{element}$, to the neutral axis (wall mid-thickness) is calculated in Table 6-1. The moment of inertia for each element, $I_{element}$, was calculated using the area of elements and the distance of elements to the neutral axis ($I_y = I_{element} + A_{element} d_{element}^2$).

The moment of inertia about the mid-thickness of the wall and the area for each element will be needed to calculate the nominal axial compressive strength, $P_{no}(T)$, and the effective stiffness of the composite section, $EI_{eff}(T)$, in the next steps of this example.

Step 4. Calculation of Compressive Strength at Elevated Temperatures

To calculate the compressive strength of the cross section at elevated temperatures, the nominal axial compressive strength, $P_{no}(T)$, and effective stiffness of the composite section, $EI_{eff}(T)$, can be calculated based on the mechanical properties of the elements through the cross section of the unit width column at elevated temperatures.

The effective stiffness of the cross section, $EI_{eff}(T)$, is calculated using Equation 6-3 based on the material mechanical properties of elements at elevated temperatures (see Step 3) for half of the cross section in Table 6-1:

$$\frac{EI_{eff}(T)}{2} = E_s(T)I_s + 0.35\sum_{i=elements} E_c(T_i)I_{ci} \quad \text{(from Eq. 6-3)}$$

$$= (1{,}270\text{ ksi})(924\text{ in.}^4)$$

$$+ 0.35\begin{bmatrix} (292\text{ ksi})(1{,}280\text{ in.}^4) + (1{,}140\text{ ksi})(1{,}080\text{ in.}^4) + (2{,}060\text{ ksi})(906\text{ in.}^4) \\ + (2{,}680\text{ ksi})(742\text{ in.}^4) + (3{,}180\text{ ksi})(594\text{ in.}^4) + (3{,}570\text{ ksi})(462\text{ in.}^4) \\ + (3{,}870\text{ ksi})(347\text{ in.}^4) + (4{,}040\text{ ksi})(249\text{ in.}^4) + (4{,}130\text{ ksi})(167\text{ in.}^4) \\ + (4{,}190\text{ ksi})(101\text{ in.}^4) + (4{,}220\text{ ksi})(52.0\text{ in.}^4) + (4{,}240\text{ ksi})(19.2\text{ in.}^4) \\ + (4{,}250\text{ ksi})(2.73\text{ in.}^4) \end{bmatrix}$$

$$= 5{,}640{,}000\text{ kip-in.}^2$$

The calculated $EI_{eff}(T)$ based on the summation of element contributions in Table 6-1 is multiplied by 2 to obtain $EI_{eff}(T)$ for the entire unit width column cross section:

$$EI_{eff}(T) = 2(5{,}640{,}000\text{ kip-in.}^2)$$

$$= 1.13\times10^7\text{ kip-in.}^2$$

Table 6-1. Obtained Temperatures and Calculated Properties for Each Element at Elevated Temperatures for Half of the Wall Thickness at 180 Min														
Material Properties	**Steel**	**Concrete**												
Thickness (in.)	0.563	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880	0.880
Temperature (°F)	1840	1450	927	694	533	406	305	229	173	134	108	92.0	83.0	79.0
$F_y(T)$ or $f'_c(T)$ (ksi)	1.80	1.09	3.61	4.64	5.19	5.39	5.54	5.66	5.76	5.85	5.91	5.95	5.97	5.98
$E_s(T)$ or $E_c(T)$ (ksi)	1270	292	1140	2060	2680	3180	3570	3870	4040	4130	4190	4220	4240	4250
Distance to N.A. $d_{element}$ (in.)	11.7	11.0	10.1	9.24	8.36	7.48	6.60	5.72	4.84	3.96	3.08	2.20	1.32	0.44
$A_{element}$ (in.^{2})	6.75	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
I_y (in.^{4})	924	1280	1080	906	742	594	462	347	249	167	101	52.0	19.2	2.73

The nominal axial compressive strength, $P_{no}(T)$, based on the calculated material mechanical properties of elements in Table 6-1 is calculated from Equation 6-2:

$$\frac{P_{no}(T)}{2} = F_y(T)A_s + \sum_{i=elements} 0.85 f'_c(T_i)A_{ci} \quad \text{(from Eq. 6-2)}$$

$$= (1.80\text{ ksi})(6.75\text{ in.}^2) + 0.85\begin{bmatrix} (1.09\text{ ksi})(10.6\text{ in.}^2) + (3.61\text{ ksi})(10.6\text{ in.}^2) + (4.64\text{ ksi})(10.6\text{ in.}^2) \\ +(5.19\text{ ksi})(10.6\text{ in.}^2) + (5.39\text{ ksi})(10.6\text{ in.}^2) + (5.54\text{ ksi})(10.6\text{ in.}^2) \\ +(5.66\text{ ksi})(10.6\text{ in.}^2) + (5.76\text{ ksi})(10.6\text{ in.}^2) + (5.85\text{ ksi})(10.6\text{ in.}^2) \\ +(5.91\text{ ksi})(10.6\text{ in.}^2) + (5.95\text{ ksi})(10.6\text{ in.}^2) + (5.97\text{ ksi})(10.6\text{ in.}^2) \\ +(5.98\text{ ksi})(10.6\text{ in.}^2) \end{bmatrix}$$

$$= 612\text{ kips}$$

The calculated $P_{no}(T)$ based on the summation of element contributions for half of the cross section in Table 6-1 is multiplied by 2 to obtain $P_{no}(T)$ for the entire unit width column cross section:

$$P_{no}(T) = 2(612\text{ kips})$$
$$= 1{,}220\text{ kips}$$

The elastic critical buckling load at elevated temperatures is calculated from Equation 6-4:

$$P_e(T) = \frac{\pi^2 EI_{eff}(T)}{L_c^2} \quad (6\text{-}4)$$

$$= \frac{\pi^2(1.13\times10^7\text{ kip-in.})}{\left[(1)(17\text{ ft})(12\text{ in./ft})\right]^2}$$

$$= 2{,}680\text{ kips}$$

From Equation 6-1, the nominal axial compressive strength of the unit width column at 180 minutes is:

$$P_n(T) = \left\{0.32^{\left[\frac{P_{no}(T)}{P_e(T)}\right]^{0.3}}\right\} P_{no}(T) \tag{6-1}$$

$$= \left[0.32^{\left(\frac{1{,}220 \text{ kips}}{2{,}680 \text{ kips}}\right)^{0.3}}\right](1{,}220 \text{ kips})$$

$$= 496 \text{ kips}$$

The design axial compressive strength of the unit width column at 180 minutes is:

$$\phi P_n(T) = 0.90(496 \text{ kips})$$

$$= 446 \text{ kips} > 330 \text{ kips}$$

The axial compressive strength of the unit width column is more than the applied axial load after 3 hr of heating. The wall will be able to resist the applied axial load for more than 3 hr under fire loading.

The design axial compressive strength of the SpeedCore wall at 180 minutes can be calculated conservatively as the axial design strength of the unit width column multiplied by the linear width of the wall divided by the unit width (tie bar spacing):

$$\phi P_n(T)_{wall} = \phi P_n(T)\left(\frac{L_{wall}}{S_{tie}}\right)$$

$$= (446 \text{ kips})\left[\frac{(11 \text{ ft})(12 \text{ in./ft})}{12 \text{ in.}}\right]$$

$$= 4{,}910 \text{ kips} > 3{,}600 \text{ kips}$$

Step 5. Design of Vent Holes

The size and spacing of vent holes can be calculated based on the vapor generation rate during a fire event. The evaporated volume/weight of water is needed to estimate the vapor generation rate. The computer program developed by Varma et al. (2020) was used to calculate the temperature profile through the wall thickness during the heating phase. The thickness of the concrete with a temperature exceeding 212°F (on a single face) was assumed as the dry concrete thickness. Table 6-2 presents the heating time, the thickness of the dry concrete, and the calculated vapor generation rate for 10 ft vent hole spacing in both the horizontal and vertical directions. The vapor generation rate in Table 6-2 is calculated using Equation 6-8.

The heating duration and dry concrete thickness associated with the maximum vapor generation rate, shown as the highlighted column in Table 6-2, are selected to design the vent holes.

$$t_{dc} = 1{,}790 \text{ sec}$$

$$T_{dc} = \frac{1.50 \text{ in.}}{12 \text{ in./ft}}$$

$$= 0.125 \text{ ft}$$

The vent hole area is calculated using Equation 6-10:

$$A = 116\frac{s_1 s_2 T_{dc} \omega \rho_w}{t_{dc} \rho_c h_c} \tag{6-10}$$

$$= 116\frac{(10 \text{ ft})(10 \text{ ft})(0.125 \text{ ft})(0.15)(62.5 \text{ lb/ft}^3)}{(1{,}790 \text{ sec})(145 \text{ lb/ft}^3)(14 \text{ ft})}(144 \text{ in.}^2/\text{ft}^2)$$

$$= 0.539 \text{ in.}^2$$

Table 6-2. Calculation of Maximum Vapor Generation Rate							
Dry Concrete Thickness, T_{dc} (in.)	0.00	0.750	1.50	2.25	3.00	3.75	4.50
Heating Duration, t_{dc} (sec)	391	952	1790	2830	4070	5570	7340
Vapor Generation Rate, m (lb/sec $\times 10^{-2}$)	0.00	6.33	6.70	6.39	5.92	5.41	4.92

The vent hole diameter is:

$$d = 2\sqrt{\frac{A}{\pi}}$$
$$= 2\sqrt{\frac{0.539 \text{ in.}^2}{\pi}}$$
$$= 0.828 \text{ in.}$$

Use vent holes with a diameter of 1 in. spaced at 10 ft along the height (vertical) and width (horizontal) of the wall.

Appendix A
Nominal Flexural Strength of SpeedCore Walls and Composite Coupling Beams (Plastic Stress Distribution Method)

Equations for the nominal flexural strength of planar and C-shaped SpeedCore walls and composite coupling beams are listed in this Appendix. These equations were developed using the plastic stress distribution method. Cases considering the effect of axial loads for SpeedCore walls are also considered. Additional cases for walls of different geometry, neutral axis location, or location of applied load are possible.

A.1 PLANAR SPEEDCORE WALLS

Flexural strength calculations for planar SpeedCore walls consider the section geometry as shown in Figure A-1.

A.1.1 Planar SpeedCore Walls Subjected to Flexure

The nominal flexural strength of a planar SpeedCore wall is calculated as follows (see Figure A-2):

$$C = \frac{2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)_c}$$ Plastic neutral axis location

$$C_1 = (t_{sc} - 2t_p)t_p F_y$$ Flange plate compression force

$$C_2 = 2t_p C F_y$$ Web plate compression force

$$C_3 = 0.85 f_c'(t_{sc} - 2t_p)(C - t_p)$$ Concrete compression force

$$T_1 = (t_{sc} - 2t_p)t_p F_y$$ Flange plate tensile force

$$T_2 = 2t_p (L_{wall} - C) F_y$$ Web plate tensile force

Nominal flexural strength of planar SpeedCore wall:

$$M_{P.wall} = C_1\left(C - \frac{t_p}{2}\right) + C_2\left(\frac{C}{2}\right) + C_3\left(\frac{C - t_p}{2}\right) + T_1\left(L_{wall} - C - \frac{t_p}{2}\right) + T_2\left(\frac{L_{wall} - C}{2}\right)$$

A.1.2 Planar SpeedCore Walls Subjected to Tension

The nominal flexural strength of a planar SpeedCore wall subjected to tension load is calculated as follows (see Figure A-3):

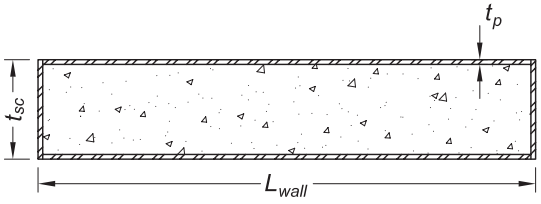


Fig. A-1. Geometry of planar SpeedCore wall.

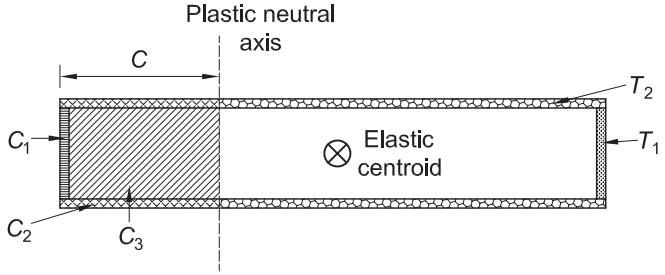


Fig. A-2. Plastic stress distribution and component forces in planar SpeedCore wall.

Note: The applied axial load, $T_{r.wall}$, is positive in these equations.

$$C_T = \frac{-T_{r.wall} + 2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$ Plastic neutral axis location

$$C_{1.T} = (t_{sc} - 2t_p)t_p F_y$$ Flange plate compression force

$$C_{2.T} = 2t_p C_T F_y$$ Web plate compression force

$$C_{3.T} = 0.85 f_c'(t_{sc} - 2t_p)(C_T - t_p)$$ Concrete compression force

$$T_{1.T} = (t_{sc} - 2t_p)t_p F_y$$ Flange plate tensile force

$$T_{2.T} = 2t_p(L_{wall} - C_T)F_y$$ Web plate tensile force

Nominal flexural strength of planar SpeedCore wall:

$$M_{PT.wall} = C_{1.T}\left(C_T - \frac{t_p}{2}\right) + C_{2.T}\left(\frac{C_T}{2}\right) + C_{3.T}\left(\frac{C_T - t_p}{2}\right) + T_{1.T}\left(L_{wall} - C_T - \frac{t_p}{2}\right) + T_{2.T}\left(\frac{L_{wall} - C_T}{2}\right) + T_{r.wall}\left(C_T - \frac{L_{wall}}{2}\right)$$

A.1.3 Planar SpeedCore Walls Subjected to Compression

The nominal flexural strength of a planar SpeedCore wall subjected to compression load is calculated as follows (see Figure A-4):

Note: The applied axial load, $P_{r.wall}$, is positive in these equations.

$$C_C = \frac{P_{r.wall} + 2t_p L_{wall} F_y + 0.85 f_c'(t_{sc} - 2t_p)t_p}{4t_p F_y + 0.85 f_c'(t_{sc} - 2t_p)}$$ Plastic neutral axis location

$$C_{1.C} = (t_{sc} - 2t_p)t_p F_y$$ Flange plate compression force

$$C_{2.C} = 2t_p C_C F_y$$ Web plate compression force

$$C_{3.C} = 0.85 f_c'(t_{sc} - 2t_p)(C_C - t_p)$$ Concrete compression force

$$T_{1.C} = (t_{sc} - 2t_p)t_p F_y$$ Flange plate tensile force

$$T_{2.C} = 2t_p(L_{wall} - C_C)F_y$$ Web plate tensile force

Nominal flexural strength of planar SpeedCore wall:

$$M_{PC.wall} = C_{1.C}\left(C_C - \frac{t_p}{2}\right) + C_{2.C}\left(\frac{C_C}{2}\right) + C_{3.C}\left(\frac{C_C - t_p}{2}\right) + T_{1.C}\left(L_{wall} - C_C - \frac{t_p}{2}\right) + T_{2.C}\left(\frac{L_{wall} - C_C}{2}\right) + P_{r.wall}\left(\frac{L_{wall}}{2} - C_C\right)$$

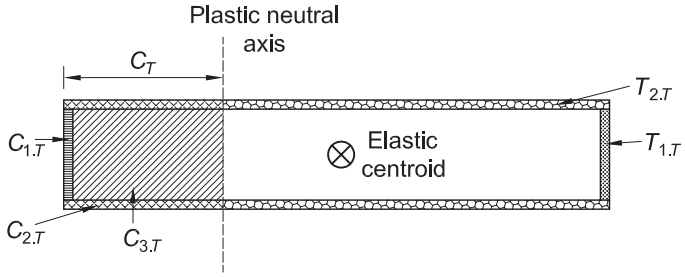


Fig. A-3. Plastic stress distribution and component forces in planar SpeedCore wall with applied tensile force.

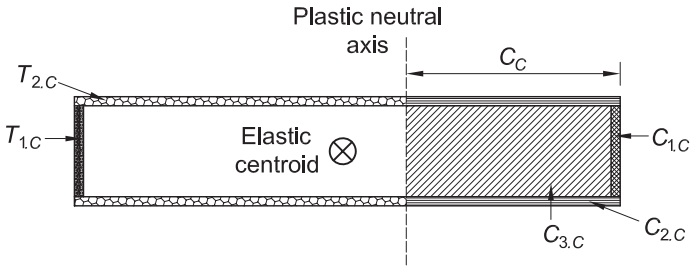


Fig. A-4. Plastic stress distribution and component forces in planar SpeedCore wall with applied compression force.

A.2 C-SHAPED SPEEDCORE WALLS

Flexural strength calculations for C-shaped SpeedCore walls consider the section geometry as shown in Figure A-5.

Additional geometric properties used for C-shaped SpeedCore wall calculations:

$W_c = W_{wall} - 3t_p$ — Length of flange wall without perpendicular steel plates

$t_{c.w} = t_{sc.w} - 2t_p$ — Width of concrete in web wall

$t_{c.f} = t_{sc.f} - 2t_p$ — Depth of concrete in flange wall

A.2.1 C-Shaped SpeedCore Walls with PNA in Flanges

Assuming the plastic neutral axis is located in the flange of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall is calculated as follows (see Figure A-6):

$$C = \frac{4t_p W_{wall} F_y + 2t_p t_{c.f} F_y + 2t_{c.f}(W_{wall} - t_p)0.85f'_c - 2t_p(L_{wall} - 4t_p)F_y}{8t_p F_y + 2t_{c.f}\,0.85f'_c}$$ — Plastic neutral axis location

$C_1 = 4t_p(W_{wall} - C)F_y$ — Flange walls, edge plates, compression force

$C_2 = 2t_{c.f} t_p F_y$ — Flange walls, end plates, compression force

$C_3 = 2t_{c.f}(W_{wall} - C - t_p)0.85f'_c$ — Concrete compression force

$T_1 = 4t_p C F_y$ — Flange walls, edge plates, tension force

$T_2 = t_p(L_{wall} - 4t_p)F_y$ — Web wall, inside steel plate, tension force

$T_3 = t_p(L_{wall} - 4t_p)F_y$ — Web wall, outside steel plate, tension force

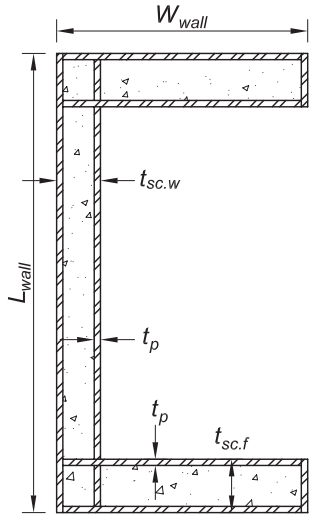


Fig. A-5. Geometry of C-shaped SpeedCore wall.

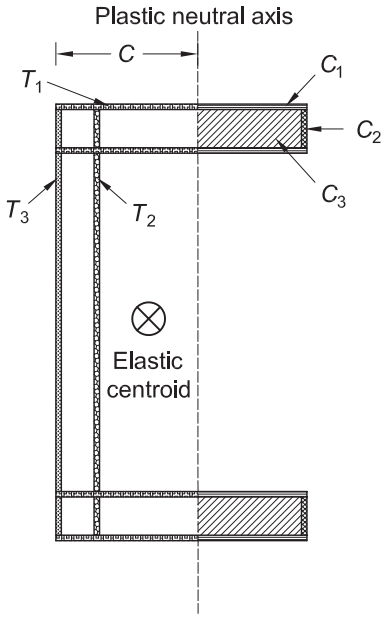


Fig. A-6. Plastic stress distribution and component forces in C-shaped SpeedCore wall (bent about minor axis subjecting flange wall tips to compression).

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{P.wall} = C_1\left(\frac{W_{wall}-C}{2}\right)+C_2\left(W_{wall}-C-\frac{t_p}{2}\right)+C_3\left(\frac{W_{wall}-C-t_p}{2}\right)+T_1\left(\frac{C}{2}\right)+T_2\left(C-t_{sc.w}+\frac{t_p}{2}\right)+T_3\left(C-\frac{t_p}{2}\right)$$

A.2.2 C-Shaped SpeedCore Walls with PNA in Web

Assuming the plastic neutral axis is located in the web of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall is calculated as follows (see Figure A-7):

$$C = \frac{(L_{wall}-4t_p)t_p 0.85f_c' + 4W_{wall}t_pF_y + 2t_pt_{c.f}F_y}{8t_pF_y + (L_{wall}-4t_p)0.85f_c'}$$ Plastic neutral axis location

$C_1 = 4t_pCF_y$ Flange walls, edge plates, compression force

$C_2 = t_p(L_{wall}-4t_p)F_y$ Web wall, exterior plate, compression force

$C_3 = (L_{wall}-4t_p)(C-t_p)0.85f_c'$ Concrete compression force

$T_1 = 4(W_{wall}-C)t_pF_y$ Flange walls, edge plates, tension force

$T_2 = t_p(L_{wall}-4t_p)F_y$ Web wall, interior steel plate, tension force

$T_3 = 2t_pt_{c.f}F_y$ Flange wall, end plates, tension force

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{P.wall} = C_1\left(\frac{C}{2}\right)+C_2\left(C-\frac{t_p}{2}\right)+C_3\left(\frac{C-t_p}{2}\right)+T_1\left(\frac{W_{wall}-C}{2}\right)+T_2\left(t_{sc.w}-C-\frac{t_p}{2}\right)+T_3\left(W_{wall}-C-\frac{t_p}{2}\right)$$

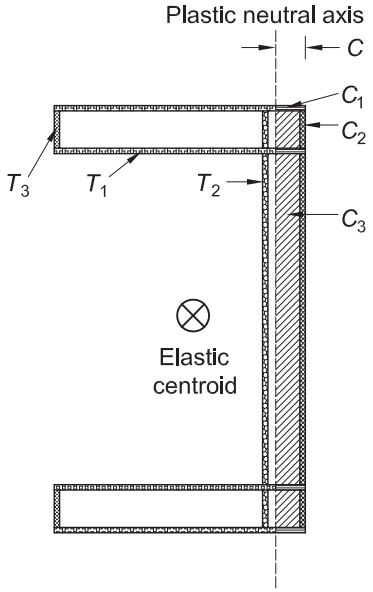


Fig. A-7. Plastic stress distribution and component forces in C-shaped SpeedCore wall (bent about minor axis subjecting web wall to compression).

A.2.3 C-Shaped SpeedCore Walls with PNA in Flanges Subjected to Tension

Assuming the plastic neutral axis is located in the flange of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall subjected to tension load is calculated as follows (see Figure A-8):

Note: The applied axial load, $T_{r.wall}$, is positive in these equations.

$$C_T = \frac{T_{r.wall} + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y + 0.85 f_c' 2t_{c.f} (W_{wall} - t_p) - 2t_p (L_{wall} - 4t_p) F_y}{8t_p F_y + 0.85 f_c' 2t_{c.f}} \quad \text{Plastic neutral axis location}$$

$$C_{1.T} = 4(W_{wall} - C_T) t_p F_y \quad \text{Flange walls, edge plates, compression force}$$

$$C_{2.T} = 2t_{c.f} t_p F_y \quad \text{Flange wall, end plates, compression force}$$

$$C_{3.T} = 0.85 f_c' 2t_{c.f} (W_{wall} - C_T - t_p) \quad \text{Concrete compression force}$$

$$T_{1.T} = 4C_T t_p F_y \quad \text{Flange walls, edge plates, tension force}$$

$$T_{2.T} = (L_{wall} - 4t_p) t_p F_y \quad \text{Web wall, interior steel plate, tension force}$$

$$T_{3.T} = (L_{wall} - 4t_p) t_p F_y \quad \text{Web wall, exterior steel plate, tension force}$$

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{PT.wall} = C_{1.T}\left(\frac{W_{wall} - C_T}{2}\right) + C_{2.T}\left(W_{wall} - C_T - \frac{t_p}{2}\right) + C_{3.T}\left(\frac{W_{wall} - C_T - t_p}{2}\right) + T_{1.T}\left(\frac{C_T}{2}\right) + T_{2.T}\left(C_T - t_{sc.w} + \frac{t_p}{2}\right) + T_{3.T}\left(C_T - \frac{t_p}{2}\right) + T_{r.wall}(\bar{y} - C_T)$$

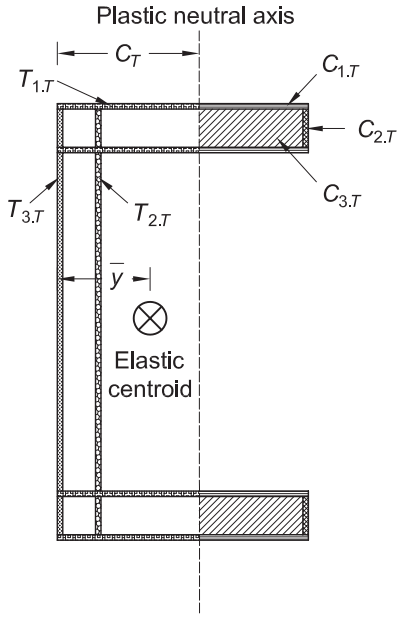


Fig. A-8. Plastic stress distribution and component forces in C-shaped SpeedCore wall with applied tension force (bent about minor axis subjecting flange wall tips to compression).

A.2.4 C-Shaped SpeedCore Walls with PNA in Flanges Subjected to Compression

Assuming the plastic neutral axis is located in the flange of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall subjected to compression load is calculated as follows (see Figure A-9):

Note: The applied axial load, $P_{r.wall}$, is positive in these equations.

$$C_C = \frac{P_{r.wall} - 2t_p(L_{wall} - 4t_p)F_y - t_{c.w}(L_{wall} - 4t_p)0.85f'_c + 2t_{c.f}t_{sc.w}0.85f'_c + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y}{8t_p F_y + 2t_{c.f}0.85f'_c}$$

Plastic neutral axis location

$C_{1.C} = 4C_C t_p F_y$ — Flange walls, edge plates, compression force

$C_{2.C} = t_p(L_{wall} - 4t_p)F_y$ — Web wall, exterior plate, compression force

$C_{3.C} = t_p(L_{wall} - 4t_p)F_y$ — Web wall, interior plate, compression force

$C_{4.C} = 0.85f'_c t_{c.w}(L_{wall} - 4t_p)$ — Web wall, concrete compression force

$C_{5.C} = 0.85f'_c 2t_{c.f}(C_C - t_{sc.w})$ — Flange walls, concrete compression force

$T_{1.C} = 4(W_{wall} - C_C)t_p F_y$ — Flange walls, edge plates, tension force

$T_{2.C} = 2t_p t_{c.f} F_y$ — Flange wall, end plates, tension force

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{PC.wall} = C_{1.C}\left(\frac{C_C}{2}\right) + C_{2.C}\left(C_C - \frac{t_p}{2}\right) + C_{3.C}\left(C_C - t_{sc.w} + \frac{t_p}{2}\right) + C_{4.C}\left(C_C - \frac{t_{sc.w}}{2}\right) + C_{5.C}\left(\frac{C_C - t_{sc.w}}{2}\right) + T_{1.C}\left(\frac{W_{wall} - C_C}{2}\right) + T_{2.C}\left(W_{wall} - C_C - \frac{t_p}{2}\right) + P_{r.wall}(\bar{y} - C_C)$$

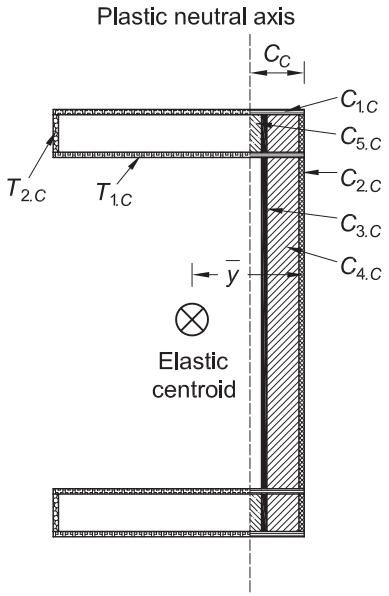


Fig. A-9. Plastic stress distribution and component forces in C-shaped SpeedCore wall with applied compression force (bent about minor axis subjecting web wall to compression resulting in PNA in flange concrete).

A.2.5 C-Shaped SpeedCore Walls with PNA in Web Plate Subjected to Compression

Assuming the plastic neutral axis is located in the web steel plate of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall subjected to compression load is calculated as follows (see Figure A-10):

Note: The applied axial load, $P_{r.wall}$, is positive in these equations.

$$C_C = \frac{P_{r.wall} - t_p(L_{wall} - 4t_p)F_y + (t_{c.w} + t_p)(L_{wall} - 4t_p)F_y}{8t_pF_y + 2(L_{wall} - 4t_p)F_y} + \frac{-t_{c.w}(L_{wall} - 4t_p)0.85f'_c + 4t_p W_{wall} F_y + 2t_p t_{c.f} F_y + t_{sc.w}(L_{wall} - 4t_p)F_y}{8t_pF_y + 2(L_{wall} - 4t_p)F_y}$$ Plastic neutral axis location

$$C_{1.C} = 4C_C t_p F_y$$ Flange walls, edge plates, compression force

$$C_{2.C} = t_p(L_{wall} - 4t_p)F_y$$ Web wall, exterior plate, compression force

$$C_{3.C} = (C_C - t_{c.w} - t_p)(L_{wall} - 4t_p)F_y$$ Web wall, interior steel plate, compression force

$$C_{4.C} = 0.85f'_c t_{c.w}(L_{wall} - 4t_p)$$ Concrete compression force

$$T_{1.C} = 4(W_{wall} - C_C)t_p F_y$$ Flange walls, edge plates, tension force

$$T_{2.C} = (t_{sc.w} - C_C)(L_{wall} - 4t_p)F_y$$ Web wall, interior steel plate, tension force

$$T_{3.C} = 2t_p t_{c.f} F_y$$ Flange wall, end plates, tension force

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{PC.wall} = C_{1.C}\left(\frac{C_C}{2}\right) + C_{2.C}\left(C_C - \frac{t_p}{2}\right) + C_{3.C}\left(\frac{C_C - t_{c.w} - t_p}{2}\right) + C_{4.C}\left(C_C - t_p - \frac{t_{c.w}}{2}\right) + T_{1.C}\left(\frac{W_{wall} - C_C}{2}\right) + T_{2.C}\left(\frac{t_{sc.w} - C_C}{2}\right) + T_{3.C}\left(W_{wall} - C_C - \frac{t_p}{2}\right) + P_{r.wall}(\bar{y} - C_C)$$

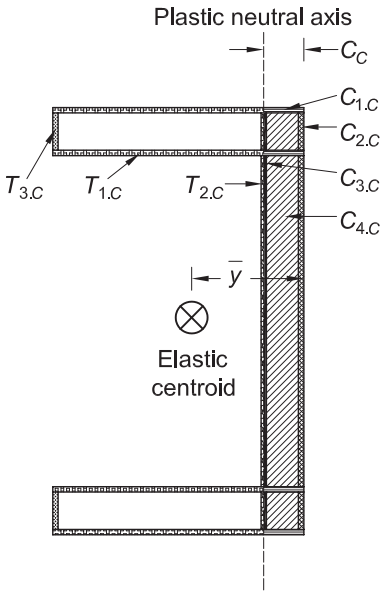


Fig. A-10. Plastic stress distribution and component forces in C-shaped SpeedCore wall considering applied compression force (bent about minor axis subjecting web wall to compression resulting in PNA in web steel).

A.2.6 C-Shaped SpeedCore Walls Bent about the Major Axis

Assuming the plastic neutral axis is located in the web of the C-shaped SpeedCore wall, the nominal flexural strength of the C-shaped SpeedCore wall is calculated as follows (see Figure A-11):

$$C = \frac{2L_{wall}t_pF_y + 0.85f_c'(t_{c.w}t_{sc.f} - t_{c.f}W_c)}{4t_pF_y + 0.85f_c't_{c.w}}$$ Plastic neutral axis location

$C_1 = W_c t_p F_y$ Flange wall, exterior steel plate, compression force

$C_2 = W_c t_p F_y$ Flange wall, interior steel plates, compression force

$C_3 = t_{sc.f} t_p F_y$ Flange wall, end plates, compression force

$C_4 = 2Ct_pF_y$ Web wall, steel plates, compression force

$C_5 = 0.85f_c'W_ct_{c.f}$ Flange wall, concrete compression force

$C_6 = 0.85f_c'(C - t_{sc.f})t_{c.w}$ Web wall, concrete compression force

$T_1 = W_c t_p F_y$ Flange wall, exterior steel plate, tension force

$T_2 = W_c t_p F_y$ Flange wall, interior steel plate, tension force

$T_3 = t_{sc.f} t_p F_y$ Flange wall, end plate, tension force

$T_4 = 2(L_{wall} - C)t_pF_y$ Web wall, steel plates, tension force

Nominal flexural strength of C-shaped SpeedCore wall:

$$M_{P.wall} = C_1\left(C - \frac{t_p}{2}\right) + C_2\left(C - t_{sc.f} + \frac{t_p}{2}\right) + C_3\left(C - \frac{t_{sc.f}}{2}\right) + C_4\left(\frac{C}{2}\right) + C_5\left(C - \frac{t_{sc.f}}{2}\right) + C_6\left(\frac{C - t_{sc.f}}{2}\right)$$
$$+ T_1\left(L_{wall} - C - \frac{t_p}{2}\right) + T_2\left(L_{wall} - C - t_{sc.f} + \frac{t_p}{2}\right) + T_3\left(L_{wall} - C - \frac{t_{sc.f}}{2}\right) + T_4\left(\frac{L_{wall} - C}{2}\right)$$

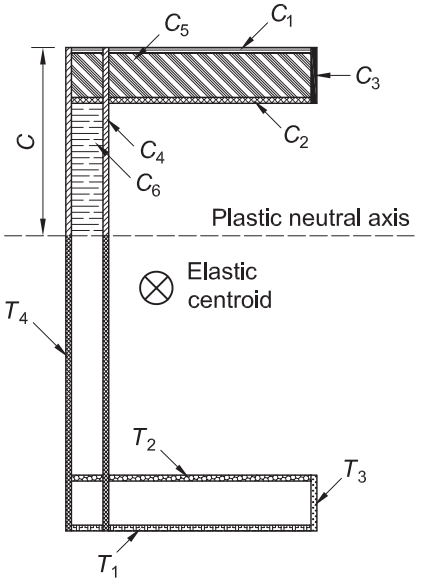


Fig. A-11. Plastic stress distribution and component forces in C-shaped SpeedCore wall (bent about major axis).

A.3 COMPOSITE COUPLING BEAMS

Flexural strength calculations for the composite coupling beam use the section geometry as shown in Figure A-12.

The nominal flexural strength of a composite coupling beam is calculated as follows (see Figure A-13):

$$C_{CB} = \frac{2t_{pw.CB}h_{CB}F_y + 0.85f'_c t_{c.CB}t_{pf.CB}}{4t_{pw.CB}F_y + 0.85f'_c t_{c.CB}}$$ Plastic neutral axis location

$$C_1 = (b_{CB} - 2t_{pw.CB})t_{pf.CB}F_y$$ Flange plate, compression force

$$C_2 = 2t_{pw.CB}C_{CB}F_y$$ Web plate, compression force

$$C_3 = 0.85f'_c t_{c.CB}(C_{CB} - t_{pf.CB})$$ Concrete compression force

$$T_1 = t_{pf.CB}(b_{CB} - 2t_{pw.CB})F_y$$ Web plate, tension force

$$T_2 = 2t_{pw.CB}(h_{CB} - C_{CB})F_y$$ Flange plate, tension force

Nominal flexural strength of composite coupling beam:

$$M_{Pn.CB} = C_1\left(C_{CB} - \frac{t_{pf.CB}}{2}\right) + C_2\left(\frac{C_{CB}}{2}\right) + C_3\left(\frac{C_{CB} - t_{pf.CB}}{2}\right) + T_1\left(h_{CB} - C_{CB} - \frac{t_{pf.CB}}{2}\right) + T_2\left(\frac{h_{CB} - C_{CB}}{2}\right)$$

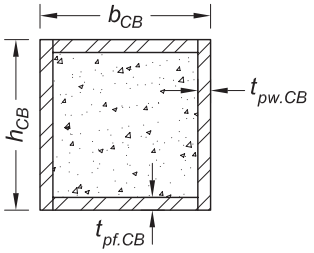


Fig. A-12. Geometry of composite coupling beam.

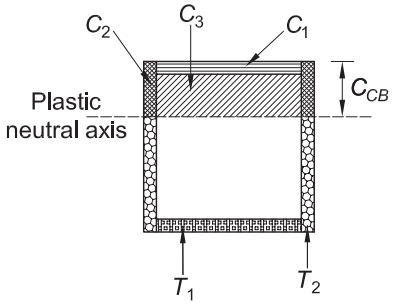


Fig. A-13. Plastic stress distribution and component forces in composite coupling beam.

Symbols

A	Area of cross-sectional element
A_c	Area of concrete in SpeedCore wall, in.^{2}
A	Vent hole area, ft^2
A_{ci}	Area of concrete element, in.^{2}
A_{cw}	Area of concrete section in the direction of shear, in.^{2}
$A_{c.CB}$	Area of concrete in coupling beam, in.^{2}
$A_{f.e}$	Effective area of flange plate, in.^{2}
$A_{f.g}$	Gross area of flange plate, in.^{2}
$A_{f.n}$	Net area of flange plate, in.^{2}
$A_{f.CB}$	Area of coupling beam flange plate, in.^{2}
$A_{f.SR}$	Net shear area of coupling beam flange plate for shear rupture, in.^{2}
$A_{f.SY}$	Gross shear area of coupling beam flange plate for shear yielding, in.^{2}
A_g	Gross area of composite member, in.^{2}
A_s	Area of steel in SpeedCore wall, in.^{2}
A_s	Areas of steel plates, in.^{2}
A_s	Areas of steel section, in.^{2}
$A_{s.CB}$	Area of steel in coupling beam, in.^{2}
$A_{s.CB.min}$	Minimum area of steel required in coupling beam, in.^{2}
$A_{s.min}$	Minimum steel required in SpeedCore wall, in.^{2}
$A_{s.max}$	Maximum steel required in SpeedCore wall, in.^{2}
A_{sw}	Area of steel plates in the direction of in-plane shear, in.^{2}
A_{sr}	Area of continuous reinforcing bars, in.^{2}
$A_{sw.CB}$	Area of steel web of coupling beam in the direction of shear, in.^{2}
$A_{uncr.CB}$	Area of uncracked concrete in the coupling beam, in.^{2}
A_v	Shear area of the steel portion of a composite member, in.^{2}
$A_{w.SR}$	Net shear area of wall web plate for shear rupture, in.^{2}
$A_{w.SY}$	Gross shear area of wall web plate for shear yielding, in.^{2}
C	Plastic neutral axis location in SpeedCore wall, in.
C_C	Plastic neutral axis location in SpeedCore wall in compression, in.
C_{CB}	Plastic neutral axis location in the coupling beam, in.
$C_{CB.exp}$	Plastic neutral axis location in the coupling beam considering expected strength, in.
C_d	Deflection amplification factor
C_{exp}	Expected plastic neutral axis location in SpeedCore wall, in.
C_T	Plastic neutral axis location in SpeedCore wall in tension, in.
$C_{1.C}$	Compression force in the steel plate perpendicular to the web in C-shaped SpeedCore wall, in.
$C_{1.T}$	Compression force in flange edge plate in C-shaped SpeedCore wall, in.
$C_{2.C}$	Compression force in the outside steel plate of the web in C-shaped SpeedCore wall, in.
$C_{2.T}$	Compression force in flange end plate in C-shaped SpeedCore wall, in.
C_3	Coefficient for calculation of effective rigidity of filled composite compression member
$C_{3.C}$	Compression force in the inside steel plate of the web in C-shaped SpeedCore wall, in.
$C_{3.CB}$	Coefficient for calculation of effective rigidity of filled composite beam
$C_{3.T}$	Compression force in the concrete in C-shaped SpeedCore wall, in.
$C_{4.C}$	Compression force in the concrete in the web of C-shaped SpeedCore wall, in.
$C_{5.C}$	Compression force in the concrete in the flange of C-shaped SpeedCore wall, in.
$C_{8.8}$	Coefficient for eccentrically loaded weld groups, linearly interpolated from AISC *Manual* Table 8-8
$C_{1\text{-}8.3}$	Electrode strength coefficient from AISC *Manual* Table 8-3
D_L	Floor dead load, ksf
EA_{eff}	Effective axial stiffness of SpeedCore wall, kips

$EA_{uncr.CB}$ Uncracked axial stiffness of concrete in coupling beam, kips

E_c Modulus of elasticity of concrete, ksi

$E_c(T)$ Modulus of elasticity of concrete at elevated temperatures, ksi

EI_{eff} Effective flexural stiffness of SpeedCore wall, kip-in.^{2}

$EI_{eff}(T)$ Effective flexural stiffness of SpeedCore wall at elevated temperatures, kip-in.^{2}

$EI_{eff.CB}$ Effective flexural stiffness of coupling beam, kip-in.

$EI_{eff.CB}$ Effective flexural stiffness of coupling beam, kip-in.^{2}

$EI_{eff.min}$ Effective flexural stiffness of SpeedCore wall about the minor axis, kip-in.^{2}

$EI_{C.walls}$ Flexural stiffness of compression wall calculated from cross-sectional analysis, kip-in.^{2}

$EI_{T.walls}$ Flexural stiffness of tension wall calculated from cross-sectional analysis, kip-in.^{2}

E_s Modulus of elasticity of steel, ksi

$E_s(T)$ Modulus of elasticity of steel at elevated temperatures, ksi

F_{cr} Critical buckling stress for steel elements of filled composite members, ksi

F_{EXX} Filler metal classification strength, ksi

F_u Specified minimum tensile strength of steel, ksi

F_y Specified minimum yield stress of steel, ksi

$F_y(T)$ Specified minimum yield stress of steel at elevated temperature, ksi

$GA_{v.eff}$ Effective shear stiffness of SpeedCore wall, kips

$GA_{v.CB}$ Effective shear stiffness of coupling beam, kips

G_c Shear modulus of elasticity of concrete, ksi

G_s Shear modulus of elasticity of steel, ksi

H Total building height, ft

H_{wall} Total wall height, ft

I_c Moment of inertia of the concrete section about the elastic neutral axis of the composite section, in.^{4}

$I_{c.CB}$ Moment of inertia of the concrete section about the elastic neutral axis in coupling beam, in.^{4}

I_{ci} Moment of inertia of a concrete element about the elastic neutral axis of the composite section, in.^{4}

$I_{c.min}$ Moment of inertia of the concrete section about the minor axis in coupling beam, in.^{4}

I_e Building Importance factor

I_p Moment of inertia of the steel faceplates, in.^{4}

I_s Moment of inertia of the steel section about the elastic neutral axis of the composite section, in.^{4}

$I_{s.CB}$ Moment of inertia of the steel section about the elastic neutral axis in coupling beam, in.^{4}

$I_{s.min}$ Moment of inertia of the steel section about the minor axis in SpeedCore wall, in.^{4}

I_{sr} Moment of inertia of the reinforcing bars about the elastic neutral axis of the composite section, in.^{4}

I_t Moment of inertia of the steel tie bar, in.^{4}

K Effective length factor

K_c Stiffness of concrete

K_d Discharge coefficient

$K_{s.con}$ Rotational stiffness of wall-to-foundation connection, kip-in./rad

L Length of the unit width column or height of composite plate shear wall at a floor, in.

L Length of member, in.

L_c Effective length of the member, in.

L_{cb} Clear span coupling beam length, in.

L_{eff} Effective distance between wall centroids, in.

L_f Building length, ft

$L_{f.w}$ Length of flange weld, in.

$L_{H.weld}$ Horizontal length of web weld, in.

L_{req} Required weld length, in.

$L_{V.weld}$ Vertical length of web weld, in.

L_{wall} Length of planar SpeedCore wall, in.

L_{wall} C-shaped SpeedCore web wall length, in

M_{ck} Cracking moment, kip-in.

$M_{C.weld}$ Moment demand on C-shaped weld, kip-in.

M_m Molar weight of water, lb/mol

M_n Nominal flexural strength, kip-in.

$M_{n.CB}$ Nominal flexural strength of coupling beam, kip-in.

$M_{nC.wall}$ Nominal flexural strength of SpeedCore wall in compression, kip-in.

$M_{nT.wall}$ Nominal flexural strength of SpeedCore wall in tension, kip-in.

$M_{n.wall}$ Nominal flexural strength of SpeedCore wall, kip-in.

M_p Plastic flexural strength, kip-in.

$M_{PC.wall}$ Plastic flexural strength of SpeedCore wall in compression, kip-in.

$M_{p.exp}$ Expected flexural strength, kip-in.

$M_{p.exp.CB}$ Expected plastic flexural strength of coupling beam, kip-in.

$M_{p.exp.wall}$ Expected plastic flexural strength of uncoupled SpeedCore wall, kip-in.

$M_{Pn.CB}$ Plastic flexural strength of coupling beam, kip-in.

M_{pr} Probable flexural strength, kip-in.

$M_{PT.wall}$ Plastic flexural strength of SpeedCore wall in tension, kip-in.

$M_{P.wall}$ Plastic flexural strength of SpeedCore wall, kip-in.

M_r Required flexural strength, kip-in.

$M_{r.CB}$ Required moment for coupling beam, kip-in.

$M_{r.wall}$ Required flexural strength for SpeedCore wall, kip-in.

M_u Maximum moment demand along the member length, kip-in.

$M_{u.CB}$ Flexural design demands for the coupling beam, kip-in.

$M_{UC.wall}$ Portion of overturning moment resisted by SpeedCore wall in compression, kip-in.

$M_{UT.wall}$ Portion of overturning moment resisted by SpeedCore wall in tension, kip-in.

M_{walls} Total factored moment in SpeedCore walls, kip-in.

M_{web} Moment in the web of the coupling beam relative to the centroid of the web, kip-in.

M_{yc} Yield moment in compression, kip-in.

M_{yt} Yield moment in tension, kip-in.

P Hydrostatic pressure on steel plates associated with concrete pour height (allowable pressure), lb/ft-s^2

P_{Axial} Percentage of axial compression force (gravity load) level, kips

P_e Euler buckling load (compressive strength), kips

$P_e(T)$ Euler buckling load at elevated temperatures, kips

P_n Compressive strength at ambient temperature, kips

P_n Nominal tensile strength, kips

$P_n(T)$ Nominal compressive strength for flexural buckling at elevated temperatures, kips

$P_{n.C}$ Compressive strength of the SpeedCore wall, kips

$P_{n.T}$ Tensile strength of the SpeedCore wall, kips

P_{no} Nominal compressive strength of SpeedCore wall, kips

$P_{no}(T)$ Compressive strength of SpeedCore wall at elevated temperatures, kips

P_r Required axial force, kips

$P_{r.wall}$ Required compression force in compression SpeedCore wall, kips

P_u Ultimate axial load, kips

$P_{weld.T}$ Design weld strength to resist tension, kips

$P_{weld.V}$ Design weld strength to resist eccentric shear, kips

R Fire resistance rating, hr

R Seismic response modification coefficient

$\overline{R}$ Ideal gas constant, lb-ft^2/s^2-K-mol

R_c Factor to account for expected strength of concrete

R_t Ratio of the expected tensile strength to the specified minimum tensile strength, F_u

R_y Ratio of the expected yield stress to the specified minimum yield stress, F_y

S Largest clear spacing of the tie bars, in.

S_{tie} Tie spacing, in.

$S_{tie.top}$ Tie spacing above the flexural yielding zone, in.

T Temperature, °F

T	Maximum vapor temperature, K
$T_{C.weld}$	Tensile demand on C-shaped weld, kips
T_{dc}	Dry concrete thickness, ft
T_{flange}	Required strength of flange plate connection, kips
T_{web}	Tensile force in coupling beam web, kips
$T_{r.wall}$	Required tensile force in tension SpeedCore wall, kips
$T_{1.C}$	Tensile force in the steel in the wall perpendicular to the web in C-shaped SpeedCore wall, in.
$T_{1.T}$	Tensile force in the steel in the wall perpendicular to the web in C-shaped SpeedCore wall, in.
$T_{2.C}$	Tensile force in the flange end plate in C-shaped SpeedCore wall, in.
$T_{2.T}$	Tensile force in the inside steel web plate in C-shaped SpeedCore wall, in.
$T_{3.T}$	Tensile force in the web in C-shaped SpeedCore wall, in.
U	Shear lag factor
V_{amp}	Amplified base shear, kips
V_{base}	Base shear calculated by analysis, kips
$V_{C.weld}$	Shear demand on C-shaped weld, kips
V_c	Shear strength of concrete infill, kips
V_n	Nominal shear strength, kips
$V_{n.CB}$	Nominal shear strength of coupling beam, kips
$V_{n.exp}$	Expected shear strength of coupling beam calculated using expected yield strength, R_yF_y, for steel and expected compressive strength, R_cf_c', for concrete, kips
$V_{n.Mp,exp}$	Shear strength of coupling beam corresponding to the expected moment, kips
$V_{n.wall}$	Nominal in-plane shear strength of SpeedCore wall, kips
V_r	Required shear strength, kips
$V_{r.CB}$	Required shear force for coupling beam, kips
$V_{r.wall}$	Required shear strength for SpeedCore wall, kips
V_s	Shear strength of the webs of the coupling beam, kips
V_u	Maximum shear demand along the member length, kips
V_{web}	Shear in web of coupling beam corresponding to plastic flexural strength, kips
W_c	Length of flange wall without perpendicular steel plates, in.
W_f	Building width, ft
W_{wall}	C-shaped SpeedCore flange wall length, in.
a	Ratio of force eccentricity to vertical weld length, in.
b	Largest unsupported length of plate between rows of steel anchors or ties, in.
b_{CB}	Width of coupling beam, in.
$b_{c.CB}$	Clear width of coupling beam flange plate, in.
c_g	Horizontal centroid of the weld group, in.
d	Vent hole diameter, in.
d	Member length in the direction of bending, in.
d_{tie}	Effective diameter of the tie bar, in.
e	Eccentricity, in.
e_x	Horizontal eccentricity of shear from vertical weld, in.
f_c'	Specified compressive strength of concrete, ksi
$f_c'(T)$	Specified compressive strength of concrete at elevated temperatures, ksi
g	Gravitational acceleration, 32.2 ft/s^2
h	Story height
$h_{c.CB}$	Clear width of the coupling beam web plate, in.
h_c	Concrete pouring height, ft
h_{CB}	Depth of coupling beam, in.
h_{typ}	Typical story height, ft
h_1	First-story height, ft
k	Ratio of horizontal weld length to vertical weld length
m	Vapor generation rate, lb/s
n	Number of stories
r_1	Ratio of limiting moment to $M_{P,wall}$
r_2	Ratio of limiting moment to $M_{p.exp.wall}$
s_1	Horizontal spacing of vent holes, ft
s_2	Vertical spacing of vent holes, ft
t_c	Width of concrete in SpeedCore wall, in.

$t_{c.CB}$ Width of concrete in coupling beam, in.

$t_{c.f}$ Thickness of concrete in SpeedCore flange wall, in.

$t_{c.w}$ Thickness of concrete in SpeedCore web wall, in.

t_{dc} Heating duration associated with the selected dry concrete thickness, s

t_p Steel plate thickness in SpeedCore wall, in.

$t_{pf.CB}$ Thickness of steel flange plate in coupling beam, in.

$t_{pw.CB}$ Thickness of steel web plate in coupling beam, in.

t_{sc} Thickness of SpeedCore wall, in.

$t_{sc.f}$ Thickness of SpeedCore flange wall, in.

$t_{sc.w}$ Thickness of SpeedCore web wall, in.

w_c weight of concrete per unit volume, lb/ft^3

y Distance from the centroid of a cross-sectional element to the elastic neutral axis, in.

$\overline{y}$ Elastic centroid of SpeedCore wall, in.

Ω_0 Overstrength factor

α Ratio of the flexural stiffness of the steel plate to the flexural stiffness of the tie bar

γ Specific heat ratio

γ_1 Amplification factor accounting for increase in lateral loading from the formation of the earliest plastic hinges to the formation of plastic hinges in all coupling beams over the full wall height

ε_c Strain at maximum concrete stress, in./in.

ε_{cr} Buckling strain of steel, in./in.

ε_y Yield strain of steel, in./in.

ε_u Ultimate strain of steel, in./in.

ρ Seismic redundancy factor

ρ_c Concrete density, lb/ft^3

ρ_w Water density, lb/ft^3

ϕ_b Resistance factor for flexure

ϕ_c Resistance factor for compression

ϕ_t Resistance factor for tension

ϕ_v Resistance factor for shear

ω Concrete water content, % by volume

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